CHAPTER

20 Introduction to the Lower Limb

The lower limbs/extremities of the body are specialized for transmission of body weight and locomotion. A brief description of comparative anatomy of limbs immensely facilitates the understanding of their structure and function (see page 2, Textbook of Anatomy: *Upper Limb and Thorax* by Vishram Singh).

PARTS/REGIONS

For descriptive purposes, the lower limb is divided into six parts or regions (Fig. 20.1):

- Gluteal region.
- Thigh or femoral region.
- Knee or knee region.
- Leg or leg region.
- Ankle or talocrural region.
- Foot or foot region.

Gluteal Region

The gluteal region lies on the back and side of the pelvis. It consists of two parts: the rounded prominent posterior region called **buttock** and lateral less prominent region called **hip** or **hip region**.

The **buttock** is bounded superiorly by the *iliac crest*, medially by the *intergluteal cleft*, and inferiorly by the *gluteal fold*. The bulk of it forms the gluteal muscles. The **hip region** overlies the hip joint and greater trochanter of the femur.

Thigh or Femoral Region

The femoral region lies between the gluteal, abdominal, and perineal regions proximally and the knee region distally. The bone of the thigh is **femur**, which articulates with the hip bone at the hip joint and upper end of the tibia and patella at the knee joint.

The abdomen is demarcated from the thigh anteriorly by the inguinal ligament and medially by the ischiopubic ramus. This junction between the abdomen and thigh is termed **inguinal region** or **groin**. The gluteal fold is the upper limit of the thigh posteriorly. The lower part of the back of thigh and back of the knee is termed **ham** (*L. poples*).

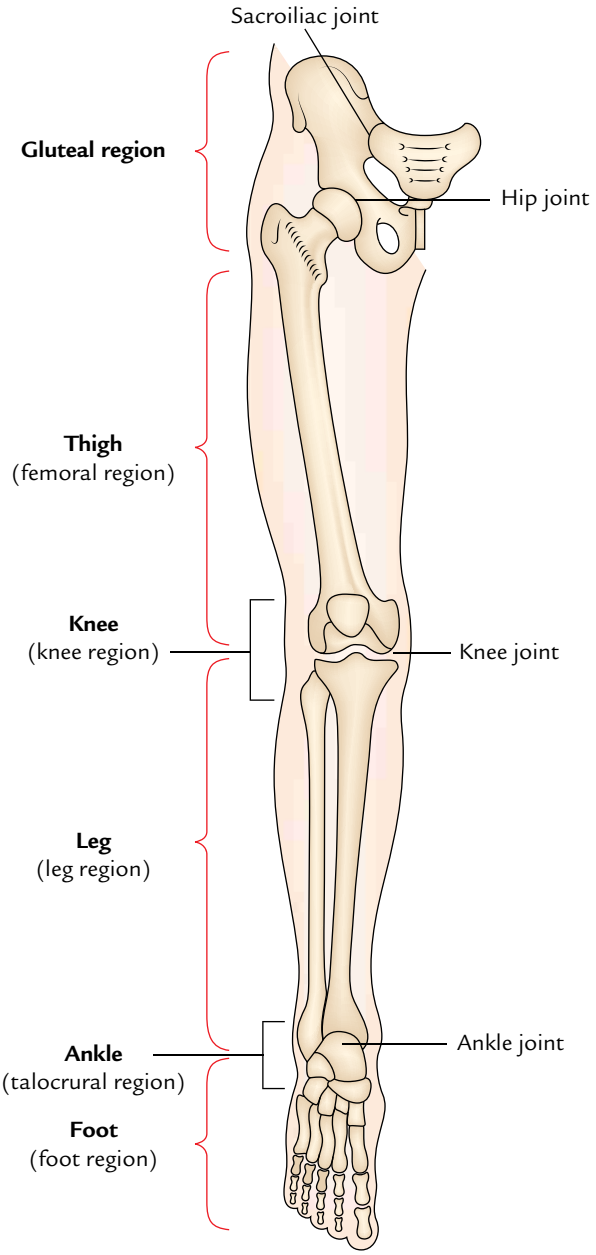


Fig. 20.1 Parts/regions of the lower limb.

Knee or Knee Region

The knee region includes prominences (condyles) of the femur and tibia, head of fibula, and patella (knee cap). The **patella** lies in front of the distal end of femur and medial and lateral femorotibial joints. The posterior aspect of the knee presents a well defined hollow called **popliteal fossa**.

Leg or Leg Region

The leg region lies between the knee and ankle regions. The leg contains two bones: a long and thick medial bone—the **tibia**, and long and narrow lateral bone—the **fibula**. The prominence on the posterior aspect of the leg formed by a large triceps surae muscle is called **calf** (*L. sura*).

Ankle or Talocrural Region

The ankle region includes distal part of the leg, and prominences (malleoli) of tibia and fibula. The ankle joint is located between the malleoli and the talus.

Foot or Foot Region

The foot is the distal part of the lower limb. It contains 7 tarsals, 5 metatarsals, and 14 toe bones (phalanges). The superior surface of the foot is called **dorsum of the foot** and its inferior surface is called **sole of the foot**. The sole is homologous with the palm of the hand. The foot provides a platform for supporting the body weight when standing and plays a key role in locomotion.

N.B. The foot has undergone maximum changes during evolution. In the lower primates (e.g., apes and monkeys), it is a prehensile organ and grasp the boughs with their feet. They can oppose their great toes over the lesser toes. In humans, the great toe comes to lie in line with other toes and loses its power of opposition. It is greatly enlarged to provide principal support to the body. Its four lesser toes because of loss of prehensile function have become vestigial and reduced in size. The tarsal bones have become large, strong, and wedge-shaped to provide stable support.

The four major parts of the lower limb are summarized in Table 20.1. The basic structure of the lower limb is similar to that of upper limb, but it is modified to support the weight of the body and for locomotion.

The lower limb joins the **pelvic girdle** at the hip joint. This girdle is formed by the two hip bones which join anteriorly at the pubic symphysis and posteriorly with the sacrum at the sacroiliac joints. The lower limb is supported by the heavy bones. The **femur** (largest and strongest bone of the body) articulates superiorly with the pelvis at the hip joint and inferiorly with the upper end of tibia at the knee joint. The tibia and fibula articulate with each other at proximal, intermediate, and inferior tibiofibular joints. The distal ends of these bones articulate with the talus to form the ankle joint.

Table 20.1 Parts of the lower limb

Parts	Bones	Joints
Gluteal region (on side and back of pelvis)	Hip bone	Sacroiliac joint
Thigh (from hip to knee)	Femur	• Hip joint • Knee joint
Leg (from knee to ankle)	• Tibia • Fibula	• Tibiofibular joints • Ankle joint
Foot (from heel to toes)	• Tarsals (7) • Metatarsals (5) • Phalanges (14)	• Intertarsal joints (e.g., subtalar talocalcaneonavicular, transverse tarsal, etc.) • Intermetatarsal joints • Metatarsophalangeal joints • Interphalangeal joints

BONES

The bones of the lower limb are described in detail in Chapter 21.

Transmission of the Body Weight through Lower Limb

The body weight is transferred from the vertebral column through sacroiliac joints to the pelvic girdle and from pelvic girdle through the hip joints to the femurs (Fig. 20.2).

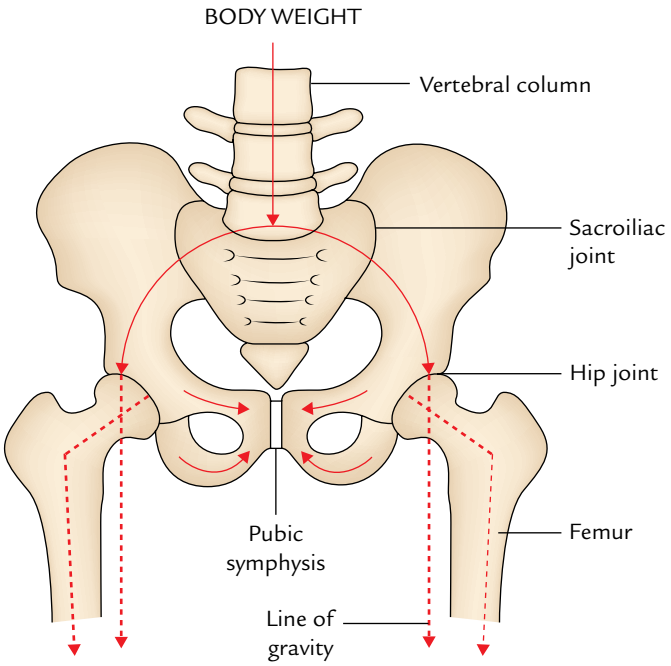


Fig. 20.2 Transmission of body weight from vertebral column to the femurs.

To support the erect posture efficiently, the femurs are oblique being directed inferomedially within the thighs so that during standing the knees are close to each other and placed directly below the trunk. As a result, the centre of gravity falls to the vertical lines of the supporting legs and feet.

A little description of comparative anatomy will make it easy to understand. In quadrupeds, the femurs are vertical and knees are apart with trunk suspended between the limbs (Fig. 20.3).

The body weight is transferred from the knee joint to the ankle joint by the tibia. The fibula does not articulate with the femur, hence does not transfer any weight. At the ankle, the weight born by the tibia is transformed to the talus—the keystone of the longitudinal arch of the foot. The longitudinal arch formed by the tarsals and metatarsals, evenly distribute the weight between the heel and foot when standing. In this way the foot provides a flexible but stable platform to support the body. The weight-bearing points of the foot are medial process of the calcaneus posteriorly and six equal pillars anteriorly formed by the heads of four lateral metatarsals and two sesamoid bones (one on each side of V-shaped ridge) below the head of first metatarsal (Fig. 20.4).

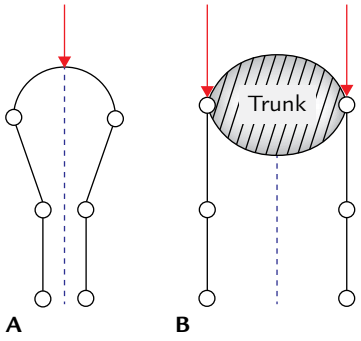


Fig. 20.3 Comparison of arrangement of the lower limb bones in humans (A) and quadrupeds (B).

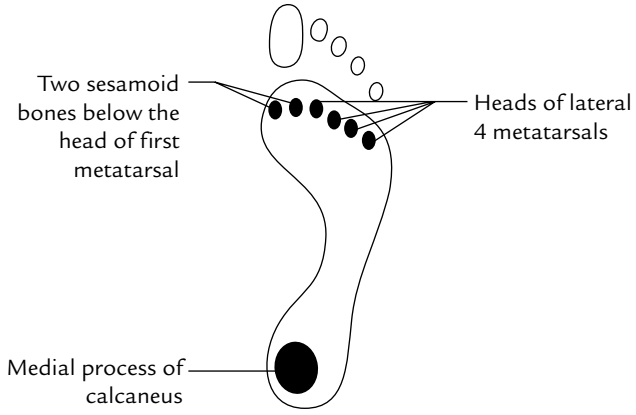


Fig. 20.4 Weight-bearing points of the foot.

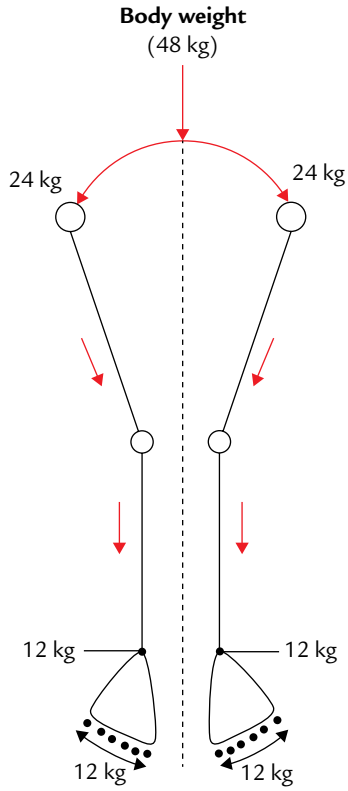


Fig. 20.5 Distribution of weight to the feet.

Mode of Weight Distribution to the Feet (Fig. 20.5)

The body weight of a person weighing say 48 kg when standing (in fundamental position) is distributed as follows:

- 24 kg goes to each lower limb.
- In each limb, 12 kg goes to the medial process of calcaneus and 12 kg to the metatarsals.
- The metatarsals present six points of equal contact with the ground, viz., heads of lateral four metatarsals and two sesamoid bones below the head of first metatarsal each supporting 2 kg. Thus, the great toe supports the double load of body weight as compared to the other toes of the foot.

N.B. The line of gravity of the body passes through the cervical vertebrae, anterior to the thoracic vertebrae, through the lumbar vertebrae, anterior to the sacrum, posterior to the hip joints, and anterior to the knee and ankle joints (Fig. 20.6).

MUSCLES

The muscles of the lower limb are strong and movements produced by them are coarse than those produced by the muscles of the upper limb.

The major muscles in the gluteal region are gluteus maximus, gluteus medius, and gluteus minimus. The gluteal muscles mainly rotate the thigh laterally.

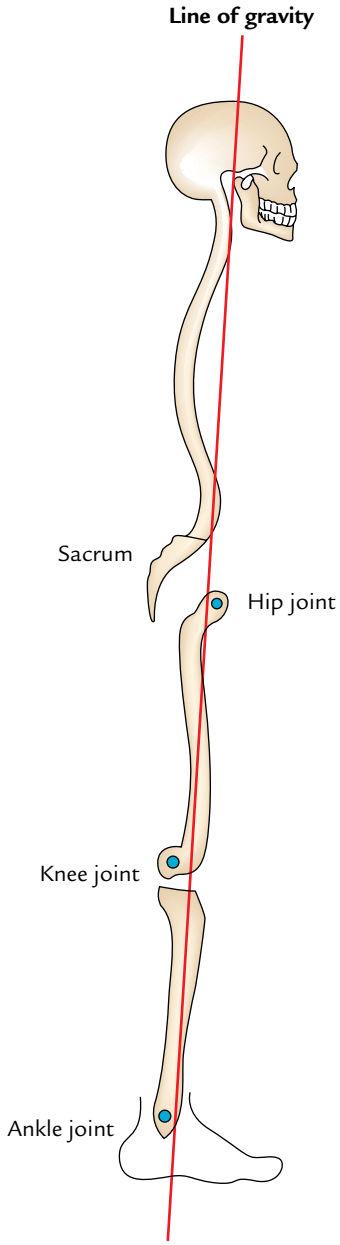


Fig. 20.6 Line of gravity.

In the thigh and leg, the deep fascia forms intermuscular septa and divide these regions into underlying compartments. The thigh is divided into **anterior, medial, and posterior compartments**. The muscles in the anterior compartment are primarily extensor of the knee joint. The main muscle in this compartment is **quadriceps femoris**, which consists of four parts: vastus lateralis, vastus medialis, vastus intermedius, and rectus femoris. The muscles of medial compartment of the thigh are adductors of the hip joint. The muscles of posterior compartment of the thigh are called **hamstring muscles**. They are extensors of the hip joint and flexors of the knee joint.

The leg is divided into **anterior, lateral,** and **posterior compartments.** The muscles of the anterior compartment are primarily dorsiflexors of the ankle and extensors of the toes. The muscles of lateral compartment plantarflex and evert the foot. The muscles of posterior compartment are divided into superficial and deep groups by **transverse fascia of the leg**. The superficial muscles, e.g., gastrocnemius, and soleus, which together form the **triceps surae** which is inserted by a long thick tendon (**tendoachillis**) into calcaneus. They act on calcaneus to plantarflex the foot. The muscles of deep group flex the toes and plantarflex the ankle.

The foot has 20 individual muscles, out of which 18 are located in the sole and only two are located on the dorsum of the foot. The muscles of the sole are arranged into four intricate layers. Despite their layered arrangement, the muscles of sole function primarily as a group during the support phase of the stance, maintaining the arches of the foot.

N.B. The **antigravity muscles** of the lower limb are far better developed than those of the upper limb because they lift up the whole body to attain the erect posture and in walking up the staircase. These muscles are: gluteus maximus (extensor of the hip), quadriceps femoris (extensor of the knee), and gastrocnemius, and soleus (plantarflexors of the ankle). They have extensive origin and bulky muscle belly.

Clinical correlation

Intramuscular injection: The muscles commonly used for intramuscular injection in the lower limb are: (a) **gluteus medius** of gluteal region in adults (most common site) and (b) **vastus lateralis** muscle of the thigh region in children.

NERVES

The nerves of the lower limb are derived from **lumbar plexus** of nerves (L1 – L4) within the posterior abdominal wall and **sacral plexus** of nerves (L4 – S4) in the posterior pelvic wall. The nerves of the lower limb are: femoral, obturator, sciatic, tibial, and common peroneal nerves (Fig. 20.7).

1. **Femoral nerve:** It is present on the front of the thigh and innervates the anterior thigh muscles.
2. **Obturator nerve:** It is present on the medial side of the thigh and innervates the adductors of the thigh.
3. **Sciatic nerve** (largest and thickest nerve in the body): It is present in the gluteal region and the back of the thigh. It supplies muscles on the back of the thigh. In the lower part of the back of thigh, it divides into tibial and common peroneal nerves.
4. **Tibial nerve:** It is present on the back of the leg and supplies all the muscles on the back of the leg. At the ankle,

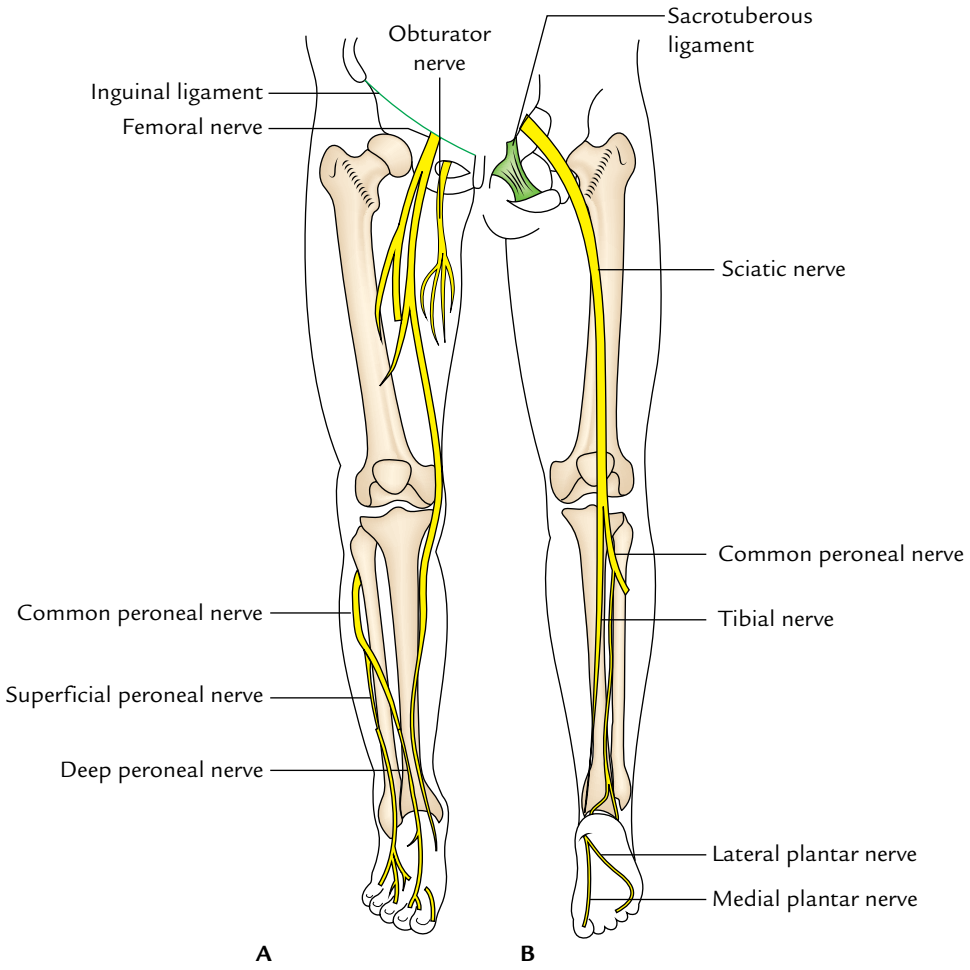


Fig. 20.7 Nerves of the lower limb, an overview: A, on the anterior aspect; B, on the posterior aspect.

behind medial malleolus, it divides into **medial and lateral plantar nerves**, which together supply all the muscles of the sole.

5. **Common peroneal nerve:** At the lateral side of the neck of fibula, it divides into the deep and superficial peroneal nerves. The **deep peroneal nerve** is present in the anterior compartment of the leg and supplies all the anterior leg muscles.

The **superficial peroneal nerve** is present in the lateral compartment of the leg and supplies all the lateral leg muscles.

In addition to the muscles, all the nerves (vide supra) also supply joints and provide cutaneous innervation to the lower limb.

ARTERIES (Fig. 20.8)

The blood to the lower limb is mostly supplied by the **femoral artery** and its branches.

1. **Femoral artery:** It begins at the midpoint of inguinal ligament as the continuation of the external iliac artery. It enters the front of thigh and quickly gives off **deep femoral artery** (**profunda femoris artery**) which supplies all the structures of the thigh.
2. **Popliteal artery:** It is the continuation of femoral artery into the popliteal fossa. The popliteal artery descends through the popliteal fossa and at the head of fibula it divides into the **anterior** and **posterior tibial arteries**.
3. **Anterior tibial artery:** It lies in the anterior compartment of the leg which it supplies and continues into the dorsum of the foot as *dorsalis pedis artery*.
4. **Dorsalis pedis artery:** It lies on the dorsum of the foot which it supplies.
5. **Posterior tibial artery:** It lies in the posterior compartment of the leg which it supplies. At the ankle, behind the medial malleolus it divides into *medial and lateral plantar arteries*.
6. **Medial and lateral plantar arteries:** They are the terminal branches of posterior tibial artery and supply the sole of the foot.

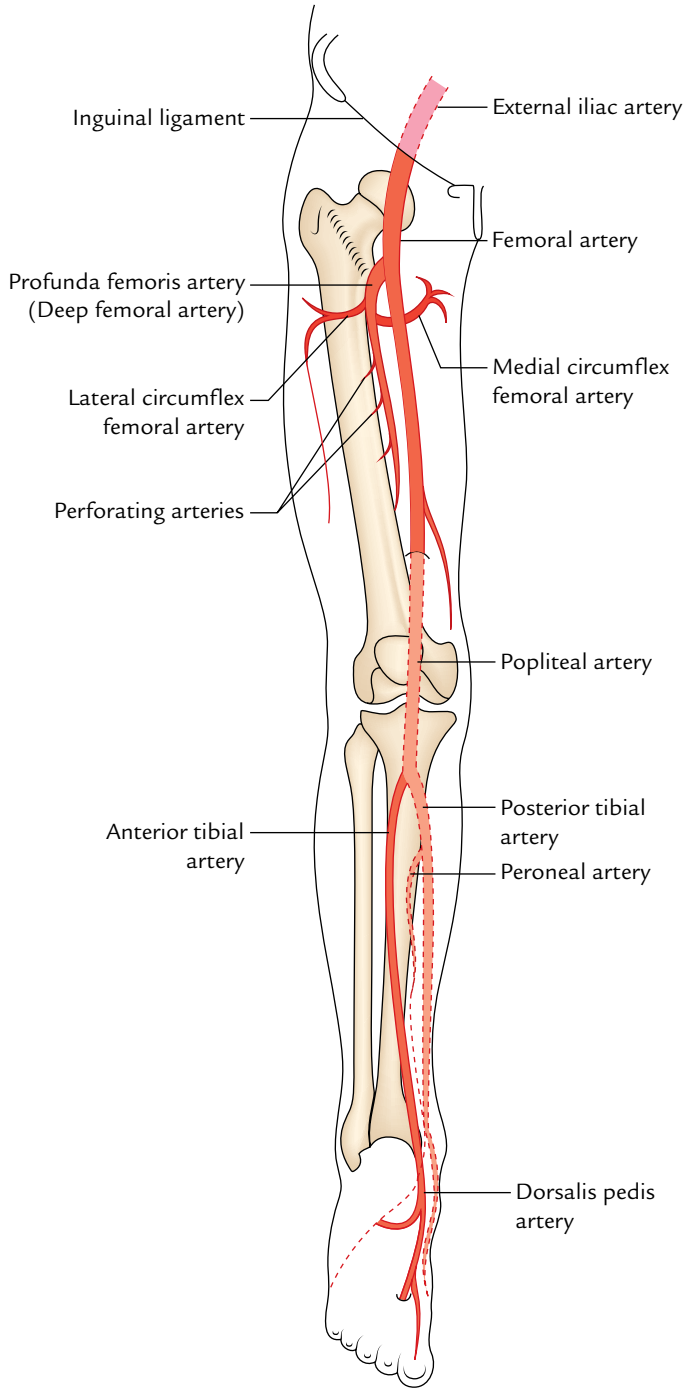


Fig. 20.8 Arteries of the lower limb, an overview.

N.B. In addition to the femoral artery the following arteries also contribute a little to the blood supply of the lower limb:

1. Superior gluteal artery.
2. Inferior gluteal artery.
3. Obturator artery.

They are all branches of the internal iliac artery.

Clinical correlation

- **Sites where arterial pulse can be palpated/felt in the lower limb:**
  1. **Femoral pulse** can be felt at the midinguinal point (i.e., the point midway to anterior superior iliac spine and pubic symphysis.
  2. **Popliteal pulse** can be felt behind the knee deep in the popliteal fossa with knee flexed.
  3. **Posterior tibial pulse** can be felt behind the medial malleolus midway between it and tendocalcaneus.
  4. **Dorsalis pedis pulse** can be felt on the dorsum of the foot just medial to the tendon of extensor hallucis longus.
- **Acute arterial occlusion:** It is mostly caused by embolism or thrombosis. It usually occurs in the femoral artery where it gives off the profunda femoris artery. The clinical features include five P's: Pain, Paresthesia, Pallor, Paralysis, and Pulselessness.
- **Chronic arterial occlusive disease:** It is mostly caused by the atherosclerosis. It commonly involves femoral or popliteal artery (50% of cases). However, in the diabetic patients, it may also involve anterior tibial, posterior tibial, and peroneal arteries. Clinically, it presents as **intermittent claudication**, i.e., profound leg pain in the leg on walking for some distance which is relieved by taking rest for 5–10 minutes and reoccurs on walking again for the same distance.

VEINS

The veins of the lower limb are classified into three types: **superficial, perforating, and deep veins.** The superficial veins communicate with the deep veins via perforating veins.

1. **Deep veins:** They accompany the arteries of the leg and finally drain into **femoral vein**, which continues as external iliac vein. The named deep veins are: **venae comitantes** of anterior and posterior tibial arteries, **popliteal vein, and femoral vein.** The deep veins are subject to muscle compression (musculovenous pump) to facilitate the venous return.
2. **Superficial veins:** They drain into deep veins through *perforating veins.* The named superficial veins are **great and small saphenous veins**, which originate from dorsal venous arch of the foot. The great saphenous vein ascends anterior to medial malleolus, continues along the medial side of leg and thigh before emptying into the *femoral vein* in the femoral triangle. The small saphenous vein ascends behind the lateral malleolus in the calf and empties into the popliteal vein in the popliteal fossa.

3. **Perforating veins (perforators):** They connect and direct blood of superficial veins into the deep veins. The presence of valves in them prevent the back flow from the deep veins.

N.B. The venous blood of the lower limb is drained against gravity therefore, all the lower limb veins have valves to overcome the effect of gravity.

Clinical correlation

Varicose veins: The superficial veins (great and small saphenous veins) draining skin continuously shunt their blood into the deep veins by perforating veins.

The incompetence of valves of perforating veins allows backflow of blood into superficial veins. This backflow causes dilatation **of the superficial veins** and their tributaries called **varicose veins**.

LYMPHATICS

The lymphatics of the lower limb are classified into two types—superficial and deep.

1. **Superficial lymphatics:** They lie superficial to deep fascia and drain into the **superficial inguinal lymph nodes** present just below the inguinal ligament and near the terminal part of great saphenous vein. The lymph from buttock, perineum, and lower abdomen also drain into these lymph nodes.
2. **Deep lymphatics:** They accompany the arteries and drain into **deep inguinal lymph nodes** present within the femoral sheath, which in turn drain into **iliac lymph nodes** within the pelvis and **aortic lymph nodes** in the lower posterior abdominal wall.

N.B. Lymph from glans penis/glans clitoris drain into deep inguinal lymph nodes located in the femoral canal which is termed lymph node of Cloquet /Rosenmüller.

Golden Facts to Remember

► Most important function of the lower limb	Transmission of the body weight and locomotion
► Most important digit of the foot	Great toe
► Commonest fracture in the lower limb	Fracture neck of femur
► Most commonly dislocated bone in the lower limb	Patella
► Most common peripheral neuropathy in the lower limb	Compression of the common peroneal nerve (against the neck of fibula)
► Most commonly used nerve in the body for grafting	Sural nerve
► Most commonly used vein in the body for grafting	Great saphenous vein
► Commonest site of venous thrombosis in the body	Deep veins of the lower limb
► Most common lower limb injuries	Injuries of the knee, leg, and foot
► Longest muscle in the body	Sartorius
► Thickest nerve in the body	Sciatic nerve
► Largest bone in the body	Femur
► Largest sesamoid bone in the body	Patella
► Strongest ligament in the body	Iliofemoral ligament
► Strongest tendon in the body	Tendocalcaneus
► Largest synovial cavity in the body	Synovial cavity of the knee joint

CHAPTER

21 Bones of the Lower Limb

Each lower limb contains 31 bones:

- **Hip bone/innominate bone** the bone of gluteal region (1).
- **Femur**, the bone of thigh (1).
- **Patella** or **knee cap** (1).
- **Tibia and fibula**, the bones of leg (2).
- **Tarsal bones** (7).
- **Metatarsals**, (5). } Bones of foot
- **Phalanges,** (14).

HIP BONE (OS COXAE, OS INNOMINATUM)

The hip bone is a large irregular flat bone in the region of hip. It is formed by the fusion of three primary bones—the **ilium**, the **ischium,** and the **pubis**.

PARTS

The hip bone presents upper and lower expanded parts and a middle constricted part which carries *a cup-shaped hollow* (*acetabulum*) on the outer aspect. The upper expanded part is called *ilium*. The lower expanded part presents an oval or triangular foramen called *obturator foramen*. The part anteromedial to this foramen is called **pubis**, and the part posteroinferior to it is called **ischium**.

ANATOMICAL POSITION AND SIDE DETERMINATION

The side of the hip bone is determined by holding it in such way that:

- The acetabulum is directed laterally.
- The obturator foramen lies below the acetabulum.
- The flat expanded ilium is directed upward, thin pubis is directed anteromedially, and thick ischium is directed posteroinferiorly.
- The pubic tubercle and anterior superior iliac spine lie in the same coronal plane (Fig. 2.10).
- The ischial tuberosity occupies the lowest position.

FEATURES AND ATTACHMENTS (Figs 21.1 and 21.2)

Ilium

It projects upward from the acetabulum as an expanded fan-shaped plate of bone. It presents the following features:

- Two ends—upper and lower.
- Three borders—anterior, posterior, and medial.
- Three surfaces—iliac fossa, gluteal, and sacropelvic.

Ends

Upper end (Fig. 21.3)

1. It is represented by a curved and thickened border called **iliac crest.** The **highest point of iliac crest** lies at the level of interval between L3 and L4 vertebrae.
2. In the horizontal plane, the iliac crest is concave inward anteriorly and convex inward posteriorly. The anterior and posterior ends of the iliac crest are termed anterior superior iliac spine and posterior superior iliac spine, respectively.
   - **Anterior superior iliac spine** provides attachment to the lateral end of inguinal ligament above and sartorius below.
   - **Posterior superior iliac spine** provides attachment to the sacrotuberous ligament.
3. The iliac crest is divided into a **larger ventral segment** (anterior two-third) and a **smaller dorsal segment** (posterior one-third).
   - **Ventral segment**, presents an outer lip, an intermediate area, and an inner lip.

(a) *Outer lip*

(i) It presents **tubercle of the iliac crest** 5 cm behind the anterior superior iliac spine.

(ii) It provides attachment to the following structures:

- **Tensor fasciae latae**, originates in front of the tubercle of iliac crest.
- **External oblique muscle**, inserted in its anterior two-third.

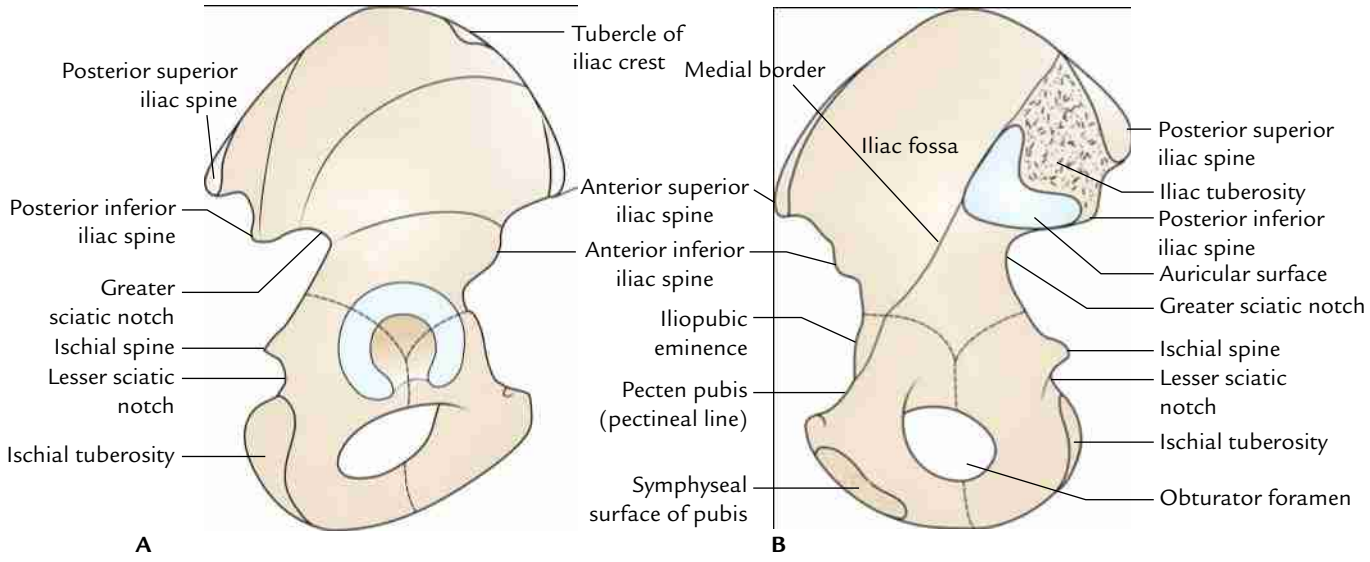


Fig. 21.1 General features of the ilium: A, outer surface; B, inner surface.

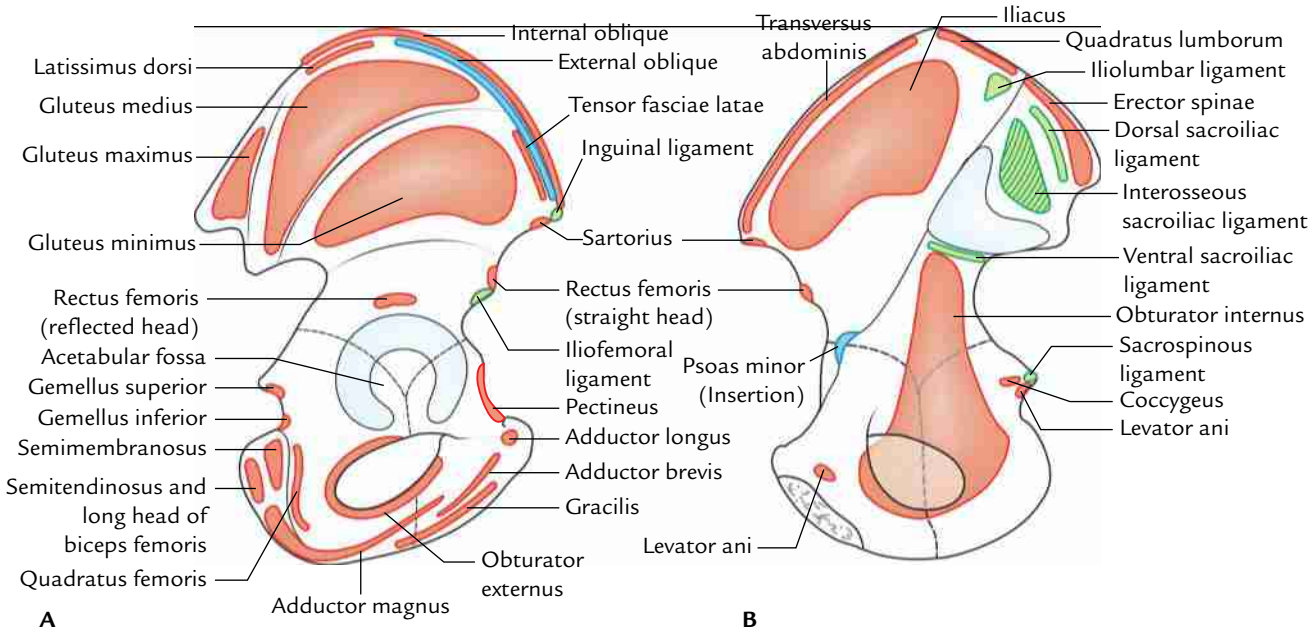


Fig. 21.2 Special features (attachments) of the ilium: A, outer surface; B, inner surface.

- **Latissimus dorsi,** originates just behind its highest point.

(b) *Intermediate area:* It provides origin to the internal oblique muscle along its whole extent.

(c) *Inner lip*

(i) Its anterior two-third provides attachment to **transversus abdominis muscle.**

(ii) Its posterior one-third provides attachment to **quadratus lumborum.**

- **Dorsal segment** is divided by a ridge into inner and outer sloping surfaces:
  - *Outer sloping* surface gives origin to gluteus maximus muscle.
  - *Inner sloping* surface gives origin to erector spinae muscle.

Lower end

The lower end of ilium forms the upper two-fifth of acetabulum.

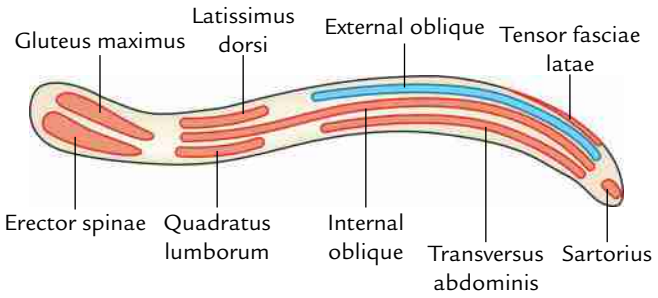


Fig. 21.3 Attachments on the right iliac crest (as seen from above).

Borders of the Ilium

Anterior border

1. It extends from the anterior superior iliac spine to the acetabulum.
2. Its upper half forms a notch and its lower half projects forward as the **anterior inferior iliac spine**, which provides attachment to the straight head of *rectus femoris* in the upper part and *iliofemoral ligament* in the lower part.

Posterior border

1. It extends from the posterior superior iliac spine to the upper end of the posterior border of ischium.
2. It presents from above downward, posterior inferior iliac spine, greater sciatic notch, ischial spine, and lesser sciatic notch.
3. The greater and lesser sciatic notches are converted into greater and lesser sciatic foramina by the sacrospinous and sacrotuberous ligaments (see page 204).

   The structures passing through greater and lesser sciatic foramina are described in Chapter 24, pages 355–356.
4. The dorsal aspect of ischial spine is crossed from lateral to medial side by three structures:
   - *P*udendal nerve.
   - *I*nternal pudendal vessels.
   - *N*erve to obturator internus.

Mnemonic: **PIN** structures.

Medial border

1. It extends downward, forward, and medially from the iliac crest to the iliopubic eminence.
2. It separates the iliac fossa from the sacropelvic surface.
3. Lower one-third of the medial border forms a rounded **arcuate line.**

Surfaces of the Ilium

Iliac fossa

1. It is a shallow concave surface on the inner aspect of ilium, below the iliac crest, and in front of the medial border.
2. Iliacus muscle arises from its upper two-third.

Gluteal surface

1 It is the outer aspect of the ilium which is convex in front and concave behind.
2. It is divided into four areas by **three gluteal lines:** posterior, anterior, and inferior. All these lines radiate from the upper margin of the greater sciatic notch. *The posterior gluteal* line meets the dorsal segment of the iliac crest, *anterior gluteal* line meets the iliac crest close to the tubercle of the iliac crest, and *inferior gluteal line* meets the upper end of anterior inferior iliac spine.
3. The gluteal surface provides origin to four muscles:
   (a) *Gluteus maximus*, behind the posterior gluteal line.
   (b) *Gluteus medius*, between the anterior and the posterior gluteal lines.
   (c) *Gluteus medius*, between the anterior and the inferior gluteal lines.
   (d) Reflected head of rectus femoris, between the inferior gluteal line and upper margin of acetabulum.

Sacropelvic surface

It is situated behind the medial border, on the inner side of ilium. It is divided into two parts—sacral part dorsally and pelvic part ventrally.

Sacral part of the sacropelvic surface

1. The sacral part presents the **tuberosity of the ilium** behind and the **auricular surface** in front.
2. The **iliac tuberosity** is a rough area below the dorsal part of iliac crest and provides attachment to three ligaments. From behind forward these are:
   (a) iliolumbar, (b) dorsal sacroiliac, and (c) interosseous sacroiliac.
3. The **auricular surface** is an ear-shaped articular surface and articulates with the corresponding surface of the sacrum to form a **sacroiliac joint**.

Pelvic part of the sacropelvic surface

1. It is smooth and continuous with the pelvic surface of the ischium to form the lateral wall of lesser pelvis.
2. It presents **pre-auricular sulcus** just in front of auricular surface, along the lateral border of greater sciatic notch.
3. Most of this surface provides origin to **obturator internus.**

Ischium

It is the thick posteroinferior part of hip bone and consists of a body and a ramus.

Body

It is very thick and lies below and posterior to the acetabulum. It presents two ends—upper and lower; three borders—anterior, posterior, and lateral; and three surfaces—femoral, dorsal, and pelvic.

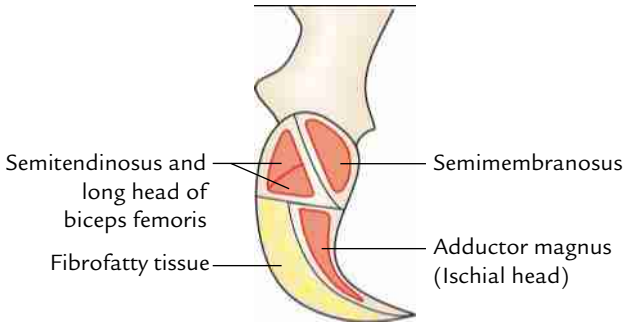


Fig. 21.4 Attachments on the left ischial tuberosity (posterior aspect).

Ends

Upper end: It forms posteroinferior two-fifth of the acetabulum, where it unites with the ilium and pubis.

Lower end

1. It forms the ischial tuberosity (Fig. 21.4) and gives off ramus which forms an acute angle with body.
2. Ischial tuberosity is rough and divided by a horizontal ridge into an upper quadrilateral area and lower triangular area.
3. The upper quadrilateral area is subdivided by an oblique ridge into upper lateral and lower medial parts.
4. The lower triangular area is subdivided by a longitudinal ridge into outer and inner parts.
5. The ischial tuberosity provides origin to four muscles:
   (a) Semimembranosus arises from the lateral part of the upper quadrangular area.
   (b) Semitendinosus along with long head of biceps femoris arises from the medial part of the upper quadrilateral area.
   (c) Adductor magnus (ischial head), arises from the outer part of lower triangular area.
6. The inner part of the lower triangular area is covered by fibrofatty tissue and transmits the body weight in sitting position.
7. Medial margin of the ischial tuberosity is sharp and provides attachment to the lower end of sacrotuberous ligament.

Borders

Anterior border: It forms the posterior margin of the obturator foramen and gives attachment to the **obturator membrane.**

Posterior border

1. It is continuous above with the posterior border of ilium and ends below the upper end of ischial tuberosity.
2. It presents a triangular projection called **ischial spine.** The tip of spine provides attachment to the sacrospinous ligament, and levator ani and coccygeus muscles arise from its pelvic surface.
3. Ischial spine forms the lower boundary of greater sciatic notch and the lower boundary of lesser sciatic notch.
4. The **gemellus superior** and **gemellus inferior** muscles arise from the upper and lower ends of the lesser sciatic notch, respectively.

Lateral border

1. It forms the lateral margin of the ischial tuberosity.
2. Its upper part is notched close to the lower margin of the acetabulum and lodges the **tendon of obturator externus.**

Surfaces

Femoral surface

1. It is directed forward and laterally, and is situated between the anterior and lateral borders.
2. **Obturator externus** arises from this surface along the obturator foramen.
3. **Quadratus femoris** arises from this surface along the lateral border of the ischial tuberosity.

Dorsal surface

1. It is continuous above with the gluteal surface of ilium.
2. It is divided by a transverse groove into the upper and lower parts.
3. *Tendon of obturator internus* along with *two gemelli* traverses through the groove between the upper and lower parts.

Pelvic surface

It is smooth and lies between the anterior and posterior borders. It gives origin to the **obturator internus.**

Ramus of the Ischium

It arises from the lower part of the body and runs upward, forward, and medially, and meets with the inferior ramus pubis to form **conjoint ischiopubic ramus.**

Conjoint ischiopubic ramus

The conjoint ischiopubic ramus presents two borders, upper and lower, and two surfaces—outer and inner.

1. *Upper border* forms the lower margin of obturator foramen and provides attachment to the *obturator membrane.*
2. *Lower border*
   (a) It forms the lateral boundary of the subpubic angle.
   (b) It is everted. The eversion is more marked in males than females due to attachment of crus of penis.
3. *Outer surface* is concave and provides linear attachment to four muscles. From above downward these are:
   (a) Obturator externus.
   (b) Adductor magnus.
   (c) Adductor brevis.
   (d) Gracilis.

4. *Inner surface* is convex and subdivided by the upper and lower oblique bony ridges into three areas—upper, middle, and lower.
   (a) **Upper ridge** provides attachment to the superior fascia of **urogenital diaphragm**.
   (b) **Lower ridge** provides attachment to the perineal membrane (inferior fascia of urogenital diaphragm) and falciform process of sacrotuberous ligament.
   (c) **Upper area** provides origin to the obturator internus.
   (d) **Middle area** provides attachments to the sphincter urethrae and deep transversus perinei muscles.
   (e) **Lower area** provides attachments from before backward to crus of penis, clitoris, ischiocavernosus, and superficial transversus perinei muscles.

Pubis (Pubic Bone/OS Pubis)

It is the anteroinferior part of the hip bone and consists of three parts—body, superior ramus, and inferior ramus.

Body

It is quadrilateral in outline and flattened anteroposteriorly. It presents the following features: pubic crest, pubic tubercle, and three surfaces (anterior, posterior, and medial).

Pubic crest and pubic tubercle

The pubic crest is the thickened upper border of the body of pubis and its lateral end projects forward to form tubercle called **pubic tubercle**.

Surfaces

Anterior surface

1. It is directed downward, forward, and laterally.
2. It provides attachments to the following four muscles:
   (a) **Adductor longus** arises by a rounded tendon from the angle between the pubic crest and pubic symphysis.
   (b) **Gracilis** arises from its lower part and extends further on the ischiopubic ramus.
   (c) **Adductor brevis** arises lateral to gracilis and extends further on the ischiopubic ramus.
   (d) **Obturator externus** arises from this surface adjacent to the obturator foramen.

Posterior surface

1. It is smooth and forms the anterior wall of bony pelvis, hence also called pelvic surface.
2. It is directed upward and backward, and is related to the urinary bladder.

Medial surface (symphyseal surface): It articulates with the medial surface of the opposite pubic bone to form secondary cartilaginous joint called symphysis pubis.

Superior Ramus of Pubis

It arises from the superolateral angle of the body of pubis and extends laterally above the obturator foramen and ends at **iliopubic eminence** where it unites with the ilium. It is triangular in cross section and presents three borders—posterior (pectineal line), anterior (obturator crest), and inferior; and three surfaces—pectineal, pelvic, and obturator.

Borders

Pectineal line (posterior border)

1. It is also called pecten pubis.
2. It extends laterally as a sharp line from the pubic tubercle to the posterior part of iliopubic eminence where it becomes continuous with arcuate line.
3. Pectineal line provides attachment to a large number of structures:
   (a) **Medial part** from before backward provide attachments to:
      – Lacunar ligament.
      – Reflected part of inguinal ligament.
      – Conjoint tendon.
      – Fascia transversalis.
   (b) **Lateral part** from behind forward provides attachment to:
      – Pectineal ligament (ligament of Cooper).
      – Pectineus muscle.
      – Pectineus fascia.
      – Psoas minor (if present).

Obturator crest (anterior border): It extends from the pubic tubercle to the acetabular margin near the **acetabular notch**.

Inferior border: It forms the upper margin of obturator foramen and provides attachment to the obturator membrane leaving a gap for the *obturator canal*.

Surfaces

Pectineal surface

1. It is a triangular area between the obturator crest and the pectineal line.
2. It extends from the pubic tubercle to iliopubic eminence.
3. Pectineus muscle arises from the upper part of this surface.

Pelvic surface: It lies between the pectineal line and inferior border of the superior ramus and continues medially with the pelvic surface of the body of pubis.

It is featureless and covered by the peritoneum.

N.B. The pelvic surface of the superior ramus of pubis is crossed subperitoneally by the obliterated umbilical artery.

Obturator surface

1. It is situated between the obturator crest and the inferior border.

2. Obturator nerve and vessels traverse the obturator groove seen on this surface.

Inferior Ramus of Pubis

It extends downward and laterally from the lower part of the body of pubis to join the ramus of ischium and form the conjoint ischiopubic ramus.

OBTURATOR FORAMEN

1. It is a large gap in the lower part of the hip bone situated between the pubis and ischium.
2. It is oval and larger in males and triangular and smaller in females.
3. Obturator membrane is attached along to its margin except where it bridges the obturator groove to convert it into **obturator canal.**

The obturator canal transmits the obturator nerves and vessels.

ACETABULUM (LATIN ACETABULUM: SHALLOW VINEGAR CUP)

1. It is a large cup-shaped cavity on the outer surface of the middle constricted part of the hip bone.
2. It is directed laterally and slightly anteroinferiorly.
3. All the three parts of the hip bone contribute in its formation as follows:
   (a) *Pubis* forms its anterior one-fifth.
   (b) *Ischium* forms its posterior two-fifth.
   (c) *Ilium* forms its superior two-fifth.
4. Its margin is deficient inferiorly to form the **acetabular notch**, which is bridged by the *transverse acetabular ligament*, and the notch is converted into acetabular foramen.
5. The acetabulum receives the head of femur to form the hip joint.
6. The fibrocartilaginous rim, the **acetabular labrum,** is attached to the margins of the acetabulum.
7. The acetabulum presents a horseshoe-shaped articular surface, the **lunate surface,** in its periphery and non-articular area in its central part, the **acetabular fossa.** The acetabular fossa is filled with the pad of fat and the lunate surface is covered by a plate of hyaline cartilage.

The differences between the male and female hip bones are given in Table 21.1.

OSSIFICATION

The hip bone ossifies in cartilage by three primary centres and eight secondary centres (Table 21.2).

Table 21.1 Differences between the male and female hip bones

Features	Female	Male
Greater sciatic notch	Wider (75°)	Narrower (< 50°)
Ischial spine	Not inverted	Inverted
Ischiopubic ramus	Thin and not everted	Thick and everted
Obturator foramen	Triangular	Oval
Acetabular diameter	Less than 5 cm	More than 5 cm
Distance between the pubic tubercle and anterior acetabular margin	Less than transverse diameter of acetabulum	Equal to the transverse diameter of acetabulum
Pre-auricular sulcus	More conspicuous	Less conspicuous
Iliac fossa	Shallower	Deeper

Table 21.2 Ossification of the hip bone

Centres	Age of appearance	Age of fusion
Primary centres		
• 1 for ilium	2nd month of IUL	18th year
• 1 for ischium	3rd month of IUL	
• 1 for pubis	4th month of IUL	
Secondary centres		
• 2 for iliac crest	Puberty	20–25 years (except acetabular cartilage which ossifies at the age of 17 years)
• 2 for Y-shaped acetabular cartilage		
• 1 for ischial tuberosity		

FEMUR

The femur is the longest and strongest bone of the body, present in the thigh (Latin *femur* = thigh). It is about 18 inches (45 long), i.e., about one-fourth of the height of the individual. At the upper end it articulates with the hip bone to form the hip joint, and at the lower end it articulates with the patella and tibia. The femur transmits body weight from the hip bone to the tibia in standing position.

PARTS

The femur is the long bone and consists of three parts: upper end, shaft, and lower end.

The **upper end** consists of the **head**, **neck**, and **greater and lesser trochanter**.

The **shaft** of the femur is gently convex anteriorly with maximum convexity in the middle third where the shaft is narrowest.

The **lower end** of the femur is enlarged to form **medial and lateral condyles**. Both condyles project backward and are separated by the **intercondylar fossa**. The most prominent points on the condyles are called **epicondyles**.

N.B. The shape of femur departs markedly from other long bones in the sense that the long axis of head and neck projects superomedially at an angle to the long axis of the shaft of bone. This angle allows greater mobility at the hip joint but imposes great strain on the neck during transmission of weight, making it vulnerable to fracture.

ANATOMICAL POSITION AND SIDE DETERMINATION

The side of femur is determined by holding bone vertically in such a way that:

- Its head faces upward, medially, and slightly forward.
- Long axis of the shaft is directed downward and medially with convexity of the shaft facing anteriorly.
- Lower surfaces of both the condyles are in the same horizontal plane.
- **Trick-device**: The anatomical position can be achieved by balancing the neck of femur on your index finger (Fig. 21.5).

N.B.

- *Angle of inclination of femur:* It is an angle between the long axes of neck and shaft of the femur hence also called **neck–shaft angle.** The normal neck–shaft angle is 125° in adults and 160° in children (Fig. 21.6). It is less in female.
- *Angle of femoral torsion:* It is an angle between long axis of the head and neck of femur, and transverse axis of the femoral condyles. It measures about 7° in males and 12° in females (Fig. 21.7).

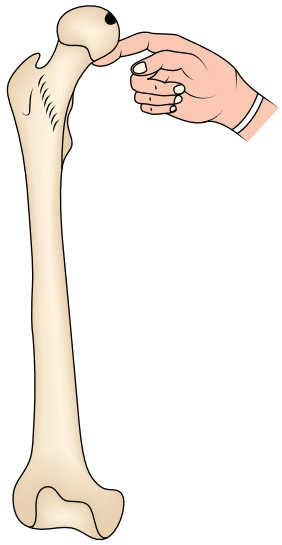


Fig. 21.5 Balancing the femur on the index finger to keep it in the anatomical position.

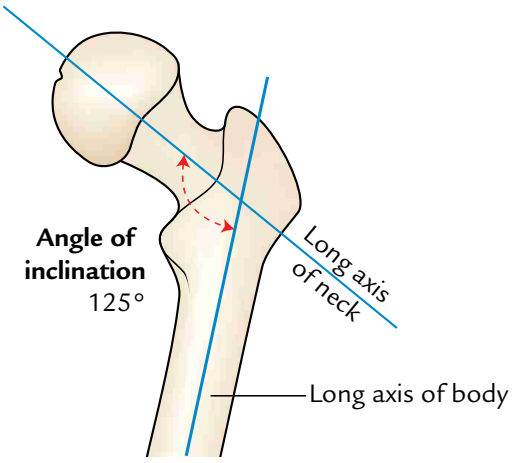


Fig. 21.6 Angle of inclination of the femur.

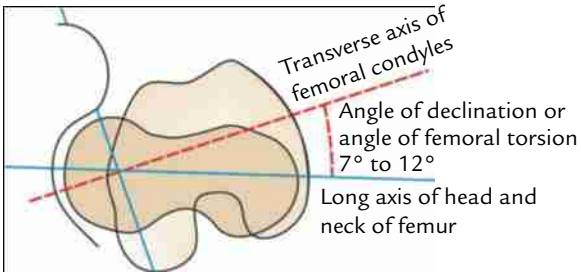


Fig. 21.7 Angle of the femoral torsion (angle of declination of the femur).

FEATURES AND ATTACHMENTS (Figs 21.8 and 21.9)

Upper End

Head

1. It forms about two-third of a sphere and articulates with the acetabulum of the hip bone to form the hip joint.
2. It presents a small pit, the **fovea**, just below and behind the centre, which provides attachment to the ligament of the head of femur (**ligamentum teres femoris**).

Neck

1. It is 5 cm long and connects the head with the shaft.
2. It is directed upward, medially, and slightly forward.
3. The angle between its lower border and the medial border of shaft is called *neck–shaft angle*.
4. The neck presents two borders—upper and lower, and two surfaces—anterior and posterior.

Upper border

1. It is concave and horizontal.
2. It meets the shaft near the greater trochanter.

Lower border

1. It is straight and oblique.
2. It meets the shaft near the lesser trochanter.

Anterior surface

1. It is flat and bears a number of oblique bony ridges.
2. It meets the shaft at **intertrochanteric line**.
3. It is completely intracapsular.

Intertrochanteric line

1. It continues downward and medially below the lesser trochanter on the posterior aspect of femur as **spiral line**.
2. It provides attachment to two ligaments and two muscles:
   (a) Capsule of the hip joint.
   (b) Iliofemoral ligament (strongest ligament in the body).
   (c) Vastus lateralis to its upper end.
   (d) Vastus medialis to its lower end.

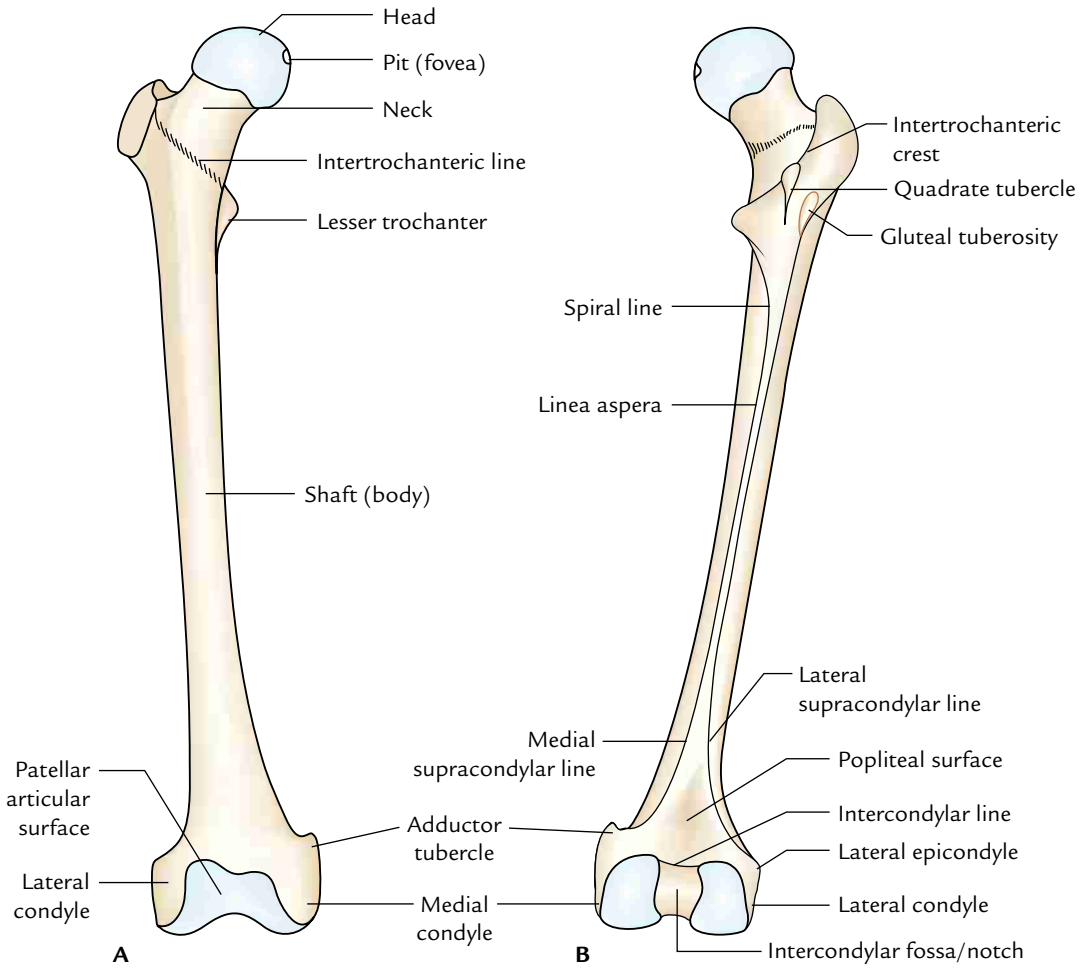


Fig. 21.8 General features of the femur: A, anterior view; B, posterior view.

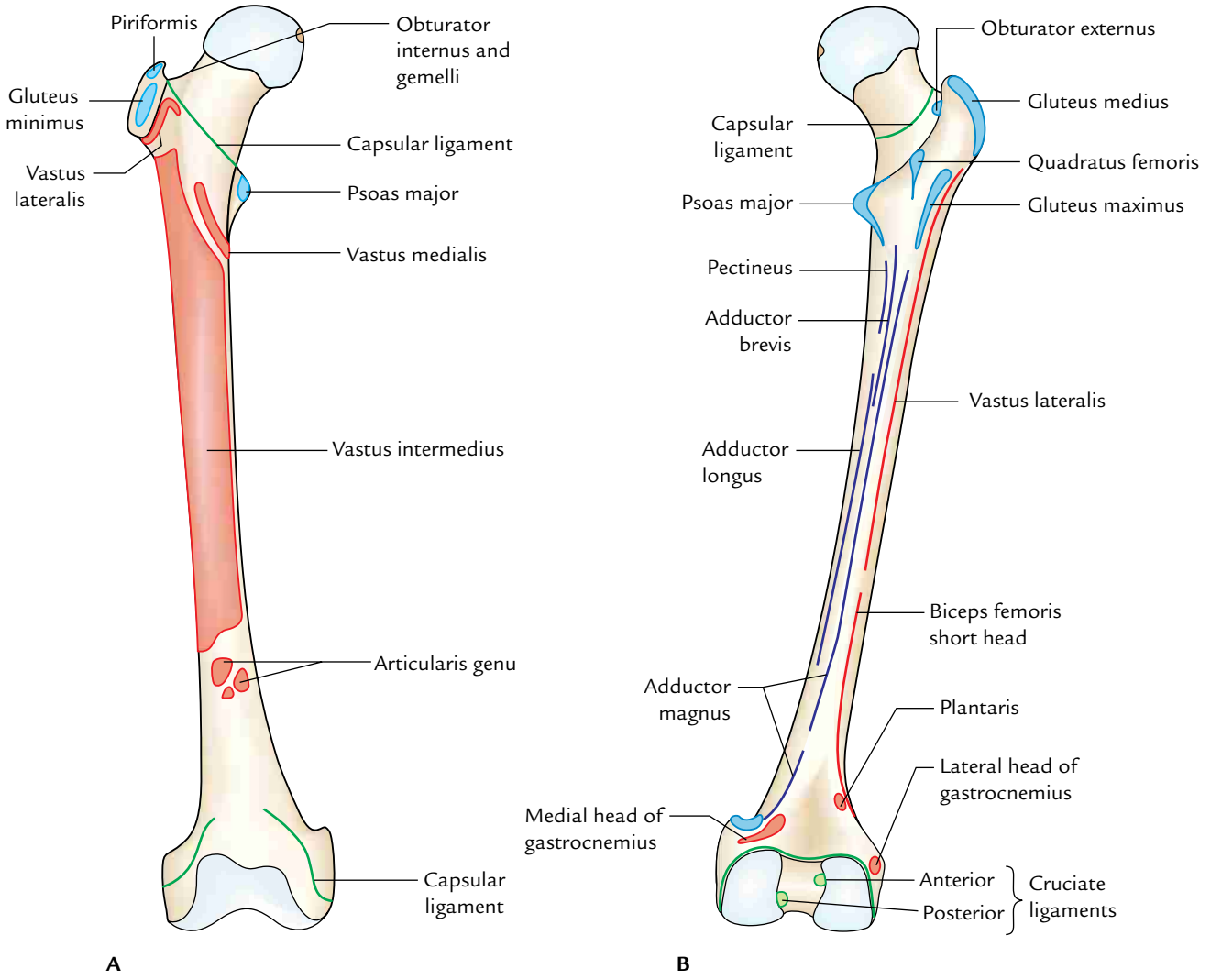


Fig. 21.9 Special features (attachments) of the femur: A, anterior view; B, posterior view.

Posterior surface

1. It is convex from above downward and concave from side-to-side.
2. It meets the shaft at **intertrochanteric crest**.
3. Its medial half is intracapsular.
4. It presents a faint groove, which lodges the tendon of obturator externus.

Intertrochanteric crest

1. It extends from the posterosuperior angle of greater trochanter to the tip of lesser trochanter.
2. It presents a rounded tubercle near the middle called **quadrate tubercle**, which provides insertion to the quadratus femoris.

N.B. The neck connects the head to the shaft about 5 cm away from the hip joint. This enables the lower limb to swing clear of the pelvis. Thus, the neck of femur is akin to the clavicle in the upper limb.

Greater Trochanter

1. It is a quadrilateral elevation, projecting upward from the lateral aspect of the junction of neck and shaft.
2. It presents one border (upper) and three surfaces (anterior, medial, and lateral).

Upper border

Its posterior part presents the apex or tip of greater trochanter, which provides attachment to the piriformis.

Anterior surface

It provides attachment to the gluteus minimus on a ridge in its lateral part. Its medial part is related to a synovial bursa—trochanteric bursa of gluteus minimus.

Medial surface

It presents two depressions:

1. A depression (**trochanteric fossa**) at its junction with the neck for insertion of obturator externus.
2. A shallow depression above and in front of trochanteric fossa for insertion of obturator internus along with the gemellus superior and gemellus inferior.

Lateral surface

It is quadrilateral and divided diagonally by an oblique ridge into the upper and lower triangular areas.

1. The ridge provides attachment to the *gluteus medius muscle.*
2. The triangular areas—anterior and posterior to the ridge are related to the trochanteric bursae of the gluteus medius and gluteus maximus, respectively.

Lesser Trochanter

1. It is a conical projection arising from the posteromedial surface of the neck–shaft angle. It is directed medially.
2. Its apex provides attachment to the psoas major.
3. Iliacus is attached to its base on the front.

N.B. *Attachment of fibrous capsule at the upper end of femur:* Anteriorly it is attached to the intertrochanteric line but posteriorly it is attached about 1 cm medial to the intertrochanteric crest. Therefore, the lateral part of the posterior surface of the neck of femur is extracapsular and its medial half is intracapsular.

Shaft

Features

Basically the shaft of femur presents three surfaces—anterior, medial, and lateral and three borders—medial, lateral, and posterior.

The middle third of posterior border is represented by a thick ridge called **linea aspera**, which acts as a buttress to resist the compressive forces and prevents the anterior bowing of the shaft. The linea aspera presents inner and outer lips and an intervening intermediate area.

When the upper end of linea aspera is traced upward, its inner and outer lips diverge and enclose a triangular **posterior surface** in the upper one-third of the shaft. The inner (medial) lip becomes continuous with the **spiral line** and the outer (lateral) lip becomes continuous with the **gluteal tuberosity,** which extends up to the root of greater trochanter.

When the lower end of linea aspera is traced downward, its inner and outer lips, diverge below, and enclose a triangular **popliteal surface** in the lower one-third of the shaft. The inner (medial) lip becomes continuous with the **medial supracondylar line,** which ends at the **adductor tubercle.** The outer (lateral) lip becomes continuous with **lateral supra condylar line.** Thus, strictly speaking, the shaft presents five surfaces—anterior, medial, lateral, upper posterior, and lower posterior (popliteal surface).

The borders and surfaces of the shaft of femur in upper one-third, middle one-third, and lower one-third are given in Table 21.3.

Attachments

The shaft of femur provides attachment to the following muscles and intermuscular septa:

1. **Attachments of intermuscular septa:**
   (a) *Medial intermuscular septum* is attached to the medial lip of linea aspera.
   (b) *Lateral intermuscular septum* is attached to the lateral lip of the linea aspera.
2. **Muscular attachments:**
   (a) *Vastus intermedius* arises from the upper three-fourth of anterior surface and adjoining lateral surface.
   (b) *Articularis genu* arises by few small slips from the lower one-fourth of anterior surface immediately below the vastus intermedius.

Table 21.3 Borders and surfaces of the shaft of femur

Part of femur	Borders	Surfaces
Middle 1/3rd of the shaft	• Medial border • Lateral border • Posterior border (linea aspera)	• Anterior surface • Medial surface • Lateral surface
Upper 1/3rd of the shaft	• Medial border • Lateral border • Spiral line • Gluteal tuberosity	• Anterior surface • Medial surface • Lateral surface • Posterior surface
Lower 1/3rd of the shaft	• Medial border • Lateral border • Medial supracondylar line • Lateral supracondylar line	• Anterior surface • Medial surface • Lateral surface • Popliteal surface

(c) *Vastus lateralis* arises in a linear fashion from the upper part of intertrochanteric line, anterior and inferior borders of greater trochanter, lateral margin of gluteal tuberosity, and lateral lip of linea aspera.
(d) *Vastus medialis* arises in a linear fashion from the lower part of intertrochanteric line, spiral line, and the medial lip of linea aspera and the upper one-fourth of medial supracondylar line.
(e) *Gluteus maximus* is inserted into the gluteal tuberosity.
(f) *Adductor longus* is inserted into the medial lip of linea aspera.
(g) *Pectineus* is inserted into a line extending from the lesser trochanter to the upper end of linea aspera.
(h) *Adductor brevis* is inserted into a line extending from the lesser trochanter to the upper part of linea aspera.
(i) *Adductor magnus* is inserted into the medial margin of gluteal tuberosity, linea aspera, medial supracondylar line, and adductor tubercle.
(j) *Gastrocnemius:* The **medial head** arises from the popliteal surface just above the medial condyle. The **lateral head** arises mainly from the lateral condyle but also extends over the lower end of lateral supracondylar line.
(k) *Plantaris* arises from the lower end of lateral supracondylar line just above the lateral head of gastrocnemius.

N.B. *Arrangement of nine structures attached to the linea aspera:* From medial to lateral side, these are (Fig. 21.10):

- Vastus medialis.
- Medial intermuscular septum.
- Adductor brevis (in the upper) and adductor longus (in the lower part).
- Adductor magnus.
- Short head of biceps femoris.
- Posterior intermuscular septum.
- Vastus lateralis.
- Vastus intermedius.

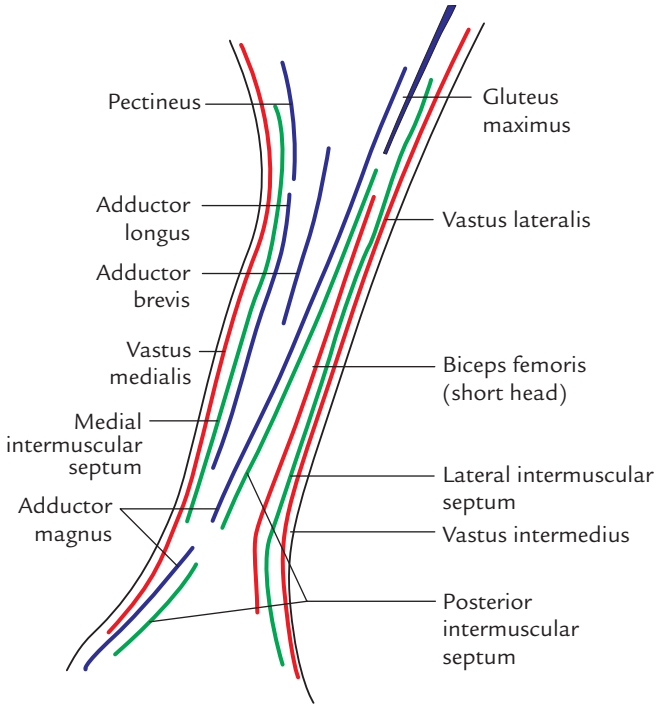


Fig. 21.10 Structures attached to the linea aspera (magnified view).

Lower End

Medial Condyle

1. Its most prominent point is called medial epicondyle, which provides attachment to the upper end of medial collateral ligament.

   A projection posterosuperior to the medial epicondyle is called **adductor tubercle,** which provides insertion to the ischial head of adductor magnus.
2. Its lateral surface forms medial boundary of intercondylar fossa.

Lateral Condyle

1. It is stouter and stronger than the medial condyle but less prominent.
2. Its lateral surface presents a prominence called **lateral epicondyle** which provides attachment to the fibular collateral ligament of the knee joint.
3. Smooth impression above and behind the lateral epicondyle gives origin to the lateral head of gastrocnemius.
4. A groove below and behind the lateral epicondyle provides attachment to popliteus in its anterior part. Tendon of popliteus occupies the posterior part of this groove during full flexion at the knee joint.
5. Its medial surface forms the lateral boundary of intercondylar fossa.

Intercondylar Fossa (Intercondylar Notch)

1. It is a deep notch, which separates two condyles posteriorly.
2. It is limited posteriorly above by intercondylar line.
3. It presents medial and lateral walls and floor.
4. Medial wall of the fossa provides attachments to the upper end of **posterior cruciate ligament** in its anteroinferior part.
5. Lateral wall of the fossa provides attachment to the upper end of **anterior cruciate ligament** in its posterosuperior part.

Mnemonic: LAMP = Lateral condyle provides attachment to Anterior cruciate ligament and Medial condyle to Posterior cruciate ligament.

Articular Surface

The lower end of femur presents a V-shaped articular surface occupying the anterior, inferior, and posterior surfaces of both condyles.

1. The apex of 'V' is called **patellar surface** which occupies the anterior surfaces of two condyles and articulates with the patella.
2. The two links of 'V' form tibial surfaces which occupy the inferior and posterior surfaces of two condyles to come in contact with corresponding articular surfaces of tibial condyles.
3. The **patellar surface** is saddle-shaped. Its lateral portion is broader and extends to a higher level than the medial portion, corresponding to articular surface of the patella.

N.B. *Attachment of fibrous capsule at the lower end of femur:* Posteriorly, it is attached to the intercondylar line and articular margins. Anteriorly, it is deficient for communication of suprapatellar bursa with the synovial cavity of knee joint. Laterally and medially, it is attached along a line 1 cm above the articular margins. It is important to note that the groove for popliteus is intracapsular.

Clinical correlation

- **Fracture neck of femur:** It is very common in elderly particularly in women due to osteoporotic changes in the neck.
- **Types of fracture:** The fracture may be intracapsular (*subcapital*, *transverse cervical*) or extracapsular (*basal, intertrochanteric*, and *subtrochanteric*). In intracapsular fracture, the retinacular vessels—the chief source of blood supply to the head—are injured. This leads to delayed healing or non-union of fracture, or even avascular necrosis of the head of femur (for details, see *Clinical and Surgical Anatomy*, 2nd Edition by Vishram Singh).

In intracapsular fracture of the neck of femur, the affected limb is shortened and characteristically held in laterally rotated position with the toes pointing laterally.

OSSIFICATION

The femur ossifies from five centres: one primary and four secondary centres. The primary centre appears in the midshaft. Three secondary centres appear in the upper end and one secondary centre in the lower end. The age of appearance and fusion of these centres is given in Table 21.4.

Clinical correlation

Medicolegal significance of ossification centre at the lower end of femur: The secondary centre at the lower end of femur is unique in the sense that it appears during birth/just before birth (ninth month of IUL). It is of medicolegal importance because its appearance in radiograph indicates maturity of the fetus (for details, see *Textbook of Clinical Embryology* by Vishram Singh).

PATELLA (KNEE CAP)

The patella is the largest sesamoid bone, found in the tendon of quadriceps femoris. It is situated in front of the knee joint, hence it is also called knee cap. It is a flattened and triangular bone with the base facing upward, and the apex downward. Its anterior aspect is convex and rough, whereas its posterior surface presents a large articular surface divided into small medial part and large lateral part.

ANATOMICAL POSITION AND SIDE DETERMINATION

Hold the patella in such a way that:

- Its apex faces downward and its base faces upward.
- Its articular surface faces posteriorly. The large lateral part of articular surface determines the side.

Trick-device: Keep the articular surface of patella on the table-top in such a way that its base is directed toward you and its apex away from you. Now observe the tilt of patella. The patella, as a rule, always tilts toward the side it belongs to.

Table 21.4 Ossification of the femur

Centres	Time of appearance	Time of fusion	
Primary centre In mid shaft	7th to 8th week of IUL		
Secondary centres • 1 for head • 1 for greater trochanter • 1 for lesser trochanter	• 1 year • 3 years • 13 years	Three separate epiphysis	18th year
For lower end 1	9 months (at the time of birth)		20 years

FEATURES AND ATTACHMENTS (Figs 21.11 and 21.12)

The patella presents the following features:

- Apex.
- Three borders: Superior, medial, and lateral.
- Two surfaces: Anterior and posterior.

Apex

It is directed downward and provides attachment to the ligamentum patellae, which connects the patella to the tibial tubercle.

Borders

Superior border (also called base)

It provides attachment to rectus femoris in the anterior part and vastus intermedius in its posterior part.

Lateral border

It provides attachment to vastus lateralis in its upper one-third and lateral patellar retinaculum in the lower two-third.

Medial border

It provides attachment to vastus medialis in the upper two-third and medial patellar retinaculum in its lower one-third.

The lateral and medial patellar retinaculum are the expansions of the tendons of vastus lateralis and vastus medialis, respectively.

Surfaces

Anterior surface

1. It is rough, convex, longitudinally striated, and presents numerous vascular foramina.
2. It is subcutaneous and **subcutaneous prepatellar bursa** intervenes between it and skin.

Posterior surface

1. Its lower one-fourth is rough and non-articular, while its upper three-fourth is smooth and articular.
2. Small non-articular part near the apex is divided into two areas: lower and upper.
   (a) *Lower area* provides attachment to **ligamentum patellae.**
   (b) *Upper area* is related to **infrapatellar pad of fat.**

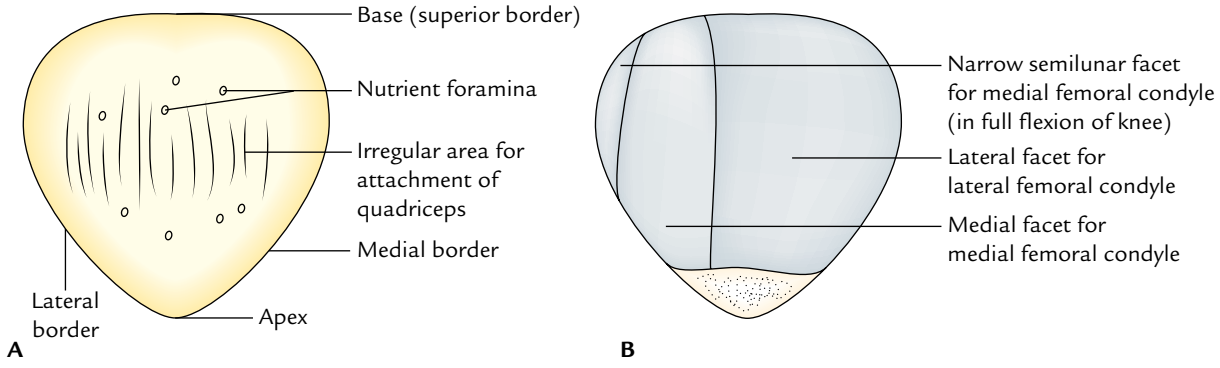


Fig. 21.11 General features of the right patella: A, anterior aspect; B, posterior aspect.

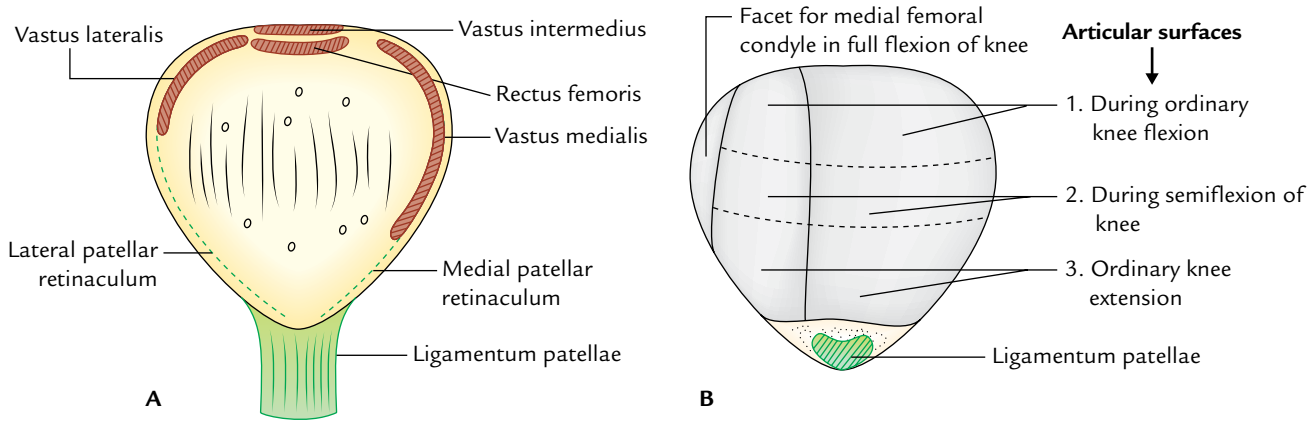


Fig. 21.12 Special features of the right patella: A, anterior aspect; B, posterior aspect.

3. The large articular area is primarily divided by a vertical ridge into a larger lateral and a smaller medial area in order to fit with the reciprocal articular surfaces of the femur. The vertical ridge itself occupies the groove on the patellar surface of femoral lower end in the position of extension.
   (a) Larger lateral articular area lies in contact with the lateral femoral condyle in all positions of the knee.
   (b) Small medial articular area is further divided by another vertical ridge into a large lateral part and a narrow medial part called **medial strip**. The medial strip comes in contact with the under surface of medial femoral condyle during full flexion at the knee.

   Leaving aside this medial strip, whole of articular surface of the patella is subdivided into upper, middle, and lower areas by two faint horizontal lines.

The patellar surface of femur comes in contact with the upper area in ordinary flexion, with the middle area in semiflexion, and with the lower area in ordinary extension.

Clinical correlation

- **Dislocation of the patella:** The patella has natural tendency to dislocate laterally due to upward and lateral pull by the quadriceps. However, it is countered by three factors:
  (a) More forward projection of lateral femoral condyle.
  (b) More prolonged insertion of vastus medialis to the medial border of the patella than that of vastus lateralis on the lateral border.
  (c) Medial pull exerted by medial patellar retinaculum.
- **Fracture of the patella:** (a) A direct blow on the patella fractures it into two or more pieces, and (b) a sudden and forceful contraction of quadriceps femoris causes a transverse fracture of the patella. The patella being a sesamoid bone is devoid of the periosteum, hence when fractured bony union does not take place.

OSSIFICATION

The patella ossifies from several centres which appear during 3–6 years and fuse quickly to form a single centre. The ossification of patella is completed at puberty.

Clinical correlation

Abnormal ossification of the patella. Abnormal ossification centres in the patella may form two or three separate pieces of the patella (called bipartite or tripartite patella). This condition can be differentiated from fracture of the patella as it is bilateral and the pieces have smooth margins.

TIBIA (SYNONYM SHINBONE)

The tibia is the second largest bone in the body situated on the medial side in the leg. The tibia is the pre-axial bone and homologous with the lateral bone of the forearm, the radius. It transmits weight of the body from femur to the foot.

PARTS

The tibia is a long bone and consists of three parts: upper end, lower end, and intervening shaft.

The **upper end** is expanded and bears prominent **medial and lateral condyles** and **tibial tuberosity**. The medial condyle is larger than the lateral condyle. The two condyles articulate with the condyle of the femur.

The **shaft** has three borders—anterior, medial, and lateral; and three surfaces—medial, lateral, and posterior. The anterior border is the sharpest and takes a sinuous course from the tibial tuberosity and becomes continuous below with the anterior border of the medial malleolus.

The **lower end** is small and projects medially and inferiorly as **medial malleolus**.

ANATOMICAL POSITION AND SIDE DETERMINATION

The side of tibia is determined by holding the bone vertically in such a way that:

1. Its expanded end with condyles faces upward.
2. Its tibial tuberosity and sharp sinuous anterior border of the shaft faces anteriorly.
3. Medial malleolus is on the medial side.
4. Superior surface of the upper end (tibial plateau) lies in a horizontal plane.

FEATURES AND ATTACHMENTS (Figs 21.13 and 21.14)

Upper End

It presents two condyles (medial and lateral), an intercondylar area and tibial tuberosity.

Medial Condyle

It is larger than the lateral condyle and presents four surfaces: superior, posterior, anterior, and medial.

Superior surface (Fig. 21.15)

1. It has a smooth oval articular surface which articulates with the femoral condyle.
2. It is slightly concave in the centre and flattened at the periphery. The peripheral flattened part is covered by fibrocartilaginous plate, the **medial meniscus**.

Posterior surface

It presents a deep transverse groove for the insertion of tendon of semimembranosus.

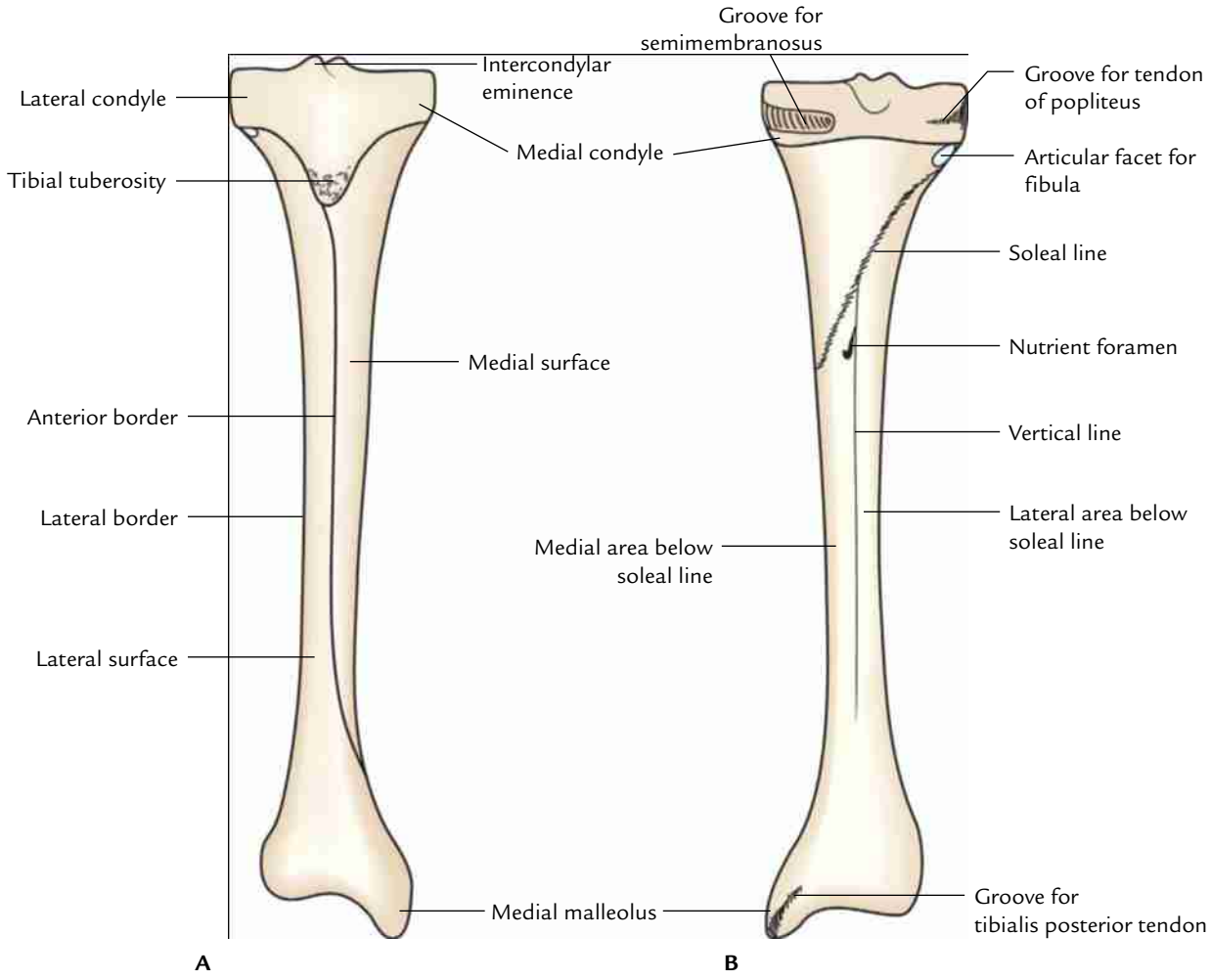


Fig. 21.13 General features of the tibia: A, anterior view; B, posterior view.

Anterior and medial surfaces

The anterior and medial surfaces are marked by the vascular foramina and adjoining margin provides attachment to the medial patellar retinaculum.

Lateral Condyle

Like medial condyle the lateral condyle also presents four surfaces:

Superior surface

1. It presents a circular articular surface for articulation with the lateral femoral condyle.
2. It is slightly concave in the centre and flattened at the periphery. The flattened peripheral part is covered by a plate of fibrocartilage—the **lateral meniscus**.

Posterior surface

1. Inferolaterally this surface presents a circular smooth *articular facet for the head of fibula.*
2. Between the articular facet for the fibula and the margin of superior surface there is a shallow *groove for the tendon of popliteus.*

Anterior and lateral surfaces

1. The anterior surface has a flat *triangular facet for the attachment of the iliotibial tract.*
2. Anterior and lateral surfaces provide attachment to the lateral patellar retinaculum.

Intercondylar Area

1. It is the rough area on the superior surface of the upper end of tibia between the articular surfaces of two condyles.
2. The middle of intercondylar area is narrow and marked by an elevation called **intercondylar eminence.** The medial and lateral parts of the eminence are more prominent and constitute the medial and lateral **intercondylar tubercles.**

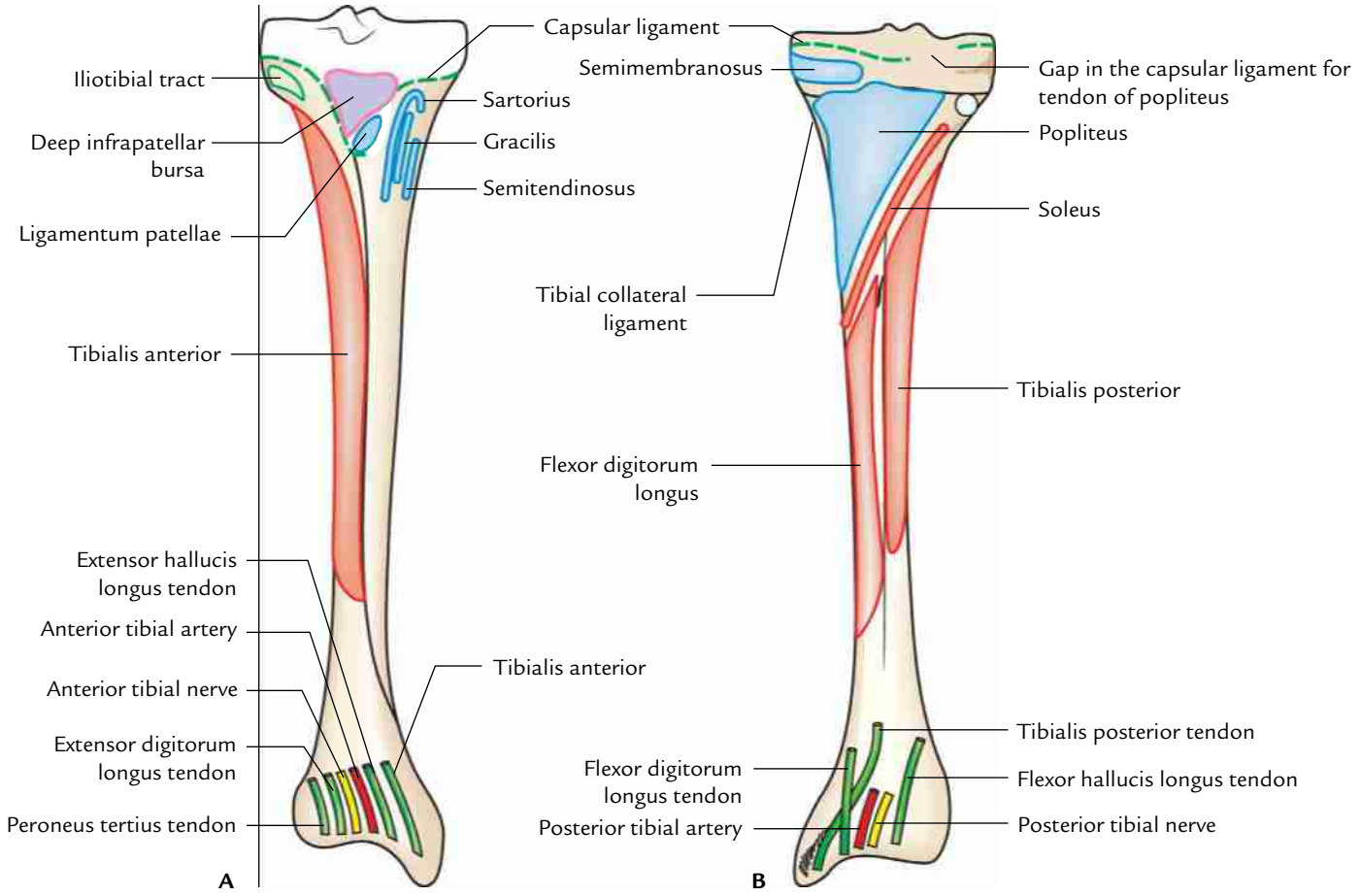


Fig. 21.14 Special features (attachment) of the tibia: **A**, anterior aspect; **B**, posterior aspect.

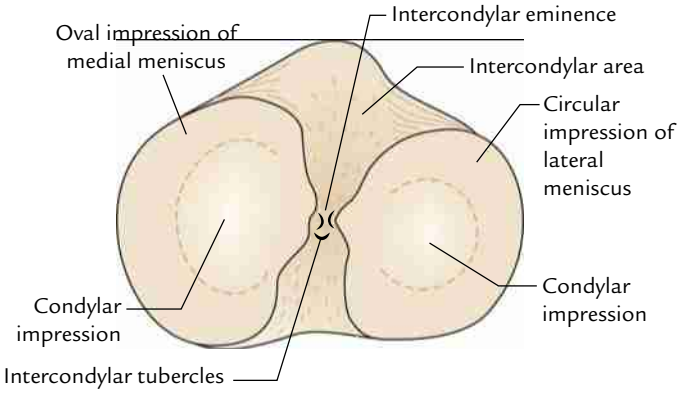


Fig. 21.15 Features on the upper aspect of the tibia.

3. From before backward, the intercondylar area provides attachments to six structures (Fig. 21.16):
   (a) Anterior horn of Medial meniscus.
   (b) Anterior Cruciate ligament.
   (c) Anterior horn of Lateral meniscus.
   (d) Posterior horn of Lateral meniscus.
   (e) Posterior horn of Medial meniscus.
   (f) Posterior Cruciate ligament.

Mnemonic: Medical College Lucknow, Lucknow Medical College.

Tibial Tuberosity

1. It is a projection at the apex of rough triangular area located on the anterior aspect upper end of the tibia.
2. It is divided into an upper smooth and a lower rough part.
3. Upper smooth part provides attachment to the ligamentum patellae.
4. Lower rough part is related to the **subcutaneous infrapatellar bursa** which separates it from the skin.
5. Rough triangular area above the tuberosity is related to the **deep infrapatellar bursa**.

Shaft

It presents three borders—anterior, medial, and lateral; and three surfaces—lateral, medial, and posterior.

Borders

Anterior border (shin of the tibia)

1. It is the sharpest, subcutaneous, and sinuous in course.
2. It extends from the tibial tuberosity to the anterior border of medial malleolus.

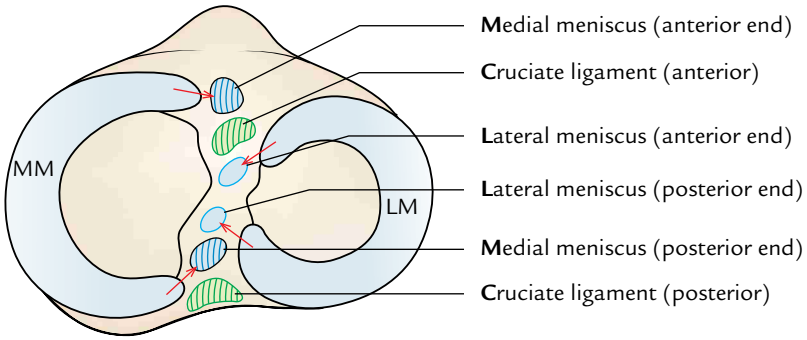


Fig. 21.16 Structures attached in intercondylar area on the superior aspect of the tibia (MM = medial meniscus, LM = lateral meniscus).

Medial border

1. It extends from the medial condyle to the posterior border of medial malleolus.
2. The soleal line joins the medial border at the junction of its upper one-third with the lower two-third.

Lateral (interosseous) border

1. It is sharp and extends from the lateral condyle below the facet for the head of fibula to the anterior border of fibular notch.
2. It provides attachment to the interosseous membrane which unites the tibia and fibula.
3. Its distal part divides to enclose a triangular notch for the attachment of interosseous **inferior tibiofibular ligament**.

Surfaces

Medial surface

1. It faces anteromedially and lies between the anterior and medial borders.
2. It is almost entirely subcutaneous.
3. Its upper part close to the medial border receives insertions of three muscles. From before backward, these are: sartorius, gracilis, and semitendinosus.

Mnemonic: Girl between two surgeons – 'SGS'.

Lateral surface

1. It is directed anterolaterally and lies between anterior and interosseous borders.
2. Its upper two-third provides origin to the **tibialis anterior**.
3. Its lower one-third is crossed by the following structures from medial to lateral sides while passing from the leg to the foot:
   (a) **T**ibialis anterior.
   (b) Extensor **h**allucis longus.
   (c) Anterior tibial **a**rtery.
   (d) Deep peroneal **n**erve/anterior tibial nerve.
   (e) Extensor **d**igitorum longus.
   (f) **P**eroneus tertius.

Mnemonic: The Himalayas Are Not Dry Plateaus.

Posterior surface

1. It lies between the medial and lateral borders.
2. A rough bony ridge extends from the fibular facet to the junction of upper and middle thirds of the medial border called **soleal line**. This line provides attachments to the following structures from above downward:
   (a) Fascia covering popliteus.
   (b) Fascia covering soleus.
   (c) Soleus (origin).
   (d) Deep transverse fascia.
3. The posterior surface of the tibial shaft is subdivided by the soleal line into an upper small triangular area and a lower large area.
4. A triangular area above the soleal line provides insertion to popliteus.
5. A large area below the soleal line is divided into a medial and a lateral part by a vertical ridge.
6. The **nutrient foramen** of the tibia is located at the upper end of this vertical ridge. Nutrient canal is directed downward. Nutrient artery is a branch of the posterior tibial artery and the **largest nutrient artery of the body**.
7. The flexor digitorum longus arises from the medial area below the soleal line.
8. The tibialis posterior arises from the lateral area below the soleal line.
9. The lower one-fourth of the posterior surface beneath the flexor retinaculum is related to the following structures from medial to lateral:
   (a) **T**ibialis posterior.
   (b) Flexor **d**igitorum longus.
   (c) Posterior tibial **a**rtery.
   (d) Tibial **n**erve.
   (e) Flexor **h**allucis longus.

Mnemonic: The Doctors Are Not Hunters.

Lower End

The lower end presents five surfaces (anterior, posterior, medial, lateral, and inferior) and a downward projection from the medial part, the medial malleolus.

Surfaces

1. **Anterior surface** has an upper smooth and featureless part and lower rough and grooved part.
2. **Posterior surface** in its medial part presents a *groove for the tendon of tibialis posterior*.
3. **Medial surface** is subcutaneous and is continuous with the medial surface of the medial malleolus.
4. **Lateral surface** presents a **fibular notch**. Anterior and posterior margins of the fibular notch provide attachment to **anterior and posterior tibiofibular ligaments.** The upper part of the fibular notch is rough and provides attachments to the **interosseous tibiofibular ligament**.
5. **Inferior surface** is smooth and articulates with the body of the talus.

Medial Malleolus

1. Its tip lies at a higher level than that of the lateral malleolus and provides attachment to the *deltoid ligament*.
2. Its lateral surfaces present a comma-shaped articular facet for articulation with the similar facet on the medial surface of the body of the talus.
3. Its posterior surface presents a *vertical groove for the tendon of tibialis posterior*.
4. Medial margin of the groove is prominent and provides attachment to the flexor retinaculum.

CAPSULAR ATTACHMENTS AT THE UPPER AND LOWER ENDS OF TIBIA

1. **At the upper end**, the capsule is attached to the margins of tibial condyles medially, laterally, and posteriorly. Anteriorly it is attached to the sides of triangular area on the anterior aspect of condyles blending with the medial and lateral patellar retinacula and to the tibial tubercle where it blends with the ligamentum patellae.

   Behind the lateral condyle, the capsule is deficient for the passage of tendon of popliteus.
2. **At the lower end**, the capsule is attached to the margins of articular surfaces where it extends on to the lower end of fibula.

OSSIFICATION

The tibia ossifies from three centres—one primary and two secondary.

1. **Primary centre** appears in the middle of shaft at the age of seventh week of IUL.
2. **Secondary centres.**
   (a) For the upper end:
   Appearance: At birth or shortly after birth.
   Fusion with the shaft: 20 years.
   (b) For the lower end:
   Appearance: 2 years.
   Fusion with the shaft: 18 years.

N.B. The upper epiphysis extends anteriorly as a tongue-like process to form the upper part of tibial tuberosity.

Clinical correlation

- **Osteomyelitis of the upper end of tibia:** The upper end of tibia is the commonest site of **acute osteomyelitis**, but knee joint remain unaffected because the capsule of knee joint is attached near to the margins of articular surfaces proximal to the epiphyseal line.
- **Fracture of tibia:** The tibia is commonly fractured at the junction of upper 2/3rd and lower 1/3rd of its shaft. (The tibial shaft is narrowest at the junction of upper two-third and lower one-third, hence the commonest site of fracture.)

  The lower one-third of the tibial shaft is bare area (hence devoid of any muscular attachment) and have low blood supply; for this reason, the fractures in the lower 1/3rd of the shaft of tibia show delayed union or non-union (for details, see *Clinical and Surgical Anatomy*, 2nd Edition by Vishram Singh).

FIBULA

The fibula is the lateral bone of the leg and is homologous with the ulna of forearm. In Latin, the term fibula means "pin"; hence the lateral bone of leg is rightly named fibula because it is a long pin-like bone. It is a long slender post-axial bone of the leg and does not take part in the transmission of the body weight.

It subserves two important functions:

1. It provides attachments—muscles.
2. Its lower end along with the lower end of tibia forms a socket (**tibiofibular mortise**), to hold the talus in place.

PARTS

The fibula is a long bone and consists of three parts: upper end, lower end, and intervening shaft.

- The **upper end** (head) is round and presents a circular articular facet. An upward projection posterolateral to this facet is called styloid process.
- The **shaft** is described to have anterior, interosseous, and posterior borders; and medial, lateral, and posterior surfaces. However, only interosseous border is clear-cut; other borders and surfaces spiral so that it is difficult to

ascertain. None of them remains strictly in the position implied by its name. Therefore, making an attempt to identify them is just wastage of time.

- The **lower end** is flattened and bears a *triangular articular facet* on its medial surface for articulation with the talus. Behind and below this is a roughened fossa called malleolar fossa.

SIDE DETERMINATION AND ANATOMICAL POSITION

The side of fibula can be determined by holding it vertically in such a way that:

- Its round end called head is directed upward.
- Its relatively flattened end is directed downward.
- A triangular articular facet on its lower end faces medially.
- A depression at the lower end (malleolar fossa) lies behind and below the triangular articular facet at this end.

FEATURES AND ATTACHMENTS (Figs 21.17 and 21.18)

Upper End

It presents head and neck.

Head

It is round and presents the following three features:

1. An **oval or circular articular facet** on its superior aspect for articulation with the lateral condyle of the tibia.
2. A **styloid process** posterolateral to the articular facet which provides attachment to the *fibular collateral ligament.*
3. A sloping surface in front of the styloid process for C-shaped insertion of biceps femoris.

Neck

It is a constriction below the head, connecting it with the shaft. The common peroneal nerve is related to the posterolateral aspect of neck and anterior tibial artery on its medial aspect.

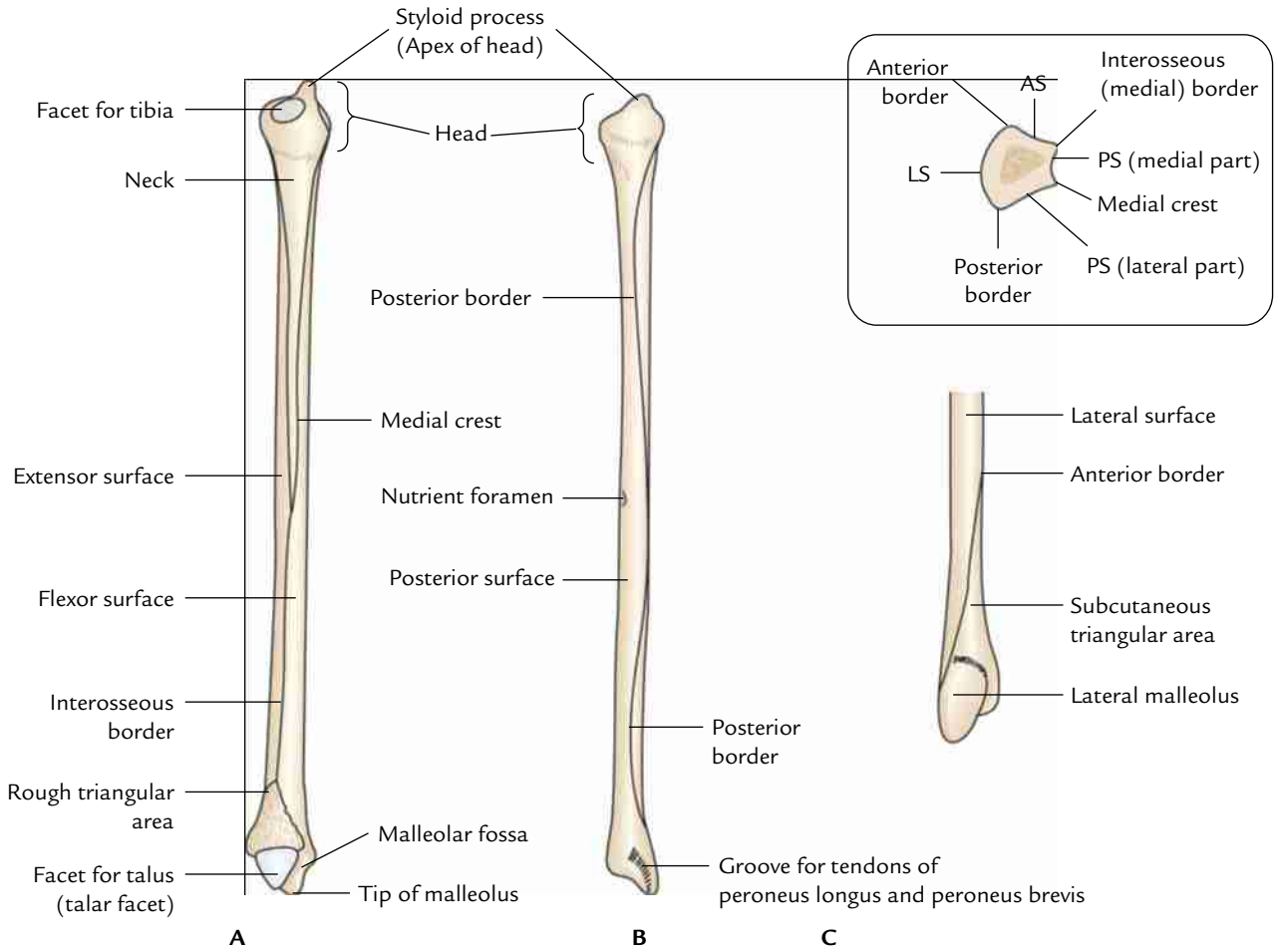


Fig. 21.17 General features of the right fibula: A, medial aspect; B, posterior aspect; C, lateral aspect of lower part. Figure in the inset is the transverse section of the lower part of shaft showing borders and surfaces (AS = anterior surface, PS = posterior surface, LS = lateral surface).

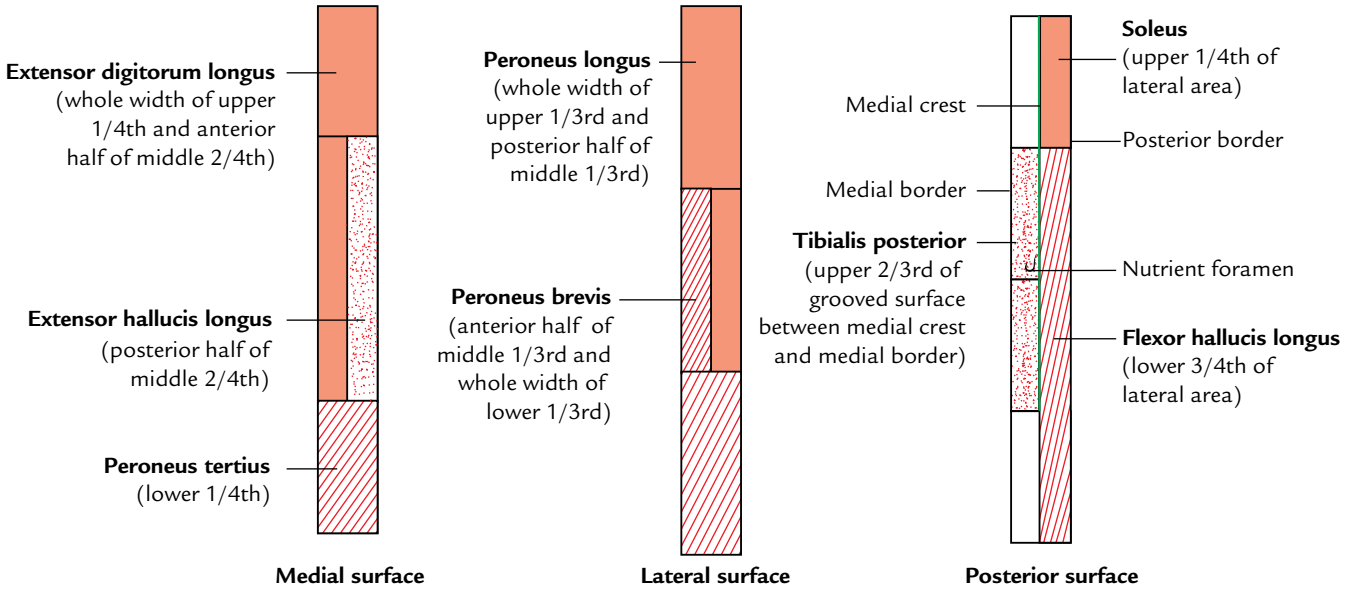


Fig. 21.18 Schematic diagram to show the attachments of muscles on the different surfaces of the fibula.

Shaft

It presents three borders—anterior, posterior, and medial; and three surfaces—medial, lateral, and posterior.

Borders

Anterior border

1. It provides attachment to the anterior intermuscular septum of the leg in its upper three-fourth.
2. It begins just below the anterior aspect of the head. Inferiorly it splits to enclose a triangular area, which continues on the lateral surface of lateral malleolus.
3. Superior extensor retinaculum is attached to the anterior margin of the triangular area and superior peroneal retinaculum to the posterior margin of triangular area.

Posterior border

1. It extends from the posterior aspect of head to the lateral margin of groove on the posterior surface of the lateral malleolus.
2. Posterior intermuscular septum of the leg is attached to its upper three-fourth.

Interosseous medial border

1. It lies close and just medial to the anterior border (separated from it only by 3 mm in the upper part).
2. Inferiorly it ends at the upper end of the roughened area for the interosseous ligament.
3. Interosseous membrane is attached along its whole length except at the upper end to leave a gap for the passage of anterior tibial vessels.

Surfaces

Medial (extensor) surface

1. It is narrow and lies between the anterior and interosseous borders.
2. It gives origin to *extensor digitorum* longus in upper three-fourth (whole width of its upper fourth and anterior half of its middle two-fourth).
3. *Extensor hallucis longus* arises from the posterior half of the middle two-fourth medial to the *extensor digitorum longus*.
4. Its lower one-fourth provides origin to the *peroneus tertius*.

Lateral (peroneal) surface

1. It lies between the anterior and posterior borders.
2. *Peroneus longus* arises from the upper two-third (whole width of upper one-third and the posterior half of the middle one-third).
3. *Peroneus brevis* arises from the anterior half of its middle one-third and whole width of its lower one-third.

Posterior (flexor) surface

1. It is extensive and lies between the interosseous and posterior borders.
2. Its upper two-third is divided into medial concave and flattened lateral parts by a sharp vertical ridge **medial crest**.
3. Fascia covering the tibialis posterior is attached to the medial crest.
4. Medial concave part gives origin to tibialis posterior.

5. Lateral flattened part gives origin to the soleus in upper one-fourth and to the flexor hallucis longus in lower three-fourth.
6. Peroneal artery descends along medial crest.
7. Nutrient artery, a branch of the peroneal artery, enters the nutrient foramen present just above the middle of the posterior surface. Nutrient canal is directed downward.

Lower End (Fig. 21.19)

The lower end of fibula is expanded anteroposteriorly to form **lateral malleolus,** which presents four surfaces—anterior, posterior, medial, and lateral.

1. **Anterior surface** is rough and round. It provides attachment to the anterior talofibular ligament. A notch at its lower border provides attachment to the calcaneofibular ligament.
2. **Posterior surface** presents a groove, which lodges tendons of peroneus brevis and peroneus longus, the latter being superficial to the former.
3. **Medial surface** presents a triangular articular surface in front and a depression (malleolar fossa) below and behind it.

   The upper part of malleolar fossa provides attachment to the posterior tibiofibular ligament and its lower part to the **posterior talofibular ligament.**
4. **Lateral surface** is triangular and subcutaneous.

OSSIFICATION

The fibula ossifies from three centres: one primary and two secondary.

1. Primary centre appears in the middle of the shaft: at the age of eighth week of IUL.

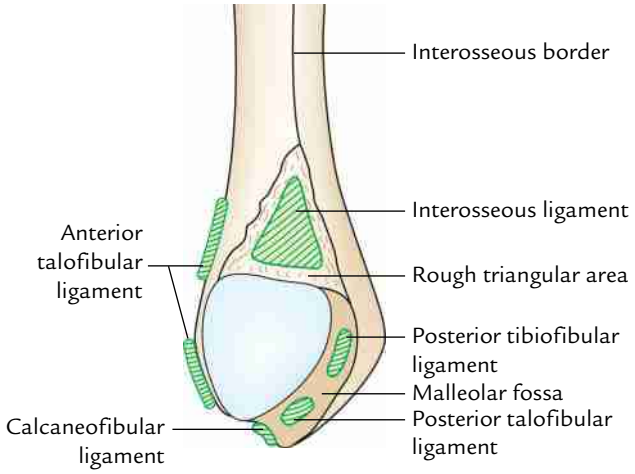


Fig. 21.19 Special features of the lower end of right fibula (medial aspect).

2. Secondary centres:
   (a) For the upper end:
   Appearance: 3–4 years.
   Fusion with the shaft: 20 years.
   (b) For the lower end:
   Appearance: 1–2 years.
   Fusion with the shaft: 18 years.

N.B. In fibula, **law of union of epiphysis** is violated [cf. according to this law, epiphyseal (secondary) centre which appears first unites last with the diaphysis. In fibula, epiphyseal centre for lower end appears first and also unites first]. The explanation to this violation is that epiphyseal centre in the lower end of fibula appears earlier because it is pressure epiphysis and in the upper end later because it is traction epiphysis. Since the growing end fibula is its upper end (as evidenced by the direction of nutrient foramen) it unites with the diaphysis last although its epiphyseal centre also appears last.

Clinical correlation

- **Bone grafts:** Since the fibula does not take part in the transmission of body weight, it is a common source of bone for grafting.
- **Fibular fracture:** The fibula is commonly fractured, 2 to 5 cm proximal to the distal end of the lateral malleolus. It is often associated with '*fracture dislocation of the ankle joint*'.

SKELETON OF THE FOOT (Figs 21.20 and 21.21)

The skeleton of the foot from behind forward consists of the following bones:

- Tarsals.
- Metatarsals.
- Phalanges.

TARSAL BONES

These are short bones which together form **tarsus.** These are arranged in three rows:

(a) *Proximal row* consists of talus and calcaneus.
(b) *Middle row* consists of navicular.
(c) *Distal row* consists of three cuneiforms (medial, intermediate, and lateral) and cuboid.

N.B. *Identification of bones in the skeleton of the foot:*

- *Calcaneus* (heel bone) is the largest and most proximal bone.
- *Talus* is the second largest bone and lies above the calcaneus like a rider, hence highest bone in the skeleton of foot.

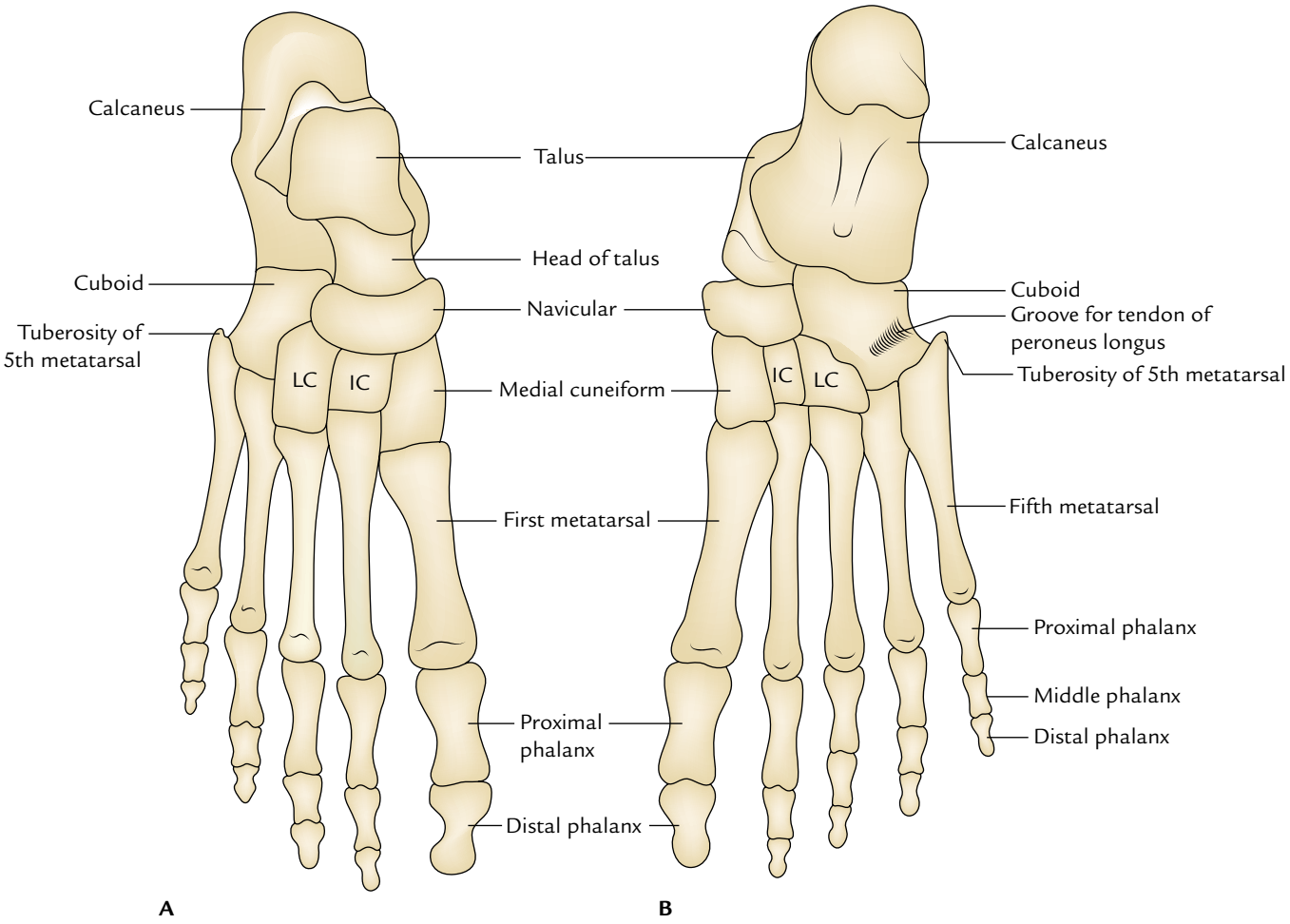


Fig. 21.20 General features of the skeleton of the foot: **A**, dorsal aspect; **B**, plantar aspect (IC = intermediate cuneiform, LC = lateral cuneiform).

- *Navicular* is boat-shaped and lies in front of the head of talus.
- *Cuboid* is cubical in shape in front of the lateral part of calcaneum.
- *Cuneiforms* are small wedge-shaped bones and arranged from side to side in front of navicular.

Talus (Latin Talus = Ankle Bone)

The talus is the second largest bone of the foot situated on the upper surface of the anterior two-third of the calcaneus. It forms the connecting link between the bones of the foot and the leg. It participates in the formation of three joints—ankle (talocrural), talocalcaneal (subtalar), and talocalcaneonavicular (pretalar).

N.B. Talus is the only bone of the foot, which is devoid of any muscular attachment.

Side Determination and Anatomical Position

The side of talus is determined by keeping it on the table-top in such a way that:

1. Its rounded head is directed forward.
2. Its trochlear articular surface (which is convex anteroposteriorly and concave from side to side) faces upward.
3. Triangular articular surface on the side of body faces laterally while the comma-shaped articular surface on the side of body is directed medially.

Features and Attachments (Fig. 21.22)

The talus presents a head, a neck, and a body.

Head

1. It is directed forward, medially, and slightly downward.
2. Its *anterior surface* presents a convex articular surface which articulates with the concavity of navicular bone.

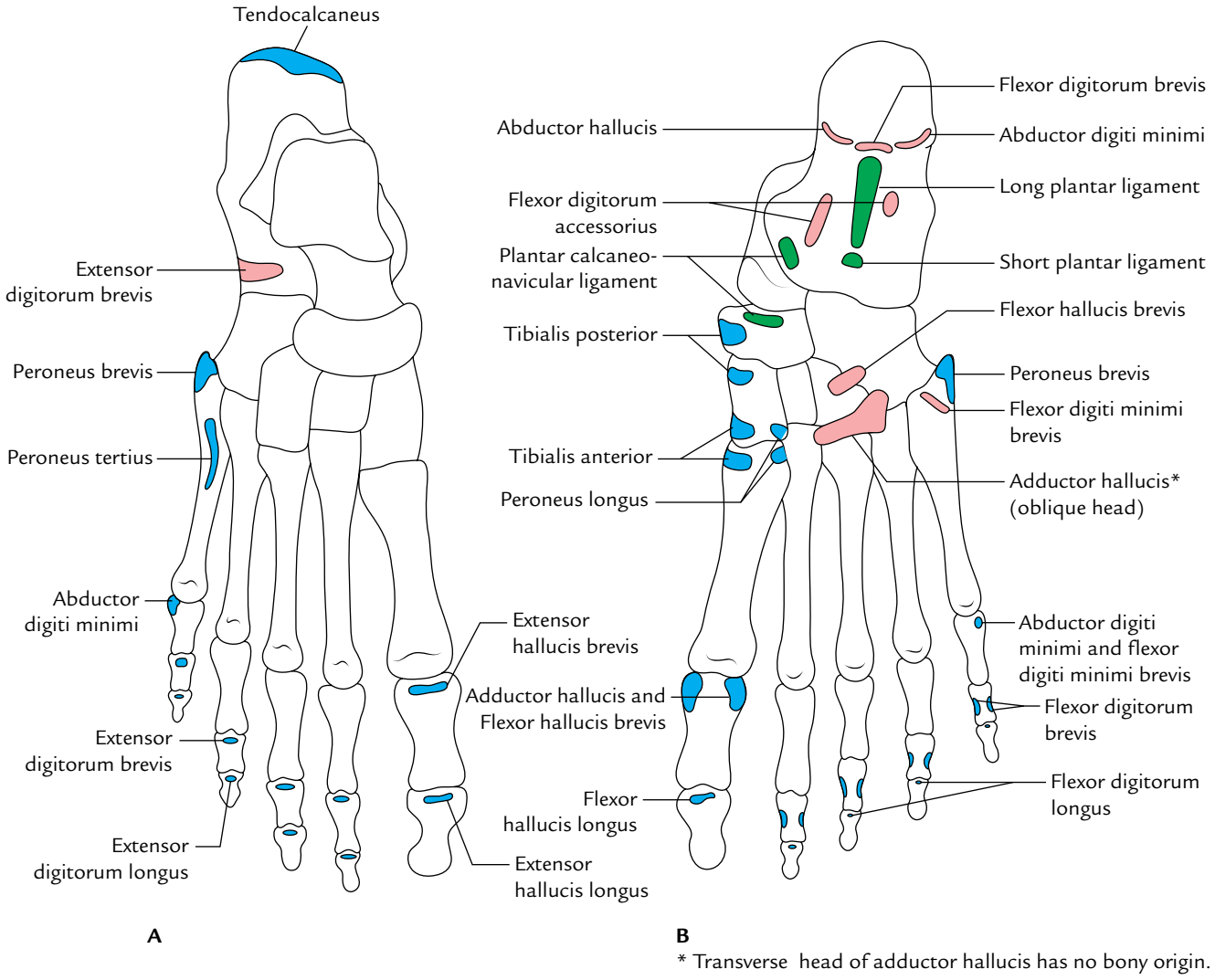


Fig. 21.21 Special features of the skeleton of the foot: **A**, dorsal aspect; **B**, plantar aspect.

3. Its *inferior surface* presents three articular facets:
   (a) **Posterior facet** is largest and articulates with the middle facet of calcaneus and forms a plane synovial subtalar joint.
   (b) **Middle facet** articulates with a spring ligament.
   (c) **Anterolateral facet** articulates with the anterior facet of calcaneus.

Neck

1. It is a constriction between the head and the body.
2. It projects forward and medially from the body making an angle of 150° with the body (neck–body angle).
3. Distal part of dorsal surface of neck provides attachment to the **dorsal talonavicular ligament**.
4. Lateral surface of the neck provides attachment to the **anterior talofibular ligament**.
5. Plantar surface of the neck presents a narrow deep groove called **sulcus tali** which forms **sinus tarsi** with the corresponding groove of calcaneus.

Body

The body of talus is cuboidal in shape. It presents five surfaces—superior, inferior, medial, lateral, and posterior.

Superior surface: It presents trochlear articular surface being wide anteriorly. It articulates with the distal surface of the tibia.

Inferior surface: It presents a large oval concave facet which articulates with the posterior facet of calcaneus to form the **subtalar joint**.

Medial surface: Its upper part presents a *comma-shaped* articular facet which articulates with the medial malleolus.

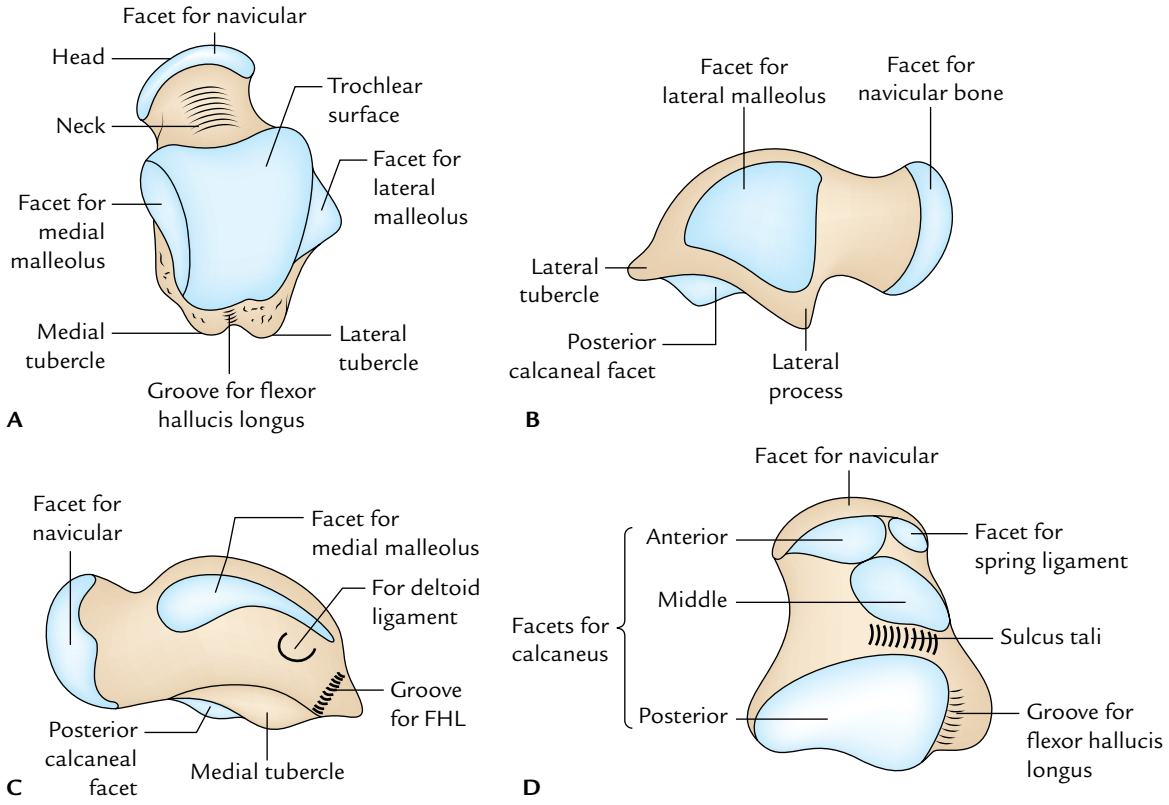


Fig. 21.22 External features of the right talus: A, dorsal aspect; B, lateral aspect; C, medial aspect; D, plantar aspect (FHL = flexor hallucis longus).

Lateral surface: It presents a *triangular articular facet* to articulate with lateral malleolus.

Posterior surface: It is narrow and projects backward as the **posterior process of talus.** It presents an oblique groove bounded by a small medial and large lateral tubercles. The groove lodges the tendon of flexor hallucis longus.

Clinical correlation

Fracture neck of talus: It sometimes occurs due to forceful dorsiflexion of foot. The arteries supplying the talus enter through the front part of the neck and pass backward. These arteries are damaged in the fracture neck of talus and cause delayed union or necrosis of the posterior segment.

Ossification

The talus ossifies from one centre which appears during the sixth month of intrauterine life.

Calcaneus (Latin Calcaneus = Heel)

The calcaneus is the largest and strongest bone of the foot. It is situated below the talus and extends behind it. It is directed forward and laterally with an upward inclination.

Side Determination and Anatomical Position

The side of calcaneus is determined by holding it in its anteroposterior axis in such a way that:

1. Its narrow end bearing concavo-convex facet faces anteriorly.
2. Its concave surface bearing shelf-like projection faces medially.
3. Surface bearing large convex articular facet faces dorsally.

Features and Attachments (Fig. 21.23)

The calcaneum presents six surfaces—anterior, posterior, superior, plantar, lateral, and medial.

Anterior surface

It is the smallest and bears a concavo-convex articular facet to articulate with cuboid.

Posterior surface

It is divided into the following three parts:

(a) *Upper one-third* is smooth and related to a synovial bursa.
(b) *Middle one-third* provides attachment to tendocalcaneus and plantaris.
(c) *Lower one-third* is subcutaneous and weight bearing during standing posture.

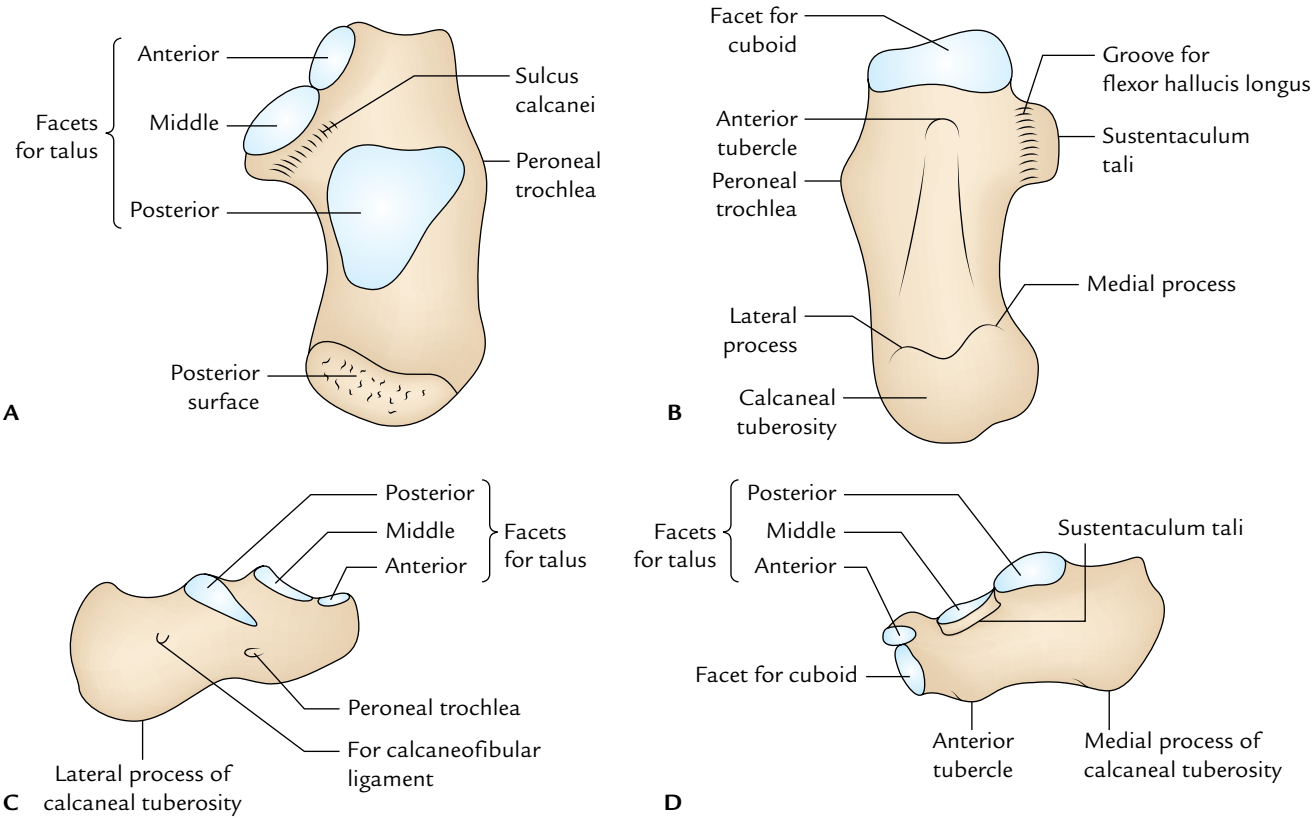


Fig. 21.23 General features of the right calcaneus: A, dorsal aspect; B, plantar aspect; C, lateral aspect; D, medial aspect.

Superior (dorsal) surface

1. It is divided into the following three areas:
   (a) Posterior one-third (non-articular).
   (b) Middle one-third (articular).
   (c) Anterior one-third (partly articular and partly non-articular).
2. *Posterior one-third* is rough and covered with pad of fibrofatty tissue deep to tendocalcaneus.
3. *Middle one-third* bears a large oval facet—the posterior facet for talus.
4. *Anterior one-third* more anteriorly presents two articular facets called middle and anterior facets for talus.
5. Non-articular part of anterior one-third behind the articular facets is divided into medial narrow and lateral wide parts:
   (a) Medial narrow part is grooved and called sulcus calcanei, which provides attachment to the interosseous talocalcanean ligament.
   (b) Lateral part is large and provides attachments to:
      (i) Extensor digitorum brevis.
      (ii) Inferior extensor retinaculum.
      (iii) Stem of bifurcate ligament.

Plantar (inferior) surface

It is rough and marked by three tubercles.

1. A small tubercle at the anterior end called *anterior tubercle.*
2. An elevation at its posterior end called *calcaneal tuberosity* presenting a **large medial and small lateral tubercles.**
3. Triangular area between the three tubercles provides attachment to the **long plantar ligament.**

A groove in front of the anterior tubercle gives attachment to **short plantar ligament.**

Lateral surface

1. It is almost flat and subcutaneous.
2. It presents a small elevation in its anterior part called **peroneal trochlea (tubercle)** which lies 2 cm below the tip of lateral malleolus.
3. Peroneal trochlea lies between the groove for the tendon of peroneus brevis above and the groove for the tendon of peroneus longus below.

Medial surface

1. It is concave from above downward.

2. A shelf-like projection called **sustentaculum tali** projects from its upper anterior part.
3. Superior surface of sustentaculum tali bears a facet for talus while its inferior surface bears a groove for the tendon of flexor hallucis longus.
4. Tendon of flexor digitorum longus is related to the medial surface of sustentaculum tali.
5. Medial surface of sustentaculum tali provides attachments to the following structures from anterior to posterior:
   (a) Spring ligament.
   (b) A slip from tibialis posterior tendon.
   (c) Deltoid ligament (superficial fibres).
   (d) Medial talocalcaneal ligament.

Clinical correlation

Calcanean fracture: It occurs when a person falls on his heals from a considerable height. The calcaneum breaks into a number of pieces and fracture lines usually run vertically.

Navicular Bone (Latin Navicular = A Little Ship)

The navicular bone is boat-shaped. It is flattened anteroposteriorly and presents six surfaces—anterior, posterior, dorsal, plantar, medial, and lateral.

1. **Anterior surface** is convex and possesses three articular facets for three cuneiforms. The medial facet is the largest.
2. **Posterior surface** is deeply concave and articulates with the head of talus.
3. **Dorsal surface** is convex and rough, and provides attachment to the dorsal talonavicular ligament.
4. **Plantar surface** presents a groove in its medial part through which passes the tendon of tibialis posterior. Its lateral part provides attachment to the **spring ligament.**
5. **Medial surface** projects downward to form **tuberosity of navicular bone**, which provides insertion to the major part of the tendon of tibialis posterior.
6. **Lateral surface** is rough and may present a facet for cuboid.

Side Determination and Anatomical Position

The side of navicular bone is determined by holding it in such a way that:

1. Its deep concave surface faces posteriorly.
2. A prominent projection, the navicular tuberosity, is directed medially.
3. A groove adjacent to tuberosity faces inferiorly.

Cuboid Bone

The cuboid bone as the name implies is cuboid in shape. It is situated in front of calcaneum and behind the fourth and fifth metatarsals.

It presents six surfaces: anterior, posterior, dorsal, plantar, medial, and lateral.

1. **Anterior (distal) surface** presents medial quadrangular and lateral triangular articular facets, which articulate with the bases of fourth and fifth metatarsals, respectively.
2. **Posterior (proximal) surface** presents a concavo-convex articulate facet (saddle-shaped articular facet) for articulation with the anterior surface of the calcaneum.
3. **Dorsal (superior) surface** is rough and flat. It is directed upward and laterally, and provides attachment to dorsal ligaments of the foot.
4. **Plantar (inferior) surface** presents an oblique groove in its distal part for the passage of tendon of peroneus longus.
5. **Medial surface** presents two articular facets—a large distal oval facet and a small proximal oval facet which articulate with the navicular and lateral cuneiform bones, respectively.
6. **Lateral surface** is small and grooved by the tendon of peroneus longus.

Cuneiform Bones (Latin Cuneiform = Wedge-Shaped)

These bones are wedge-shaped and three in number—medial (first cuneiform), intermediate (second cuneiform), and lateral (third cuneiform).

The medial cuneiform is the largest and the intermediate cuneiform is the smallest.

1. Proximal surfaces of cuneiforms articulate with the navicular bone.
2. Distal surface of medial cuneiform articulates with the base of first metatarsal.
3. Distal surface of intermediate cuneiform articulates with the base of second metatarsal.
4. Distal surface of lateral cuneiform articulates with the base of third metatarsal.

N.B. The base of second metatarsal is most fixed/secured because it fits into the **mortise** formed by three cuneiforms. For this reason, the **axial line of foot** passes through the second toe and movements of abduction or adduction of toes are mentioned with reference to the second toe.

METATARSAL BONES

These are five miniature long bones. The five metatarsal bones together constitute the **metatarsus**.

They are numbered from medial to lateral sides as first, second, third, fourth, and fifth.

Identification

First metatarsal

1. It is the shortest, thickest, and strongest, and is adapted for weight transmission.

2. Proximal surface of its base presents a kidney-shaped articular surface.

Second metatarsal

1. It is the longest metatarsal bone.
2. Proximal surface of its base has a triangular concave articular surface.

Third metatarsal

1. Proximal surface of its base has a flat triangular articular facet.
2. The lateral side of its base has two facets while the medial side has one facet.

Fourth metatarsal

1. The proximal surface of its base has a quadrilateral facet, which articulates with the cuboid.
2. The lateral side of base has one facet while its medial side has one facet divided into two parts—proximal and distal.

Fifth metatarsal

The lateral side of its base projects proximally and slightly laterally to form a large tuberosity (styloid process).

Features and Attachments

Each metatarsal consists of three parts: distal end (head), shaft (body), and proximal end (base).

Head

1. It articulates with the base of corresponding proximal phalanx to form the metatarso-phalangeal (MP) joint.
2. Each side of the head presents a tubercle dorsally to provide attachment to the collateral ligament of MP joint.

Shaft

1. Its plantar aspect is concave from before backward.
2. Sides of the shaft provide attachments to the *interossei muscles.*
3. Plantar aspect of the fifth metatarsal gives origin to the **flexor digiti minimi**.

Base

The shapes of articular surfaces on the proximal surfaces of bases of metatarsals and their articulations are given in Table 21.5.

1. Base of the first metatarsal provides insertion to:
   (a) Tendon of *tibialis anterior* inferomedially.
   (b) Tendon of *peroneus longus* inferolaterally.
2. Plantar aspects of bases of middle three (second to fourth) metatarsals provide insertion to slips of *tibialis posterior* and origin to the oblique head of a*dductor hallucis.*

Table 21.5 Shapes of the proximal surfaces of the bases of the metatarsals and their articulations

Metatarsal	Shape of the articular surface	Articulation
1st	Kidney-shaped	Medial cuneiform
2nd	Triangular	Intermediate cuneiform
3rd	Triangular	Lateral cuneiform
4th	Quadrangular	Cuboid
5th	Triangular	Cuboid

3. Base of the fifth metatarsal provides the following attachments:
   (a) *Peroneus brevis* is inserted on the styloid process of base.
   (b) *Peroneus tertius* is inserted on the dorsal aspect of the base.
   (c) *Flexor digiti minimi brevis* originates from the plantar aspect of the base.

Ossification

Each metatarsal ossifies from two centres—**one primary** and **one secondary.**

1. **Primary centre** for shafts of all the metatarsals appears during 9th week of IUL except for the shaft of first metatarsal which appears during 10th week of IUL.
2. **Secondary centre** appears during 3rd to 4th year in the heads of all the metatarsals except in the first metatarsal where it appears in the base. All of them join with the shaft during 17–20 years.

The differences between the metatarsals and metacarpals are given in Table 21.6.

Clinical correlation

- **Fracture of the tuberosity (styloid process) of the fifth metatarsal:** It usually occurs due to forced inversion of the forefoot, when the tendon of peroneus brevis pulls off the styloid process leading to its avulsion. This fracture is also called **Jones' fracture** after the name of an orthopaedic surgeon Sir Robert Jones, who himself sustained this injury.
- **March fracture:** It typically occurs in the shaft or neck of second and third metatarsals due to aggressive prolonged march past by the soldiers.

PHALANGEAL BONES

The phalangeal bones are miniature long bones. They are 14 in number in each foot—two for the great toe and three for each of the other four toes.

Table 21.6 Differences between metatarsals and metacarpals

Metatarsals	Metacarpals
More massive and more strong	Less massive and less strong
Head is smaller than base (except first)	Head is larger or equal to the base (except first)
Dorsal surface of the shaft presents no flattening	Dorsal surface presents triangular flattening
Shaft narrows from the base to the head	Shaft thickens from the base to the head
Concave surface of the shaft faces downward	Concave surface of the shaft faces forward
Numbered (1 to 5) from medial to lateral side	Numbered (1 to 5) from lateral to medial side

The phalanges in the great toe are proximal and distal, and phalanges in other toes are proximal, middle, and distal.

1. The base of the proximal phalanx presents a concave facet which articulate with the head of metatarsal bone to form the metacarpophalangeal joint.
2. The distal end of the proximal phalanx presents a pulley-like articular surface and articulates with the middle phalanx in the lateral four toes and with terminal phalanx of first toe. Thus, lateral four toes possess two interphalangeal (IP) joints, proximal and distal, and the great toe possesses only one IP joint.
3. Both, proximal and distal articular surfaces of the middle phalanx are pulley-shaped.
4. The distal phalanx of each toe bears a rough tuberosity on plantar aspect of its distal end.

Ossification

1. Each phalanx ossifies from two centres: a primary centre for the shaft and a secondary centre for the base.
2. The time of appearance of **primary centres** is as follows:
   (a) For proximal phalanges: 12th week of IUL.
   (b) For middle phalanges: 15th week of IUL.
   (c) For distal phalanges: 9th week of IUL.
3. The time of appearance of **secondary centres** is as follows.
   (a) For proximal phalanges: 2 years.
   (b) For middle phalanges: 4 years.
   (c) For distal phalanges: 8 years.

The fusion of epiphysis (base) with diaphysis (shaft) occurs in about the 18th year.

Golden Facts to Remember

► Longest and strongest bone in the body	Femur
► Largest sesamoid bone in the body	Patella
► Third trochanter	Prominent gluteal tuberosity
► Commonest site of the fracture femur	Neck of femur
► Commonest site of the fracture tibia	Shaft of tibia at the junction of middle one-third and lower one-third
► Largest nutrient artery in the body	Nutrient artery of tibia
► Largest and strongest bone of the foot	Calcaneum
► First tarsal bone to ossify	Calcaneum
► Most common accessory bone of the foot	Accessory navicular bone (navicular secundarium)
► Shortest and strongest metatarsal bone	First metatarsal
► Longest metatarsal bone	Second metatarsal

Clinical Case Study



A 71-year-old woman fell in her bathroom. After the fall, she could not get up or even move her right leg. She was taken to the orthopedic surgeon who on examination noted that toes of the patient's right foot were pointing laterally. On measuring the length, the right lower limb was found shorter as compared to the left limb. The X-ray revealed **subcapital fracture of the neck of right femur**.

Questions

1. What is the commonest cause of the fracture neck of femur in elderly women following a trivial fall?
2. How is the length of the lower limb measured?
3. Give the reason of shortening of the lower limb in this case.
4. Give the anatomical basis of the lateral rotation of the thigh in fracture neck of femur.

Answers

1. Osteoporosis due to old age and deficiency of estrogen.
2. It is measured from the anterior superior iliac spine to the medial malleolus.
3. Following fracture neck, the shaft of femur is pulled up by the contraction of hamstring and adductor muscles.
4. The thigh is laterally rotated by the contraction of gluteus maximus and short lateral rotators of the thigh (e.g., piriformis, obturator internus, gemelli, and quadratus femoris). Further, in the fracture neck of femur, the psoas major becomes lateral rotator due to shift in its axis of action.

CHAPTER

22 Front of the Thigh

The thigh is the part of the lower limb between the hip and knee joints. For descriptive purposes, the thigh is divided into three regions—front of the thigh, medial side of the thigh, and back of the thigh.

The front of the thigh corresponds to the back of arm.

The study of the front of thigh is of immense clinical and surgical importance.

SURFACE LANDMARKS (Fig. 22.1)

1. **Fold of the groin** is a shallow curved groove extending from the pubic tubercle to the anterior superior iliac spine. It corresponds to the underlying inguinal ligament and separates the anterior abdominal wall from the front of the thigh.
2. **Anterior superior iliac spine** is palpated at the lateral end of the fold of the groin.
3. **Pubic tubercle** is a small bony projection felt at the medial end of the fold of groin.
4. **Greater trochanter** lies a hand's breadth below the tubercle of the iliac crest and forms a prominence in front of the hollow on the side of the hip.
5. **Midinguinal point** is a point midway between the anterior superior iliac spine and the pubic symphysis.
6. **Midpoint of inguinal ligament** is a point midway between the anterior superior iliac spine and the pubic tubercle.
7. **Medial and lateral condyles of femur and tibia** form large bony masses on the medial and lateral sides of the knee, respectively. The most prominent points on the condyles are called epicondyles.
8. **Fleshy swelling above the medial condyle of the femur** is formed by the lower part of the vastus medialis muscle.
9. **Patella** (**knee cap**) is easily felt as a triangular bone in front of the knee. It is freely mobile when the knee is extended but becomes rigid when the knee is flexed.
10. **Tibial tuberosity** is easily felt as a bony prominence on the front of the upper end of the tibia.

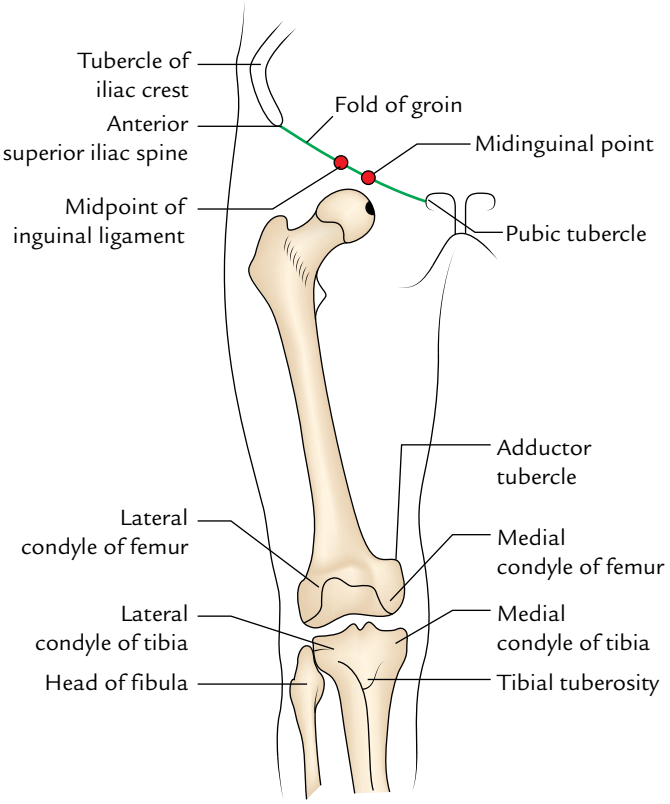


Fig. 22.1 Bony landmarks of the thigh.

11. **Ligamentum patellae** can be felt as a strong fibrous band stretching between patella and tibial tuberosity.
12. **Adductor tubercle** can be felt just above the medial condyle of the femur and on deep pressure a cord-like tendon of adductor magnus is felt above the tubercle.

SUPERFICIAL FASCIA

The superficial fascia on the front of the thigh near the inguinal region consists of two layers: thick superficial fatty

layer and thin deep membranous layer, which are continuous above with the corresponding layers of the anterior abdominal wall.

The membranous layer of the superficial fascia is attached in a linear fashion to the deep fascia of thigh (fascia lata) along a horizontal line which extends laterally from the pubic tubercle for about 8 cm. The line of fusion of the membranous layer of superficial fascia to the fascia lata in front of the thigh corresponds to the flexion crease of the hip joint called **Holden's line**.

The superficial fascia on the front of the thigh contains:

1. Cutaneous nerves.
2. Cutaneous arteries.
3. Termination of saphenous vein and its tributaries.
4. Superficial inguinal lymph nodes.

Cutaneous Nerves (Fig. 22.2)

The skin on the front of the thigh is supplied by seven cutaneous nerves which are derived from the lumbar plexus (described in Chapter 12, p. 196). These are:

1. Ilioinguinal nerve (L1).
2. Femoral branch of the genitofemoral nerve (L1, L2).
3. Lateral cutaneous nerve of the thigh (L2, L3).
4. Intermediate cutaneous nerve of the thigh (L2, L3).
5. Medial cutaneous nerve of the thigh (L2, L3).

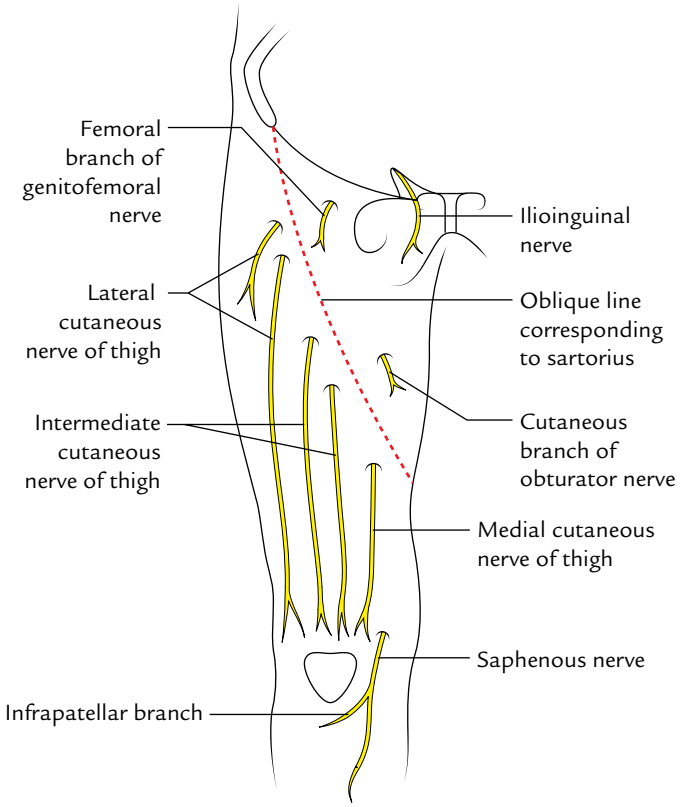


Fig. 22.2 Cutaneous nerves on the front of the thigh.

6. Saphenous nerve (L3, L4).
7. Cutaneous branch of the obturator nerve (L2, L3).

The **ilioinguinal nerve** (L1) is the collateral branch of the iliohypogastric nerve. It emerges through the superficial inguinal ring and supplies the skin at the root of the penis, and the anterior one-third of the scrotum in male; and mons pubis and anterior one-third of labium majus in female; and the skin of the upper medial aspect of the thigh.

The **femoral branch of the genitofemoral nerve** (L1) pierces the femoral sheath and the overlying deep fascia 2 cm below the midinguinal point, and supplies the skin over the femoral triangle.

The **lateral cutaneous nerve of the thigh** (L2, L3) is a direct branch of the lumbar plexus. It enters the thigh by passing behind or through the lateral end of the inguinal ligament, a centimeter medial to the anterior superior iliac spine, and divides into anterior and posterior branches. It supplies the skin on the anterolateral side of the upper thigh.

Clinical correlation

The lateral cutaneous nerve of the thigh is sometimes compressed as it passes through the inguinal ligament, causing pain and paresthesia (altered sensations) in the upper lateral aspect of the thigh leading to a clinical condition called '**meralgia paresthetica**'. The surgical treatment of this condition requires division of the inguinal ligament and releasing the nerve from the compression.

The **intermediate cutaneous nerve of the thigh** (L2, L3) is a branch of the anterior division of the femoral nerve. It passes vertically downward and pierces the deep fascia at the junction of the upper one-third and middle one-third of the thigh. It divides into two or more branches and supplies the skin on the front of the thigh as far down as the knee.

The **medial cutaneous nerve of the thigh** (L2, L3) is a branch of the anterior division of the femoral nerve. It divides into anterior and posterior branches, which run medially across the femoral vessels, pierce the fascia lata at the mid-thigh, and supply the skin on the medial side of the thigh.

The **saphenous nerve** (L3, L4) is a branch of the posterior division of the femoral nerve. It pierces deep fascia on the medial side of the knee and runs downward in front of the great saphenous vein. It gives *infrapatellar branch* before piercing the deep fascia, which runs downward and laterally to supply skin over the ligamentum patella.

The **cutaneous branch of the obturator nerve (L2, L3)** is a small twig which arises from the anterior division of the obturator nerve and supplies the lower one-third of the medial aspect of the thigh.

N.B. The lateral, intermediate, and medial cutaneous nerves of the thigh pierce the deep fascia of thigh (fascia lata) along an oblique line that roughly corresponds to the course of the sartorius muscle.

Patellar Plexus

It is a plexus of nerves present in front of the patella, ligamentum patellae, and the upper end of the tibia, and supplies the skin over these structures. It is formed by the branches from (a) the lateral cutaneous nerve of thigh, (b) the intermediate cutaneous nerve of thigh, (c) medial cutaneous nerve of thigh, and (d) infrapatellar branch of the saphenous nerve.

Cutaneous Arteries

These are three small arteries arising from the femoral artery, a little below the inguinal ligament (Fig. 22.3).

1. The **superficial external pudendal artery** arises medially from the femoral artery, pierces the cribriform fascia, runs medially in front of the spermatic cord, and supplies the external genitalia.
2. The **superficial epigastric artery** arises anteriorly from the femoral artery, pierces the cribriform fascia, to ascend anterior to the inguinal ligament and runs towards the umbilicus. It supplies the lower part of the anterior abdominal wall.
3. The **superficial circumflex iliac artery** (smallest branch) arises near the superficial epigastric artery, pierces the fascia lata lateral to the saphenous opening, to run upward and laterally below the inguinal ligament.

Long (Great) Saphenous Vein (Fig. 22.3)

It is the largest and longest superficial vein of the lower limb.

It begins on the dorsum of the foot from the medial end of the dorsal venous arch, runs upward in front of the medial malleolus, and ascends on the medial side of the leg. It enters the medial aspect of the thigh by passing behind the knee (posterior to the medial condyle). In the thigh, it inclines forward to reach the saphenous opening where it pierces the cribriform fascia and opens into the femoral vein by hooking the inferior crescentic margin of the saphenous opening.

Before piercing the cribriform fascia, it receives three tributaries: superficial epigastric vein, superficial external pudendal vein, and superficial circumflex iliac vein (Fig. 22.3).

The saphenous vein is described in detail in Chapter 33.

Superficial Inguinal Lymph Nodes

The superficial inguinal lymph nodes are variable in their number and size. They are divided into two sets—horizontal and vertical. The two sets are arranged in a T-shaped manner (Fig. 22.4).

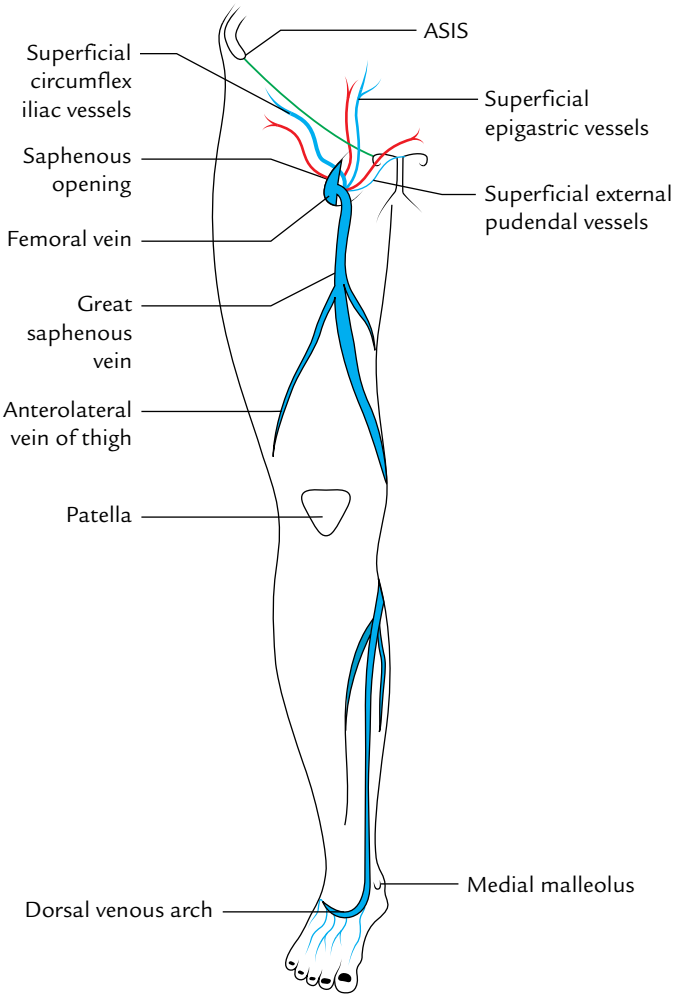


Fig. 22.3 Superficial vessels on the front of the thigh (ASIS = anterior superior iliac spine).

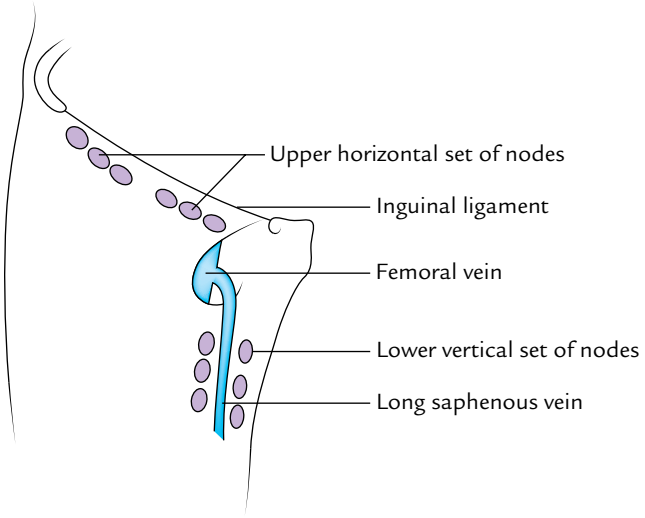


Fig. 22.4 Superficial inguinal lymph nodes.

The nodes of the upper horizontal set (five or six in number) are located a short distance below the inguinal ligament. The nodes of this set can be subdivided into the upper lateral and upper medial groups. The nodes of the lower vertical set (four or five in number) lie along the upper part of the long saphenous vein.

The efferent lymph vessels from the superficial inguinal nodes pass through the saphenous opening and drain into the deep inguinal nodes.

DEEP FASCIA OF THE THIGH

The deep fascia of the thigh is very strong and envelops the thigh like a sleeve. It is called **fascia lata** because it encloses a wide area of the thigh (Latin *Latus*: broad). Its attachments are as follows:

1. *Superiorly,* on the front of the thigh, it is attached to the anterior superior iliac spine, inguinal ligament, and pubic tubercle. Laterally it is attached to the iliac crest; posteriorly (through the gluteal fascia) to the sacrum, coccyx, and sacrotuberous ligament; and medially it is attached to the pubis, pubic arch, and ischial tuberosity.
2. *Inferiorly* on the front and sides of the knee, it is attached to subcutaneous bony prominences and the capsule of the knee joint.

MODIFICATIONS OF DEEP FASCIA OF THE THIGH

The deep fascia of the thigh presents two modifications–**iliotibial tract** and **saphenous opening.**

Iliotibial Tract (Fig. 22.5)

The fascia lata is thickened on the lateral aspect of the thigh to form about 2 inches wide band called **iliotibial tract.** Superiorly along the iliac crest, this tract splits into two layers to enclose the two muscles (tensor fasciae latae and gluteus maximus) and forms a single thickened sheet, the **gluteal aponeurosis** between them which covers the gluteus medius. The superficial lamina is attached to the tubercle of the iliac crest and the deep lamina to the capsule of the hip joint. Inferiorly, the tract is attached to a smooth area on the anterior surface of the lateral condyle of the tibia.

The upper part of iliotibial tract provides insertion to two muscles: the gluteus maximus (except deep fibres of its lower half) and the tensor fasciae latae.

Functional Significance

The iliotibial tract stabilizes the knee both in extension and in partial flexion; hence it is used constantly during walking and running. On leaning forward with slightly flexed knees the iliotibial tract is the main support of the knee against gravity and prevents the individuals from falling forward.

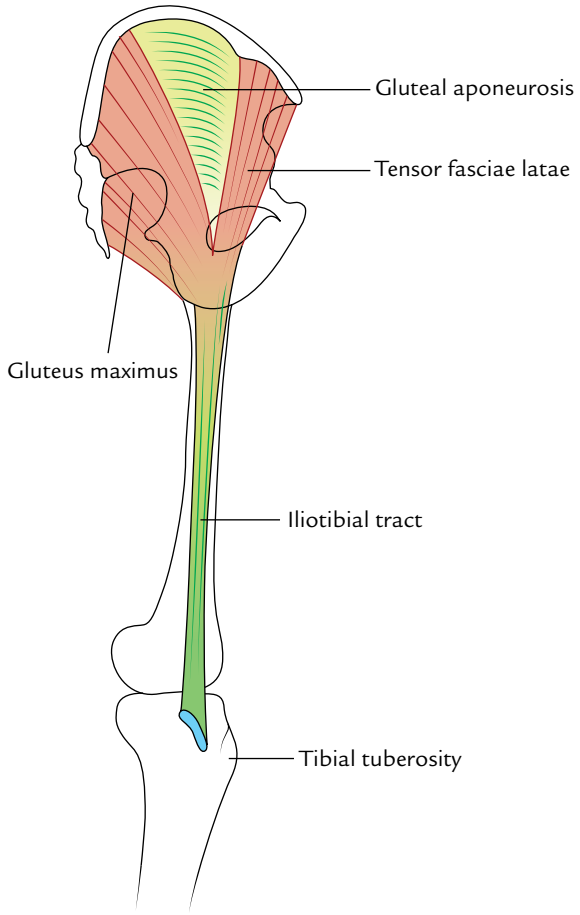


Fig. 22.5 Attachment of the iliotibial tract. Note superficial portion of gluteus maximus and tensor fasciae latae insert into the iliotibial tract from opposite directions.

Saphenous Opening (Fig. 22.6)

This is an oval opening in the fascia lata in the upper medial part of the front of the thigh. The centre of the opening is about 4 cm below and lateral to the pubic tubercle. Its vertical length measures about 3–4 cm. The opening is bounded inferolaterally by a sharp crescentic (falciform) margin. It is formed by the superficial stratum of the fascia lata, which lies in front of the femoral sheath. The medial margin of the opening is ill-defined and formed by the deep stratum which lies at a deeper level and becomes continuous with the fascia overlying the pectineus (pectineal fascia). It lies behind the femoral sheath.

The saphenous opening is closed by a membrane of areolar tissue—the cribriform fascia which is pierced by number of structures making it sieve-like, hence the name cribriform.

N.B. In the region of saphenous opening, the fascia lata is twisted to form an oval opening and presents two strata, superficial and deep.

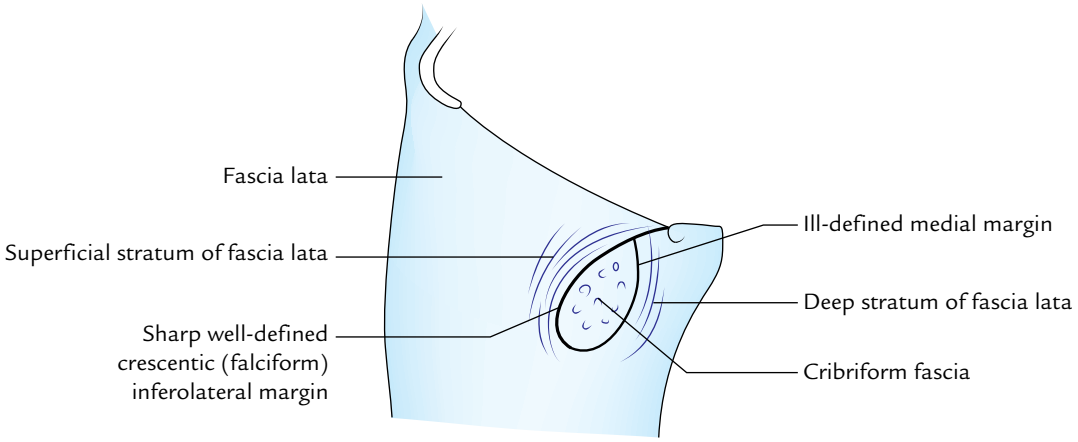


Fig. 22.6 The saphenous opening. Note the saphenous opening resembles a double-breasted jacket.

Structures passing through the saphenous opening are as follows:

1. Great saphenous vein.
2. Superficial epigastric and superficial external pudendal vessels.
3. Few lymph vessels connecting superficial and deep inguinal lymph nodes.

Clinical correlation

The fascia lata is attached to the under surface of the inguinal ligament. When the thigh is extended, it pulls the abdominal wall downward and makes it tense. Therefore, in order to relax the abdomen (for palpation of abdominal contents) the patient is asked to draw the legs up to overcome the pull of the fascia lata on the abdominal wall.

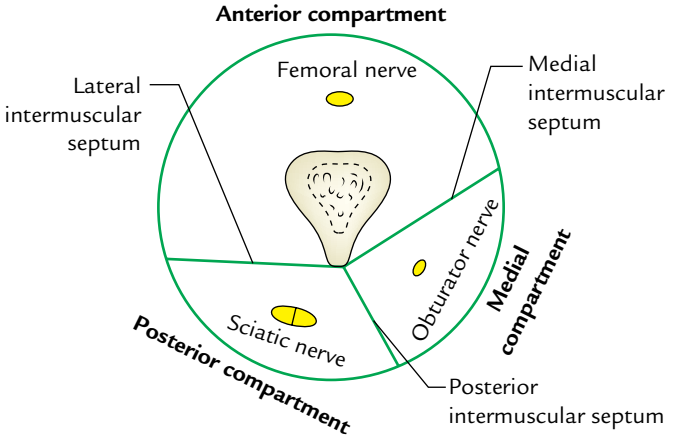


Fig. 22.7 Fascial compartments of the thigh.

FASCIAL COMPARTMENTS OF THE THIGH (Fig. 22.7)

The fascia lata encloses the entire thigh like a sleeve/stocking. Three intermuscular fascial septa (lateral, medial, and posterior) pass from the inner aspect of the deep fascial sheath of the thigh to the linea aspera of the femur and divide the thigh into three compartments—anterior, medial, and posterior.

The *anterior (extensor) compartment,* lies between lateral and medial intermuscular septa; the *Medial (adductor) compartment*, lies between medial and posterior septa; and the *posterior (flexor) compartment,* lies between posterior and lateral septa.

Each compartment contains muscles, nerves, and arteries.

N.B. The lateral intermuscular septum is the strongest and extends from the iliotibial tract to the lateral lip of linear aspera. The medial intermuscular septum is attached to the medial lip of the linear aspera. The posterior intermuscular septum is poorly defined.

ANTERIOR COMPARTMENT OF THE THIGH

The extensor compartment of the thigh is bounded laterally by lateral intermuscular septum and medially by medial intermuscular septum.

CONTENTS

1. **Muscles:** Quadriceps femoris, articularis genu, sartorius, and tensor fasciae latae.
2. **Nerve:** Femoral nerve.
3. **Artery:** Femoral artery.

MUSCLES

Quadriceps Femoris

This muscle is so named because it consists of four parts: rectus femoris, vastus lateralis, vastus medialis, and vastus intermedius. It is supplied by the femoral nerve. It forms most of the bulk on the anterior aspect of the thigh and is the most powerful extensor of the knee joint.

N.B. All the parts of quadriceps cross only one joint, i.e., knee joint, except rectus femoris which crosses two joints, i.e., hip and knee joints.

The origin, insertion, and nerve supply of various parts of quadriceps femoris are described in the following text.

Rectus Femoris (Fig. 22.8)

It is a long straight muscle with fusiform bipinnate belly in the upper two-third and flat tendon in the lower one-third (Latin *rectus* = straight).

Origin

It arises from the ilium by two heads:

(a) *Straight head* arises from the upper part of the anterior inferior iliac spine.

(b) *Reflected head* arises from a groove above the acetabulum.

Insertion

The two heads meet at an acute angle and form a fusiform belly, which occupies the front of the thigh. In the lower one-third of the thigh, the muscle ends in a flat tendon which descends vertically downward to be inserted into the base of the patella.

Nerve supply

It is by posterior division of the femoral nerve.

Actions

It extends the knee and flexes the hip.

N.B. The rectus femoris is also called the *"kicking muscle"* because it extends the knee and flexes the hip—the actions required during kicking.

Vastus Lateralis (Fig. 22.8)

Origin

It takes a linear and aponeurotic origin from the upper part of the intertrochanteric line, anterior and inferior borders of the greater trochanter, lateral lip of gluteal tuberosity, and upper half of the lateral lip of linea aspera. It also arises from the lateral intermuscular septum.

Insertion

It is inserted (a) by a broad aponeurosis into the tendon of rectus femoris, lateral part of the base of patella, upper one-third of the lateral border of the patella, and (b) retinacular fibres to the front of the lateral condyle of tibia which replaces the anterolateral aspect of the capsule of the knee joint.

Nerve supply

It is by the posterior division of the femoral nerve.

Actions

Extension of the knee joint.

Vastus Medialis

It covers the medial surface of the shaft of femur.

Origin

It takes a linear origin from the lower part of intertrochanteric line, spiral line, medial lip of linea aspera, and upper two-third of the medial supra condylar line. It also arises from the medial intermuscular septum.

Insertion

It is inserted by a broad aponeurosis into the tendon of rectus femoris, medial part of the base, and upper two-third of the medial border of the patella. From its lower attachment it sends a fibrous expansion, the medial **patellar retinaculum**, to the front of the medial condyle of the tibia which replaces the anteromedial part of the capsule of the knee joint.

Nerve supply

It is by the nerve to vastus medialis, which is the thickest muscular branch of the posterior division of the femoral nerve.

Actions

Extension of the knee joint.

N.B. The lower fibres of vastus medialis are attached far more down on the medial border of patella than those of the vastus lateralis on the lateral border. Therefore, they prevent the natural tendency of patella to dislocate laterally during the extension of the knee joint.

The role of vastus medialis is indispensable for the stability of the patella.

Vastus Intermedius (Fig. 22.8)

Origin

It arises from the anterior and lateral aspects of the upper three-fourth of the shaft of femur.

Insertion

Into the base of patella deep to the tendon of rectus femoris.

Nerve supply

It is by the posterior division of the femoral nerve.

Action

Extension of the knee joint.

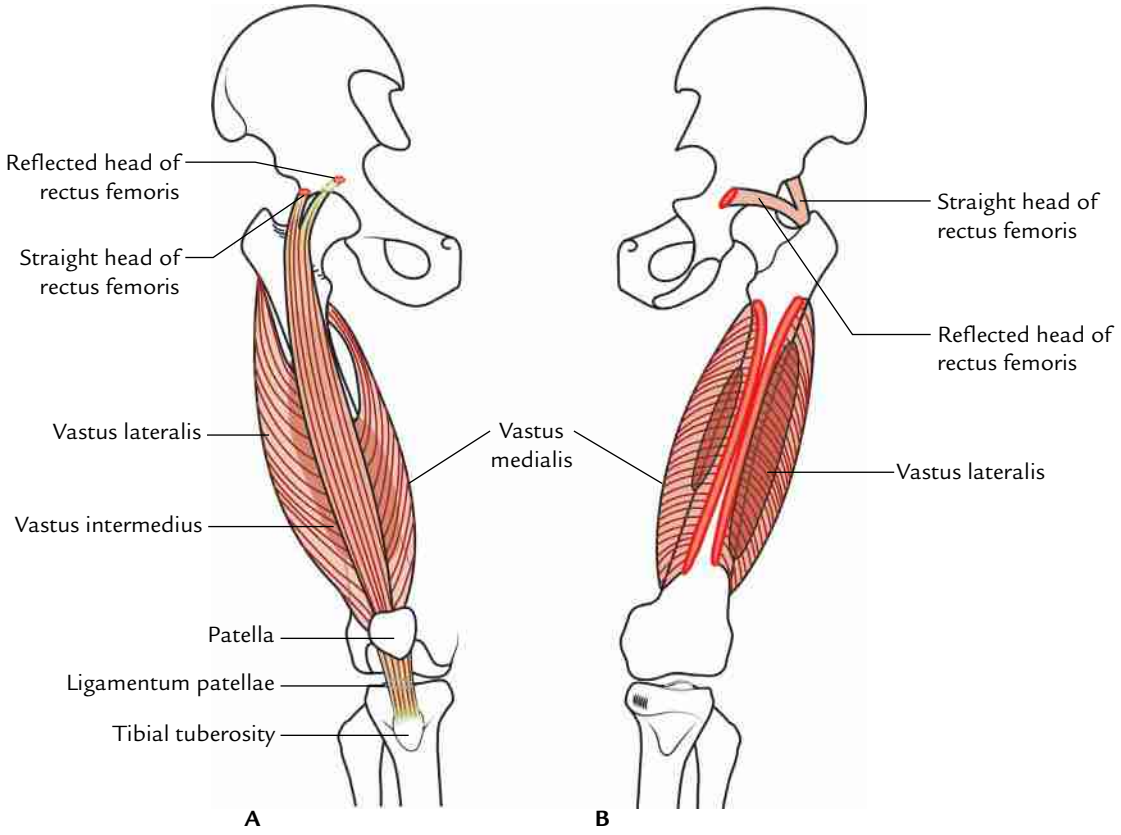


Fig. 22.8 Origin and insertion of quadriceps femoris: A, anterior view; B, posterior view.

Articularis Genu

It consists of three or four muscular slips which are considered to be a detached part of the vastus intermedius.

Origin

It arises from the anterior surface of the lower part of the shaft of the femur, few centimeters above the patellar articular margin.

Insertion

Into the upper part of the synovial membrane of the knee joint.

Nerve supply

It is by a twig from the nerve to vastus intermedius.

Actions

The articularis genu pulls up the synovial membrane upward to prevent its damage when the knee is extended.

Sartorius (Fig. 22.9)

It is the longest muscle in the body, which crosses the front of the thigh obliquely from the lateral to the medial side.

Origin

It arises from the anterior superior iliac spine and upper half of the notch immediately below it.

Insertion

The muscle spirals obliquely across the thigh from lateral to medial side to reach the posterior aspect of the medial condyle of femur, where its tendon runs forward to be inserted into the upper part of the medial surface of the shaft of tibia in front of the insertions of gracilis and semitendinosus. The insertion of sartorius is inverted hockey-stick-shaped.

Nerve supply

It is by the anterior division of the femoral nerve.

Actions

It flexes both the hip and the knee joints, and adducts and rotates the thigh laterally to bring the lower limb into the sitting position of a tailor (Latin *sartor*: tailor)/"palthi-position" adopted by Indian Hindus during meal.

Tensor Fasciae Latae (Fig. 22.9)

It is a short thick muscle, which lies at the junction of the gluteal region and the upper part of the front of the thigh.

Origin

It arises from the outer lip of the iliac crest extending from the anterior superior iliac spine to the tubercle of the crest.

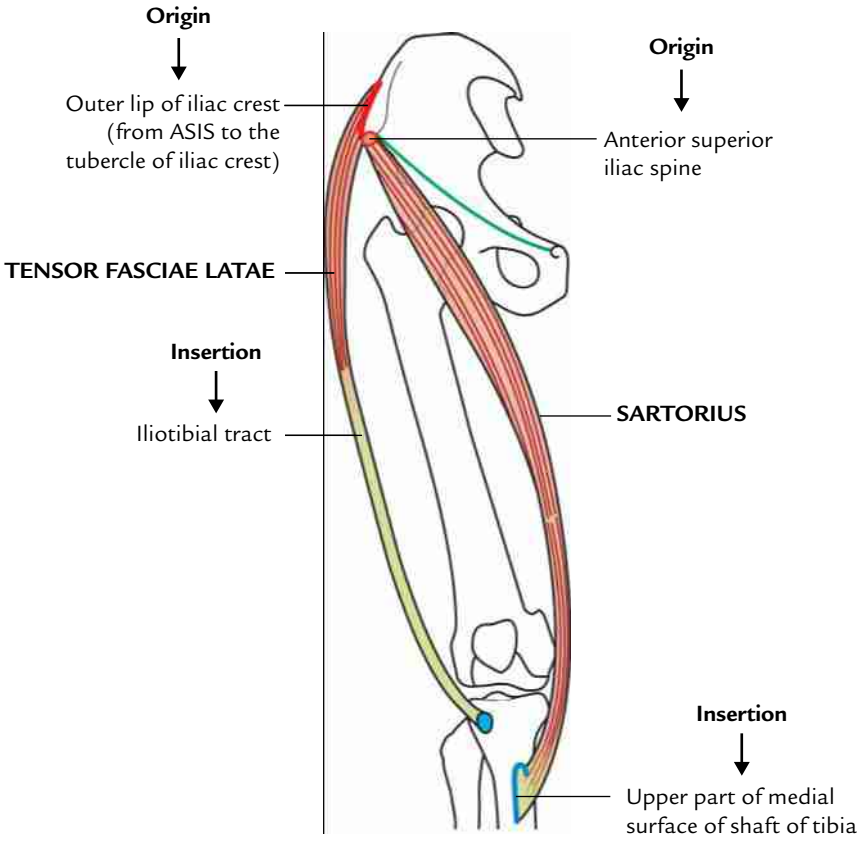


Fig. 22.9 Origin and insertion of the tensor fasciae latae and sartorius (ASIS = anterior superior iliac spine).

Insertion

The muscle passes downward and slightly backward and inserted into the iliotibial tract 3–5 cm below the level of the greater trochanter.

Nerve supply

It is by the superior gluteal nerve.

Actions

It abducts the hip joint and maintains the extended position of the knee joint through the iliotibial tract.

N.B. Morphologically, the *tensor fasciae latae* is a muscle of the gluteal region which has migrated on the lateral aspect of the thigh during the course of evolution but retains its nerve supply. It is supplied by the superior gluteal nerve—the nerve of gluteal region.

The origin, insertion, nerves supply, and actions of the muscles of the front of the thigh are summarized in Table 22.1.

FEMORAL TRIANGLE

It is a triangular depression on the front of the upper one-third of the thigh below the inguinal ligament. Its apex is directed downward.

Boundaries

These are as follows (Fig. 22.10):

Lateral: Medial border of sartorius.
Medial: Medial border of adductor longus.
Base: Inguinal ligament.

Apex: It is formed by the meeting point of the medial borders of adductor longs and sartorius.

Floor: It is gutter-shaped and muscular. From lateral to medial side it is formed by the following muscles (Fig. 22.11):

1. Iliacus.
2. Psoas major (tendon).
3. Pectineus.
4. Adductor longus.

Roof: It is formed by the fascia lata having saphenous opening (Fig. 22.12). The superficial fascia overlying the roof contains superficial branches of the femoral artery and accompanying veins, upper part of great saphenous vein, superficial inguinal lymph nodes, femoral branch of the genitofemoral nerve, and branches of ilioinguinal nerve.

Contents

The main contents of the femoral triangle are as follows (Fig. 22.13):

1. Femoral artery and its branches.
2. Femoral vein and its tributaries.
3. Femoral nerve.
4. Deep inguinal lymph nodes.
5. Lateral cutaneous nerve of the thigh.

In addition to above structures, it also contains:

1. Femoral branch of the genitofemoral nerve.
2. Fibrofatty tissue.

A brief account of some contents of the femoral triangle is given in the following text.

The **femoral artery** traverses the femoral triangle from the midpoint of its base to the apex. The artery passes downward and medially. As a result, at the base of triangle, it lies lateral to the femoral vein but at its apex it lies anterior to the vein. The femoral artery gives three superficial branches (superficial epigastric, superficial circumflex iliac, and superficial external pudendal) and two deep branches (profunda femoris and deep external pudendal). The profunda femoris is the largest branch of the femoral artery. It arises from the lateral side of the femoral artery about 3.5 cm below the inguinal ligament

Table 22.1 Origin, insertion, nerve supply, and actions of the muscles of the front of the thigh

Muscle	Origin	Insertion	Nerve supply	Actions
1. Quadriceps femoris (a) Rectus femoris	• Straight head: from the upper half of the anterior inferior iliac spine • Reflected head: from the groove above the acetabulum	Base of patella	Femoral nerve	Flexion of the hip joint and extension of the knee joint
(b) Vastus lateralis	• Upper part of the intertrochanteric line • Anterior and inferior borders of greater trochanter • Lateral lip of the gluteal tuberosity • Upper half of lateral lip of linea aspera	• Base and upper 1/3rd of the lateral border of the patella • Expansion to the capsule of the knee joint and iliotibial tract	Femoral nerve	Extension of the knee joint
(c) Vastus medialis	• Lower part of the intertrochanteric line • Spiral line • Medial lip of the linea aspera • Upper 2/3rd of medial supracondylar line	• Base and upper 2/3rd of the medial border of the patella • Expansion to the capsule of the knee joint	Femoral nerve	Extension of the knee joint
(d) Vastus intermedius	Upper 3/4th of the anterior and lateral surfaces of the shaft of femur	Base of patella	Femoral nerve	Extension of the knee joint
2. Articularis genu	• Anterior surface of the lower part of the shaft of femur	Synovial membrane of the knee joint	Femoral nerve	Pulls up the synovial membrane of the knee joint during its extension
3. Sartorius	• Anterior superior iliac spine and upper half of the notch below it	Upper part of the medial surface of the shaft of the tibia	Femoral nerve	• Flexion of the hip and knee joints • Lateral rotation of the thigh
4. Tensor fasciae latae	• Outer lip of the iliac crest from anterior superior iliac spine to the tubercle of iliac crest	Iliotibial tract 3–5 cm below the greater trochanter	Superior gluteal nerve	Abduction of the hip joint

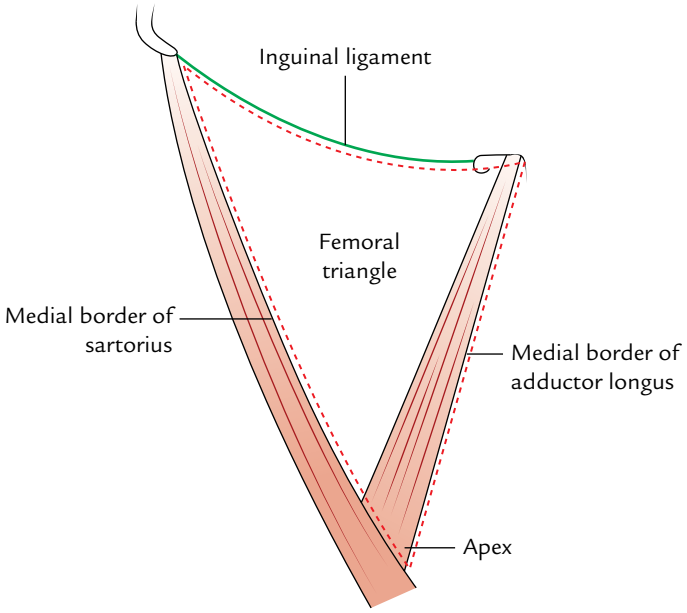


Fig. 22.10 Boundaries of the femoral triangle.

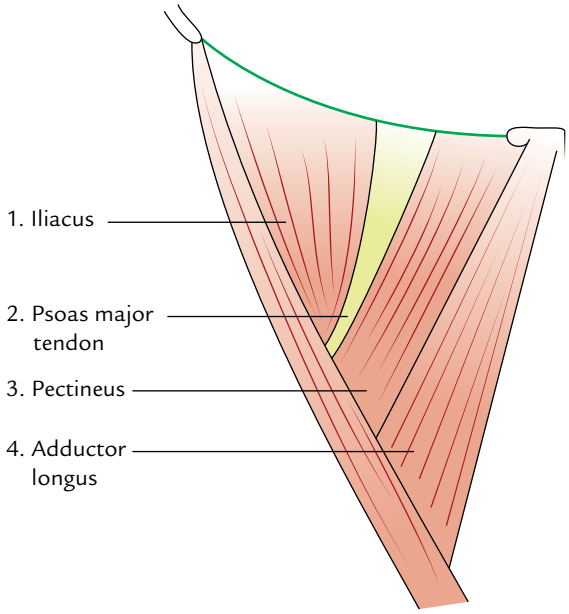


Fig. 22.11 Muscles forming the floor of the femoral triangle.

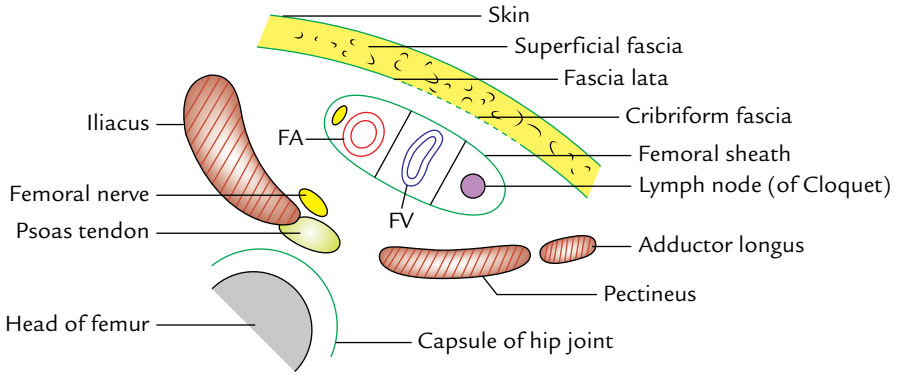


Fig. 22.12 Sagittal section through the upper part of femoral triangle showing structures forming its roof and floor (FV = femoral vein, FA = femoral artery).

and spirals medially behind the femoral vessels. It gives rise to medial and lateral circumflex femoral arteries. The medial circumflex artery disappears by passing between the psoas major and pectineus. The lateral circumflex femoral artery passes laterally between the anterior and posterior divisions of the femoral nerve.

The **deep external pudendal artery** pierces the fascia lata and passes medially behind the spermatic cord.

The **femoral vein** accompanies the femoral artery. The vein is posterior to the femoral artery at the apex and medial to it at the base of the triangle. It receives the great saphenous vein and profunda femoris vein and veins corresponding to the superficial branches of femoral artery.

The **femoral nerve** lies lateral to the femoral artery, outside the femoral sheath, in the groove between the iliacus and the psoas major. About 2.5 cm below the inguinal ligament it divides into anterior and posterior divisions which enclose lateral circumflex femoral artery between them. The anterior division gives off two cutaneous branches—intermediate and medial cutaneous nerves of the thigh. The medial cutaneous branch accompanies the lateral side of the artery; at the apex of the triangle it crosses the front of the artery from lateral to medial side. The posterior division gives rise to one cutaneous nerve—the saphenous nerve. It extends downward along the lateral side of the artery.

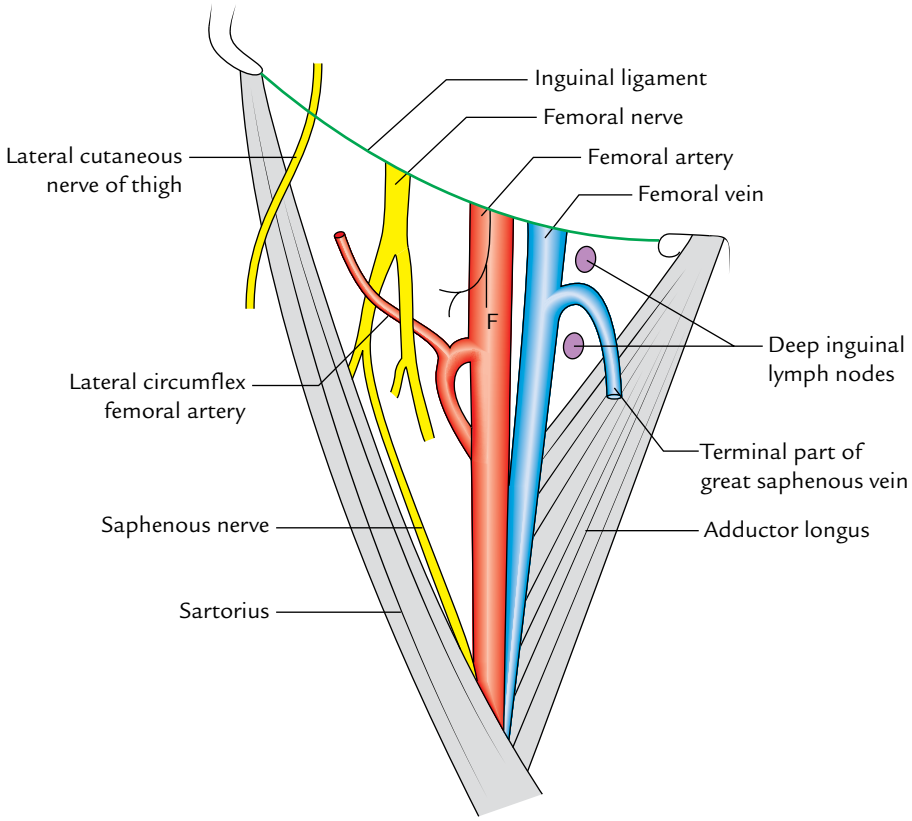


Fig. 22.13 Boundaries and contents of the femoral triangle (F = femoral branch of the genitofemoral nerve).

The **lateral cutaneous nerve of the thigh** is already described on p. 329.

The **deep inguinal lymph nodes** are usually three in number and lie medial to the upper part of the femoral vein. The lowest one is situated below the junction of great saphenous and femoral veins, the middle one in the femoral canal (the gland of Cloquet/Rosenmüller), and the highest one in the femoral ring.

Femoral Sheath (Fig. 22.14)

It is a funnel-shaped fascial sheath enclosing the upper 3–4 cm of the femoral vessels.

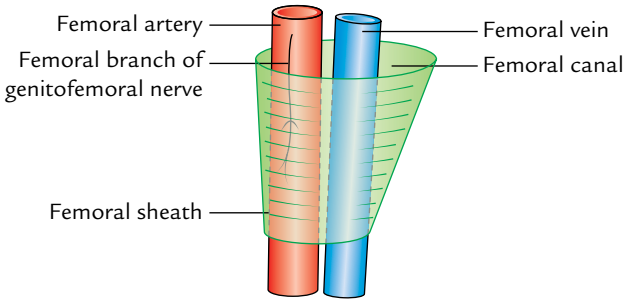


Fig. 22.14 Femoral sheath. Note the asymmetry of the femoral sheath.

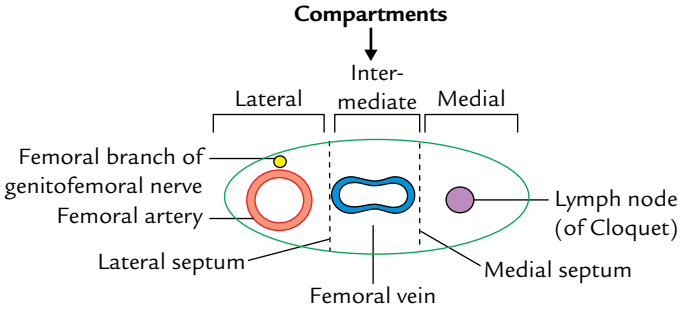


Fig. 22.15 Compartments of the femoral sheath.

The femoral sheath is divided into three compartments, *viz.* lateral, intermediate, and medial by two anteroposterior septa (Fig. 22.15).

1. The *lateral compartment* contains the femoral artery and the femoral branch of the genitofemoral nerve.
2. The *intermediate compartment* contains the femoral vein.
3. The *medial compartment* is small and known as the femoral canal.

The femoral sheath, femoral canal, and femoral hernia are described in Chapter 4, pp. 47–48.

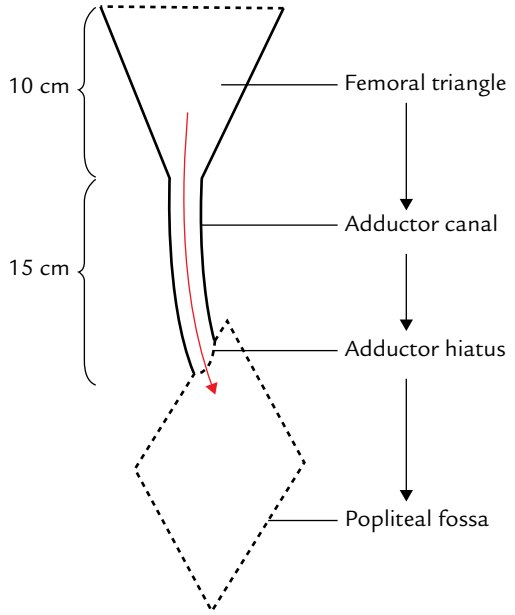


Fig. 22.16 Femoral triangle and adductor canal. (The red arrow indicates the passage of femoral artery.) Note the adductor canal is a narrow passage for the femoral vessels.

ADDUCTOR CANAL (SUBSARTORIAL CANAL/ HUNTER'S CANAL)

The adductor canal is an intermuscular tunnel situated on the medial side of the middle one-third of the thigh (Fig. 22.16).

It extends from the apex of the femoral triangle, above, to the tendinous opening in the adductor magnus, below. It provides passage to the femoral vessels from femoral triangle to the popliteal fossa (Fig. 22.16).

Boundaries (Fig. 22.17)

The adductor canal is triangular in cross section. Its boundaries are as follows:

Anterolateral wall: It is formed by vastus medialis.

Posterior (floor): It is formed by adductor longus above and adductor magnus below.

Medial (roof): It if formed by a strong fibrous membrane stretching across the anterolateral and posterior boundaries. The roof is overlapped by the sartorius muscle.

N.B. The *subsartorial plexus* of nerves lies on the roof underneath the sartorius. The plexus is formed by branches from the medial cutaneous nerve of the thigh, the saphenous nerve, and the anterior division of the obturator nerve. It supplies the overlying fascia lata and the skin.

Contents (Fig. 22.18)

These are as follows:

1. Femoral artery.
2. Femoral vein.
3. Saphenous nerve.
4. Nerve to vastus medialis.

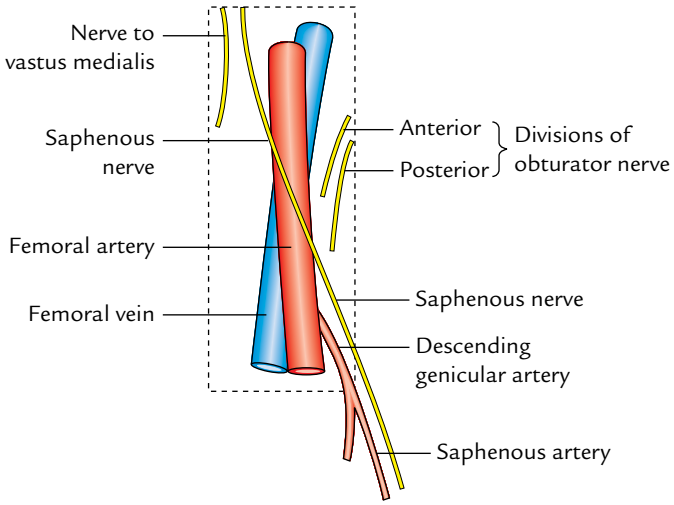


Fig. 22.18 Contents of the adductor canal.

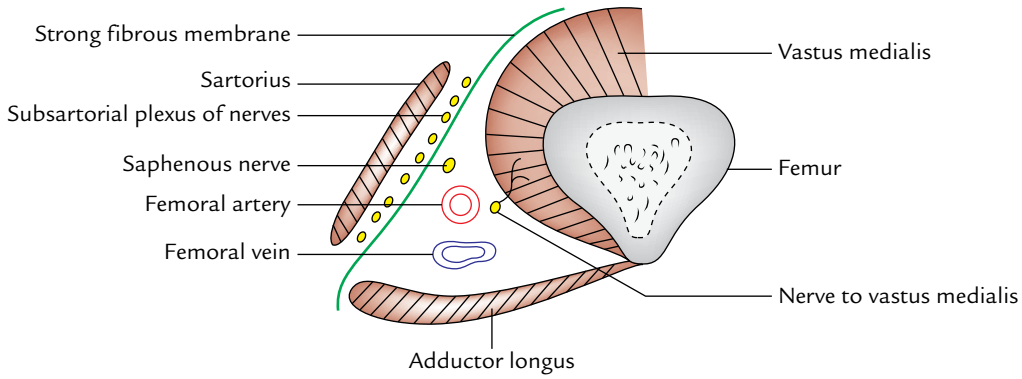


Fig. 22.17 Boundaries and contents of the adductor canal as seen in the transverse section through middle one-third of the thigh.

5. Anterior and posterior divisions of the obturator nerve (occasional).
6. Descending genicular artery, a branch of the femoral artery.

A brief account of contents is given in the following text.

The **femoral artery** enters the canal at the apex of the femoral triangle, traverses the whole length of the adductor canal, and leaves it by passing through the tendinous opening in the adductor magnus (adductor hiatus). Within the canal it gives off muscular branches and a descending genicular branch. The descending genicular artery arises before the femoral artery leaves the canal.

The **femoral vein** lies posterior to the femoral artery in the upper part and lateral to the artery in the lower part.

The **saphenous nerve** (longest cutaneous nerve of the body) crosses the femoral artery anteriorly from lateral to medial side. It leaves the canal by piercing the fibrous roof.

Just before leaving the canal it gives off **infrapatellar branch** which pierces the sartorius, and joins the patellar plexus and supplies the prepatellar skin.

The **nerve to vastus medialis** (thickest muscular branch of the femoral nerve) lies lateral to the femoral artery, and enters the vastus medialis in the upper part of the canal.

The **posterior division of obturator nerve** runs on the anterior surface of the adductor magnus and ends by supplying the knee joint.

N.B. The spiral course of the femoral vein and saphenous nerve with respect to the femoral artery in the adductor canal is due to medial rotation of the lower limb during its development.

Clinical correlation

Clinical significance of adductor canal: The femoral artery is exposed and ligated in the adductor canal during surgery for **aneurysm of the popliteal artery**. This is because the artery at this side is healthy and does not tear when tied, which may occur if the artery is tied in the popliteal fossa immediately above the popliteal aneurysm. This procedure was first performed by a famous surgeon, Dr John Hunter. The adductor canal was named after his name as **Hunter's canal**.

N.B. After ligation of the femoral artery in the adductor canal, the collateral circulation is established through arterial anastomosis around the knee joint.

FEMORAL NERVE (Fig. 22.19)

It is the chief nerve of the anterior compartment of the thigh. It is the largest branch of the lumbar plexus and arises from the dorsal divisions of the anterior primary rami of L2, L3, L4 nerves. It enters the thigh posterior to the inguinal ligament just lateral to the femoral sheath. About 2 cm below the inguinal ligament it divides into **anterior** and **posterior divisions** which are separated by the lateral circumflex femoral artery.

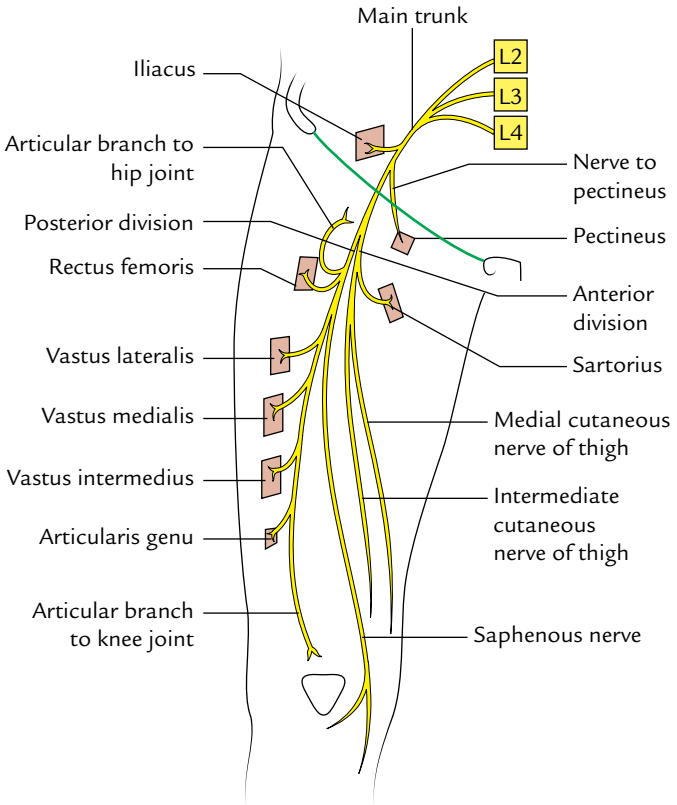


Fig. 22.19 Course, branches, and distribution of the femoral nerve.

Anterior division: It gives off two cutaneous branches and one muscular branch:

(a) The *cutaneous nerves* are: (a) medial cutaneous nerve of the thigh and (b) intermediate cutaneous nerve of the thigh.
(b) The *muscular branch* supplies the sartorius.

Posterior division: It gives off one cutaneous branch, the saphenous nerve, and four muscular branches to supply the quadriceps femoris.

The femoral nerve is described in detail on p. 468.

Clinical correlation

Effect of injury to femoral nerve: If femoral nerve is injured (very rare) by penetrating wound in the groin, the effects will be as follows:

Motor: Paralysis of the quadriceps femoris.
Sensory: Loss of sensations on the anterior and medial aspects of the thigh.

FEMORAL ARTERY

It is the chief artery of the lower limb. It is the continuation of external iliac artery and enters the femoral triangle behind the inguinal ligament at the midinguinal point. It runs downward and medially successively through the femoral triangle and adductor canal. At the lower end of the adductor canal (i.e., at the junction of middle one-third and lower one-third of the thigh), it leaves the thigh through the adductor hiatus (a tendinous opening in the adductor magnus) to enter the popliteal fossa where it continues as the **popliteal artery** (Fig. 22.20).

Surface Marking (Fig. 22.21)

When the thigh is in a position of slight flexion, abduction, and lateral rotation, the upper two-third of a line drawn from the midinguinal point to the adductor tubercle represents the femoral artery.

Branches (Fig. 22.22)

In the femoral triangle:

1. *Three superficial branches:* Superficial epigastric artery, superficial external pudendal artery, and superficial circumflex iliac artery.
2. *Three deep branches:* Profunda femoris artery, deep external pudendal artery, and muscular branches.

In the adductor canal:

1. Muscular branches.
2. Descending genicular artery.

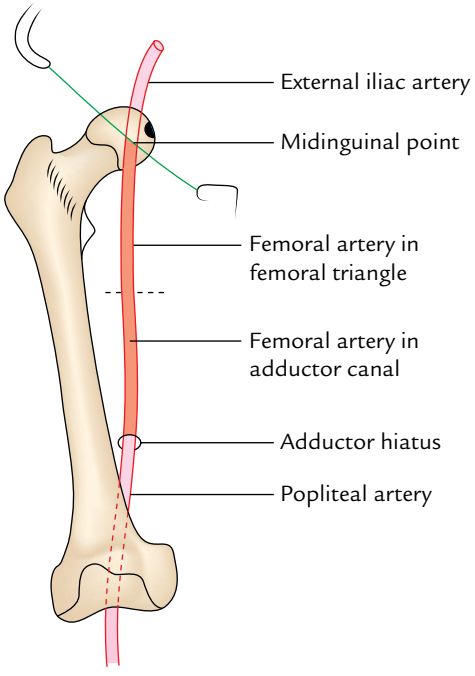


Fig. 22.20 Course and extent of the femoral artery.

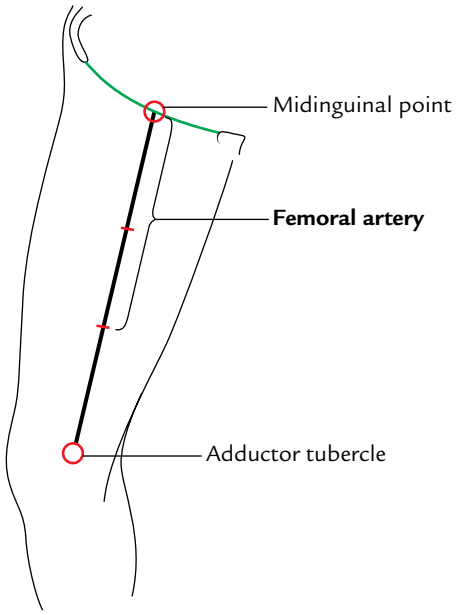


Fig. 22.21 Surface marking of the femoral artery.

The three superficial branches are already described on page 31 and profunda femoris artery is described on page 350.

The **deep external pudendal artery** arises just below the superficial external pudendal artery and passes medially deep to the spermatic cord or round ligament of the uterus and supplies the scrotum or labium majus.

The **descending genicular artery** leaves the canal by descending within the substance of vastus medialis and divides into articular and saphenous branches. The saphenous branch, also called **saphenous artery**, accompanies the saphenous nerve as it emerges through the roof of adductor canal.

N.B. *Alternative names of femoral artery:* Some vascular surgeons, call the initial part of femoral artery, proximal to the origin of profunda femoris artery as **'common femoral artery'** and its continuation distally as **'superficial femoral artery.'**

Clinical correlation

Compression, palpation and cannulation of femoral artery:

- The femoral artery can be compressed against the femoral head at the midinguinal point to control the bleeding in the distal part of the limb.
- The pulsations of the femoral artery are felt by the clinicians in the femoral triangle just below the midinguinal point.
- Since the femoral artery is quite superficial in the femoral triangle, it is the preferred artery for cannulation and injecting dye to perform procedures like angiography. It is also the preferred vessel for performing the coronary angiography and angioplasty.

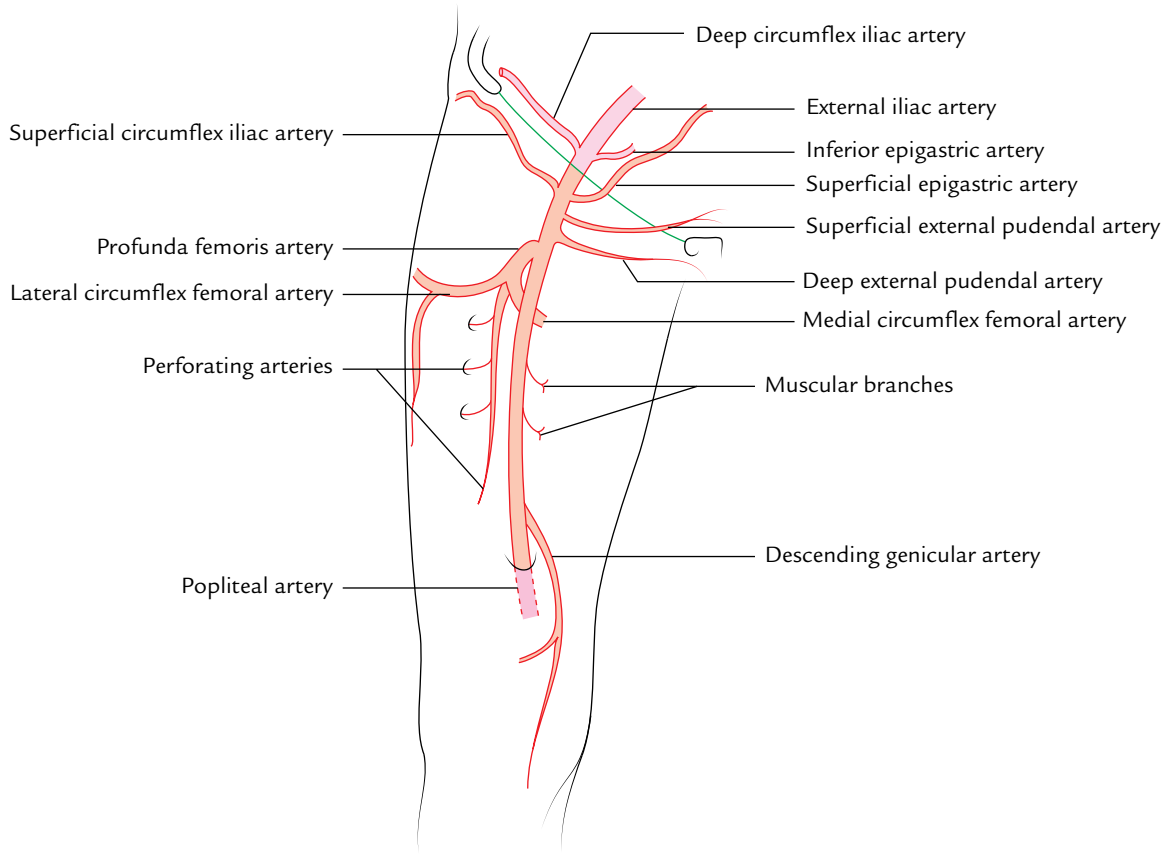


Fig. 22.22 Branches of the femoral artery.

FEMORAL VEIN

The femoral vein is the upward continuation of the popliteal vein at the adductor hiatus. Thus, it begins at the lower end of the adductor canal, ascends in adductor canal, and enters the femoral triangle, where after traversing the intermediate compartment of the femoral sheath it continues as the external iliac vein behind the inguinal ligament medial to the midinguinal point.

Tributaries

1. Great saphenous vein (longest tributary).
2. Profunda femoris vein.
3. Medial and lateral circumflex femoral veins.
4. Deep external pudendal vein.
5. Direct muscular tributaries.

Clinical correlation

Puncture and cannulation of femoral vein:

- The femoral vein is the preferred vein for intravenous infusions in infants and children and in patients with peripheral circulatory failure.
- The femoral vein is also used for inserting a catheter into the right atrial chamber and right ventricle to collect blood sample or to record pressure or to reach the pulmonary arteries.

Route to right ventricle:

Femoral vein → external iliac vein → common iliac vein → inferior vena cava → right atrium → right ventricle.

Golden Facts to Remember

► Largest muscle of the extensor compartment of the thigh	Quadriceps femoris
► Longest muscle in the body	Sartorius
► Tailor's muscle	Sartorius
► Largest branch of the femoral artery	Profunda femoris artery
► All the parts of quadriceps act on the knee joint only *except*	Rectus femoris which acts on both hip joint and knee joint
► Kicking muscle	Rectus femoris
► Commonest site of intramuscular injection in the thigh	Anterolateral aspect of the thigh into the vastus lateralis muscle
► Longest cutaneous nerve in the body	Saphenous nerve

Clinical Case Study



A 65-year-old woman visited the emergency department of a hospital and complained of abdominal pain and repeated vomiting. On questioning the patient told that the pain was severe and colicky in nature and most severe in the region of umbilicus. On examination, the abdomen was distended and excessive loud bowel sound could be heard with the stethoscope. A small lemon-sized swelling was also noticed on the upper medial aspect of the right thigh. The swelling was located inferolateral to the pubic tubercle. The patient told she is having this swelling for the last 3 or 4 months. A diagnosis of "**acute intestinal obstruction secondary to the right femoral hernia**" was made.

Questions

1. Name the hernia which is located below and lateral to the pubic tubercle. Name the passage through which the hernia has entered into the thigh.
2. What is the cause of strangulation of femoral hernia?
3. What are the boundaries of femoral ring?
4. What is the direction of femoral hernia and its importance to the surgeon?
5. Give the cause of intestinal obstruction.
6. Give the anatomical basis of pain in the umbilical region.

Answers

1. Femoral hernia: femoral ring and femoral canal.
2. Unyielding nature of the femoral ring.
3. See page 48.
4. The direction of femoral hernia is downward, forward, and upward (see pages 48–49). During manual reduction of femoral hernia, the surgeon reverses the order by pushing the hernia downward, backward, and upward.
5. Venous congestion followed by an arterial occlusion of the intestinal loop due to strangulation of hernia.
6. A loop of the small intestine was pushed into the hernial sac and pain from the small intestine was referred in the region of umbilicus.

CHAPTER

23 Medial Side of the Thigh

The adductor compartment of the thigh is well developed. Its counterpart in the arm has undergone degeneration during the course of evolution and is represented only by a weak coracobrachialis muscle of the flexor compartment of the arm.

BOUNDARIES

The medial compartment of the thigh is bounded:

Anteriorly by the anterior intermuscular septum which separates it from the anterior (extensor) compartment of the thigh.

Posteriorly by an ill-defined posterior intermuscular septum which separates it from posterior (flexor) compartment of the thigh. The posterior intermuscular septum is ill-defined and incomplete due to the presence of a composite muscle, the *adductor magnus*, consisting of two components—adductor and flexor (hamstring) belonging to adductor and flexor compartments of the thigh, respectively.

CONTENTS

- **Muscles:** Adductor longus, adductor brevis, adductor magnus, gracilis, pectineus, and obturator externus.
- **Nerve:** Obturator nerve.
- **Arteries:** Profunda femoris artery and obturator artery.

N.B. The *obturator externus* lies deep in this region and is functionally related to the gluteal region. It rotates the thigh laterally; the chief function of others is adduction of the thigh.

MUSCLES

The muscles of the medial compartment of the thigh are arranged into three layers. From superficial to deep these are (Fig. 23.1):

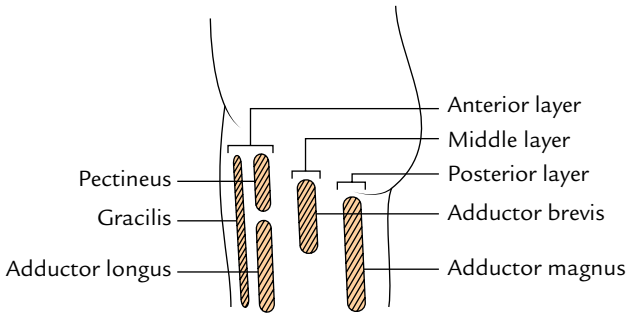


Fig. 23.1 Schematic diagram to show three layers of the adductor muscles of the thigh.

1. **Anterior (first) layer** consists of pectineus, adductor longus, and gracilis.
2. **Middle (second) layer** consists of adductor brevis.
3. **Posterior (third) layer** consists of adductor magnus.

The bony attachments of muscles of medial compartment of the thigh are shown in Figure 23.2.

N.B.

- All the muscles of the adductor compartment of the thigh are supplied by the obturator nerve.
- Two muscles of the adductor compartment of thigh, *viz.*, pectineus and adductor magnus, are composite muscles, hence they have dual innervations. The pectineus is supplied by the femoral and obturator nerves while the adductor magnus is supplied by the obturator nerve (posterior division) and tibial part of sciatic nerve.

Pectineus (Fig. 23.3)

It is a flat quadrilateral muscle which lies in the floor of femoral triangle between psoas major laterally and adductor longus medially.

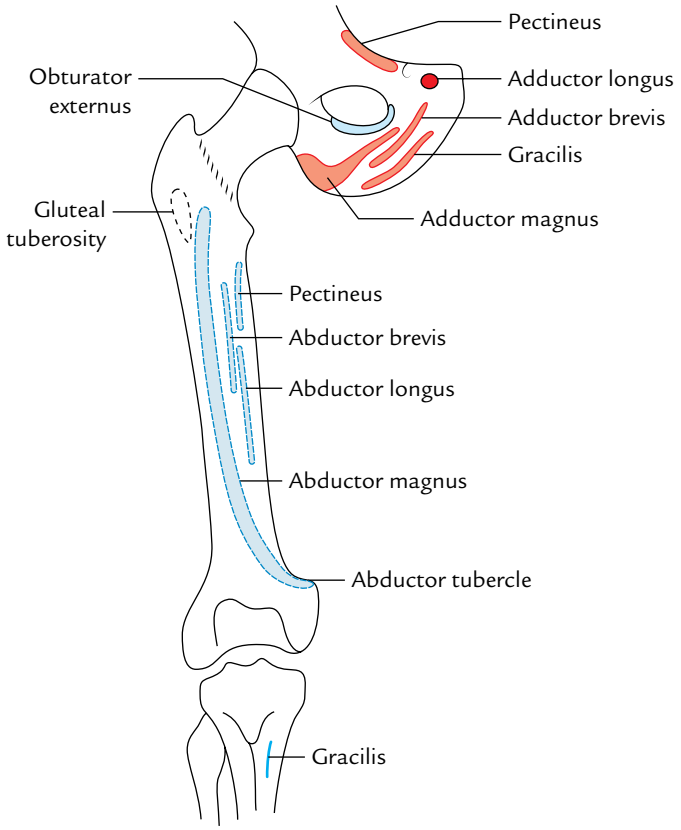


Fig. 23.2 Bony attachments of the muscles of medial compartment of the thigh. Origin is shown by the red colour and insertion is shown by the blue colour.

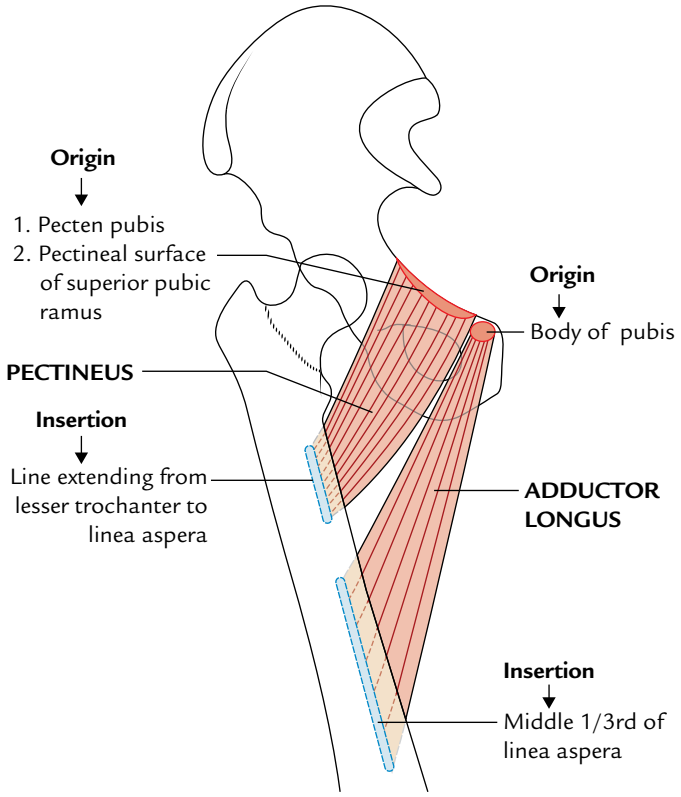


Fig. 23.3 Origin and insertion of the pectineus and adductor longus.

Origin
It arises from the pecten pubis, upper half of the pectineal surface of the superior ramus of pubis and pectineal fascia.

Insertion
The muscle slopes downward, backward, and laterally to be inserted along a line extending from the lesser trochanter to the upper end of the linea aspera.

Nerve supply
It is by two nerves. The anterior fibres are supplied by the *femoral nerve* and the posterior fibres by the *anterior division of the obturator nerve.*

Action
It adducts the thigh at the hip joint.

Adductor Longus (Fig. 23.3)
It is a triangular muscle which lies in the same plane as pectineus. It forms the floor and medial boundary of the femoral triangle.

Origin
It arises by a rounded tendon from the front of the body of pubis in the angle between the pubic crest and the pubic symphysis.

Insertion
The muscles slope downward, backward, and laterally as a broad fleshy belly to be inserted into the middle one-third of the linea aspera.

Nerve supply
The nerve supply is by the anterior division of the obturator nerve.

Action
It is a powerful adductor and medial rotator of the thigh at the hip joint.

N.B. In *horse riders*, the rounded tendon of adductor longus often gets calcified due to its friction with the horseback. This calcified tendon is called **rider's bone**. Some authorities are of view that rider's bone is a sesamoid bone, which occasionally develops in the tendon of adductor longus.

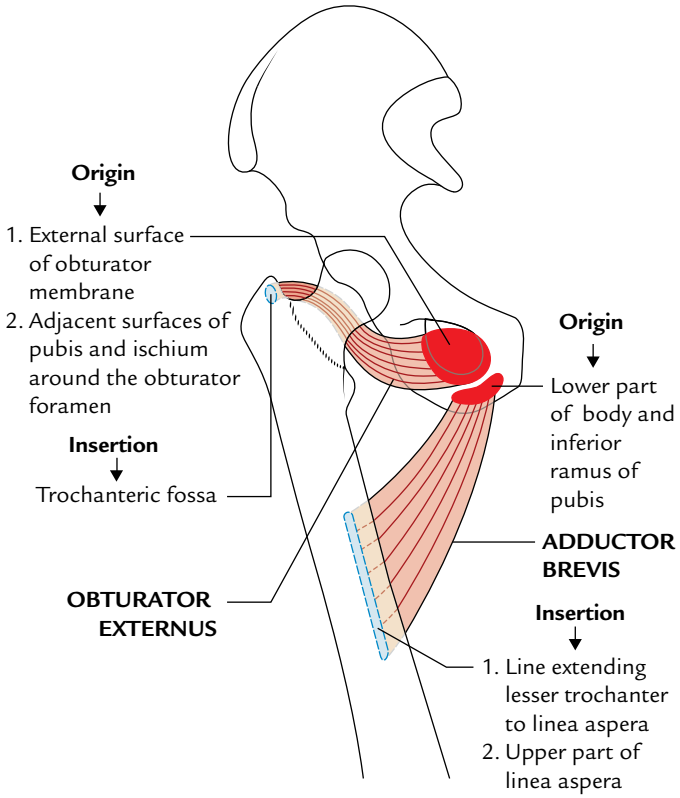


Fig. 23.4 Origin and insertion of the adductor brevis and the obturator externus.

Adductor Brevis (Fig. 23.4)

It is a triangular muscle which sometimes peeps in the floor of the femoral triangle between pectineus and adductor longus.

Origin

It arises from the anterior surface of the body of pubis and the outer surface of the inferior ramus of pubis in the interval between the gracilis and the obturator externus.

Insertion

The muscle passes downward, backward, and laterally to be inserted along a line extending from the lesser trochanter to the upper part of linea aspera, behind the pectineus and upper part of the adductor longus.

Nerve supply

It is usually supplied by the anterior division of the obturator nerve but occasionally it may also be supplied by the posterior division of the obturator nerve.

Action

It adducts the thigh at the hip joint.

Adductor Magnus (Fig. 23.5)

It is a large composite muscle consisting of two parts: adductor and hamstring.

Origin

It arises from:

(a) The **adductor part** arises from the outer surface of ischiopubic ramus, mainly from ramus of ischium.
(b) The **hamstring part** arises from the inferolateral part of the ischial tuberosity.

Insertion

(a) The fibres of *adductor part* pass obliquely downward, backward, and laterally to be inserted in linear fashion into the medial margin of gluteal tuberosity, medial lip of linea aspera, and upper part of medial supracondylar line up to the adductor hiatus.
(b) The fibres of the *hamstring part* pass vertically downward to be inserted into the *adductor tubercle* by a rounded tendon, which sends a fibrous expansion to the lower part of medial supracondylar line below the hiatus.

Nerve supply

It is by:

1. *Adductor part* by the posterior division of the obturator nerve.
2. *Hamstring part* by the tibial part of the sciatic nerve.

Actions

(a) Adduction and medial rotation of the thigh at the hip joint.
(b) Hamstring part is a weak extensor of the thigh at the hip joint.

Gracilis (Fig. 23.5)

It is a long slender muscle (G. **gracilis** = slender) which lies on the medial side of the other muscles of the adductor compartment. *It is the only muscle of the adductor compartment which is not attached to the femur.*

Origin

It arises from the medial margin of the lower half of the body pubis and adjoining anterior part of the inferior pubic ramus.

Insertion

It passes downward vertically on the medial side of thigh to the upper part of the medial surface of tibia, where it is inserted between the insertions of sartorius (in front) and semitendinosus (behind).

Nerve supply

By the anterior division of obturator nerve.

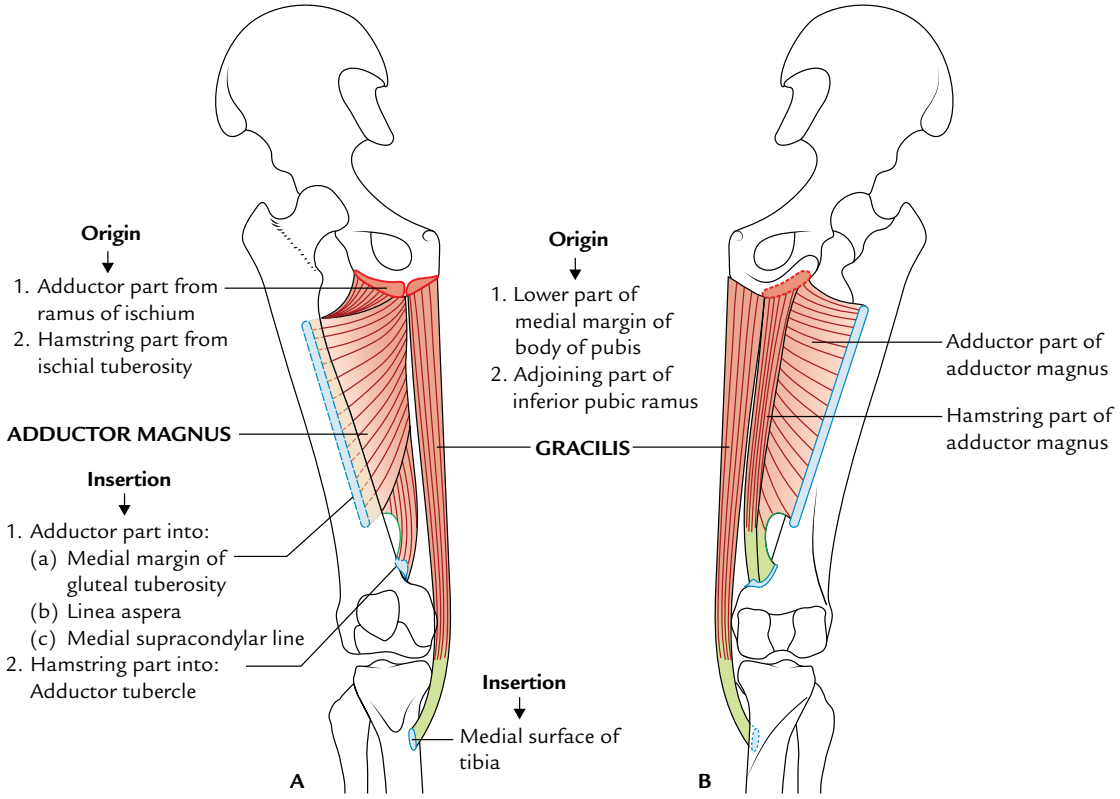


Fig. 23.5 Origin and insertion of the adductor magnus and gracilis: A, anterior view; B, posterior view.

Actions

It adducts the thigh. In addition, it is a flexor of the knee joint and a medial rotator of the leg.

N.B. The adductor muscles of the thigh have been known collectively as custodes virginitatis (i.e., **custodian of virginity**).

The origin, insertion, nerve supply, and actions of the adductor muscles of the thigh are summarized in Table 23.1.

Obturator Externus (Fig. 23.4)

It is a fan-shaped muscle lying above and lateral to the pectineus. Strictly speaking, it is not a muscle of adductor compartment but described here because of its close relationship with the structures of the adductor compartment of the thigh.

Origin

It arises from the outer surface of the anterior half of the obturator membrane and adjoining anterior and inferior margins of the obturator foramen.

Insertion

The fibres converge backward and laterally to form a tendon, which spirals upward over the inferior and posterior surfaces of the neck of femur to be inserted into the **trochanteric fossa** of greater trochanter of the femur.

Nerve supply

By the posterior division of the obturator nerve.

Actions

It is the lateral rotator of the thigh.

OBTURATOR NERVE (Fig. 23.6)

It is the chief nerve of the adductor compartment of the thigh. It arises from the lumbar plexus in the abdomen. It is formed by the ventral division of the anterior primary rami of L2, L3, L4 spinal nerves. It enters the thigh by passing through the obturator canal.

Course and Distribution (Fig. 23.6A)

While passing through the obturator canal the obturator nerve divides into anterior and posterior divisions.

1. The *anterior division* passes downwards into the thigh in front of the obturator externus. It then descends behind the pectineus and the adductor longus, and in front of the adductor brevis. The anterior division supplies the following muscles:
   (a) Pectineus.

Table 23.1 Origin, insertion, nerve supply, and actions of muscles of the adductor compartment of the thigh

Muscle	Origin	Insertion	Nerve supply	Actions
Pectineus (flat, quadrilateral muscle, composite muscle)	• Pecten pubis • Upper half of the pectineal surface of the superior ramus of the pubis • Fascia covering the pectineus	Line extending from the lesser trochanter to the linea aspera	Femoral nerve and anterior division of the obturator nerve	Adduction of the thigh
Adductor longus (triangular muscle, forming the medial part of the floor of the femoral triangle)	Body of the pubis in the angle between the pubic crest and the pubic symphysis	Middle 1/3rd of linea aspera	Obturator nerve (anterior division)	Adduction and medial rotation of thigh
Adductor brevis (triangular muscle lying behind the pectineus and adductor longus)	• Anterior surface of the body of pubis • Outer surface of the inferior ramus of the pubis between the gracilis and the obturator externus	Line extending from the lesser trochanter to the linea aspera, and upper part of the linea aspera itself just lateral to pectineus	Obturator nerve (anterior and posterior divisions)	Adduction of thigh
Adductor magnus (large composite)	(a) Hamstring part: inferolateral part of the ischial tuberosity (b) Adductor part: outer part of the ischiopubic ramus	• Medial margin of gluteal tuberosity • Linea aspera • Medial supracondylar line • Adductor tubercle	• Adductor part by the obturator nerve (posterior division) • Hamstring part by the tibial part of sciatic nerve	• Adduction and medial rotation of thigh • Weak extension of hip joint
Gracilis	Medial margin of the lower half of the body of the pubis: adjoining anterior part of inferior ramus of the pubis	Upper part of the medial surface of tibia between the insertions of sartorius (in front) and the semitendinosus (behind)	Obturator nerve (anterior division)	• Adduction of thigh • Flexion and medial rotation of leg

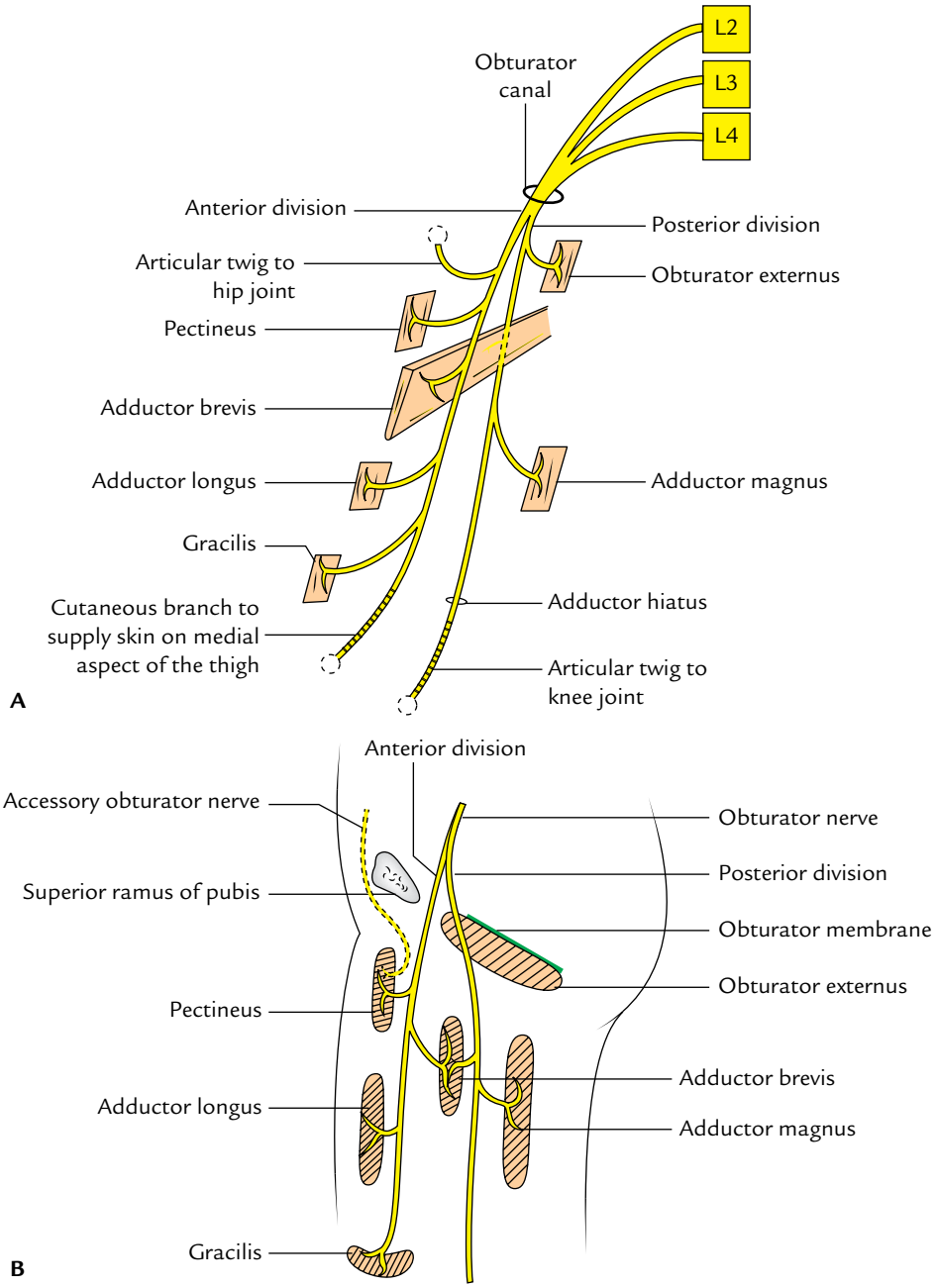


Fig. 23.6 Obturator nerve: **A**, course and distribution; **B**, accessory obturator nerve.

(b) Adductor longus.
(c) Gracilis.
(d) Adductor brevis.

The anterior division also gives an articular twig to the hip joint. Distal to the adductor longus, it enters the adductor canal where it provides a twig to the subsartorial plexus of nerves and terminates by supplying the femoral artery in the adductor canal.

2. The *posterior division* enters the thigh by piercing the anterior part of the obturator externus muscle which it supplies. It then descends behind the adductor brevis and in front of the adductor magnus. The posterior division supplies the following muscles:
(a) Obturator externus.
(b) Adductor magnus.
(c) Adductor brevis.

Its terminal part forms an articular branch called **genicular branch,** which pierces the adductor magnus or passes through hiatus for femoral vessels to reach the popliteal fossa where it runs along the popliteal vessels and pierces the oblique popliteal ligament to supply the knee joint.

Clinical correlation

- **Adductor spasm of the thigh:** Spasm of the adductors of the thigh in spastic paraplegia may be relieved by surgical division of the obturator nerve.
- **Referred pain:** In diseases of the knee joint, the pain may be referred to the hip joint along the obturator nerve (L2, L3, and L4) because it supplies both these joints.

N.B. *Accessory obturator nerve (Fig. 23.6B):* In about 30% individuals, accessory obturator nerve arises from the lumbar plexus. It is formed by the ventral divisions of the anterior primary rami of L3, L4 spinal nerves. It crosses the superior ramus of pubis deep to pectineus, which it supplies. It gives an articular twig to the hip joint and terminates by communicating with the anterior division of the obturator nerve. If small, it supplies only the pectineus muscle (Fig. 23.6A).

ARTERIES

PROFUNDA FEMORIS ARTERY (DEEP FEMORAL ARTERY; Fig. 23.7)

It is the largest branch of the femoral artery and is the chief source of blood supply to the muscles of all the three compartments of the thigh (Fig. 23.7). It arises from the lateral side of the femoral artery in the femoral triangle about 4 cm below the inguinal ligament, behind the femoral vessels, giving off medial and lateral circumflex arteries. It then passes posteriorly between pectineus and adductor longus then descends close to femur successively between adductor longus and adductor brevis, between adductor longus and adductor magnus. Here it gives off **first three perforating arteries**. Its terminal part pierces the adductor magnus as the **fourth perforating artery** to reach the back of the leg.

Branches

The branches of profunda femoris artery are summarized as follows:

- Muscular branches.
- Medial circumflex femoral artery.
- Lateral circumflex femoral artery.
- Four perforating arteries.

1. The **muscular branches** as the name implies, supply the muscles.
2. The **medial circumflex femoral artery** leaves the femoral triangle by passing posteriorly, between the pectineus and the psoas major muscles.

Then it passes successively between the obturator externus and adductor brevis and between the

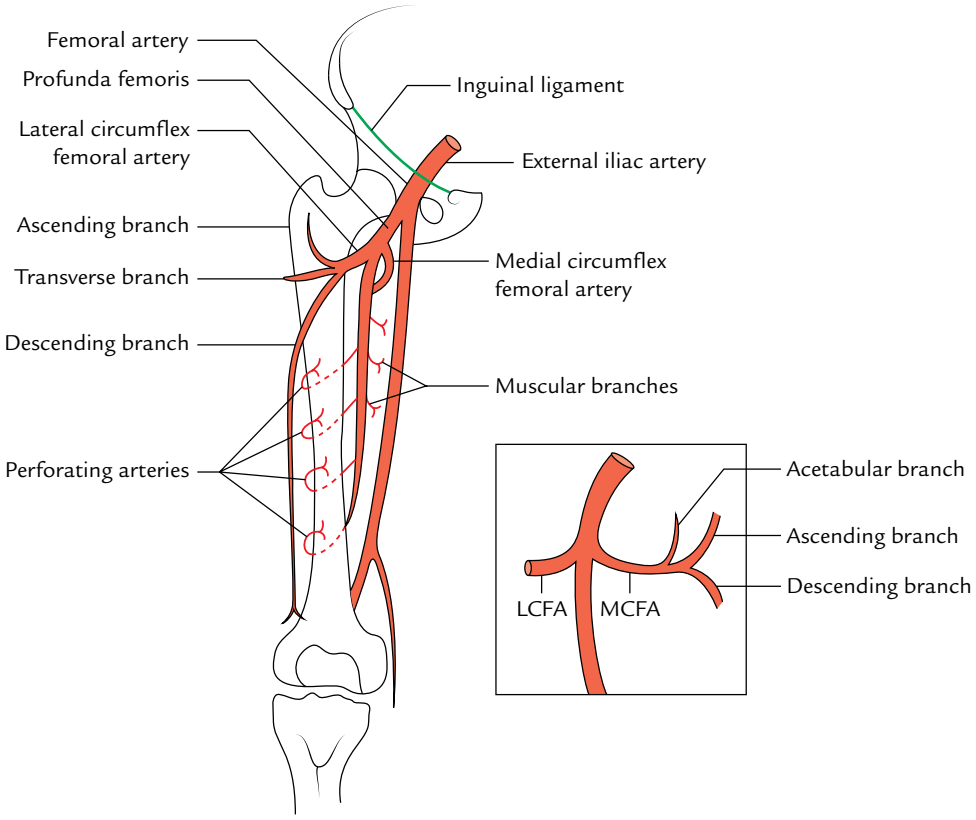


Fig. 23.7 Profunda (deep femoral) femoris artery: Figure in the inset shows branches of the medial circumflex femoral artery (LCFA = lateral circumflex femoral artery, MCFA = medial circumflex femoral artery).

quadratus femoris and upper border of the adductor magnus. Here is gives off *transverse* and *ascending branches*. The *transverse branch* takes part in the formation of *cruciate anastomosis*. The *ascending branch* passes to the trochanteric fossa and takes part in the formation of *trochanteric anastomosis*.

The *acetabular branch* of the medial circumflex femoral artery arises before the terminal branches and enters the acetabulum through acetabular notch deep to transverse acetabular ligament.

- The *posterior retinacular branches* of medial circumflex femoral artery pass through capsule of hip joint to supply the head and neck of the femur.

N.B. The medial circumflex femoral artery is especially important because it supplies most of the blood to the head and neck of femur by its posterior retinacular branches.

3. The **lateral circumflex femoral artery** is the largest branch of the profunda femoris artery. It runs laterally between the anterior and posterior divisions of the femoral nerve and divides into ascending, transverse, and descending branches. The *ascending and transverse branches* take part in the cruciate anastomosis on the back of the thigh just below the greater trochanter (Fig. 23.7). The *descending branch* runs down along the anterior border of the vastus lateralis and takes part in the anastomosis around the knee.
4. The **perforating arteries** are four in number. They are numbered from above downward as first, second, third and fourth; the fourth one being the continuation of the profunda femoris artery.

Clinical correlation

Clinical significance of profunda femoris artery: The profunda femoris artery is of great clinical importance because it is deeply located and lies in close proximity to the femoral shaft; hence, it is prone to injury in fracture of femoral shaft. The artery is also liable to injury during surgical procedure of fixing metallic screws in the femur by an orthopedic surgeon.

OBTURATOR ARTERY

It arises from the anterior division of internal iliac artery in the pelvis. It enters the adductor compartment of the thigh through the obturator canal. Just outside the obturator canal, it divides into medial and lateral branches. The lateral branch gives off an **acetabular twig,** which enters the acetabulum through a gap between acetabular notch and transverse acetabular ligament where it supplies acetabular fat and gives off a slender artery to the femoral head along the ligament of the head of femur.

N.B. *Abnormal obturator artery:* Normally a pubic branch of the obturator artery anastomoses with the pubic branch of the inferior epigastric artery. Sometimes this anastomosis is so large and well developed that the obturator artery appears to be a branch of the inferior epigastric artery.

Golden Facts to Remember

► Chief artery for the muscles of thigh	Profunda femoris artery
► Largest branch of the profunda femoris artery	Lateral circumflex femoral artery
► Largest muscle of the adductor compartment of thigh	Adductor magnus
► All the muscles of the adductor compartment of thigh are inserted into the femur *except*	Gracilis (which is inserted into the tibia)
► Muscles of the adductor compartment of thigh supplied by two different nerves	(a) Pectineus (by femoral and obturator nerve) (b) Adductor magnus (by obturator and tibial part of sciatic nerve)
► Custodes virginitatis	Adductor muscles of the thigh
► Rider's bone	Calcified tendon of adductor longus/sesamoid bone in the tendon of adductor longus

Clinical Case Study



A 55-year-old businessman came to the hospital and complained that he could walk only for about 50–60 yards before a cramp-like pain in his right leg forces him to take rest. After taking rest, pain disappears and again he could walk for the same distance. On examination, the femoral pulses were normal in both the limbs. The arteriography revealed a blockage of the right femoral artery just above the level of adductor tubercle. A diagnosis of "**intermittent claudication**" was made.

Questions

1. Which is the chief artery of the lower limb?
2. If the femoral artery is blocked above the adductor tubercle, which artery plays a major role in establishing the collateral circulation in the lower limb?
3. What is the preferred site of ligation of femoral artery?
4. Which artery is likely to be injured in fracture shaft of femur and why?
5. What will be the effects if profunda femoris artery is ruptured?

Answers

1. Femoral artery.
2. Profunda femoris artery.
3. Proximal to the origin of profunda femoris artery.
4. Profunda femoris artery, as this artery lies in close proximity to the shaft of femur.
5. Severe bleeding inside the thigh muscles, which may cause "*thigh compartment syndrome*."

CHAPTER

24 Gluteal Region

The gluteal region overlies the back and side, of the lateral half of the pelvis. It extends from the iliac crest superiorly to the gluteal fold (at the lower limit of the prominence of buttock) inferiorly. Medially it extends up to mid-dorsal line and natal cleft, and laterally up to an imaginary line joining the anterior superior iliac spine to the anterior edge of the greater trochanter. The gluteal region is one of the commonest sites of intramuscular injections.

SURFACE LANDMARKS (Fig. 24.1)

- **Buttock:** It is a round bulge on the posterior aspect of the gluteal region.
- **Gluteal fold:** It is a transverse skin crease, which forms the lower limit of the gluteal region. It is produced by a linear adherence of the skin to the deep fascia, obliquely across the inferior border of gluteus maximus.
- **Natal cleft:** It is a midline cleft between the two buttocks, which begins at the level of third spine of the sacrum and deepens inferiorly with lower sacral spines and coccyx lying in its floor.
- **Coccyx:** It lies just behind the anal orifice and can be identified by its relative mobility under pressure.

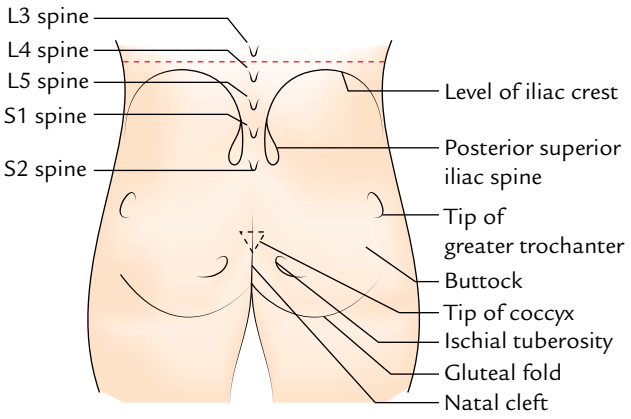


Fig. 24.1 Surface landmarks of the gluteal region.

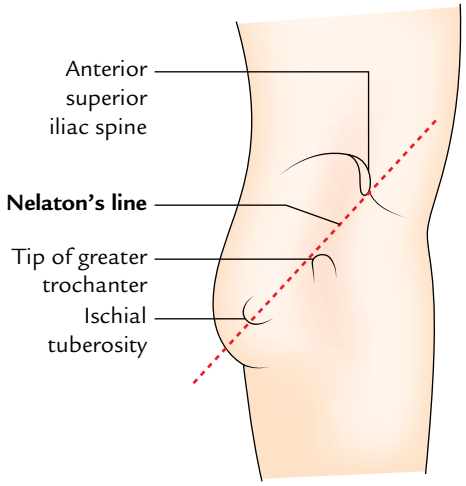


Fig. 24.2 Nelaton's line.

- **Posterior superior iliac spine (PSIS):** It lies in a skin dimple at the level of second sacral (S2) spine.
- **Ischial tuberosity:** It is a rounded bony mass on which one sits. It lies below the PSIS in the same vertical plane at a lower level than the tip of coccyx. It can be felt by pressing your fingers upward into the medial part of the gluteal fold. It is 5 cm above the gluteal fold and about same distance from the midline.
- **Tip of greater trochanter:** It lies just in front of hollow on the side of hip, about one hand's breadth below the tubercle of iliac crest.
- **Iliac crest:** It can be felt as a curved bony ridge in a groove at the lower limit/margin of the waist. The highest point of iliac crest corresponds to the interval between the spines of L3 and L4 vertebrae.

N.B.

- *Nelaton's line (Fig. 24.2):* It is the line joining the anterior superior iliac spine and the most prominent point of ischial tuberosity. It crosses the tip of greater trochanter.

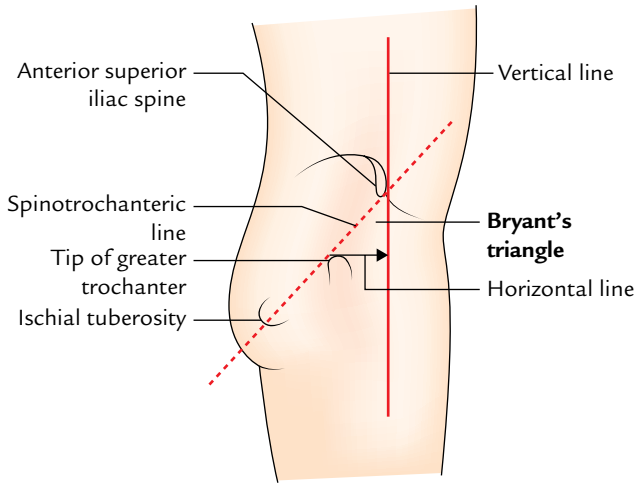


Fig. 24.3 Bryant's triangle.

- *Bryant's triangle (Fig. 24.3):* With the patient in supine position, first draw a vertical line passing downwards from the anterior superior iliac spine (ASIS) and now draw a line extending from ASIS to the tip of greater trochanter (spinotrochanteric line). Lastly draw a horizontal line from the tip of greater trochanter to the first line. The triangle thus formed is called **Bryant's triangle**.

SUPERFICIAL FASCIA

The superficial fascia in the gluteal region is thick and contains abundant subcutaneous fat particularly in adult females, which is responsible for a characteristic round contour of the buttock in them.

CUTANEOUS NERVES (Fig. 24.4)

The cutaneous nerves of the gluteal region are derived from several sources, and converge in this region from all the directions (Fig. 24.4). The cutaneous innervation of the gluteal region is divided into four quadrants—upper anterior, upper posterior, lower anterior, and lower posterior.

1. **Upper anterior quadrant** is supplied by the:
   - lateral cutaneous branch of subcostal nerve (T12), and
   - lateral cutaneous branch of iliohypogastric nerve (L1).
2. **Upper posterior quadrant** is supplied by the:
   - cutaneous branches from dorsal rami of upper three lumbar nerves (L1, L2, L3) and upper three sacral nerves (S1, S2, S3).
3. **Lower anterior quadrant** is supplied by the:
   - posterior division of lateral cutaneous nerves of the thigh (L2, L3).
4. **Lower posterior quadrant** is supplied by the:
   - posterior cutaneous nerves of the thigh (S1, S2, S3), and
   - perforating cutaneous nerves (S2, S3).

CUTANEOUS ARTERIES AND LYMPH VESSELS

The **cutaneous arteries** supplying the gluteal region are derived from the superior and inferior gluteal arteries.

The **lymph vessels** from the gluteal region drain into the lateral group of superficial inguinal lymph nodes.

DEEP FASCIA (Fig. 24.5)

The deep fascia of the gluteal region is attached above to the iliac crest and behind to the sacrum. It splits twice along the iliac crest, first time to enclose the tensor fasciae latae and second time to enclose the gluteus maximus. Between tensor fasciae latae and gluteus maximus is a thick fascial sheet called **gluteal aponeurosis** which covers the gluteus medius.

1. The layers which enclose the gluteus maximus are interconnected by numerous fibrous septa which traverse through this muscle and divide it into numerous

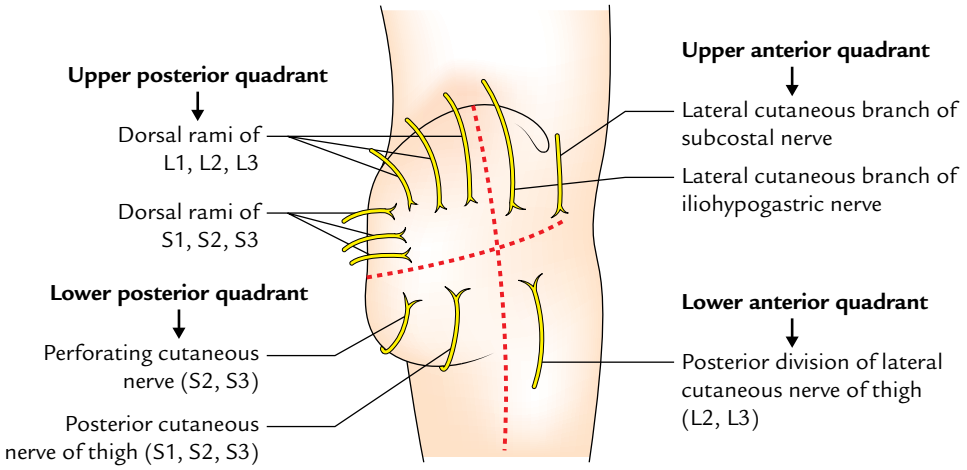


Fig. 24.4 Cutaneous nerves of the gluteal region.

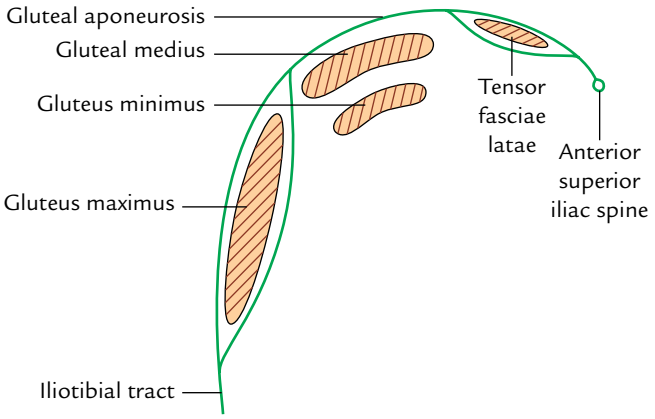


Fig. 24.5 Schematic diagram to show the deep fascia of the gluteal region.

discrete fasciculi. *Note:* Gluteus maximus is the most fasciculated muscle in the body.

2. Traced laterally, the deep fascia is continuous with the *iliotibial tract.*

GLUTEAL LIGAMENTS (Fig. 24.6)

It is of utmost importance for the students to study two important gluteal ligaments (sacrotuberous and sacrospinous ligaments) in the gluteal region before they proceed to study the deep structures in this region.

Sacrotuberous Ligament

It is broad band of fibrous tissue which extends from sides of the sacrum and coccyx to the medial side of the ischial tuberosity.

Sacrospinous Ligament

It is triangular sheet of fibrous tissue which extends from ischial spine to side of the sacrum and coccyx.

The sacrotuberous and sacrospinous ligaments are described in detail in Chapter 13, p. 204.

The sacrotuberous and sacrospinous ligaments convert the greater sciatic notch and lesser sciatic notch into greater sciatic foramen and lesser sciatic foramen, respectively.

1. The **greater sciatic foramen** is a passageway for structures leaving the pelvis and entering the gluteal region (e.g., sciatic nerve, superior and inferior gluteal vessels, etc.).
2. The **lesser sciatic foramen** is a passageway for structures entering the perineum (e.g., pudendal nerve and artery).

N.B. The greater sciatic foramen is considered as the **"door of the gluteal region"** through which all arteries and nerves enter into the gluteal region from the pelvis.

The **structures passing through the greater sciatic foramen** (Fig. 24.7) are as follows:

1. Piriformis: It emerges from the pelvis and almost completely fills the foramen. It is the key muscle of this region.
2. Structures passing below the piriformis:
   (a) Inferior gluteal nerve and vessels.
   (b) Sciatic nerve (most lateral structure).
   (c) Posterior cutaneous nerve of the thigh.
   (d) Nerve to quadratus femoris.
   (e) Pudendal nerve.
   (f) Internal pudendal vessels.
   (g) Nerve to obturator internus.

The last three structures cross the dorsal aspect of ischial spine and adjoining part of sacrospinous ligament

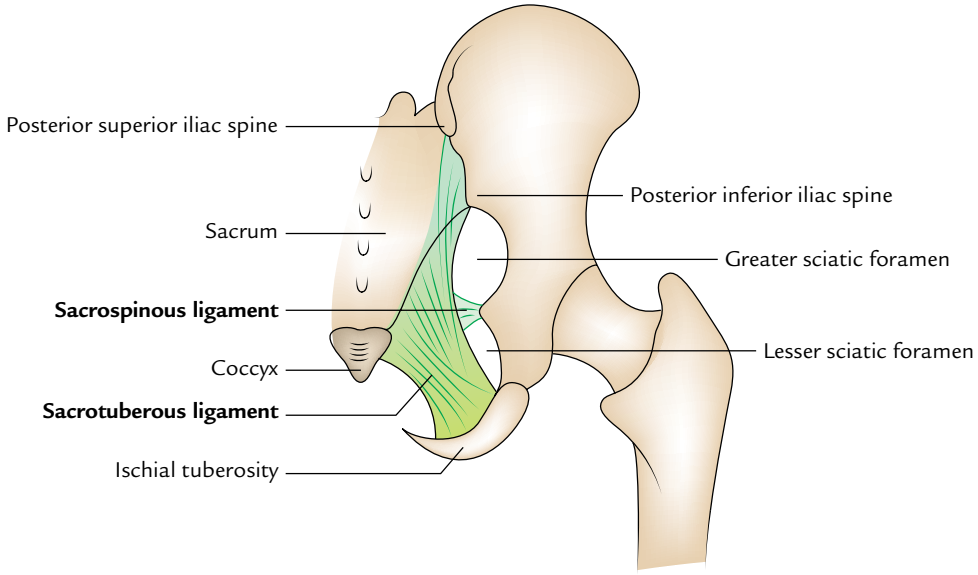


Fig. 24.6 Gluteal ligaments (sacrotuberous and sacrospinous ligaments).

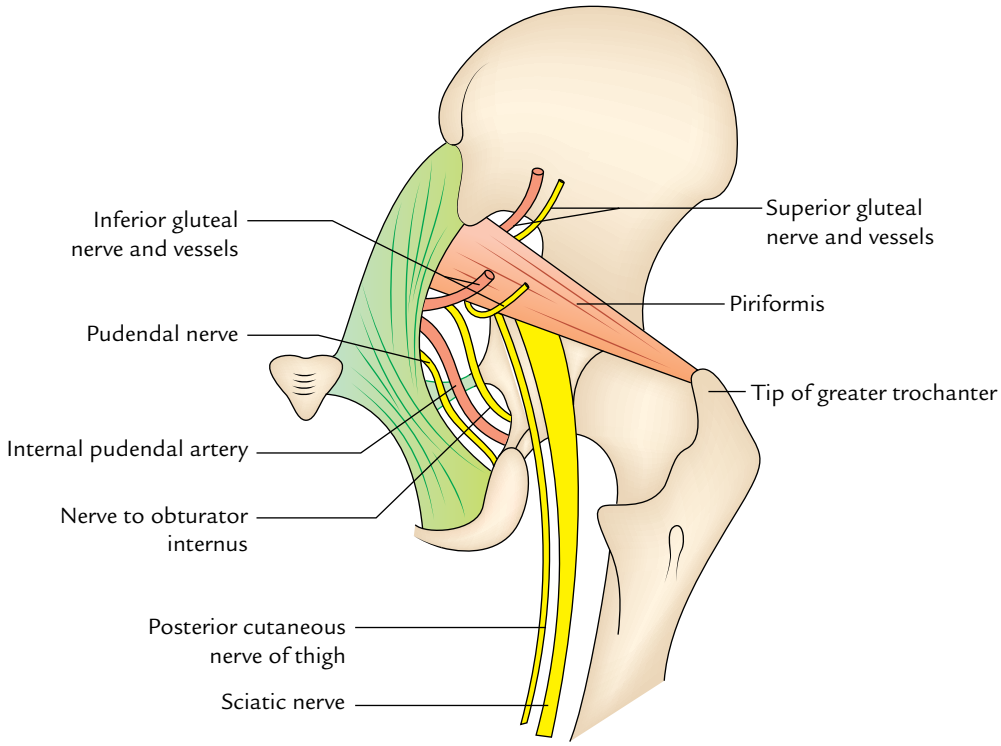


Fig. 24.7 Structures passing through the greater and lesser sciatic foramina.

(with internal pudendal artery between the nerves to obturator internus laterally and pudendal nerve medially), and curves forward to enter the perineum. The pudendal nerve and internal pudendal vessels run in the pudendal canal.

The **structures passing through the lesser sciatic foramen** are as follows:

(a) Tendon of obturator internus.
(b) Nerve to obturator internus.
(c) Internal pudendal vessels.
(d) Pudendal nerve.

MUSCLES OF THE GLUTEAL REGION

The muscles of the gluteal region are divided into two groups—major and minor.

- *Major muscles of the gluteal region:* These are four in number as given below:
  (a) Gluteus maximus.
  (b) Gluteus medius.
  (c) Gluteus minimus.
  (d) Tensor fasciae latae.
- *Minor muscles of the gluteal region:* These are six in number as follows:
  (a) Piriformis.
  (b) Superior and inferior gemelli.
  (c) Obturator internus.
  (d) Quadratus femoris.
  (e) Obturator externus.

The obturator externus is the muscle of adductor compartment of thigh but only functionally related to this region. It is described in Chapter 23, p. 347.

The *major muscles of gluteal region* are larger in size and placed superficially. They are mainly extensor, abductors, and medial rotators of the thigh.

The *minor muscles of gluteal region* are smaller in size and placed deeply under cover of the lower part of the gluteus maximus. They are lateral rotators of the thigh and help to stabilize the hip joint.

However, the major muscles are described in detail in the following text. The attachments, nerve supply and actions of muscles of the gluteal region are given in Table 24.1.

GLUTEUS MAXIMUS (Fig. 24.8)

The gluteus maximus is the largest, most coarsely fibred, and most superficial gluteal muscle. It is quadrilateral in shape and covers all of the other gluteal muscles except for the anterosuperior part of the gluteus medius—the common site for intramuscular injections (Fig. 24.9).

Table 24.1 Attachments, nerve supply, and main actions of the muscles of the gluteal region

Muscle (Fig. 24.9)	Origin	Insertion	Nerve supply	Actions
Gluteus maximus (quadrilateral muscle)	• Gluteal surface of the ilium behind posterior gluteal line • Outer slope of the dorsal segment of ilium • Dorsal surfaces of the sacrum and ilium • Sacrotuberous ligament	• 3/4th of the muscle into the iliotibial tract • 1/4th of the muscle into the gluteal tuberosity	• Inferior gluteal nerve (L5; S1, S2)	• Chief extensor of the hip joint • Assists in getting up from sitting position
Gluteus medius (fan-shaped muscle)	Gluteal surface of the ilium between anterior and posterior gluteal lines	Oblique ridge on the lateral surface of the greater trochanter	Superior gluteal nerve (L5; S1)	• Abductor of the hip joint • Prevents the sagging of pelvis on the unsupported side
Gluteus minimus (fan-shaped muscle)	Gluteal surfaces of the ilium between anterior and inferior gluteal lines	Ridge on the lateral part of the anterior and inferior gluteal lines trochanter		
Tensor fasciae latae (fusiform muscle)	Outer lip of the anterior part of iliac crest (from ASIS to tubercle)	Iliotibial tract	Superior gluteal nerve	Supports the femur on tibia during standing position
Piriformis (pear-shaped muscle, Latin *pirum* = pear)	Pelvic surface of the middle three pieces of sacrum by three digitations	Apex/tip of greater trochanter	Ventral rami of S1, S2	Lateral rotator of the thigh at hip joint
Gemellus superior	Posterior surface of the ischial spine	Medial surface of greater trochanter along with tendon of obturator internus	Nerve to obturator internus (L5; S1, S2)	Lateral rotator of the thigh at hip joint
Gemellus inferior	• Upper part of the ischial tuberosity • Lower part of greater sciatic notch	Same as that of gemellus superior	Nerve to quadratus femoris (L4; L5, S1)	Lateral rotator of the thigh at hip joint
Obturator internus (fan-shaped muscle)	Pelvic surface of the obturator membrane and surrounding bones	Medial surface of greater trochanter of femur in front of trochanteric fossa	Nerve to obturator internus (L5; S1)	Lateral rotator of the thigh at hip joint
Quadratus femoris (quadrilateral muscle)	Lateral border of the ischial tuberosity	Quadrate tubercle on the intertrochanteric crest and area below it	Nerve to quadratus femoris (L5; S1)	Lateral rotator of the thigh at hip joint

Origin

It arises from:

(a) Posterior part of the gluteal surface of ilium above and behind the posterior gluteal line.
(b) Outer sloping surface of the dorsal segment of iliac crest.
(c) Aponeurosis of erector spinae muscle.
(d) Dorsal surfaces of the lower part of sacrum and adjoining part of the coccyx.
(e) Sacrotuberous ligament.

Insertion

The insertion occurs as follows:

(a) One-fourth of the muscle (deep fibres of the lower part) is inserted into the gluteal tuberosity of femur.

(b) Three-fourth of the muscle (superficial and deep fibres of the upper part) is inserted into the iliotibial tract, which in turn inserts itself on the lateral condyle of the tibia.

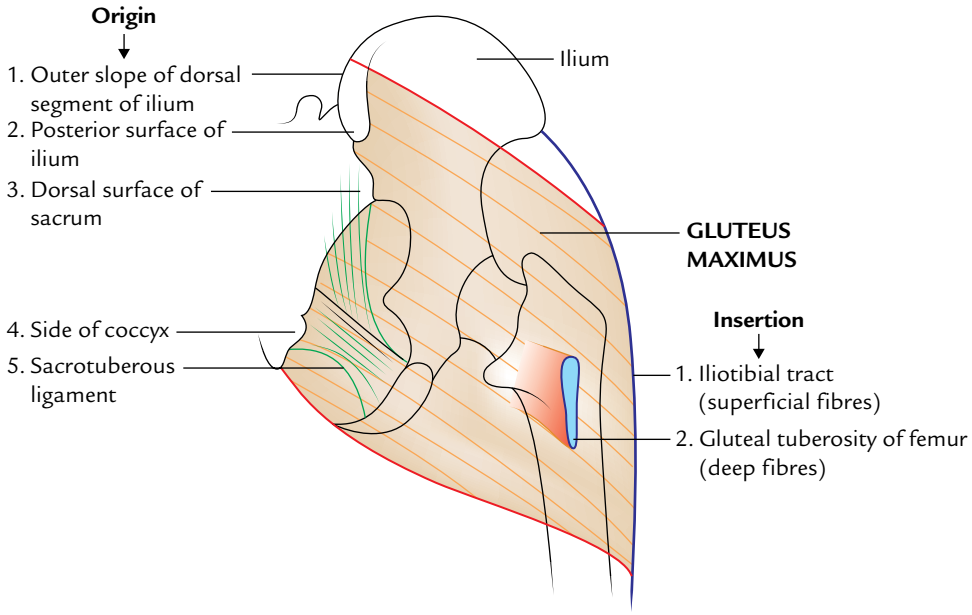


Fig. 24.8 Origin and insertion of the gluteus maximus.

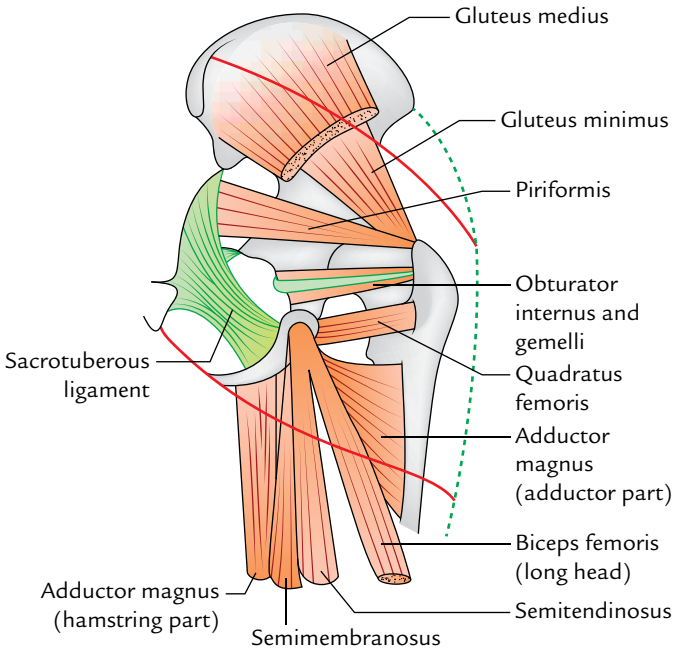


Fig. 24.9 Muscles under cover of the gluteus maximus. The upper and lower borders of gluteus maximus muscle are indicated by thick red lines.

Nerve supply

The gluteus maximus is supplied by the inferior gluteal nerve (L5; S1, S2).

Actions

1. The gluteus maximus is the chief extensor of the hip joint during standing-up from sitting position and climbing upstairs.
2. It plays a very important role in rising from a sitting position and maintaining the erect posture.

The **structures under cover of gluteus maximus** are given in Figure 24.10 and Table 24.2.

GLUTEUS MEDIUS (Fig. 24.11)

The gluteus medius is a fan-shaped muscle. Its posterior one-third is deep and covered by the gluteus maximus while its anterior two-third is superficial and not covered by the gluteus maximus. Hence, the intramuscular injection should be ideally given in this part.

Origin

Gluteus medius arises from the gluteal surface of ilium between the anterior and posterior gluteal lines and gluteal aponeurosis.

Insertion

The muscle fibres converge downward, forward, and laterally to form a flat tendon which is inserted on to the oblique ridge on the lateral surface of the greater trochanter. The oblique ridge runs downward and forward from the tip of the greater trochanter.

Between the tendon of gluteus medius and lateral surface of greater trochanter lies a bursa—the **trochanteric bursa of gluteus medius**.

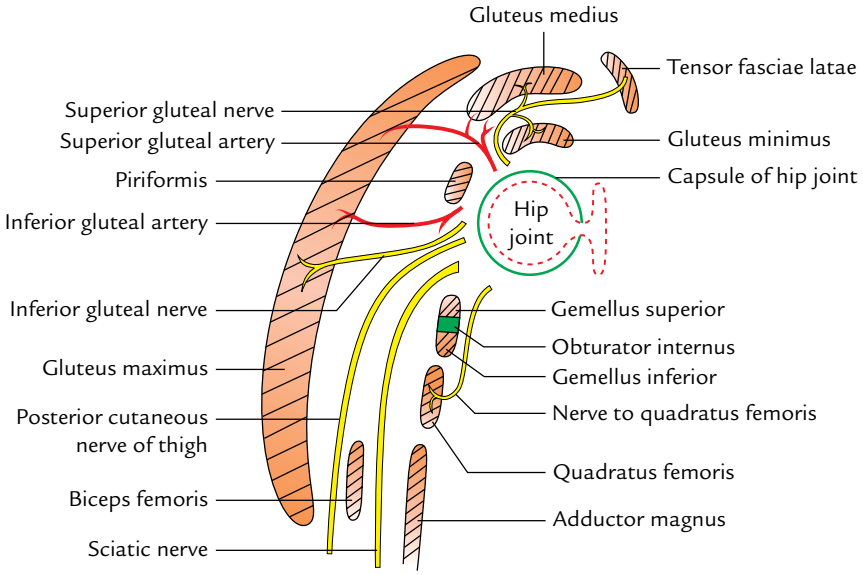


Fig. 24.10 Structures under cover of the gluteus maximus.

Table 24.2 Structures under cover of the gluteus maximus muscle

Muscles	Vessels	Nerves	Joints and ligaments	Bursae
All the muscles of the gluteal region except tensor fasciae latae	Superior and inferior gluteal vessels	Superior and inferior gluteal nerves	Hip joint	Trochanteric bursa
Reflected head of the rectus femoris	Internal pudendal vessels	Sciatic nerve	Sacroiliac joint	Ischial bursa (occasional)
Origin of hamstrings	Trochanteric arterial anastomosis	Posterior cutaneous nerve of the thigh	Sacrotuberous ligament	Gluteo-femoral bursa
Insertion of the upper fibres of the adductor magnus	Cruciate arterial anastomosis	• Nerve to quadratus femoris • Pudendal nerve • Nerve to obturator internus • Perforating cutaneous nerve	• Sacrospinous ligament • Ischiofemoral ligament	

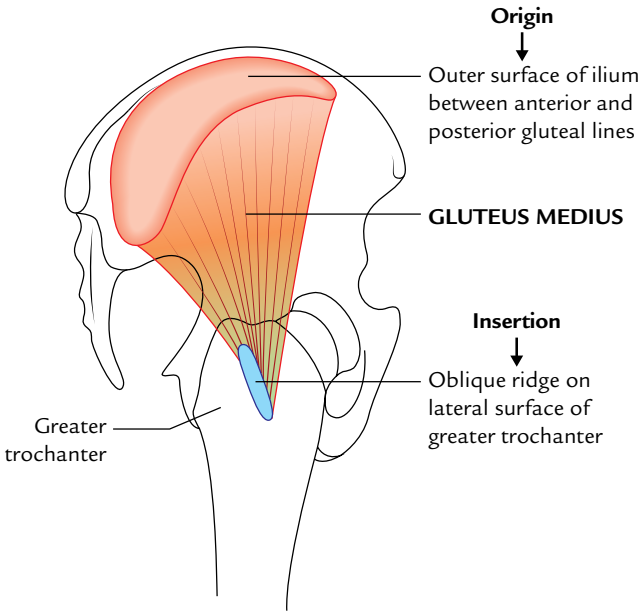


Fig. 24.11 Origin and insertion of the gluteus medius.

Nerve supply

The nerve supply is by the superior gluteal nerve.

Action

It is the main abductor of the hip joint.

GLUTEUS MINIMUS (Fig. 24.12)

The gluteus minimus is also a fan-shaped muscle and lies beneath the gluteus medius.

Origin

It arises from the gluteal surface of ilium between the anterior and inferior gluteal lines.

Insertion

The muscle fibres converge downward and slightly laterally to form a tendon, which is inserted on to the ridge in the lateral part of the anterior surface of the greater trochanter.

Between the tendon of gluteus minimus and the anterior surface of greater trochanter lies a bursa—the **trochanteric bursa of gluteus minimus.**

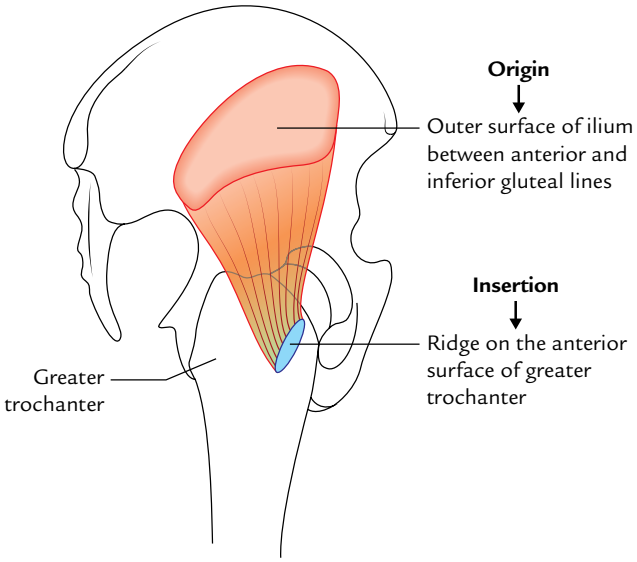


Fig. 24.12 Origin and insertion of the gluteus minimus.

Nerve supply

The gluteus minimus is supplied by the superior gluteal nerve.

Action

It is the abductor of the hip joint.

N.B. *Abductor mechanism of the hip joint:* This mechanism consists of three components: (a) power (P) provided by the gluteus medius and minimus, (b) fulcrum (F) provided by the hip joint, and (c) weight (W) by the head and neck of the femur (Fig. 24.13). If any of these components is deranged, the **gluteal-limp** occurs.

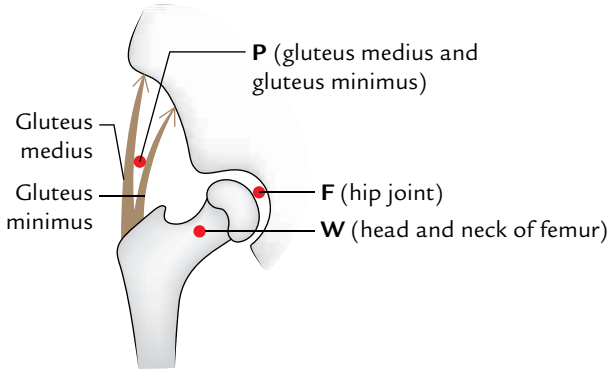


Fig. 24.13 Abductor mechanism of the hip joint (F = fulcrum, W = weight, P = power).

Clinical correlation

Trendelenburg's sign: Acting from below, both the gluteus medius and minimus prevent the unsupported side of the pelvis from sagging during walking and thus maintain the horizontal level of the pelvis provided the hip joint and neck-shaft angle of the femur are normal.

When the gluteus medius and minimus of one side are paralyzed due to injury of the superior gluteal nerve. The pelvis sags on the healthy side if that foot is off the ground. As a result, the person walks with a **lurching gait**. This is clinically known as **Trendelenburg's sign (Fig. 24.14)**.

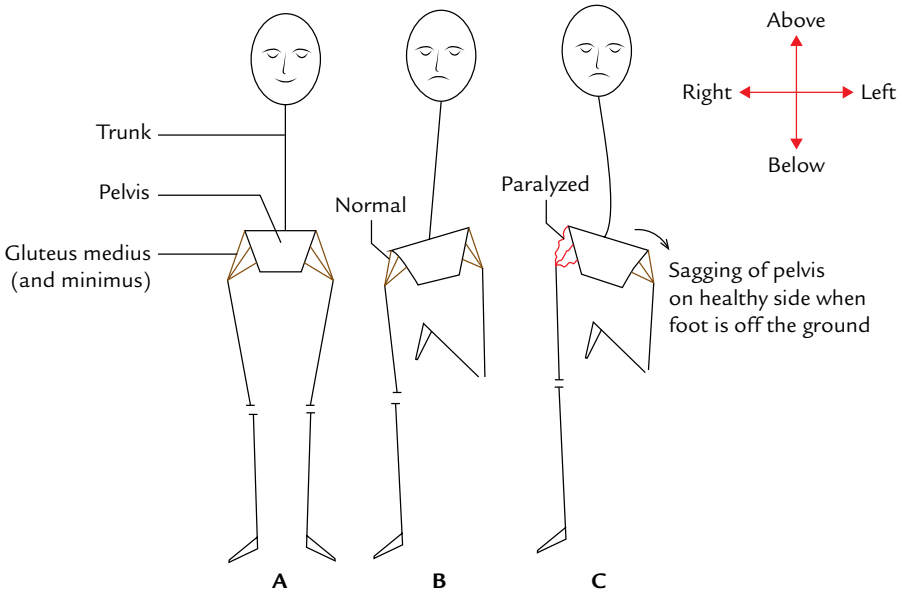


Fig. 24.14 Trendelenburg's sign: A, when both feet are supporting the body weight, horizontal level of pelvis is maintained (i.e., anterior superior iliac spines lie in the same horizontal plane); B, when only one foot (e.g., right in this figure) is supporting the body weight, the unsupported side of the pelvis is normally raised or prevented from sagging by the opposite gluteus medius and minimus muscles; C, if the gluteus medius and minimus muscles are paralyzed on one side (e.g., right side in this case), the unsupported left side of pelvis sags (this is positive Trendelenburg's sign).

TENSOR FASCIAE LATAE

This is a muscle of the gluteal region but during evolution the tensor fasciae latae muscle had migrated to the upper lateral aspect of the thigh but retains its nerve supply (e.g., superior gluteal nerve).

It tenses the fascia lata and iliotibial tract, thereby supporting the femur on tibia during standing position.

It is described with the muscles of the anterior compartment of the thigh in Chapter 22, p. 334–335.

ARTERIES OF THE GLUTEAL REGION

The arteries of the gluteal region are as follows:

1. Superior gluteal artery.
2. Inferior gluteal artery.
3. Internal pudendal artery.

SUPERIOR GLUTEAL ARTERY

The superior gluteal artery is a branch of the posterior division of internal iliac artery. It enters the gluteal region through greater sciatic foramen above the piriformis along with the superior gluteal nerve. Here it divides into the superficial and deep branches. The **superficial branch** passes between gluteus medius and maximus, and supplies both of them. The **deep branch** passes laterally between gluteus medius and minimus, and subdivides into upper and lower branches. The upper branch takes part in the formation of **spinous anastomosis** close to the anterior superior iliac spine and the lower branch takes part in the formation of **trochanteric anastomosis** (p. 370).

INFERIOR GLUTEAL ARTERY

The inferior gluteal artery is a branch of anterior division of the internal iliac artery and enters in the gluteal region below the piriformis along with the inferior gluteal nerve and continues with the posterior cutaneous nerve of the thigh. It gives off three sets of branches:

1. **Muscular branches** to the adjacent muscles.
2. **Anastomotic branches** to cruciate and trochanteric anastomoses.
3. Artery to sciatic nerve (Latin *arteria nervi ischiadici*) accompanies the sciatic nerve and sinks into its substance to supply it. It is the remnant of the **axis artery of the lower limb**.

NERVES OF THE GLUTEAL REGION

The nerves of the gluteal region are as follows:

1. Superior gluteal nerve.
2. Inferior gluteal nerve.

The other nerves in the gluteal region are:

1. Sciatic nerve.
2. Posterior cutaneous nerve of thigh.
3. Nerve to quadratus femoris.
4. Pudendal nerve.
5. Nerve to obturator internus.
6. Perforating cutaneous nerve.

SUPERIOR GLUTEAL NERVE

The superior gluteal nerve arises from the sacral plexus in the pelvis and is formed by the dorsal branches of the ventral rami of L4, L5; S1. It enters the gluteal region through the greater sciatic notch above the piriformis in company with superior gluteal artery. Here it curves upward and forward, runs between the gluteus medius and the minimus, and supplies both of them. It then comes out by passing between the anterior borders of these muscles and supplies the tensor fasciae latae from its deep surface. It also provides an articular twig to the hip joint.

INFERIOR GLUTEAL NERVE

The inferior gluteal nerve arises from the sacral plexus in the pelvis and is formed by the dorsal branches of the ventral rami of L5; S1, S2. It enters the gluteal region through the greater sciatic notch in company with inferior gluteal artery and posterior cutaneous nerve of the thigh. Here it curves upward to supply the gluteus maximus from its deep surface.

SCIATIC NERVE

It enters the gluteal region through the greater sciatic foramen below the piriformis (Fig. 24.7). (It is the most lateral structure emerging through the greater sciatic foramen below the piriformis.) It runs downward and slightly laterally under cover of gluteus maximus midway between the greater trochanter and the ischial tuberosity, and enters the back of the thigh at the lower border of the gluteus maximus. The sciatic nerve is described in detail in Chapter 25, p. 367.

Surface Marking of Sciatic Nerve in the Gluteal Region (Fig. 24.15)

To represent the course of sciatic nerve in the gluteal region, two points are located—upper and lower.

1. The **upper point** is located about 2.5 cm lateral to the midpoint of a line joining the PSIS and ischial tuberosity.
2. The **lower point** is located midway between the greater trochanter and the ischial tuberosity.

A thick (about 2 cm wide) curved line (with outward convexity) joining these two points represents the sciatic nerve.

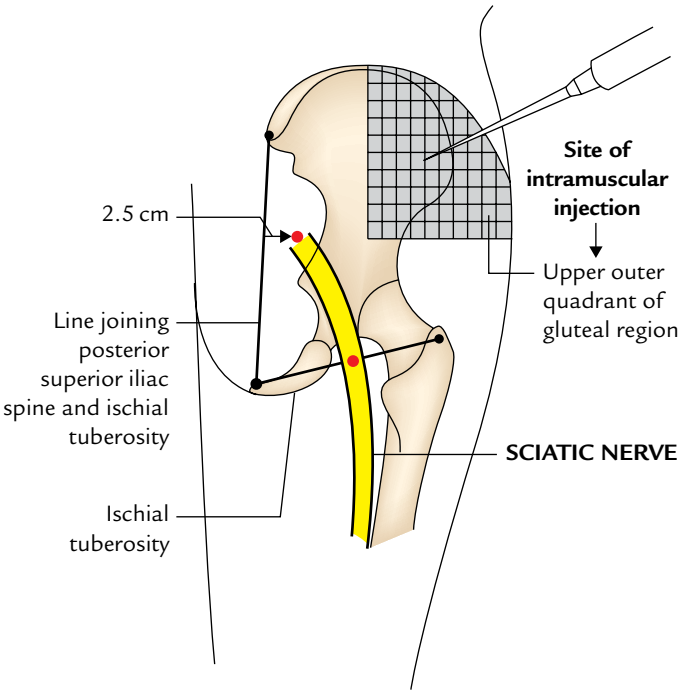


Fig. 24.15 Surface marking of the sciatic nerve in the gluteal region.

POSTERIOR CUTANEOUS NERVE OF THE THIGH

The posterior cutaneous nerve of thigh arises from the sacral plexus in the pelvis from dorsal divisions of the ventral rami of S1, S2, S3 of sacral plexus, and enters the gluteal region through the greater sciatic foramen below the piriformis and medial to the sciatic nerve. It runs downward and medially superficial to the sciatic nerve. It is continuous on the back of the thigh deep to the fascia lata.

It gives the following branches:

1. A peroneal branch to supply the skin of posterior two-third of scrotum or labium majus.
2. Gluteal branches to supply the skin of the posteroinferior quadrant of the gluteal region.

N.B. The unique feature of posterior cutaneous nerve of the thigh is that the most part of this nerve lies deep to deep fascia.

NERVE TO QUADRATUS FEMORIS

The nerve to quadratus femoris arises from ventral divisions of ventral rami of L4, L5; S1 of sacral plexus. It enters the thigh through the greater sciatic foramen below the piriformis and runs downward deep to sciatic nerve. It runs downwards deep to the tendon of obturator internus and two gemelli and supplies inferior gemellus and quadratus femoris. It also gives an articular twig to the hip joint.

PUDENDAL NERVE

The pudendal nerve arises from ventral divisions of the ventral rami of S2, S3, S4 of sacral plexus. It enters the gluteal region just to leave it. It enters the gluteal region through the greater sciatic foramen below the piriformis. It crosses the dorsal aspect of apex of the sacrospinous ligament medial to internal pudendal vessels and leaves the gluteal region by passing through the lesser sciatic foramen to enter the pudendal canal (for details see Chapter 15, page 235).

NERVE TO OBTURATOR INTERNUS

The nerve to obturator internus arises from ventral divisions of the ventral rami of L5; S1, S2 of sacral plexus. It enters the gluteal region through greater sciatic foramen below the piriformis. It crosses the dorsal aspect of ischial spine lateral to internal pudendal vessels and then passes forward through the lesser sciatic foramen deep to the fascia covering the obturator internus. It supplies obturator internus and gemellus superior.

PERFORATING CUTANEOUS NERVE

The perforating cutaneous nerve arises from the sacral plexus (S2, S3). It pierces the lower part of sacrotuberous ligament and then winds around the lower border of gluteus maximus to supply the skin of the posteroinferior quadrant of the gluteal region.

Clinical correlation

- **Intramuscular injection in gluteal region:** The gluteal region is one of the commonest sites of intramuscular injection of drugs. It is given in the **gluteus medius muscle** at this site. If given randomly it may damage the sciatic nerve. It is safe only when it is given in the upper lateral quadrant of the gluteal region or above the line extending PSIS to the upper border of greater trochanter. This line approximately corresponds to the upper border of the gluteus maximus (for details, see *Clinical and Surgical Anatomy*, 2nd Edition by Vishram Singh).
- **Sciatic nerve block:** The sciatic nerve is blocked by injecting an anesthetic agent a few centimeters below the midpoint of the line joining the PSIS and the upper border of the greater trochanter.
- **Piriformis syndrome:** It is a clinical condition characterized by pain in buttock due to compression of the sciatic nerve by the piriformis. It commonly occurs in sports that require excessive use of gluteal muscles (e.g., ice skaters, cyclists, etc.), causing hypertrophy or spasm of piriformis.

Golden Facts to Remember

► All the structures enter the gluteal region through greater sciatic foramen *except*	Tendon of obturator internus which enters the gluteal region through lesser sciatic foramen
► Largest muscle of the body	Gluteus maximus
► Key muscle of the gluteal region	Piriformis
► All the major muscles of the gluteal region are supplied by the superior gluteal nerve *except*	Gluteus maximus (which is supplied by the inferior gluteal nerve)
► Most coarsely fibred muscle of the body	Gluteus maximus
► All the gluteal muscles are covered by the gluteus maximus *except*	Anterosuperior part of the gluteus medius
► Main extensors of the hip during walking	Hamstring muscles

Clinical Case Study



A 27-year-old political party worker sustained a bullet shot injury in his right buttock in police firing during a rally for price hike.

He was admitted in the hospital and fully recovered. He was discharged from the hospital but he developed a characteristic limp during walking. There was a sagging of the left hip while taking step on the right foot. On examination the **Trendelenburg's sign** was positive.

Questions

1. Name the nerve which was injured by the bullet shot.
2. Enumerate the muscles supplied by the injured nerve.
3. What is Trendelenburg's sign?
4. Name the characteristic gait in unilateral injury and in bilateral injury of the superior gluteal nerve.

Answers

1. Superior gluteal nerve.
2. Gluteus medius, gluteus minimus, and tensor fasciae latae.
3. When one foot is off the ground, the tilting of the pelvis on that side is prevented by the contraction of gluteus medius and minimus of the opposite side. If these muscles are paralyzed due to injury of the superior gluteal nerve the pelvis will tilt or sag on the opposite healthy side if that foot is off the ground.
4. (a) Lurching gait if injury is unilateral and (b) waddling gait if injury is bilateral.

CHAPTER

25 Back of the Thigh and Popliteal Fossa

BACK OF THE THIGH

The back of the thigh extends from the gluteal fold above to the back of the knee below.

CUTANEOUS INNERVATION

The skin over the back of the thigh is supplied by the following nerves (Fig. 25.1):

1. *Posterior cutaneous nerve of the thigh* runs downward in the midline on the back of thigh deep to fascia lata and superficially to the long head of biceps femoris. On reaching the popliteal fossa, it pierces the fascial roof of the fossa and accompanies the short saphenous vein down to the middle of the calf. Its branches supply the skin on the back of thigh and proximal part of the back of leg.
2. *Cutaneous branches of the obturator nerve* supply the medial part of the back of thigh in the upper part.
3. *Medial branches of the anterior cutaneous nerve of the thigh* supply the medial part of the back of thigh in the lower part.
4. *Lateral cutaneous nerve of the thigh* sends twigs to the posterior aspect of thigh in the upper part.

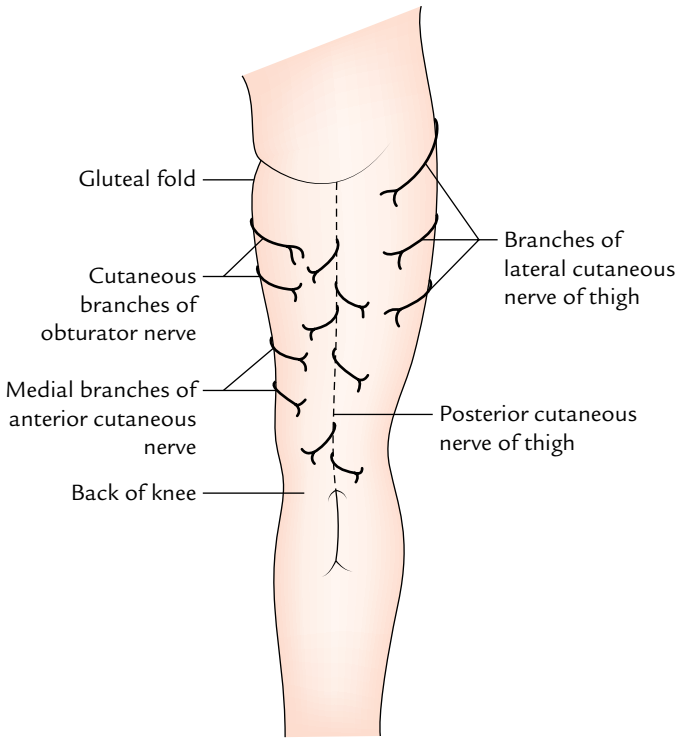


Fig. 25.1 Cutaneous nerves on the back of the thigh.

CONTENTS OF POSTERIOR COMPARTMENT OF THE THIGH

The posterior compartment of the thigh is also called flexor compartment. It is completely separated from the anterior compartment by the lateral intermuscular septum, but it is incompletely separated from the medial compartment of thigh by the ill-defined posterior intermuscular septum.

The contents of posterior compartment of the thigh are as follows:

- **Muscles:** Hamstring muscles and short head of the biceps femoris.
- **Nerve:** Sciatic nerve.
- **Arteries:** Arterial anastomoses on the back of the thigh.

MUSCLES ON THE BACK OF THE THIGH

These muscles on the back of the thigh are called the ***hamstring muscles*** (Latin *ham* = back of thigh). However conventionally the short head of the biceps femoris is not included in the hamstring group.

The hamstring muscles are:

1. Semitendinosus.
2. Semimembranosus.
3. Biceps femoris (long head).
4. Ischial head of adductor magnus.

The **characteristic features of hamstring muscles** are:

1. All arise from the ischial tuberosity.
2. All are inserted into one of the bones of the leg.
3. All are supplied by tibial part of the sciatic nerve.
4. All are flexors of the knee and extensors of the hip joint.

The bony attachments of the hamstring muscles are shown in Figure 25.2.

N.B.

- The posterior thigh muscles were named **"hamstrings"** because their tendons on the back of knee are used to hang up hams (hip and thigh regions of animals viz., pigs.)
- The adductor magnus reaches only up to the adductor tubercle of the femur, but is included amongst the hamstrings group of muscles because the **tibial collateral ligament** of the knee joint morphologically represents the degenerated tendon of this muscle, which is attached below on the tibia.

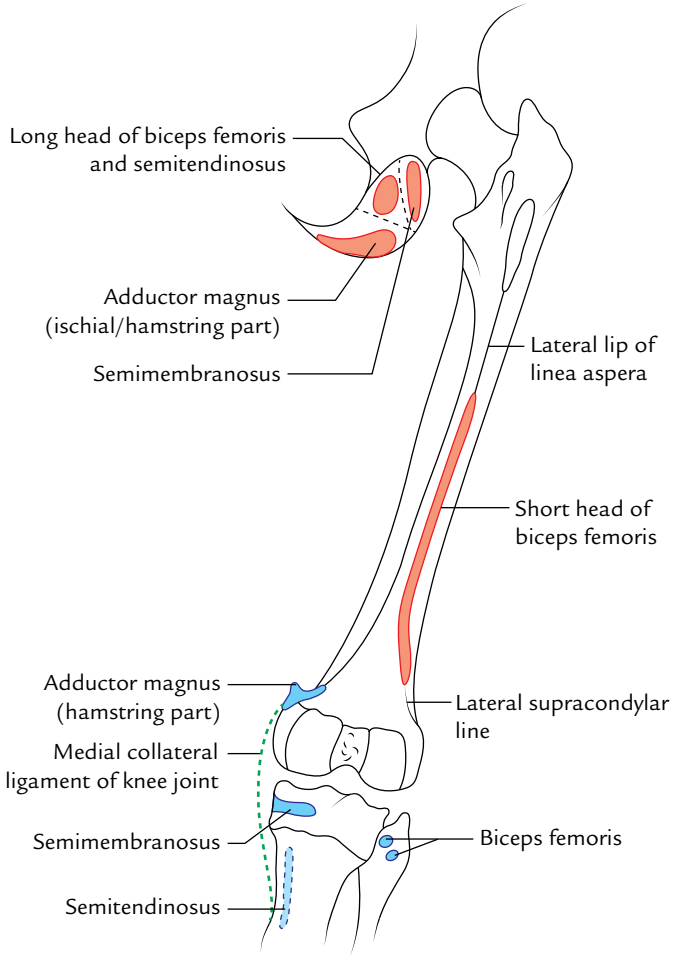


Fig. 25.2 Bony attachments of the muscles of the back.

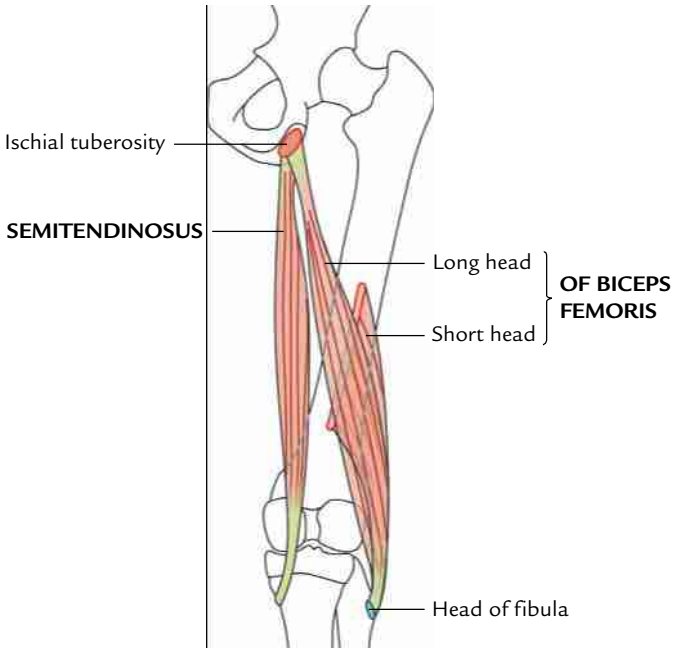


Fig. 25.3 Origin and insertion of biceps femoris and semitendinosus muscles.

Biceps Femoris (Fig. 25.3)

The biceps femoris muscle as its name implies consists of two heads—long head and short head.

Origin

1. **Long head**, arises from lower medial part of the upper quadrilateral area of the ischial tuberosity in common with the semitendinosus and also from the lower part of the sacrotuberous ligament.
2. **Short head**, arises from the lower part of the lateral lip of the linea aspera and upper two-third of the lateral supracondylar line.

Insertion

The two heads unite in the lower third of the thigh to form a conjoint tendon, which slopes downward and laterally to be inserted on to the head of fibula in front of the styloid process. Just before insertion the tendon is either folded around or split by the fibular collateral ligament.

Nerve supply

1. **Long head,** by the tibial part of the sciatic nerve.
2. **Short head,** by the common peroneal part of the sciatic nerve.

Semitendinosus (Fig. 25.3)

It is so named because its lower half is tendinous.

Origin
It arises along with the long head of biceps femoris from lower medial part of the upper quadrilateral area of the ischial tuberosity. It is fleshy in the upper part and forms a cord-like tendon in the lower part, which lies on the semimembranosus muscle.

Insertion
In the lower part of the back of thigh, it diverges medially and passes behind the medial condyle of the femur and then curves downward and forward to be inserted into the upper part of the medial surfaces of the tibia behind the insertion of sartorius and gracilis muscles.

Nerve supply
It is by tibial part of the sciatic nerve.

Semimembranosus (Fig. 25.4)
It is so called because this broad muscle is half membranous.

Origin
It arises from the upper lateral part of the quadrilateral area of the ischial tuberosity. It is membranous in the upper half and fleshy in the lower half. It lies deep to semitendinosus.

Insertion
The fleshy part converges below to form a tendon, which is inserted into a horizontal groove on the back of the medial condyle of tibia.

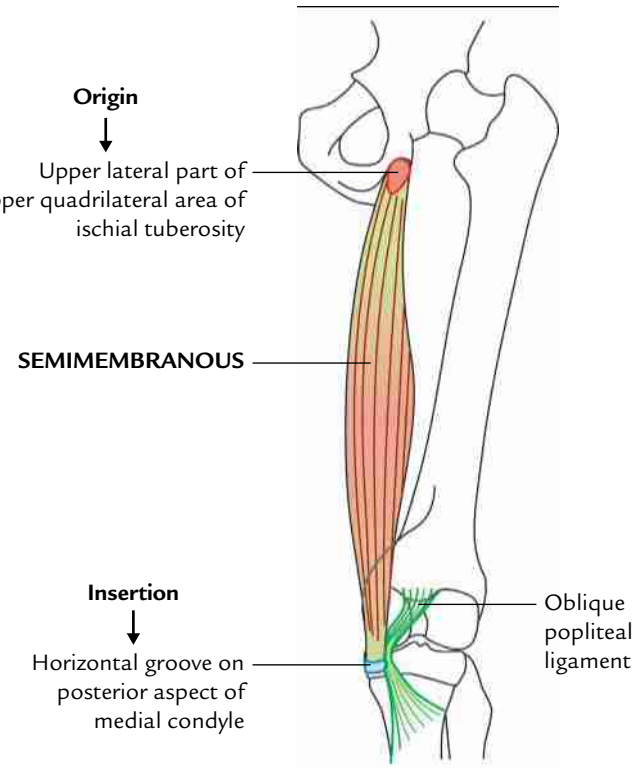


Fig. 25.4 Origin and insertion of semimembranosus muscle.

N.B.
- The tendon of insertion gives rise to the following three expansions:
  1. A fibrous band extends upward and laterally behind the posterior aspect of the capsule of knee joint to form *oblique popliteal ligament of the knee*.
  2. A fibrous expansion extends downward and laterally across the fascia covering the popliteus to be attached to the soleal line.
  3. Some fibres descend downward for attachment on the medial border of the upper part of tibia behind the tibial collateral ligament.
- A synovial bursa lines deep to the tendon of insertion of semimembranosus which may communicate with the synovial cavity of the knee joint.

Nerve supply
It is by tibial part of the sciatic nerve.

Ischial Head of the Adductor Magnus (Fig. 25.2)

Origin
It arises from the inferolateral part of the ischial tuberosity.

Insertion
It descends almost vertically downward to be inserted on the adductor tubercle.

Nerve supply
It is by tibial part of the sciatic nerve.

The origin, insertion, and nerve supply of muscles of the back of the thigh are summarized in Table 25.1.

Actions of the Hamstring Muscles
1. They are the chief flexors of the knee joint and weak extensor of the hip joint (Fig. 25.5); however, the two actions cannot be performed maximally at the same time.
2. The hamstrings are the hip extensors during walking on the flat ground when the gluteus maximus exercise minimal activity.
3. The action of hamstring muscles restricts the range of motion (ROM) of the hip flexion when the knee is extended, viz., during toe touching (Fig. 25.6).

Clinical correlation
Clinical significance of hamstring muscles:
- If hamstring muscles are paralyzed, the patient tends to fall forward because the gluteus maximus muscle cannot maintain the necessary tone to stand upright.
- In ancient times, the soldiers used to slash the back of the knees of horses of their opponents in order to cut the tendons of hamstring muscles, to bring the horse and its rider down. They also used to cut the hamstring tendons of soldiers so that they could not run. This was termed "hamstringing" the enemy.

Table 25.1 Origin, insertion, and nerve supply of the muscles on the back of thigh

Muscle	Origin	Insertion	Nerve supply
Biceps femoris (a) Long head	(a) *Long head*: From lower medial part of upper quadrilateral area of ischial tuberosity	Into the head of the fibula in front of its styloid process	(a) Long head, by the tibial part of the sciatic nerve (L5; S1, S2)
(b) Short head	(b) *Short head*: From lateral lip of the linea aspera and from the upper two-third of the lateral supracondylar line		(b) Short head by the common peroneal part of the sciatic nerve (L5; S1, S2)
Semitendinosus	From the lower medial part of upper quadrilateral area of the ischial tuberosity,	Into the upper part of the medial surface of the tibia	Tibial part of the sciatic nerve (L5; S1, S2)
Semimembranosus	From the upper lateral part of upper quadrilateral area of ischial tuberosity	Into the horizontal groove on the posterior aspect of the medial condyle of the tibia	Tibial part of the sciatic nerve (L5; S1, S2)
Ischial part of adductor magnus	From inferolateral aspect of the ischial tuberosity	Into the adductor tubercle	Tibial part of the sciatic nerve (L5; S1, S2)

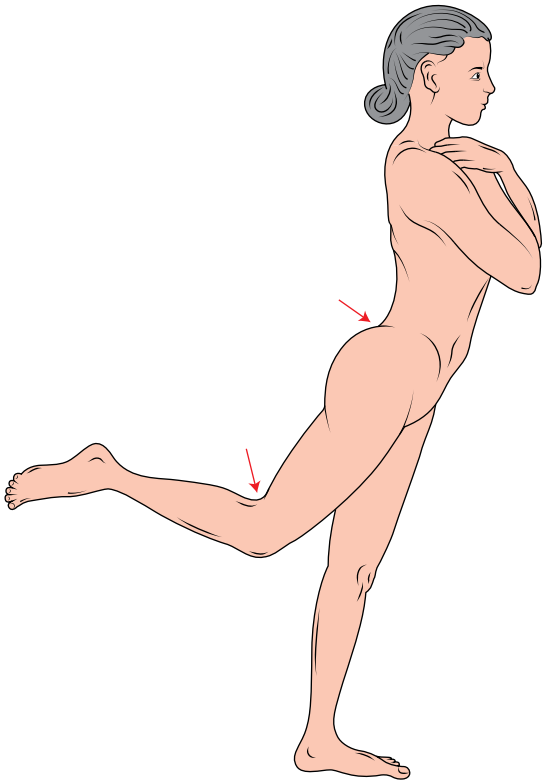


Fig. 25.5 Extension of the hip joint and flexion of the knee joint by the hamstring muscles.

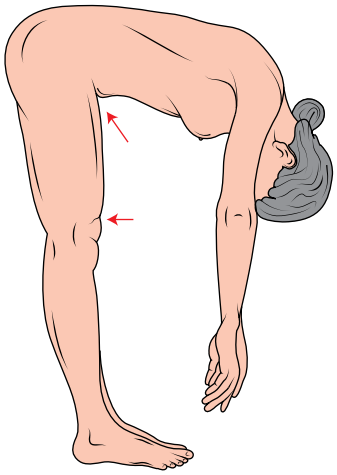


Fig. 25.6 Restriction of range of motion of hip flexion when the knee is extended (e.g., toe-touching action).

SCIATIC NERVE

The sciatic nerve is the thickest nerve in the body. It is about 1.5 to 2 cm wide at the beginning. It arises from sacral plexus in the pelvis and consists of two parts—tibial part and common peroneal part (Fig. 25.7).

The **tibial part** is formed by the ventral divisions of anterior primary rami of L4, L5; S1, S2, S3.

The **common part** is formed by the dorsal divisions of anterior primary rami of L4, L5; S1, S2.

The two parts are usually enclosed in a common sheath of the connective tissue.

Extent

It begins in the pelvis and terminates at or just above the superior angle of the popliteal fossa by dividing into tibial and common peroneal nerve.

Course

In the pelvis, it lies in front of piriformis under cover of its fascia. It enters the gluteal region through greater sciatic foramen below the piriformis. Here it runs downward with slight lateral convexity and passes between the ischial tuberosity and greater trochanter.

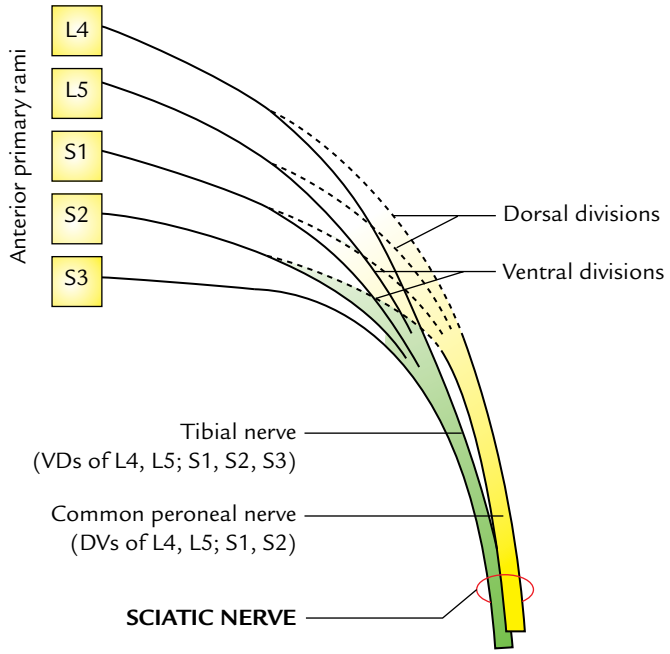


Fig. 25.7 Origin and formation of the sciatic nerve.

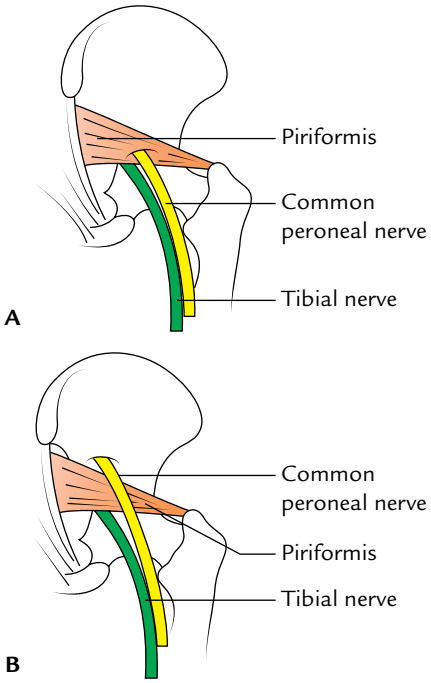


Fig. 25.8 The variations in the mode of exit of the sciatic nerve from pelvis: A, common peroneal nerve passing through the piriformis (12%); B, common peroneal nerve passing above the piriformis (0.5%).

It enters the back of thigh at the lower border of the gluteus maximus and runs vertically downward. Just above or at the superior angle of the popliteal fossa (approximately at the junction of upper two-third and lower one-third of the back of thigh) it divides into two terminal branches—tibial and common peroneal nerves.

N.B. *Variations in the mode of exit of the sciatic nerve from pelvis (Fig. 25.8):* Normally the sciatic nerve enters the gluteal region through greater sciatic foramen below the piriformis. Sometimes sciatic nerve divides into tibial and common peroneal components within the pelvis. In such cases, the mode of exit from pelvis occurs as follows:

(a) The common peroneal nerve passes through the piriformis and tibial nerve passes below the piriformis (12%).
(b) The common peroneal nerve passes above the piriformis and tibial nerve passes below the piriformis (0.5%).

Surface Markings (Fig. 25.9)

The sciatic nerve is marked on the back of thigh by joining the following three points:

1. **First point** is marked 2.5 cm lateral to the midpoint of the line joining the anterior superior iliac spine and ischial tuberosity.
2. **Second point** is marked midway between the ischial tuberosity and greater trochanter.
3. **Third point** is marked at the junction of the upper two-third and lower one-third of the back of the thigh.

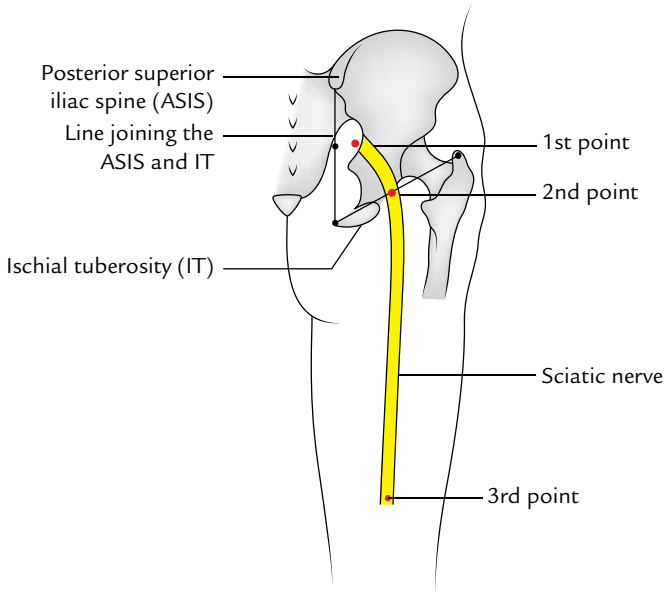


Fig. 25.9 Surface marking of the sciatic nerve.

A thick curved line (about 2 cm wide) with outward convexity joining the first and second points represents the sciatic nerve in the gluteal region and a thick straight line of the same width joining the second and third points represents the sciatic nerve in the thigh.

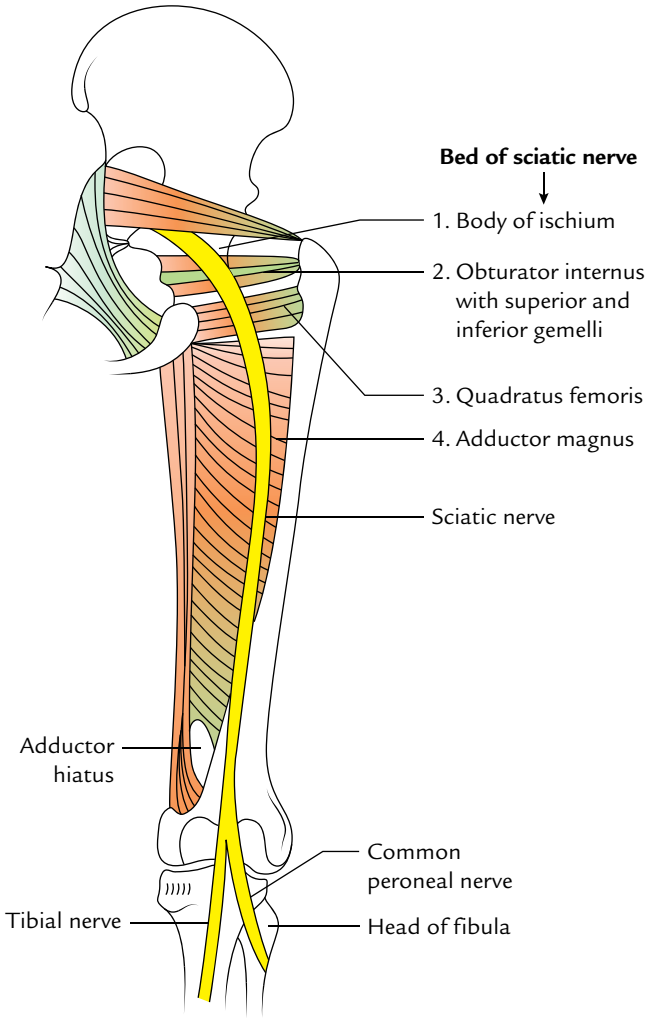


Fig. 25.10 Deep relations of the sciatic nerve (sciatic-bed).

Relations

Deep Relations (Bed of the Sciatic Nerve)

From above downward the sciatic nerve is related to (Fig. 25.10):

(a) Body of ischium (posterior surface).
(b) Tendon of obturator internus and associated gemellus superior and gemellus inferior muscles.
(c) Quadratus femoris.
(d) Adductor magnus.

Superficial Relations

From above downward, the sciatic nerve is related to:

(a) Gluteus maximus (in the gluteal region).
(b) Long head of biceps femoris (in thigh).

The sciatic nerve is accessible on the back of thigh, only in the angle between the gluteus maximus and long head of biceps femoris (Fig. 25.11).

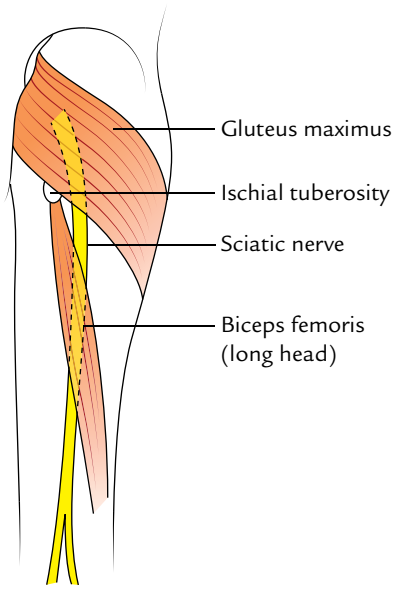


Fig. 25.11 Superficial relations of the sciatic nerve.

Branches

1. Articular branches to the hip joint arise in the gluteal region.
2. Muscular branches to the hamstring muscles arise in the lower part of the gluteal region or in the upper part of the thigh from the medial side of the nerve (Fig. 25.12).
3. Muscular branch to the short head of biceps femoris arises in the lower part of the thigh from the lateral side of the nerve (Fig. 25.12).

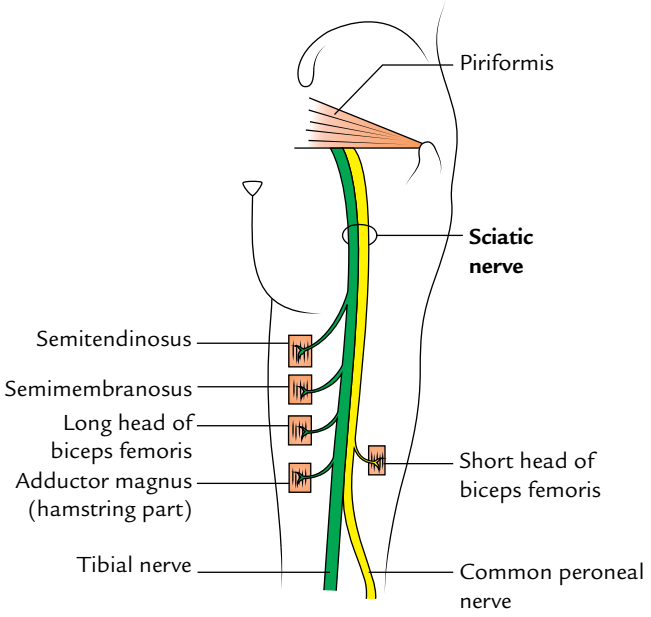


Fig. 25.12 Branches of the sciatic nerves to the muscles of the back of the thigh.

N.B.

All the muscular branches of the sciatic nerve arise from the medial side except nerve to short head of biceps femoris, which arises from the lateral side.

Therefore, the side lateral to the sciatic nerve is safe side and the side medial to it is dangerous side/unsafe side.

Clinical correlation

Sleeping foot: The sciatic nerve is uncovered on the back of thigh in the angle between the lower border of gluteus maximus and long head of biceps femoris. The temporary compression of the sciatic nerve against femur at the lower border of gluteus maximus causes paresthesia in the lower limb. It is called "**sleeping foot**", e.g., when a person sits on the hard edge of the chair for a long time.

ARTERIAL ANASTOMOSES ON THE BACK OF THE THIGH (Fig. 25.13)

The arterial anastomoses on the back of thigh are as follows:

1. **Longitudinal arterial anastomosis:** The main arterial supply to the back of the thigh is derived from the perforating branches of the profunda femoris artery. The perforating arteries pierce the adductor magnus and divide into ascending and descending branches. Close to the posterior aspect of the insertion of adductor magnus a chain of **longitudinal arterial anastomosis** is formed by the ascending and descending branches of four perforating arteries. The ascending branch of first perforating artery takes part in the formation of **cruciate anastomosis** and lowest branch anastomose with the superior muscular branch of the popliteal artery.

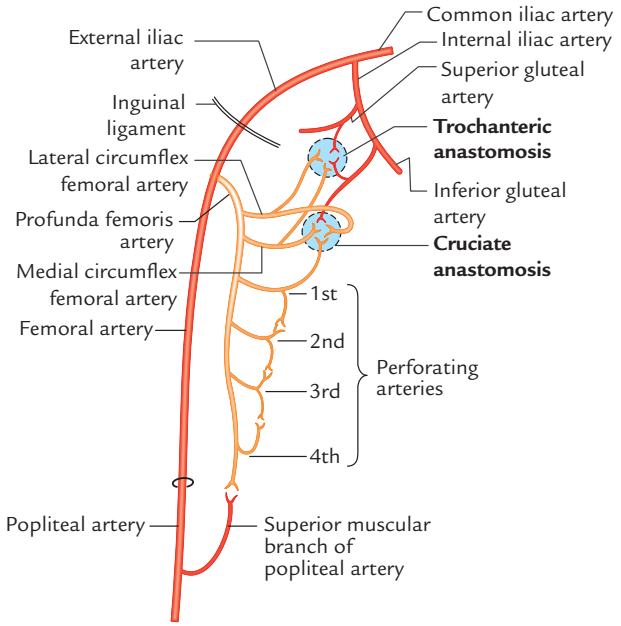


Fig. 25.13 Arterial anastomoses on the back of the thigh.

2. **Trochanteric anastomosis:** It is situated in the trochanteric fossa.

   The trochanteric anastomosis is formed by the following arteries:
   1. Ascending branch of the medial circumflex femoral artery.
   2. Ascending branch of lateral circumflex femoral artery.
   3. Descending branch of the inferior gluteal artery.
   4. Descending branch of the superior gluteal artery.
3. **Cruciate anastomosis:** It is situated at the upper part of the back of femur at the level of the lesser trochanter. The *cruciate anastomosis* is formed by the following arteries:
   1. Transverse branch of the medial circumflex femoral artery.
   2. Transverse branch of the lateral circumflex femoral artery.
   3. Ascending branch of the first perforating artery.
   4. Descending branch of the inferior gluteal artery.

Clinical correlation

Clinical significance of longitudinal arterial anastomosis on the back of thigh: It provides a collateral channels of blood supply to the lower limb bypassing the external iliac and femoral arteries, e.g., in ligation of femoral artery above the origin of profunda femoris artery it maintains an efficient blood supply through collateral circulation.

POPLITEAL FOSSA

The popliteal fossa is a diamond-shaped hollow on the back of the knee joint. It becomes prominent when the knee is flexed. This fossa is an important anatomical region because it provides passage for main vessels and nerves from the thigh to the leg.

BOUNDARIES (Fig. 25.14)

The fossa is bounded:

Superomedially: Semitendinosus and semimembranosus.

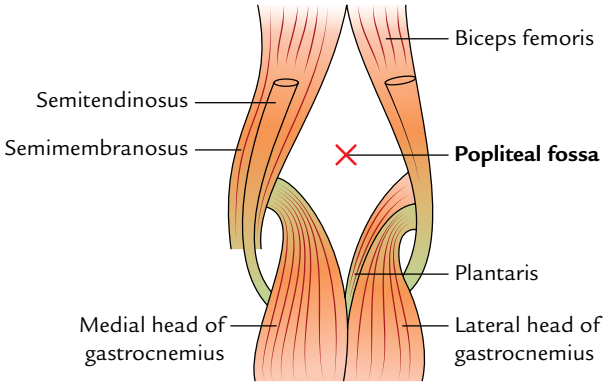


Fig. 25.14 Boundaries of the popliteal fossa.

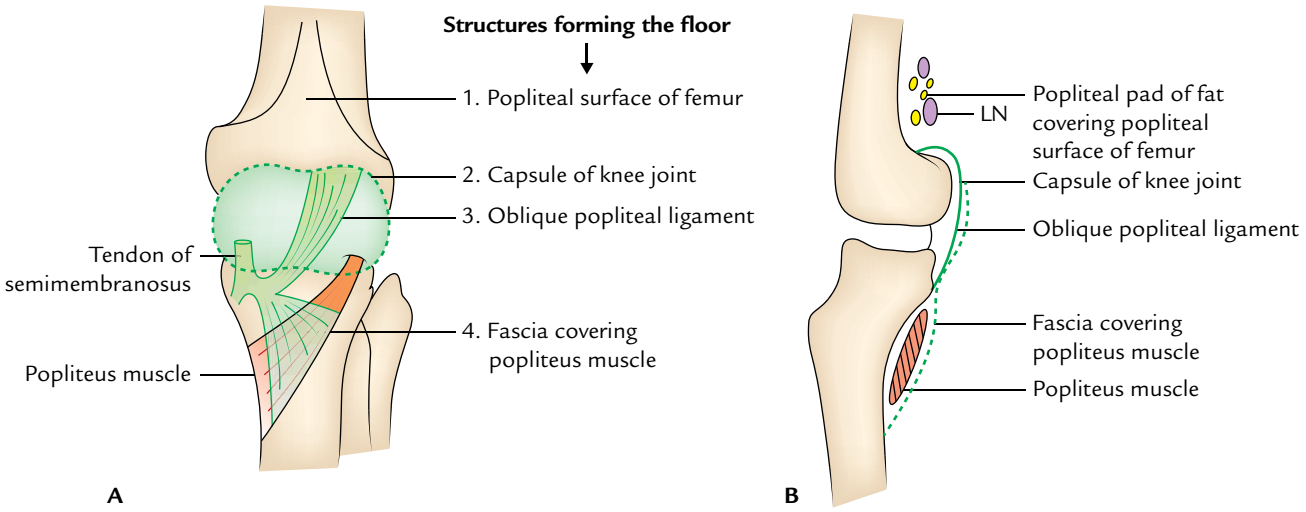


Fig. 25.15 Floor of the popliteal fossa: **A**, surface view; **B**, diagrammatic medial view (LN = popliteal lymph nodes).

Superolaterally: Biceps femoris.
Inferomedially: Medial head of gastrocnemius.
Inferolaterally: Lateral head of gastrocnemius supplemented by the plantaris.

Floor (or anterior wall; Fig. 25.15): It is formed from above downward by:

(a) The popliteal surface of the femur.
(b) The capsule of the knee joint and oblique popliteal ligament.
(c) The popliteal fascia covering the popliteus muscle.

Roof (or posterior wall): It is formed from the strong popliteal fascia. The superficial fascia over the roof contains:

(a) Short saphenous vein.
(b) Three cutaneous nerves: (i) terminal part of the posterior cutaneous nerve of the thigh, (ii) posterior division of the medial cutaneous nerve of the thigh, and (iii) sural communicating nerve). The roof is pierced by all these structures except the posterior division of medial cutaneous nerve of the thigh (Fig. 25.16).

CONTENTS (Fig. 25.17)

The main contents of the popliteal fossa are:

1. Popliteal artery and its branches.
2. Popliteal vein and its tributaries.
3. Tibial nerve and its branches.
4. Common peroneal nerve and its branches.
5. Popliteal lymph nodes.
6. Popliteal pad of fat.

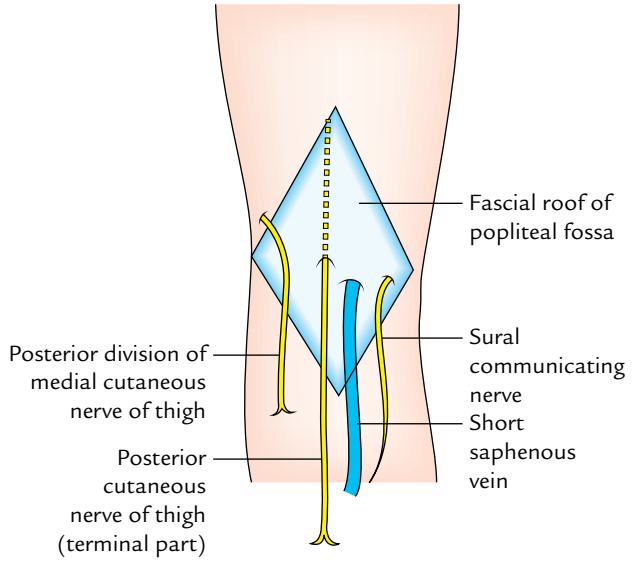


Fig. 25.16 Structures piercing the roof of the popliteal fossa.

In addition to the above-mentioned structures, the popliteal fossa also contains the following structures:

1. Posterior cutaneous nerve of the thigh (terminal part).
2. Descending genicular branch of the obturator nerve.
3. Terminal part of short saphenous vein.

N.B. *The relationship of tibial nerve, popliteal vein, and popliteal artery in the popliteal fossa:* The popliteal artery is crossed superficially by popliteal vein from the lateral to medial side; which in turn is crossed superficially by the tibial nerve from the lateral to medial side. As a result, the relative relationship of these structures differs in the upper, middle, and lower parts of the fossa as follows (Fig. 25.18):

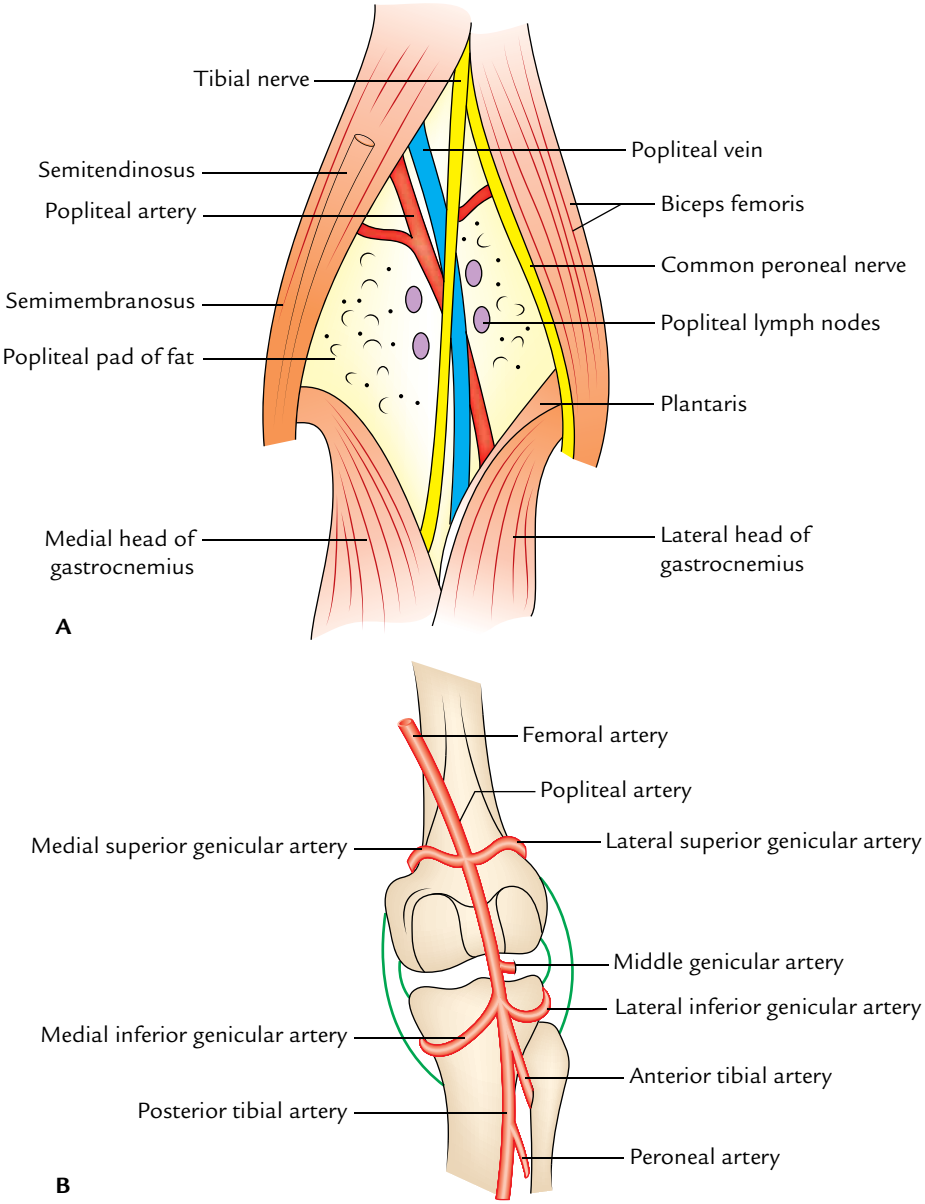


Fig. 25.17 A, Boundaries and contents of the popliteal fossa; B, genicular branches of the popliteal artery.

(a) *In the upper part of the fossa* from the lateral to medial side, the order is Nerve, Vein, and Artery (NVA).
(b) *In the middle part of the fossa* from superficial to deep, the order of arrangement is Nerve, Vein, and Artery (NVA).
(c) *In the lower part of the fossa* from the lateral to medial side, the order of arrangement is Artery, Vein, and Nerve (AVN).

SALIENT FEATURES OF THE PRINCIPAL CONTENTS

Popliteal Artery

It is the continuation of femoral artery. It begins at the adductor hiatus (an osseo-aponeurotic opening in the adductor magnus at the junction of middle one-third and lower one-third of the thigh), crosses the floor of popliteal fossa from the medial to lateral side to reach the lower border of the popliteus where it terminates by dividing into **anterior and posterior tibial arteries.**

Relations

These are as follows:

Anterior (deep): Floor of the popliteal fossa (i.e., popliteal surface of femur, posterior aspect of the knee joint, and fascia covering the popliteus muscle; Fig. 25.19).

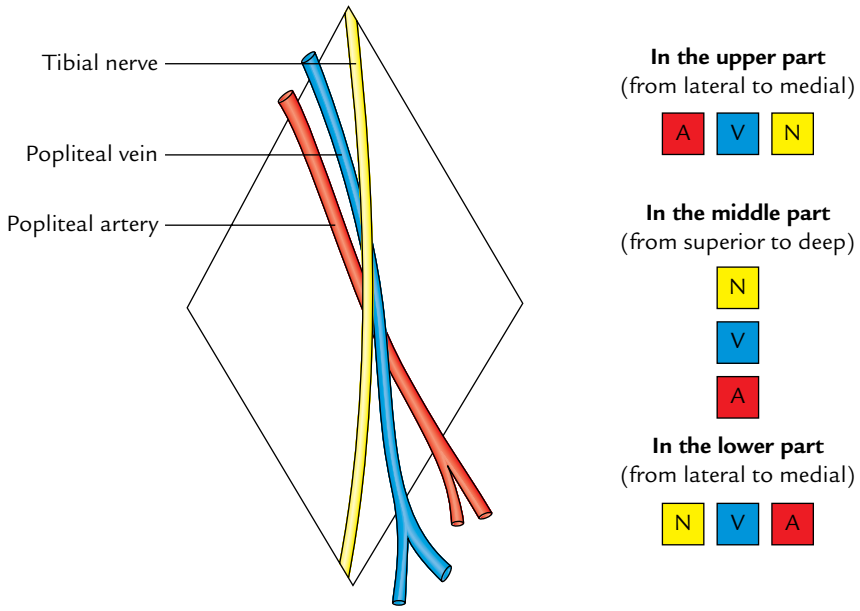


Fig. 25.18 Highly schematic diagram to show relationship of the tibial nerve, popliteal vein, and popliteal artery in the popliteal fossa. Relations of neurovascular structures at three levels (viz., upper, middle, and lower parts) are depicted on the right side.

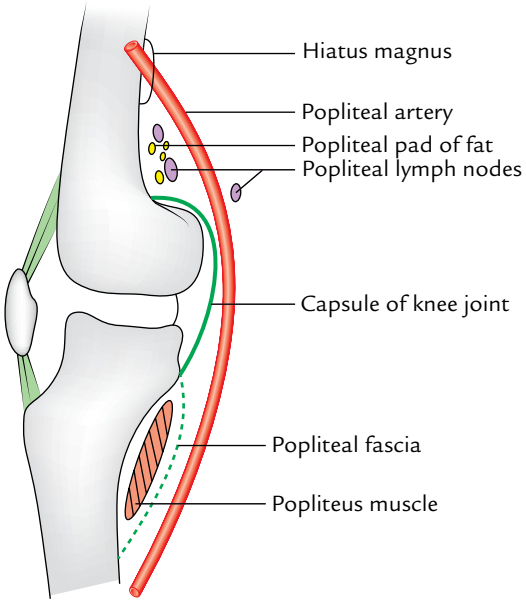


Fig. 25.19 Anterior (deep) relations of the popliteal artery.

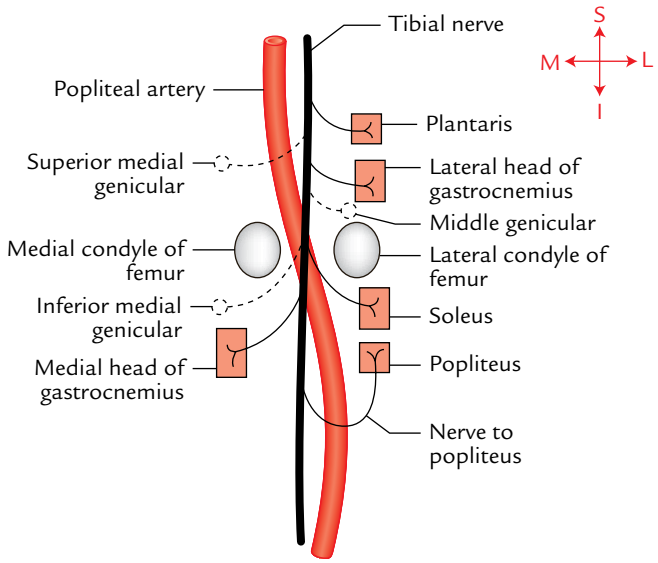


Fig. 25.20 Branches of the tibial nerve in the popliteal fossa.

Posterior (superficial): Popliteal vein, tibial nerve, fascial roof, superficial fascia, and skin (from deep to superficial).

Branches

These are divided into three groups—cutaneous, muscular, and articular (genicular).

1. **Cutaneous branches:** They pierce the roof and supply the overlying skin.
2. **Muscular branches:** They are large and several in number. The upper branches (two or three in number) supply adductor magnus and hamstring muscles. One or two of them anastomose with the fourth perforating artery.

The lower muscular branches supply the triceps surae muscles (i.e., two heads of gastrocnemius and soleus) and plantaris.

3. **Genicular (articular) branches (Fig. 25.20):** They are five in number and supply the knee joint.
   (a) *Superior medial and lateral genicular arteries:* They wind around the corresponding side of the femur immediately above the corresponding femoral condyles and take part in the formation of **genicular anastomosis.**
   (b) *Inferior medial and lateral genicular arteries:* They wind around the corresponding tibial condyles and pass deep to the corresponding collateral ligaments of the knee joint to take part in the formation of **genicular anastomosis.**
   (c) *Middle genicular artery:* It pierces the oblique popliteal ligament of the knee to supply the cruciate ligaments and synovial membrane of the knee joints.

N.B. *Genicular anastomosis* (Fig. 25.21): It is an arterial anastomosis around the knee joint formed by the branches of popliteal, anterior tibial and posterior tibial, femoral, and profunda femoris arteries.

This anastomosis maintains adequate blood supply to the knee joint and leg during flexion of the knee joint, when the popliteal artery is compressed (kinked) and blood flow in it becomes sluggish.

The anastomosis takes place as follows (Fig. 25.21):

1. *Superior medial genicular artery* anastomosis with the descending genicular branch of the femoral artery and inferior medial genicular artery.
2. *Inferior medial genicular artery* anastomosis with the superior medial genicular artery and saphenous artery—a branch of the descending genicular artery (a branch of femoral artery).
3. *Superior lateral genicular artery* anastomosis with the descending branch of the lateral circumflex femoral artery and inferior lateral genicular artery.
4. *Inferior lateral genicular artery* anastomosis with the superior lateral genicular artery, anterior and posterior recurrent branches of the anterior tibial artery, and circumflex fibular branch of posterior tibial artery.

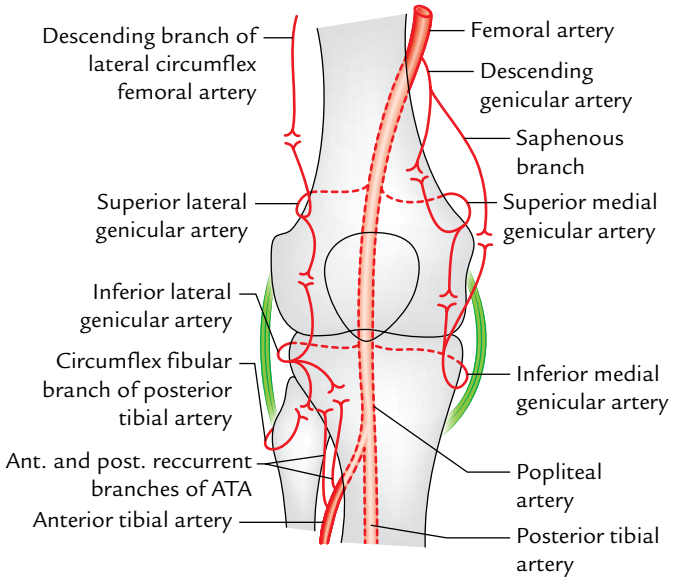


Fig. 25.21 Arterial anastomosis (genicular anastomosis) around the knee joint (ATA = anterior tibial artery).

Popliteal Vein

The popliteal vein is formed at the lower border of the popliteus by the union of veins (venae comitantes) accompanying the anterior and posterior tibial arteries. It ascends superficial to popliteal artery and crosses it from the medial to lateral side in the popliteal fossa. The popliteal vein continues as **femoral vein** at adductor hiatus.

Tributaries

The tributaries of popliteal vein are as follows:

1. Small saphenous vein.
2. Veins corresponding to the branches of popliteal artery.

Tibial Nerve (L4, L5; S1, S2, S3)

It is the *larger terminal branch of the sciatic nerve*. It extends vertically downward from the superior angle to the inferior angle of the popliteal fossa. It crosses popliteal artery superficially from the lateral to medial side with popliteal vein intervening between the artery and nerve.

Branches (Fig. 25.20)

These are as follows:

1. **Muscular branches:** They supply gastrocnemius (both heads), soleus, plantaris, and popliteus.

N.B. All the muscular branches of the tibial nerve arise from its lateral side except the nerve to medial head of gastrocnemius, which arises from its medial side. Therefore, lateral side of the tibial nerve is termed **dangerous side** and medial side as the **safe side**.

The *nerve to popliteus* crosses superficially to the popliteal artery, runs downward and laterally, and then winds around the lower border of popliteus, which it supplies from its deep surface. In addition to the popliteus, it also supplies many other structures.

Structures supplied by the nerve popliteus are:

(a) Popliteus muscle.
(b) Tibialis posterior muscle.
(c) Superior tibiofibular joint.
(d) Tibia.

(e) Interosseous membrane.
(f) Inferior tibiofibular joint.

2. **Genicular branches:** They are three in number (superior medial genicular, inferior medial genicular, and middle genicular) and accompany the arteries of the same name.
   The **middle genicular nerve** pierces the oblique popliteal ligament and supplies the interior of the knee joint.
3. **Cutaneous branch:** It is called the **sural nerve** and arises about the middle of the popliteal fossa. It runs vertically downward underneath the deep fascial roof of fossa and leaves the fossa by piercing the roof near its inferior angle.
4. **Vascular branches:** They supply the vasomotor fibres (T10–L2) to the popliteal vessels.

Common Peroneal Nerve (L4, L5; S1, S2)

It is the *smaller terminal branch of the sciatic nerve*. It appears in the popliteal fossa beneath the long head of the biceps femoris and slopes downward and laterally along the medial side of the tendon of biceps femoris up to the lateral angle of the fossa. Here it crosses superficial to the plantaris, lateral head of gastrocnemius to reach the back of the head of fibula. Finally, it winds around the posterolateral aspect of the neck of fibula, pierces the peroneus longus muscle, and terminates by dividing into **deep** and **superficial peroneal nerves**.

Branches

In the popliteal fossa, the common peroneal nerve gives rise to two cutaneous branches and three genicular branches.

Cutaneous branches

These are as follows:

1. **Sural communicating nerve:** It arises opposite to the head of the fibula, and crosses superficially to the lateral head of gastrocnemius to join the **sural nerve.**
2. **Lateral cutaneous nerve (lateral sural nerve):** It arises lower down and pierces the deep fascia to supply the skin on the upper part of the lateral side of the leg.

Genicular (articular) branches

These are superior lateral genicular, inferior lateral genicular, and recurrent genicular nerves. The first two supply the knee joint and the last one supplies the superior tibiofibular joint.

N.B.

- The common peroneal nerve can be palpated against the neck of fibula.
- The common peroneal nerve gives no muscular branch in the popliteal fossa.

Popliteal Lymph Nodes

They are usually five to six in number and embedded in the popliteal pad of fat. They are found at the following three sites:

1. One node lies at the junction of small saphenous vein and popliteal vein.
2. Few nodes lie deep to the popliteal artery.
3. Remaining nodes are situated on both sides of the popliteal artery.

The popliteal lymph nodes drain:
(a) Back and lateral side of calf of the leg.
(b) Lateral side of heel and foot.

Posterior Cutaneous Nerve of the Thigh

It pierces the fascial roof about the middle of the popliteal fossa and provides cutaneous innervation up to the middle of the back of leg.

Genicular Branch of the Obturator Nerve

It is the continuation of the posterior division of the obturator nerve. It first runs on the posterior surface of the popliteal artery and then pierces oblique popliteal ligament to supply the capsule of the knee joint.

Clinical correlation

- **Popliteal pulse:** To feel the popliteal pulse, first flex the knee to relax the popliteal fascia. Then place the fingertips of both hands in the popliteal fossa with thumbs resting on patient's patella.
  Popliteal pulse is the most difficult pulse to feel amongst all the peripheral pulses.
- **Popliteal aneurysm:** The popliteal artery is more prone to aneurysm than any other artery in the body. Clinically, popliteal aneurysm presents as a pulsatile midline swelling in the popliteal fossa.
- **Baker's cyst:** It is cystic swelling which occurs in the popliteal fossa due to inflammation of synovial bursa underneath the tendon of semimembranosus or protrusion of synovial membrane of the cavity of knee joint through the fibrous capsule of the joint.

Golden Facts of Remember

► Thickest nerve in the body	Sciatic nerve
► All the muscles on the back of the thigh are supplied by tibial part of the sciatic nerve *except*	Short head of biceps femoris being supplied by the common peroneal part of the sciatic nerve
► Most difficult peripheral pulse to feel	Popliteal pulse
► Artery most prone to aneurysm in the body	Popliteal artery

Clinical Case Study



A 55-year-old man complaining of swelling on the back of his knee joint visited the hospital. He stated that he noticed this swelling about one month back. On examination the doctor found a midline pulsatile swelling in the popliteal fossa. The diagnosis of "popliteal aneurysm" was made.

Questions

1. Name the structures from which the swelling on the back of the knee can originate.
2. How doctors differentiate swelling of the popliteal aneurysm from swelling due to Baker's cyst?
3. Give the relationship of popliteal artery, popliteal vein, and tibial nerve in the posteroanterior direction.
4. What is the most deeply located neurovascular structure in the popliteal fossa?

Answers

1. Swellings on the back of the knee can originate from the structures within and around the popliteal fossa as given in the following box.

Structure	Swelling (pathological condition)
• Skin and soft tissues	• Lipoma, sebaceous cyst
• Short saphenous vein	• Varicosity of this vein in the roof
• Popliteal artery	• Popliteal aneurysm
• Popliteal lymph nodes	• Lymphadenopathy (enlargement of lymph nodes due to inflammation)
• Bursa beneath the	• Baker's cyst semimembranosus
• Synovial cavity of the knee	• Effusion in the joint and protrusion of the synovial membrane through the capsule of the joint

2. Swelling due to popliteal aneurysm is pulsatile and located in the midline whereas swelling due to Baker's cyst is cystic and located laterally on the medial side.
3. From posterior to anterior, the relationship is as follows: tibial nerve, popliteal vein, and popliteal artery.
4. Popliteal artery.

CHAPTER

26 Hip Joint

The hip joint is a ball and socket type of synovial joint between the head of the femur and the acetabulum of the hip bone (Fig. 26.1). It is the largest **ball** and **socket type of joint** in the body. Its main functions are: (a) to support the body weight during standing and (b) to transmit the forces generated by movements of trunk femur during walking. It is multiaxial and permits same movements as shoulder joint in the upper limb because the long and narrow neck of femur acts as strut and makes an angle with the shaft (neck–shaft angle). However, its range of movements is restricted due to its role in weight bearing.

TYPE

Synovial joint of ball and socket variety

ARTICULAR SURFACES

The head of the femur articulates with the horse-shoe-shaped acetabulum of the hip bone to form the hip joint. (Fig. 26.2):

1. The *head of femur* forms more than half of a sphere. It is covered by the articular hyaline cartilage except for a small pit—the *fovea capitis* for ligamentum teres.
2. The *acetabulum* (Latin *acetabulum* = vinegar cup) presents three features: a horseshoe-shaped lunate surface, acetabular notch, and acetabular fossa. Out of these, only lunate surface is articular and covered by an articular cartilage. The depth of the acetabulum is increased by the acetabular labrum.

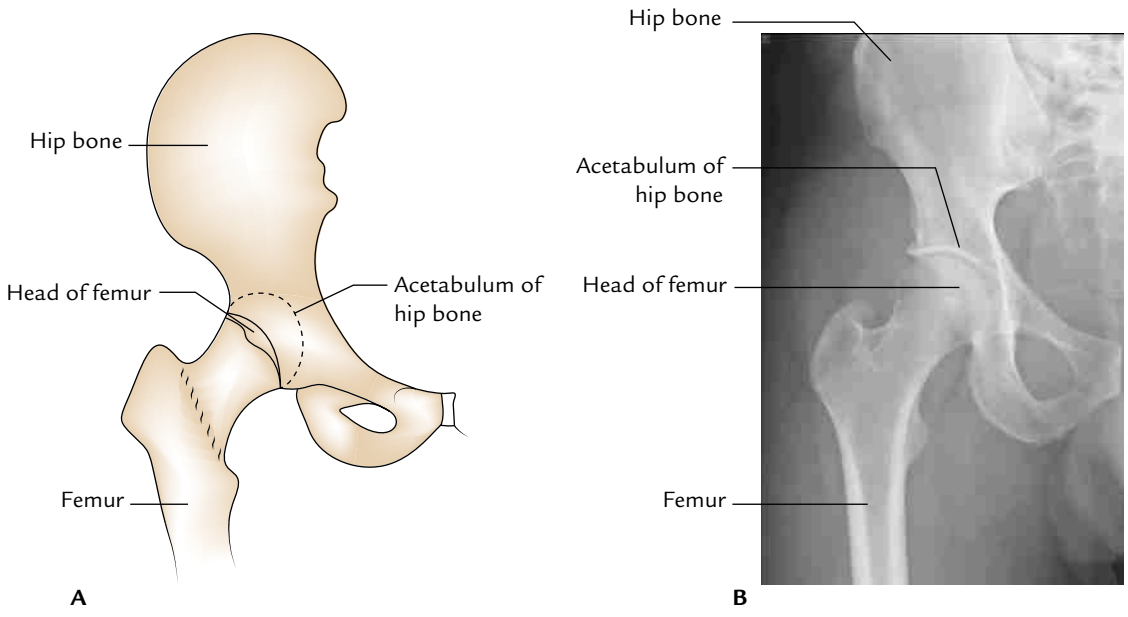


Fig. 26.1 Hip joint: A, line diagram of the right hip joint; B, radiograph of the right hip joint.

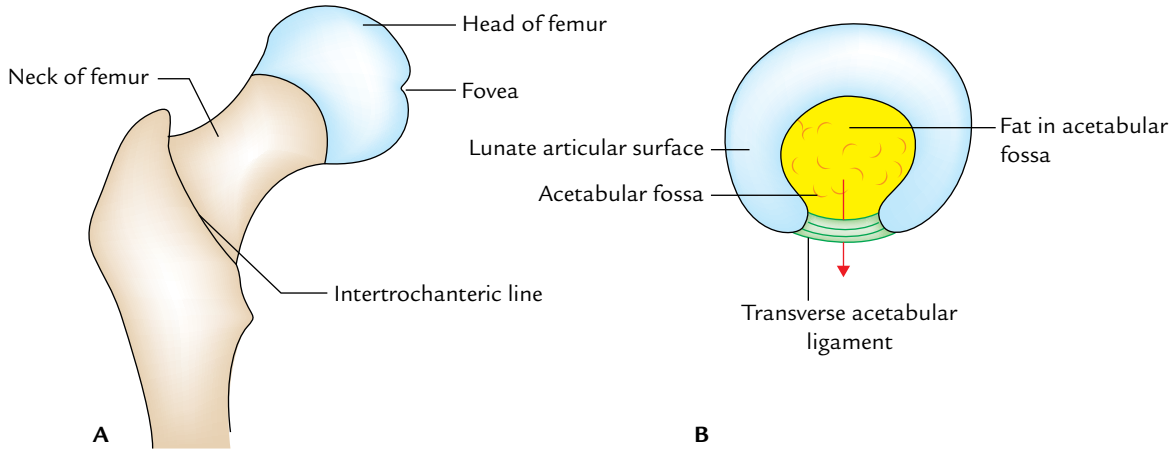


Fig. 26.2 Articular surfaces of the hip joint: A, head of femur; B, acetabulum of the hip bone. Note arrow passing through the acetabular foramen.

Though proximal and distal articular surfaces are reciprocally curved but they are not co-extensive.

LIGAMENTS

The ligaments of the hip joint are as follows:

1. Capsular ligament (joint capsule).
2. Iliofemoral ligament (strongest).
3. Pubofemoral ligament.
4. Ischiofemoral ligament.
5. Transverse acetabular ligament.
6. Acetabular labrum.
7. Ligamentum teres femoris (round ligament of the head of femur).

CAPSULAR LIGAMENT

The capsular ligament is a strong and dense fibrous sac which encloses the joint. Its attachments are as under:

1. *On the hip bone*, it is attached 5–6 mm beyond the acetabular margin, outer aspect of the acetabular labrum and transverse acetabular ligament.
2. *On the femur*, it is attached anteriorly to the intertrochanteric line and posteriorly 1 cm in front of (medial to) the intertrochanteric crest (Fig. 26.3).
   (a) The capsule is thicker anterosuperiorly, where the maximal stress occurs, particularly in the standing position. Posteroinferiorly it is thin and loosely attached.
   (b) The capsule is made up two types of fibres—inner circular fibres and outer longitudinal fibres.
   (c) The inner circular fibres form collar around the femoral neck (*zona orbicularis*). These fibres are not directly attached to the bones.

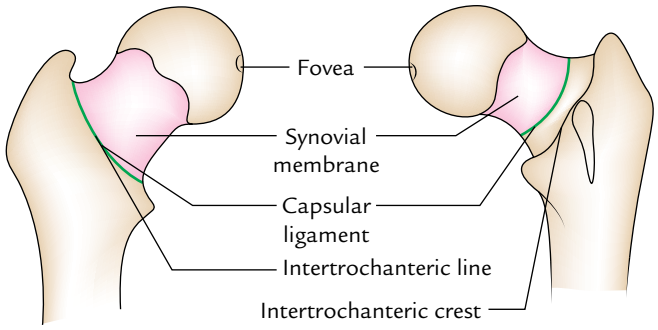


Fig. 26.3 Attachment of the capsular ligament of hip joint on the femur.

   (d) The outer longitudinal fibres are reflected along the neck toward the head to form the **retinacula**.

The **synovial membrane** lines inner aspect of the fibrous capsule, the intracapsular portion of the femoral neck, glenoid labrum (both surfaces), transverse acetabular ligament, ligamentum teres, and fat in the acetabular fossa. It is thin on the deep surface of the iliofemoral ligament where it is compressed against the head (Fig. 26.4).

ILIOFEMORAL LIGAMENT (LIGAMENT OF BIGELOW)

The iliofemoral ligament is inverted Y-shaped ligament, which lies anteriorly and intimately blended with the capsule. Its apex is attached to the lower half of the anterior inferior iliac spine and area between it and above acetabular margin. Its base is attached to the intertrochanteric line. This ligament consists of three parts—a lateral thick band of oblique fibres,

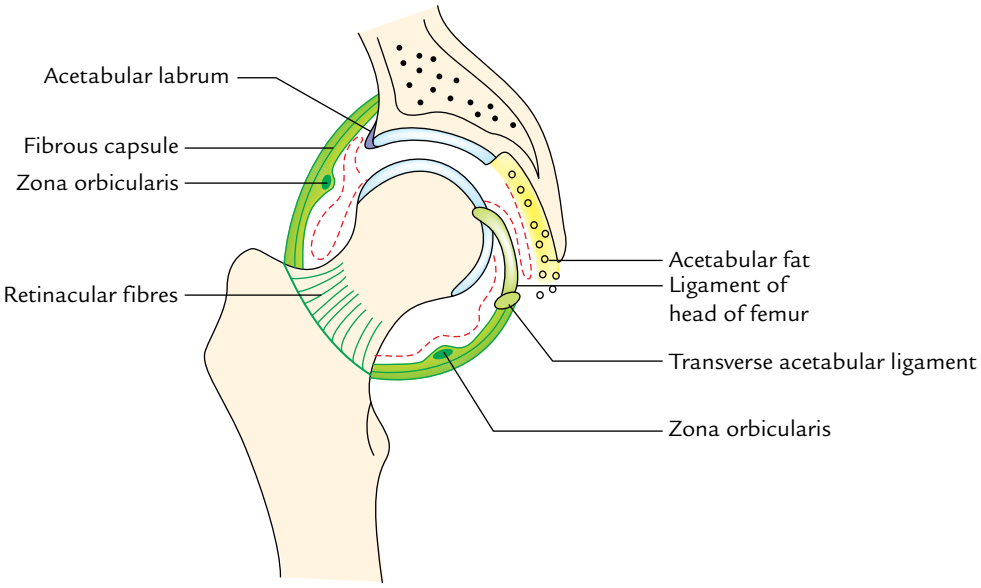


Fig. 26.4 Coronal section of the right hip joint showing the fibrous capsule and the lining of synovial membrane.

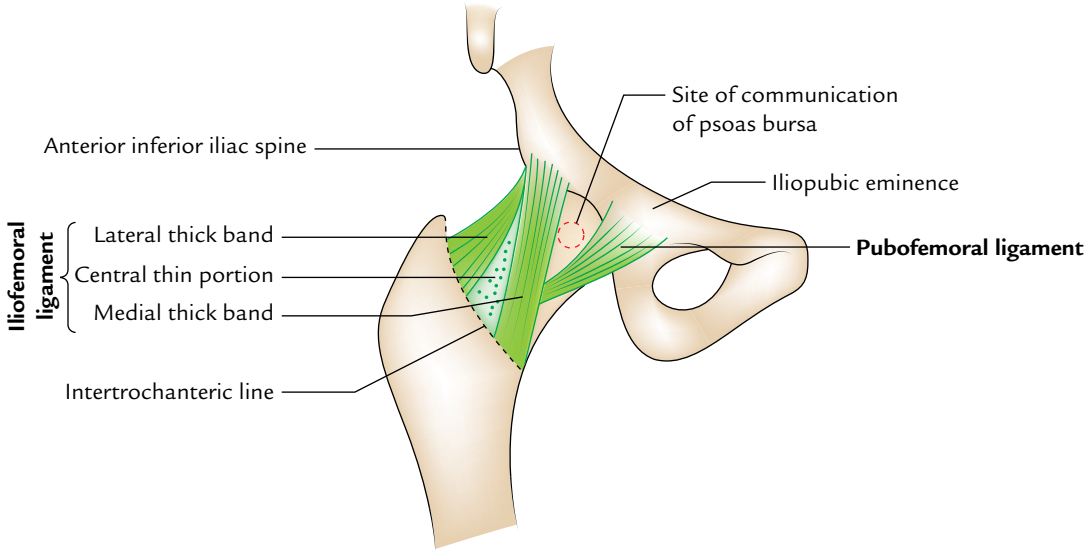


Fig. 26.5 Iliofemoral and pubofemoral ligaments.

a medial thick band of vertical fibres, and a large central thin portion (Fig. 26.5).

N.B. **Iliofemoral ligament** is the strongest ligament of body and prevents the trunk from falling backward in the standing posture.

PUBOFEMORAL LIGAMENT

The pubofemoral ligament is a triangular ligament with base above and apex below. It lies inferomedially and supports the joint on this aspect. Its base is attached to the iliopubic eminence, superior pubic ramus, and obturator crest. Inferiorly it blends with the anteroinferior part of the capsule and medial band of the iliofemoral ligament (Fig. 26.5).

ISCHIOFEMORAL LIGAMENT (Fig. 26.6)

The ischiofemoral ligament is relatively weak and supports the capsule posteriorly. Above it is attached to the ischium posteroinferior to the acetabulum. From ischium its fibres

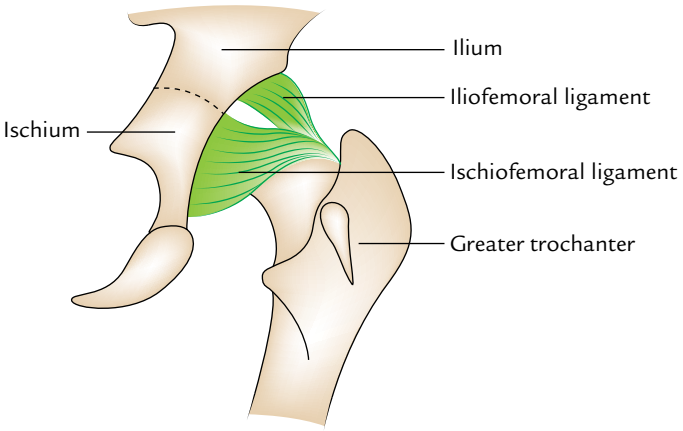


Fig. 26.6 Ischiofemoral and iliofemoral ligaments.

spiral behind the femoral neck to be attached into the greater trochanter deep to the iliofemoral ligament.

ROUND LIGAMENT OF THE HEAD OF FEMUR

This ligament is also called ligamentum teres of the head of femur. It is a flat triangular ligament with apex attached to the fovea of the head, and its base to the transverse acetabular ligament (Fig. 26.7). It is ensheathed by a conical reflection of the synovial membrane. It does not increase the stability of the joint.

It transmits arteries to the head of the femur derived from the acetabular branches of the obturator and medial circumflex femoral arteries.

N.B. Morphologically, the ligament of the head of femur represents the part of capsule that has been included within the joint.

ACETABULAR LABRUM

The acetabular labrum is a fibrocartilaginous rim attached to the acetabular margin. It is triangular in cross section. The labrum not only deepens the acetabulum (socket) but grasps the head of femur lightly to hold it in position.

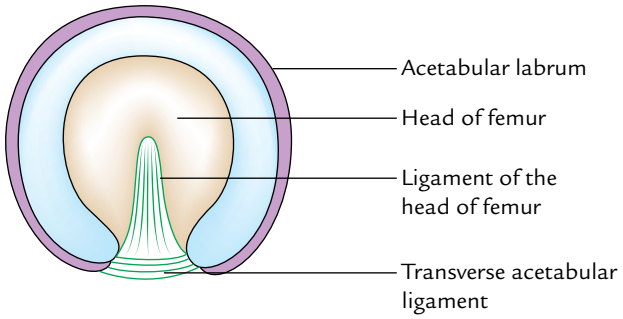


Fig. 26.7 Acetabular (glenoid) labrum, transverse acetabular ligament, and ligament of the head of femur.

TRANSVERSE ACETABULAR LIGAMENT

It is a part of acetabular labrum, which bridges the acetabular notch; however, it is devoid of cartilage cells. The acetabular notch thus becomes converted into the foramen which transmits the acetabular vessels and nerves to the hip joint.

STABILITY OF THE HIP JOINT

The stability of the hip joint is provided by the following factors which help to prevent its dislocation:

1. Depth of the acetabulum and narrowing of its mouth by the acetabular labrum.
2. Three strong ligaments (iliofemoral, pubofemoral, and ischiofemoral) strengthening the capsule of the joint.
3. Strength of the surrounding muscles, e.g., gluteus medius, gluteus minimus, etc.
4. Length and obliquity of the neck of femur.

RELATIONS (Fig. 26.8)

The relations of the hip joint are as follows:

Anteriorly:

1. Tendon of iliopsoas separated from joint by a synovial bursa, pectineus (lateral part), straight head of rectus femoris.
2. Femoral nerve in the groove between the iliacus and the psoas.
3. Femoral artery in front of the psoas tendon.
4. Femoral vein in front of the pectineus.

Posteriorly:

1. Piriformis, obturator externus, obturator internus, superior and inferior gemelli, quadratus femoris, and gluteus maximus.
2. Superior gluteal nerve and vessels above the piriformis.
3. Inferior gluteal nerve and vessels below the piriformis.
4. Sciatic nerve, posterior cutaneous nerve of the thigh, and nerve to quadratus femoris.

Superiorly:

1. Reflected head of rectus femoris medially.
2. Gluteus minimus, gluteus medius, and gluteus maximus laterally.

Inferiorly:

1. Pectineus.
2. Obturator externus.

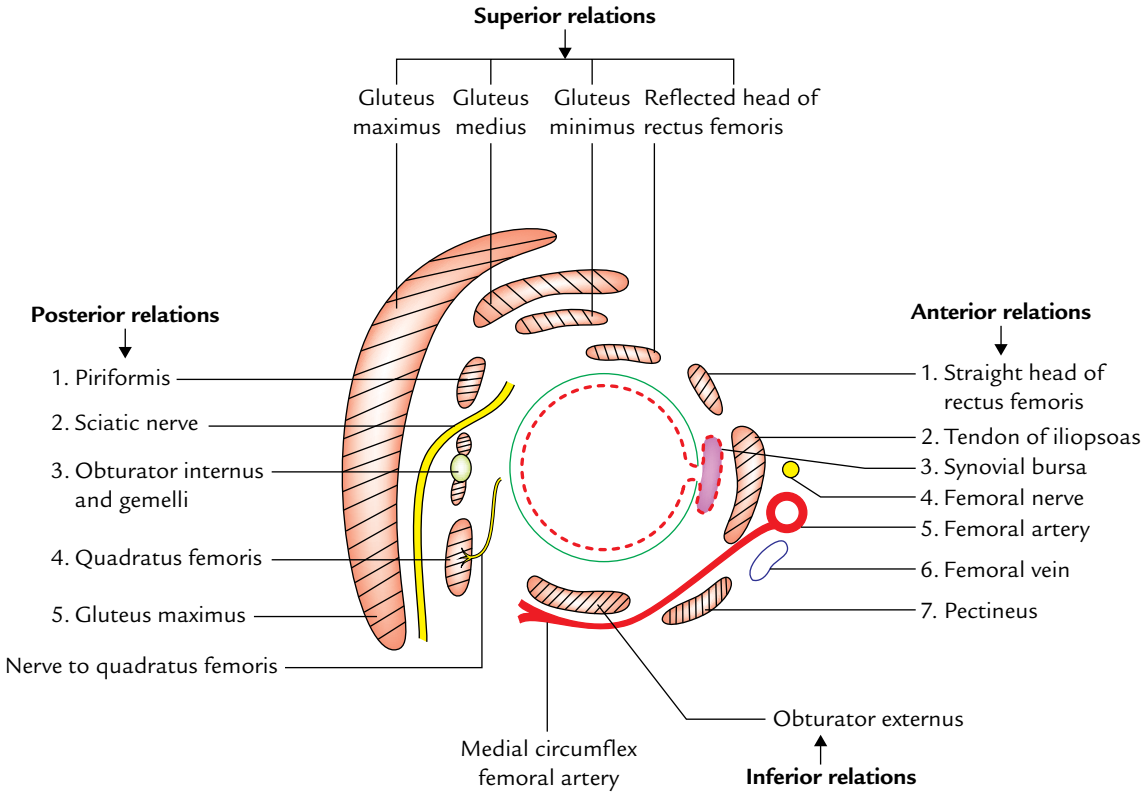


Fig. 26.8 Relations of the hip joint.

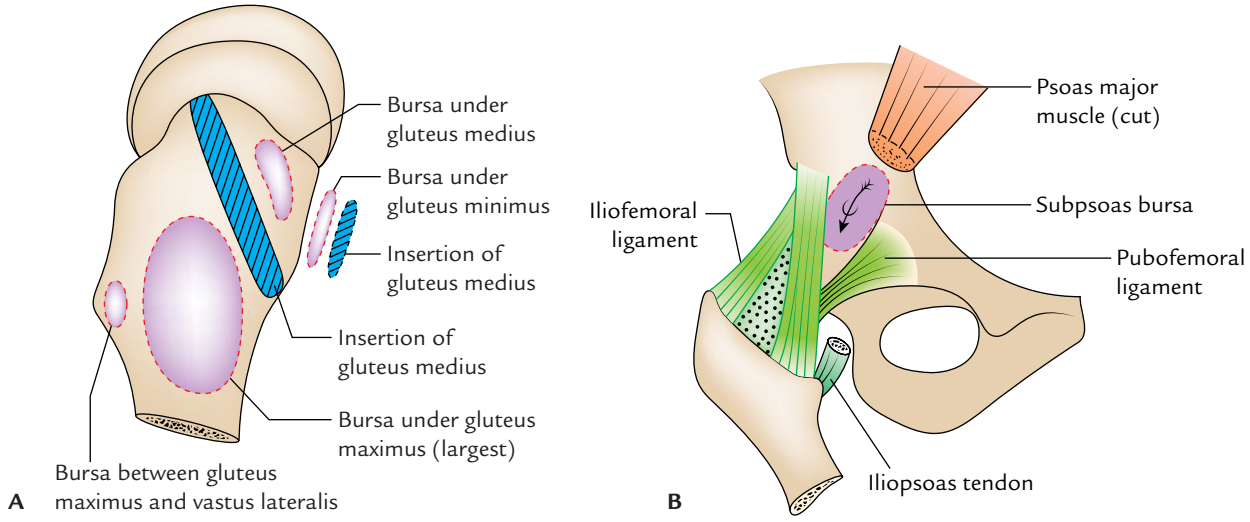


Fig. 26.9 Bursae around the hip joint: **A**, bursae on greater trochanter; **B**, subpsoas bursa communicating with the cavity of hip joint (arrow).

BURSAE AROUND THE HIP JOINT (Fig. 26.9)

These are seven in number: four under glutens maximus, one under gluteus medius, one under gluteus minimus, and one under psoas tendon as under:

1. Between gluteus maximus and smooth area of the ilium lying between the posterior curved line and the outer lip of the iliac crest.
2. Between gluteus maximus and lower part of the outer aspect of the greater trochanter (trochanteric bursa).

3. Between gluteus maximus and ischial tuberosity (ischial bursa).
4. Between the tendon of gluteus maximus and vastus lateralis (gluteofemoral bursa).
5. One bursa under the cover of gluteus medius between it and upper part of the lateral aspect of the greater trochanter.
6. One bursa under the gluteus minimus between it and anterior aspect of the greater trochanter.
7. One between the iliopubic eminence and the psoas tendon. It is called **subpsoas bursa.** In 10% individuals the psoas bursa communicates with the synovial cavity of the hip joint through a gap in the thin part of the capsule between the iliofemoral and pubofemoral ligaments.

Clinical correlation

Weaver's bottom: The subgluteal bursa between the gluteus maximus and ischial tuberosity is frequently inflamed and enlarged in people whose profession requires long periods of sitting, e.g., weavers, leading to a clinical condition called **weaver's bottom**.

ARTERIAL SUPPLY

The hip joint is supplied by the branches of the following arteries:

1. Medial circumflex femoral artery.
2. Lateral circumflex femoral artery.
3. Obturator artery.
4. Superior gluteal artery.
5. Inferior gluteal artery.

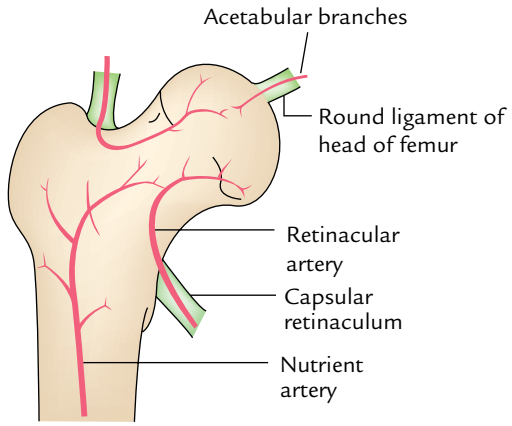


Fig. 26.10 Arterial supply of the head of femur.

N.B. The **arterial supply of the head and neck of the femur** is clinically very important. It is derived from the following three sources (Fig. 26.10):

(a) **Acetabular branches** of the obturator artery and the medial circumflex femoral arteries. These arteries reach the head through the round ligament of the head.
(b) **Retinacular vessels** (the **chief source**) arise from the medial circumflex femoral artery, run along the neck of the femur through the retinaculum of the capsule.
(c) **Nutrient artery** of the femur gives few branches to the neck and head of femur.

NERVE SUPPLY

The hip joint is supplied by the following nerves:

1. Femoral nerve via nerve to rectus femoris.
2. A branch from anterior division of obturator nerve.
3. A branch from accessory obturator nerve (if present).
4. A branch from nerve to quadratus femoris.
5. A branch from superior gluteal nerve.
6. A twig from sciatic nerve (occasional).

N.B. Four consecutive spinal segments (L2, L3, L4, L5) control the movements of the hip joints as under:

- L2 and L3 regulate flexion, adduction, and medial rotation.
- L4 and L5 regulate extension, abduction, and lateral rotation.

MOVEMENTS

The hip joint is a multiaxial joint and permits the following movements:

- Flexion and extension.
- Abduction and adduction.
- Medial and lateral rotation.
- Circumduction (combination of the above movements).

The flexion and extension movements occur around the transverse axis, medial and lateral rotation occur around the vertical axis, and abductor and adduction movements occur around the anteroposterior axis.

Range of Movements

The flexion is 110°–120°. It is limited by contact of the thigh with the abdomen and adduction is limited by contact with the opposite thigh. The range of other movements is as under:

- Extension = 15°
- Abduction = 50°

Table 26.1 Muscles producing the movements of the hip joint

Movements	Muscles producing movements
Flexion	• Psoas major and iliacus (chief flexor) • Sartorius, rectus femoris, and pectineus
Extension	• Gluteus maximus (chief extensor) • Hamstring muscles
Abduction	• Gluteus medius and minimus (chief abductors) • Tensor fasciae latae and sartorius
Adduction	• Adductor longus, adductor brevis, and adductor magnus (chief adductors) • Pectineus and gracilis
Medial rotation	• Anterior fibres of gluteus minimus and medius (chief medial rotators) • Tensor fasciae latae
Lateral rotation	Piriformis, obturator externus, obturator internus and associated gemelli, quadratus femoris (These muscles are generally termed short rotators)

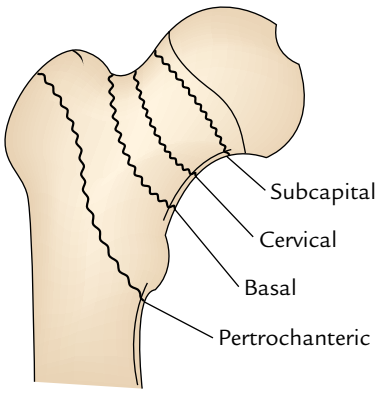


Fig 26.11 Types of fracture of the neck of femur.

- Medial rotation = 25°
- Lateral rotation = 60°

The muscles producing the various movements of the hip joint are given in Table 26.1.

Clinical correlation

- **Dislocation of the hip joint:**

(a) **Congenital dislocation:** The congenital dislocation of the hip joint is more common than any other joint in the body. It occurs due to two reasons:

(i) The joint capsule is loose at birth.

(ii) Hypoplasia of the acetabulum and femoral head: In this condition, the head of femur slips upward into the gluteal region because the upper margin of the acetabulum is developmentally deficient.

Clinically, it presents as:

- Inability of the newborn to abduct the thigh.
- Affected limb is shorter in length and externally rotated.
- Asymmetry of skin folds of the thighs.
- Lurching gait with positive Trendelenburg's sign.

(b) **Acquired dislocation:** The acquired dislocation of the hip joint is uncommon because this joint is very strong and stable. However, it may occur during an automobile accident when the hip joint is flexed, adducted, and medially rotated from the usual position of the lower limb when one is riding in a car. In this position, the joint is unstable because the femoral head is covered posteriorly by a joint capsule and not by the bone. During head on collision, the knee strikes the dashboard and dislocates the hip joint. The head of the femur is forced out of the acetabulum by tearing the capsule posteroinferiorly and lies on the lateral surface of the ilium. This causes shortening and medial rotation of the affected limb.

The dislocation of the hip may be posterior (most common), anterior (less common), or central (least common). The sciatic nerve is injured in posterior dislocation.

- **Perthes' disease (pseudocoxalgia):** It is a clinical condition characterized by destruction and flattening of the head of femur with an increased joint space in the radiograph.
- **Coxa vara and coxa valga:** The normal neck–shaft angle is about 120° in adults and 160° in children. If the neck shaft angle of the femur is reduced (e.g., fracture neck of femur, Perthes disease), it is called *coxa vara*. If the angle is increased (e.g., congenital dislocation of the hip joint), it is called **coxa valga**. This may result from Perthes disease, softening the neck due to rickets.
- **Osteoarthritis:** It is a disease of the old age. It is characterized by the growth of *osteophytes* at the articular ends which not only limits the movements but makes them grating and painful.
- **Referred pain of the hip joint:** In diseases of the hip joint such as tuberculosis, the pain is referred to the knee joint because of the common nerve supply of these two joints.
- **Aspiration of the knee joint:** It is usually done by putting a needle 5 cm below the anterior superior iliac spine, upward, backward, and medially.
- **Fractures of the neck of the femur:** Unfortunately, it is referred as **fractured hip** implying that the hip bone is broken. These fractures are usually common in individuals of more than 60 years of age especially in females because their femoral necks become weak and brittle due to **osteoporosis.**

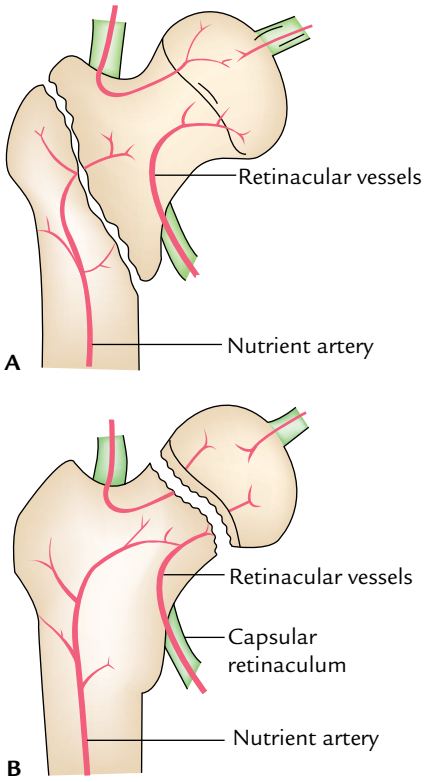


Fig. 26.12 Involvement of arteries supplying the head of femur: A, non-involvement of retinacular vessels in extracapsular pretrochanteric fracture of the neck of femur; B, involvement of retinacular vessels in intracapsular subcapital fracture of the neck of femur.

Types

The fractures of the neck of femur are of four types (Fig. 26.11):

(a) Subcapital (near the head).
(b) Cervical (in the middle).
(c) Basal (near the trochanters).
(d) Pretrochanteric fracture (just distal to two trochanters).

The retinacular vessels derived from the medial circumflex femoral artery supply most of the blood to the head and neck of the femur. A pretrochanteric fracture (extracapsular fracture) does not damage retinacular vessels, hence avascular bone necrosis of head of femur does not occur (Fig. 26.12A). Their damage in intracapsular fracture of the neck (i.e., subcapital and cervical) often lead to aseptic bone necrosis of the head of the femur (Fig. 26.12B).

The fracture of the neck of the femur often occurs due to indirect violence of the trivial nature. The person falls down and cannot get up. The affected limb is much shortened and rotated laterally. The diagnosis is generally confirmed by X-ray by observing following two lines:

(a) **Shenton's line:** In a radiograph of the hip region, the Shenton's line is represented by a continuous curved line formed by the upper border of the obturator foramen and lower margin of the neck of the femur. This curve is disrupted in fracture neck of the femur or dislocation of the hip joint.
(b) **Schoemaker's line:** It is a straight line that extends from the tip of the greater trochanter to the anterior superior iliac spine and continues upward over the anterior abdominal wall to reach the umbilicus. If greater trochanter is elevated (e.g., fracture of the neck of femur) this line passes below the umbilicus.

Golden Facts to Remember

► Strongest and most important ligament of the hip joint	Iliofemoral ligament
► Most common dislocation of the hip joint	Posterior
► Weaver's bottom	Inflamed and enlarged synovial bursa between the gluteus maximus and ischial tuberosity
► Most important source of blood supply to the head of the femur	Retinacular arteries
► Normal neck shaft angle of femur (angle of inclination)	120° in adults and 160° in children
► Unique feature of the hip joint	High degree of stability as well as mobility

Clinical Case Study



A 75-year-old woman could not stand and move her right leg after a minor fall in bathroom. On examination, it was found that her right limb was shortened and held in characteristic laterally rotated position. The X-ray revealed subcapital intracapsular fracture of the neck of femur.

Questions

1. Classify the types of fracture of the neck of the femur.
2. Give the anatomical basis of the lateral rotation and the shortening of the lower limb.
3. How is the length of the femur measured?
4. What is the possible complication of this type of fracture?
5. What are the clinical implications of the neck shaft angle?

Answers

1. The fractures of the neck of the femur are of two types: intracapsular and extracapsular. The intracapsular fractures are further divided into three subtypes: (a) subcapital, (b) transverse cervical, and (c) basal.
2. (a) The lateral rotation results from the change in the axis of the limb due to the separation of the shaft from the head of the femur. As a result of change in the axis, the psoas major becomes the lateral rotator of the thigh.
   (b) The shortening of the limb results due to upward pull of the muscles connecting the femur with the hip bone. The sudden involuntary muscular contraction causes the upward pull.
3. It is from anterior superior iliac spine to the medial malleolus.
4. Avascular necrosis of the proximal segment (e.g., head of femur) due to damage of the retinacular vessels.
5. The clinical implications of the neck–shaft angle (angle of inclination) are:
   (a) Increase in the angle causes **coxa valga,** e.g., in congenital dislocation of the hip joint. It limits the adduction movement of the hip joint.
   (b) Decrease in the angle causes **coxa vara,** e.g., in fracture neck of femur. It limits the abduction of the hip joint.

CHAPTER

27 Front of the Leg and Dorsum of the Foot

FRONT OF THE LEG

The leg is part of the lower limb that lies between the knee and ankle. In general, the structure on the front of the leg extends on to the dorsum of the foot; hence, they are studied together.

SURFACE LANDMARKS (Fig. 27.1)

Bony Landmarks

1. **Medial and lateral condyles of tibia:** These are visible and palpable landmarks at the sides of the ligamentum patellae. They are easily felt in a flexed knee with the thigh flexed and laterally rotated.
2. **Tibial tuberosity:** It is a bony prominence on the front of the upper part of tibia, 2.5 cm distal to the knee joint.
3. **Anterior border (shin) of the tibia:** It is distinct in most of its extent except in the lower part. It is easily felt as a sharp sinuously curved bony crest extending downward from the tibial tuberosity to the anterior margin of the medial malleolus.
4. **Head of the fibula:** It lies posterolaterally at the level of the tibial tuberosity. It serves as a guide to locate common peroneal nerve which winds around the posterolateral aspect of the neck of fibula where it can be rolled.
5. **Medial surface of tibia:** It is subcutaneous throughout except in the uppermost part where it provides attachment to the tendons of sartorius, gracilis, and semitendinosus. The great saphenous vein crosses the lower one-third of this surface, running obliquely upward and backward from the anterior border of medial malleolus.
6. **Medial malleolus:** It is the bony prominence on the medial side of the ankle.
7. **Lateral malleolus:** It is the bony prominence on the lateral side of the ankle. It is longer but narrower than the medial malleolus, and its tip lies about 0.5 cm below that of the medial malleolus and placed on a more posterior plane.
8. **Peroneal trochlea:** It may be felt as a little prominence about a finger breadth below the lateral malleolus.
9. **Sustentaculum tali:** It can be felt about a thumb breadth below the medial malleolus.
10. **Tuberosity of the navicular bone:** It is a bony prominence felt 1 to 1.5 inches anteroinferior to the medial malleolus. The head of the talus lies above the line joining the sustentaculum tali and tuberosity of the navicular bone.
11. **Tuberosity of the base of fifth metatarsal bone:** It is the most prominent landmark on the lateral border of the foot. It lies midway between the point of the heel and the root of the little toe.

Soft Tissue Landmarks

1. **Gastrocnemius and the underlying soleus muscle:** These together form the fleshy prominence of the back of the leg (calf). The *tendocalcaneus*, the strong and thick tendon of these muscles is easily followed downward to its attachment on calcaneus.
2. **Pulsations of posterior tibial artery:** It can be felt about 2 cm below and behind the medial malleolus.
3. **Pulsations of dorsalis pedis artery:** It can be felt on the dorsum of the foot about 5 cm distal to the malleoli and lateral to the tendon of extensor hallucis longus which becomes prominent when the foot is dorsiflexed.
4. **Tendon of tibialis anterior:** It becomes prominent when the foot is inverted. It is seen as passing downward and medially across the medial part of the anterior aspect of the ankle.
5. **Extensor digitorum brevis:** It produces an elevation on the lateral part of the dorsum of the foot when the toes are extended.
6. **First metatarso-phalangeal joint:** It is located a little in front of the centre of the ball of the big toe. The second, third, fourth, and fifth metatarso-phalangeal joints are located about 2.5 cm behind the webs of the toes.

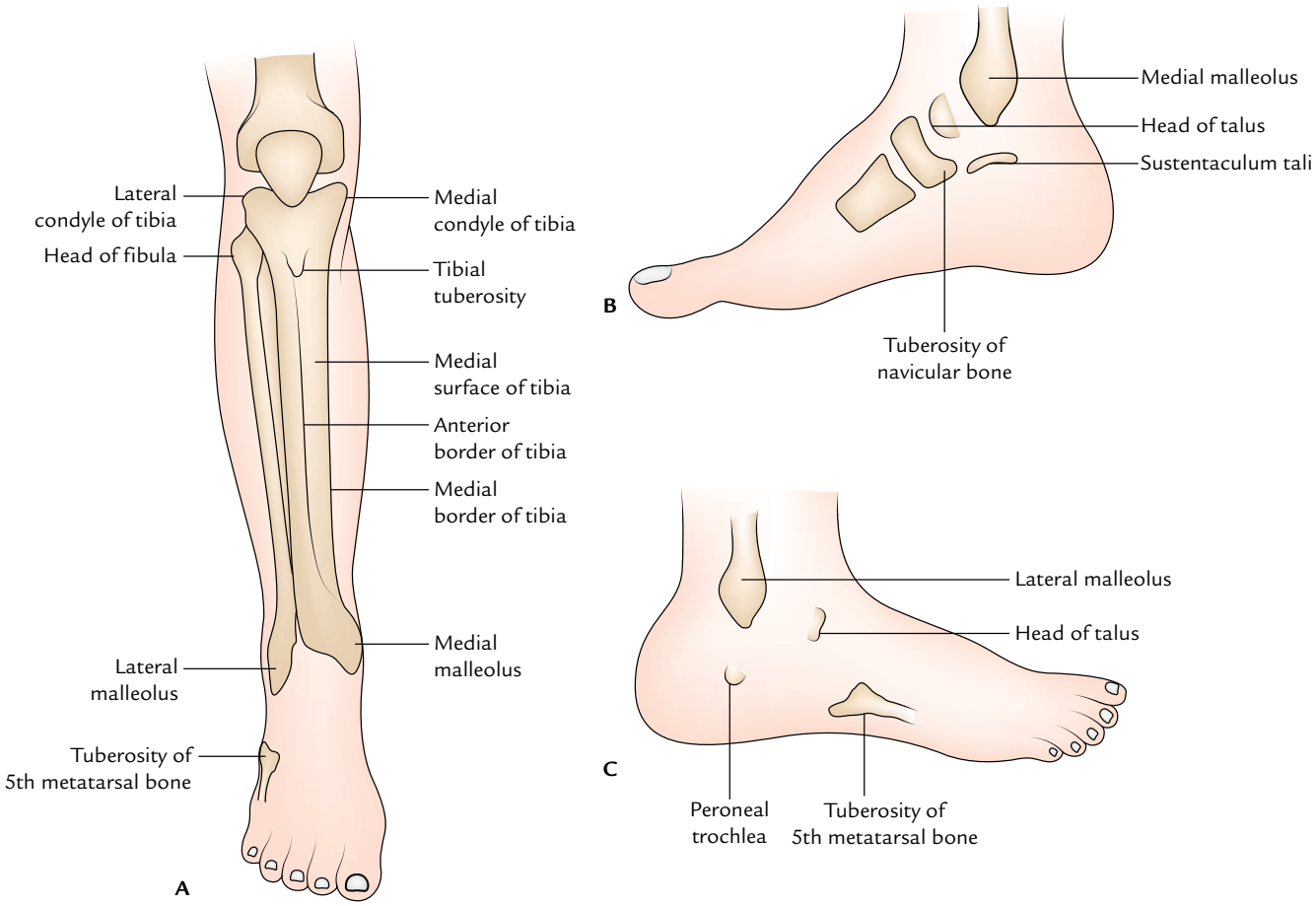


Fig. 27.1 Surface landmarks of the leg and foot: A, anterior aspect of the leg and foot; B, medial aspect of the foot; C, lateral aspect of the foot.

SUPERFICIAL FASCIA

Superficial Veins (Fig. 27.2)

The superficial veins on the dorsum of the leg and foot are as under:

1. **Dorsal venous arch:** It lies on the dorsum of the foot over the proximal parts of the metatarsals.
2. **Dorsal digital veins:** There are two dorsal digital veins in each toe. Each joins the corresponding vein of the adjacent toe to form the dorsal metatarsal vein which terminates in the dorsal venous arch. The *dorsal digital vein* on the medial side of the big toe joins the medial end of the dorsal venous arch to form the *great saphenous vein*. The *dorsal digital vein* on the lateral side of the little toe joins the lateral end of the *dorsal venous arch* to form the *small (short) saphenous vein*.
3. **Great saphenous vein:** It passes upward in front of the medial malleolus, crosses lower one-third of the medial surface of the tibia to reach the medial border of the tibia along which it ascends to reach the back of the

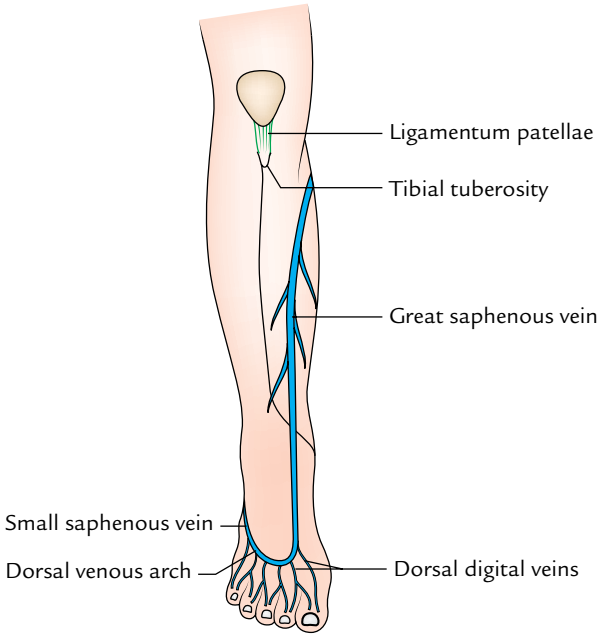


Fig. 27.2 Superficial veins on the front of leg and dorsum of the foot.

knee. The saphenous nerve runs in front of the great saphenous vein.

4. **Small saphenous vein:** It passes upward behind the lateral malleolus to reach the back of the leg where sural nerve accompanies it.

Cutaneous Nerves (Fig. 27.3)

The cutaneous nerves on the front of the leg and dorsum of the foot are as under:

1. **Infrapatellar branch of the saphenous nerve:** It pierces the deep fascia on the medial side of the knee, curves downward and forward to supply the skin over the ligamentum patellae.
2. **Saphenous nerve (L3, L4):** It pierces the deep fascia on the medial side of the knee between the sartorius and gracilis and runs downward in front of the great saphenous vein. It supplies the skin on the medial side of the leg and medial border of the foot up to the ball of the big toe.
3. **Lateral cutaneous nerve of calf (L5; S1,S2):** It is a branch of the common peroneal nerve. It pierces the deep fascia over the lateral head of the gastrocnemius and then descends to supply the skin of the upper two-third of the leg laterally.

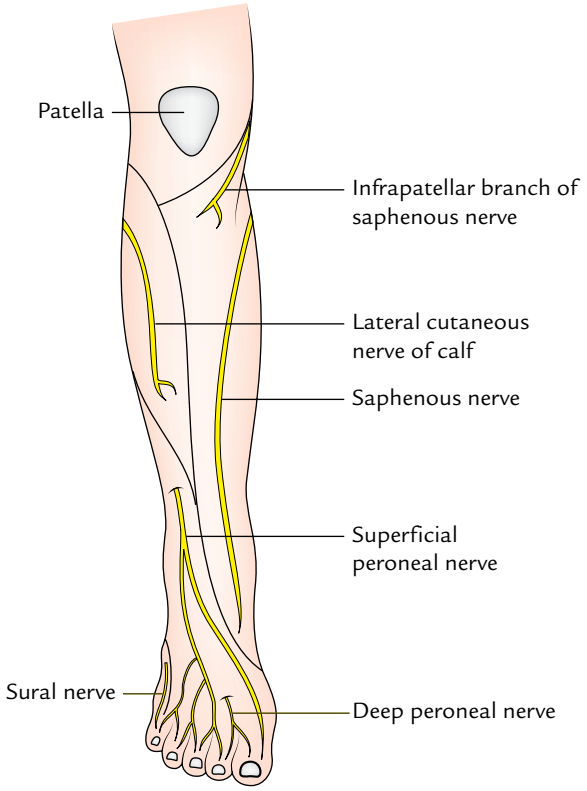


Fig. 27.3 Cutaneous nerves on the front and dorsum of the foot.

4. **Superficial peroneal nerve (L4, L5; S1):** It arises from the common peroneal nerve on the lateral side of the neck of the fibula. It pierces the deep fascia at the junction of the upper two-third and lower one-third of the lateral side of the leg and divides into medial and lateral branches. The superficial peroneal nerve supplies:
   (a) Skin over the lower one-third of the lateral side of the leg.
   (b) Whole of the skin on the dorsum of the foot except for:
      (i) Cleft between the first and second toes which is supplied by the deep peroneal nerve.
      (ii) Lateral border of the foot which is supplied by sural nerve.
      (iii) Medial border of the foot up to the ball of the big toe which is supplied by the saphenous nerve.
5. **Digital branches of the medial and lateral plantar nerves** curve upward and supply the distal parts of the dorsal aspects of the toes.

DEEP FASCIA OF THE LEG

The deep fascia of the leg is very strong and encloses the leg like a tight sleeve.

It does not cover the subcutaneous bony surfaces and is attached to their borders. In other words, it is fused with the periosteum where the bones are subcutaneous.

Two intermuscular septa, anterior and posterior, extend inward from the facial sleeve and get attached to the anterior and posterior borders of the fibula. These together with interosseous membrane divide the leg into three fascial compartments each having its own muscles, arteries, and nerves (Fig. 27.4): anterior, lateral, and posterior. In the posterior compartment, the muscles are divided into three layers by the superficial and deep transverse fascial septa.

RETINACULA

Around the ankle deep fascia is thickened to form strong fibrous bands called retinacula. They are so named because they retain the tendons around ankle in place and prevent their bowstringing during the movements of the foot. The various retinacula around the ankle are:

- **Superior and inferior extensor retinacula,** on the front of ankle
- **Superior and inferior peroneal retinacula,** on the lateral side of ankle.
- **Flexor retinaculum,** on the posteromedial aspect of the ankle.

The retinacula on the front of the ankle are described in the following text. The retinaculae on the lateral and posteromedial sides are described in the lateral and flexor compartments of the leg respectively.

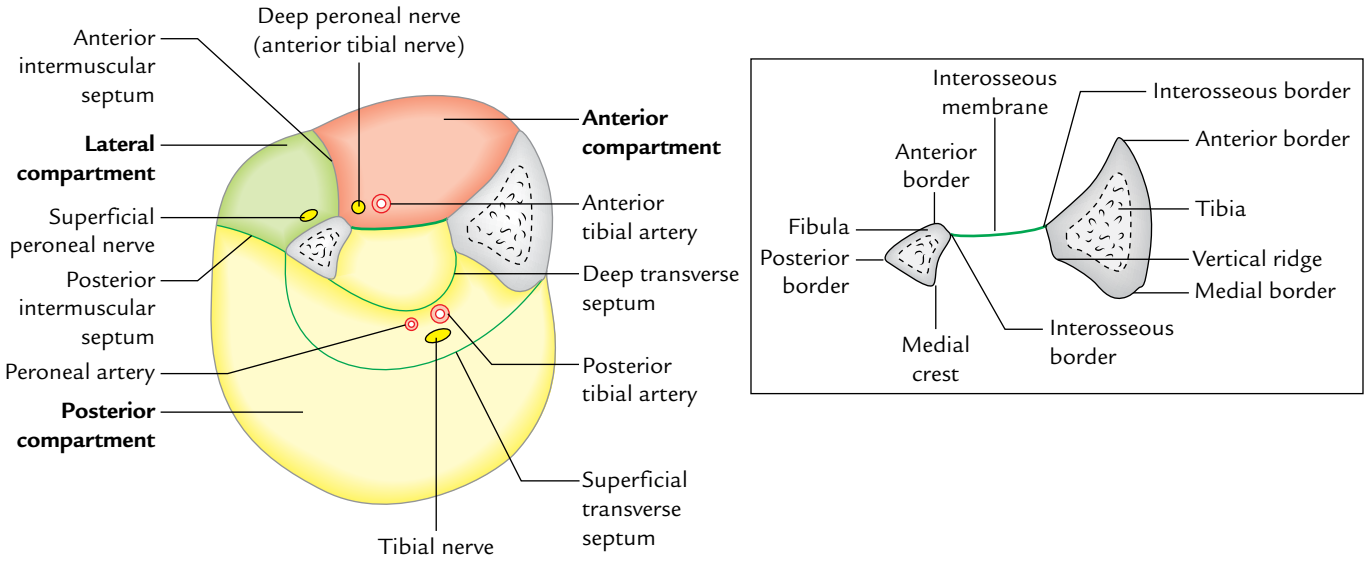


Fig. 27.4 Osseofascial compartments of the leg. Figure in the inset shows transverse section through the tibia and fibula. Note the attachment of interosseous membrane.

Retinacula in Front of the Ankle (Fig. 27.5)

Superior Extensor Retinaculum

It is a broad band of the deep fascia, just above the ankle joint. Vertically it is about 1.5 inches wide.

Attachments

1. *Medially*, it is attached to the lower part of the anterior border of the tibia.
2. *Laterally*, it is attached to the lower part of the anterior border of the fibula.

Relations

1. Medially, it splits to enclose the tendon of tibialis anterior with its synovial sheath.
2. The other structures of the anterior compartment (viz., extensor hallucis longus, deep peroneal nerve, anterior tibial artery, extensor digitorum longus, and peroneus tertius) pass deep to the retinaculum. Here the tendons of these muscles are not surrounded by the synovial sheaths, and deep peroneal nerve and anterior tibial artery lie deep to the tendon of extensor hallucis longus.

Inferior Extensor Retinaculum

It is a Y-shaped band of the deep fascia, situated in front of the ankle joint and the proximal part of the dorsum of the foot. The stem of Y lies laterally, and the upper and lower bands lie medially.

Attachments

1. The *stem* of Y is attached to the anterior nonarticular part of the superior surface of calcaneum in front of the sulcus calcanei, medial to the extensor digitorum brevis.
2. The *upper band* of Y passes upward and medially to be attached to the anterior border of medial malleolus.
3. The *lower band* of Y passes downward and medially over the foot to fuse with the deep fascia of the sole.

Relations

1. The *stem* of Y forms a loop around the tendons of extensor digitorum longus and peroneus tertius with their common synovial sheath.
2. The *upper band* of Y splits to enclose the tendons of tibialis anterior and extensor hallucis longus with their synovial sheaths. The anterior tibial artery and deep peroneal nerve pass deep to it.
3. The *lower band* of Y passes superficially to the tendons of tibialis anterior and extensor hallucis longus with their separate synovial sheaths, and also superficially to the dorsalis pedis artery and deep peroneal nerve.

Functions

1. Keeps the structures of the anterior compartment of the leg at ankle in position.
2. Prevents the tendons from springing medially when the foot is inverted. In this position, there is marked angulation of the tendon of extensor hallucis longus at the ankle.

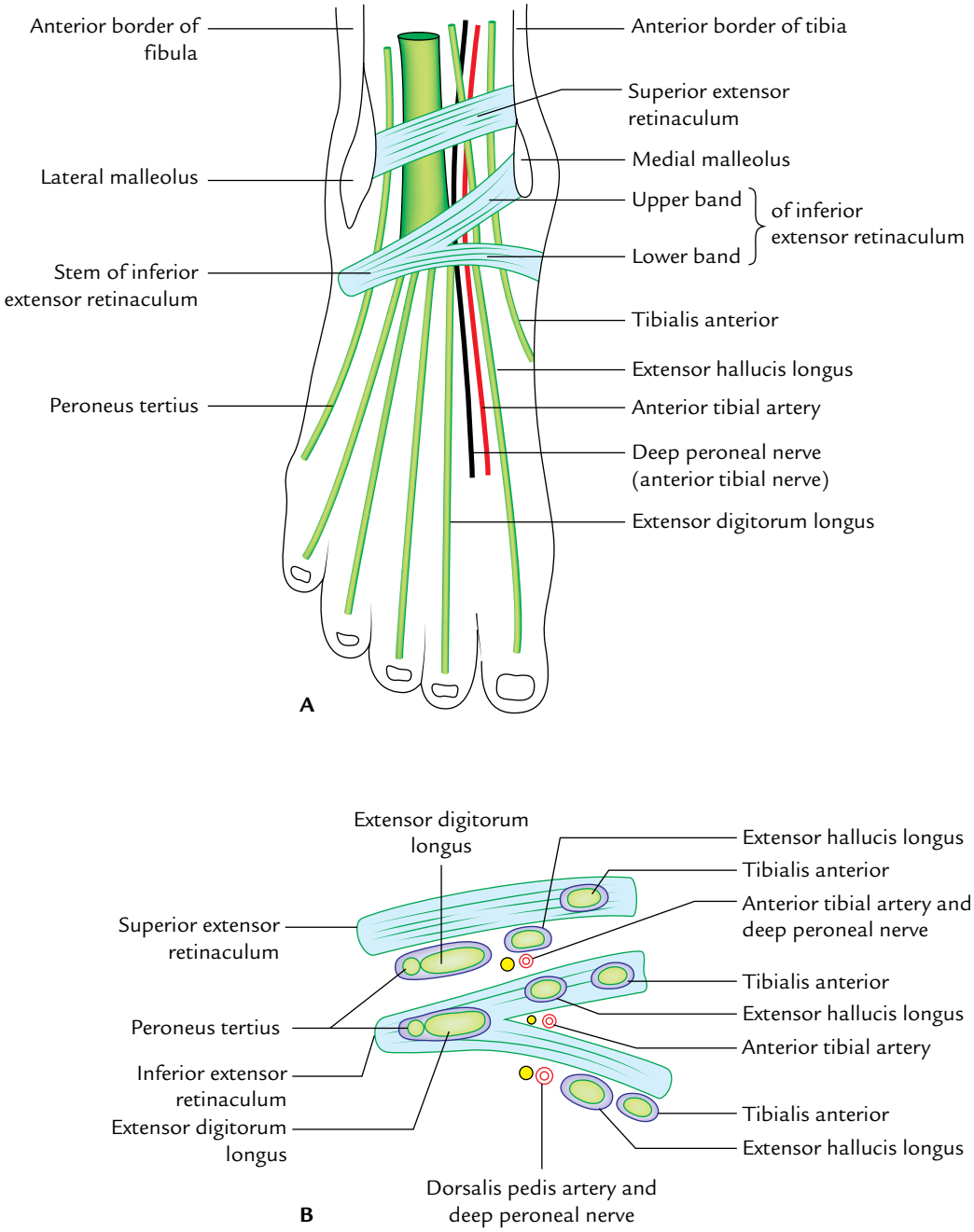


Fig. 27.5 Superior and inferior extensor retinacula on the front of the ankle: **A**, surface view showing their attachment; **B**, schematic diagram to show relations of the tendons, vessels, and nerve to these retinacula.

ANTERIOR (EXTENSOR) COMPARTMENT OF THE LEG

Boundaries

Anterior: Deep fascia of the leg.
Medial: Lateral surface of the shaft of the tibia.
Lateral: Anterior intermuscular septum.
Posterior: Interosseous membrane.

Contents (Fig. 27.6)

- **Muscles:** Four muscles (tibialis anterior, extensor hallucis longus, extensor digitorum longus, and peroneus tertius).
- **Artery:** Anterior tibial artery.
- **Nerve:** Deep peroneal nerve (anterior tibial nerve).

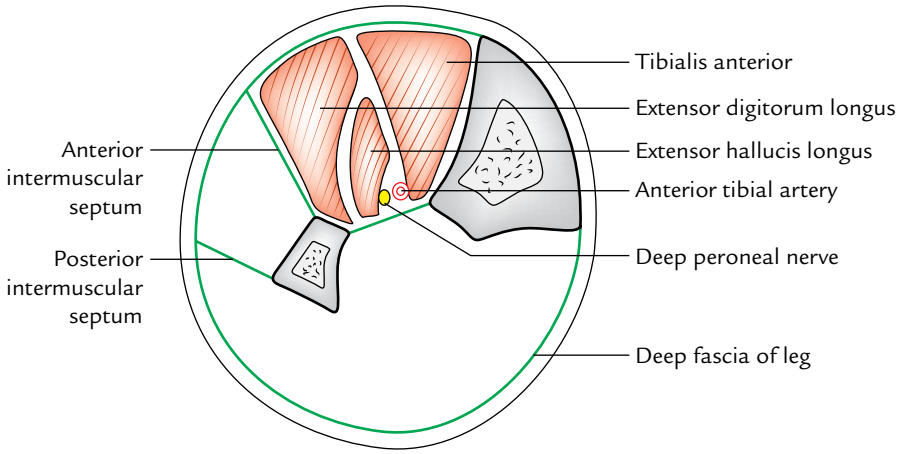


Fig. 27.6 Contents of the anterior compartment of the leg.

MUSCLES OF THE ANTERIOR COMPARTMENT OF THE LEG (Fig. 27.7)

All the muscles of the anterior compartment of the leg arise from the fibula except the tibialis anterior which arises from the tibia (Fig. 27.7). All of them are supplied by the deep peroneal nerve and dorsiflex, the foot at the ankle. The origin and insertion of the muscles of anterior compartment are shown in Figure 27.7 and Table 27.1.

The tibialis anterior being the chief muscle of the anterior compartment is described in detail below.

Tibialis Anterior (Fig. 27.8)

It is a spindle-shaped multipennate muscle. It is the most medial and superficial dorsiflexor of the foot, which lies against the lateral surface of the tibia.

Origin

It arises from:

(a) Upper two-third of the lateral surface of the tibia and adjoining part of the lateral condyle of the tibia.
(b) Adjoining part of the interosseous membrane.
(c) Overlying deep fascia of the leg.

Insertion

The muscle fibres converge below to form a tendon which is related to the lower one-third of the lateral surface of the tibia. It pierces the medial part of superior extensor retinaculum and the upper band of inferior extensor retinaculum. Now it passes medially underneath the inferior band of inferior extensor retinaculum to be inserted *on to the inferomedial side of the base of the first metatarsal bone and adjacent part of the medial cuneiform.*

Nerve supply

By the deep peroneal nerve.

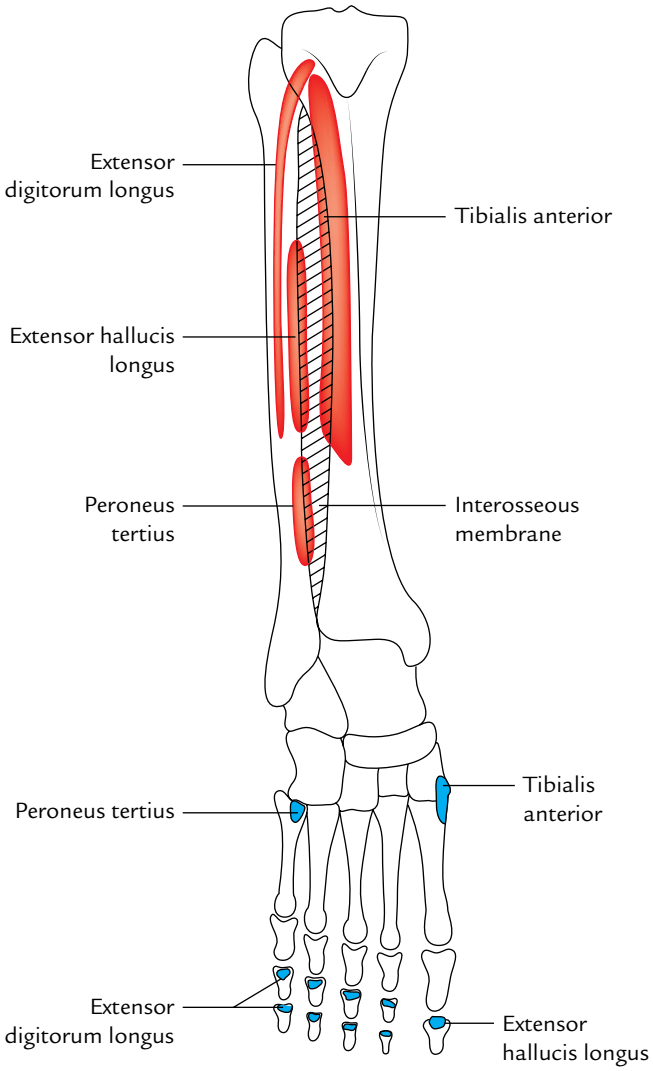


Fig. 27.7 Origin (red colour) and insertion (blue colour) of the muscles of the anterior compartment of leg.

Table 27.1 Origin, insertion, and actions of the muscles of the anterior compartment of the leg

Muscle	Origin from	Insertion into	Actions
Tibialis anterior (it has spindle-shaped multipennate belly)	• Upper 2/3rd of the lateral surface of the tibia • Adjacent part of the interosseous membrane • Its own covering deep fascia • Distal part of the lateral condyle of the tibia	• Medial surface of the medial cuneiform • Adjoining medial surface of the base of the 1st metatarsal	• Dorsiflexion of the ankle • Inversion of the foot • Maintenance of the medial longitudinal arch of the foot
Extensor hallucis longus (Fig. 27.9)	• Middle 2/4th (posterior part) of the medial surface of the shaft of fibula • Adjacent part of the interosseous membrane	Base (dorsal surface) of the distal phalanx of the big toe	• Extension of the phalanges of the big toe • Dorsiflexion of the foot
Extensor digitorum longus (Fig. 27.10)	• Whole of the upper 1/4th and medial part of the middle 2/4th of the anterior surface of the fibula	Middle and distal phalanges of the lateral four toes by four tendons	• Dorsiflexion of the foot • Extension of MP*, PIP**, and DIP*** joints of the lateral four toes
Peroneus tertius	• Lower 1/4th of the anterior surface of the fibula • Adjacent part of the interosseous membrane	Dorsal surface (medial part) of the base of the 5th metatarsal bone	• Dorsiflexion of the foot • Eversion of the foot

*MP: metatarsophalangeal, **PIP = proximal interphalangeal, ***DIP = distal interphalangeal.

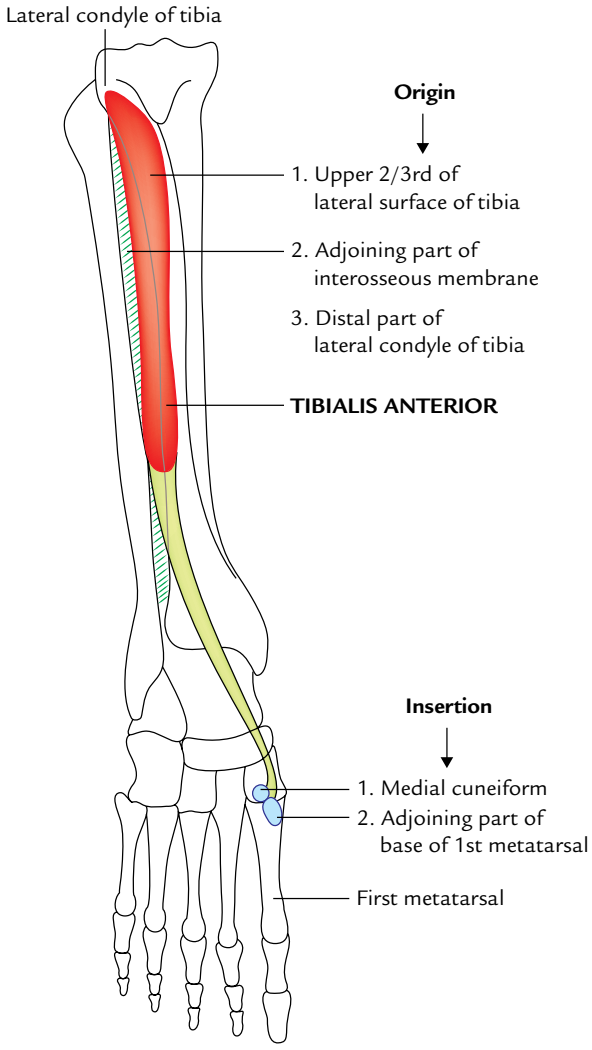


Fig. 27.8 Origin and insertion of the tibialis anterior.

Actions

1. It is the chief dorsiflexor of the foot at the ankle joint.
2. It maintains the medial longitudinal arch.
3. It acts as an inverter of the foot at the midtarsal and subtalar joints.

Clinical testing

To test tibialis anterior clinically, ask the patient to stand on heels or dorsiflex the foot against resistance. If normal, its tendon can be seen and palpated.

Clinical correlation

Anterior tibial compartment syndrome/shin splints (Fresher's syndrome): It occurs due to overexertion of the muscles of the anterior compartment especially when the untrained persons who lead a sedentary life are asked to walk or run for long distances. Shin splints also occur in trained runners who do not warm-up.

Muscles of the compartment swell within a tight compartment due to sudden overuse which may impede venous return leading to accumulation of more fluid inside the compartment with unyielding walls. The pressure tends to compress the anterior tibial artery, and reducing the blood supply to the muscles, leading to ischemia and pain. It is frequently seen in freshers (e.g., newly admitted medical students/newly recruited army personnel) who are made to run excessively. Hence, this condition is also referred to as army **fresher's syndrome**.

ANTERIOR TIBIAL ARTERY (Fig. 27.11)

The anterior tibial artery is the main artery of the anterior compartment of the leg. It corresponds to the posterior interosseous artery of the forearm. The blood supply to the

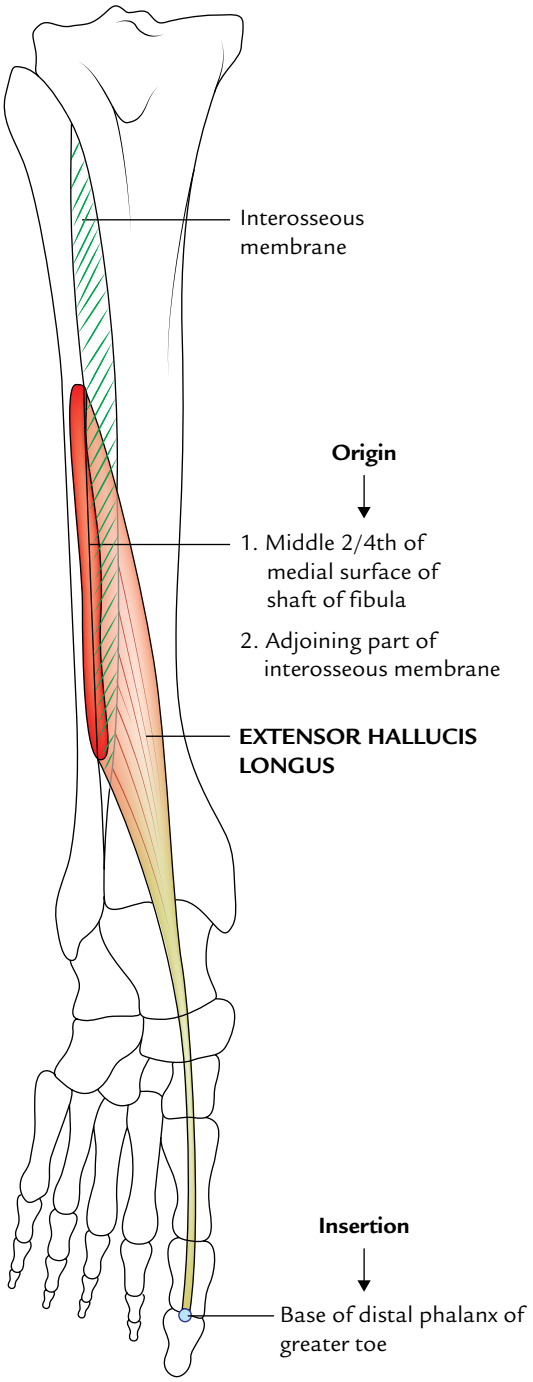


Fig. 27.9 Origin and insertion of the extensor hallucis longus.

anterior compartment of the leg is reinforced by the perforating branch of peroneal artery. Therefore, the size of peroneal artery is inversely proportional to that of the anterior tibial artery.

The anterior tibial artery is accompanied by two venae comitantes.

Origin

Anterior tibial artery is the smaller terminal branch of popliteal artery given at the lower border of popliteus muscle.

Course

1. It begins in the back of the leg at the lower border of popliteus.
2. It enters the anterior compartment of the leg by passing forward between the two heads of the tibialis posterior, through an opening in the upper part of the interosseous membrane. In the anterior compartment, it runs vertically downward to a point midway between the medial and lateral malleoli, where it enters the foot and changes its name to **dorsalis pedis artery,** which ends near the web between the big and second toes.

Relations (Fig. 27.12)

- In the upper one-third of the leg it lies between the tibialis anterior and extensor digitorum longus.
- In the middle one-third of the leg it lies between the tibialis anterior and extensor hallucis longus.
- In the lower one-third of the leg it lies between extensor hallucis longus and extensor digitorum longus.

It is crossed from the lateral to medial side by the tendon of extensor hallucis longus. As a result, the deep peroneal nerve lies lateral to it in its upper one-third and lower one-third, and anterior to it in its middle one-third.

Branches

1. *Anterior and posterior tibial recurrent arteries:* They take part in the arterial anastomosis around the knee joint.
2. *Muscular branches* to adjacent muscles.
3. *Anterior medial and anterior lateral malleolar arteries:* They take part in the anastomosis around the ankle joint.

N.B. *Variations of the anterior tibial artery:* The anterior tibial artery may fail to grow more than a short way down the leg. In such cases, the dorsalis pedis artery arises from the perforating branch of the peroneal artery or perforating branch of the posterior tibial artery.

DEEP PERONEAL NERVE (ANTERIOR TIBIAL NERVE)

It is the nerve of anterior compartment of the leg and dorsum of the foot. It corresponds to the posterior interosseous nerve of the forearm.

Origin

It is one of the two terminal branches of the common peroneal nerve at the neck of the fibula.

Course and Relations (Fig. 27.12)

- It *begins* on the lateral side of the neck of fibula, under cover of the upper fibres of peroneus longus muscle.

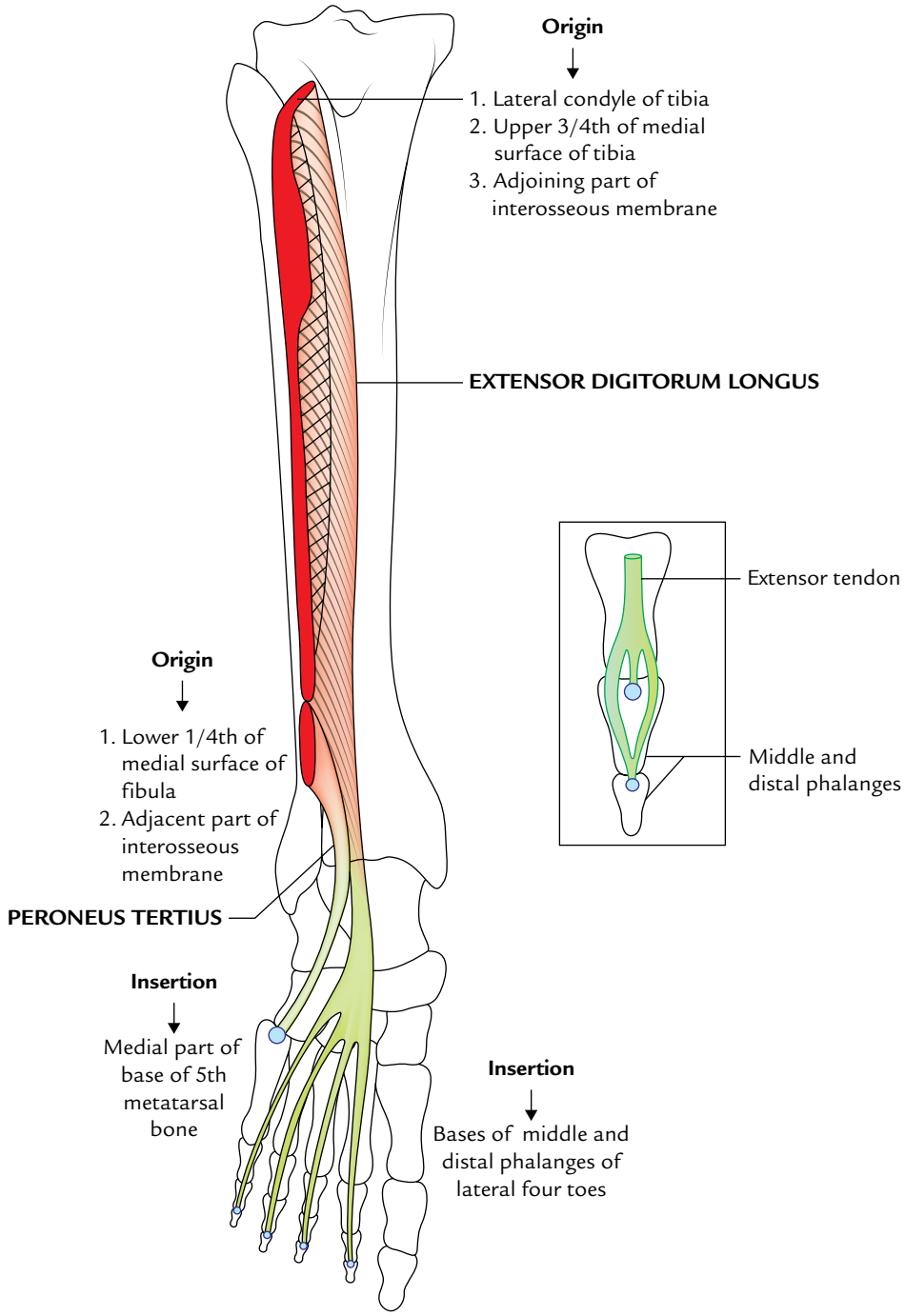


Fig. 27.10 Origin and insertion of the extensor digitorum longus and peroneus tertius muscles. Figure in the inset shows the detailed mode of insertion of extensor tendon.

- It *enters the anterior compartment* of the leg by piercing the anterior intermuscular septum. It pierces extensor digitorum longus and descends in this compartment with the anterior tibial artery.
- In the leg, it accompanies the anterior tibial artery. The nerve lies lateral to artery in its upper one-third and lower one-third and anterior to artery in the middle one-third. It is said that in the middle one-third the nerve hesitates to cross the artery from lateral to medial side, so it goes back to the lateral side of the artery. Hence, deep peroneal nerve is also referred to as **nervus hesitans.**
- The nerve ends in front of the ankle by dividing into the lateral and medial terminal branches.

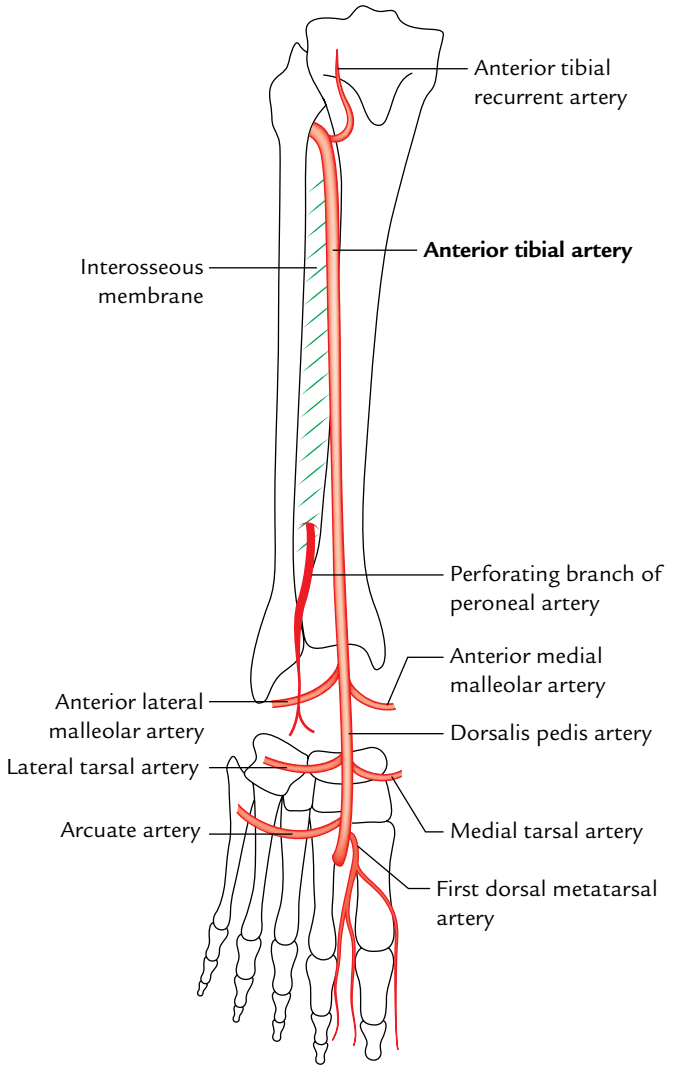


Fig. 27.11 Anterior tibial artery.

The *lateral terminal branch* runs laterally and ends in a pseudoganglion deep to the extensor digitorum brevis. Branches from the pseudoganglion supply the extensor digitorum brevis and tarsal and metatarsal joints on the lateral side of the foot.

The *medial terminal branch* runs forward and ends by supplying the skin of the adjacent sides of big and second toes (first interdigital cleft) and the first dorsal interosseous muscle (Fig. 27.13).

Branches

1. *Muscular branches* supply all the four muscles of the anterior compartment of the leg, extensor digitorum brevis on the dorsum of the foot and first dorsal interosseous muscle of the sole of the foot.
2. *Cutaneous branch* supplies the skin of the first interdigital cleft.

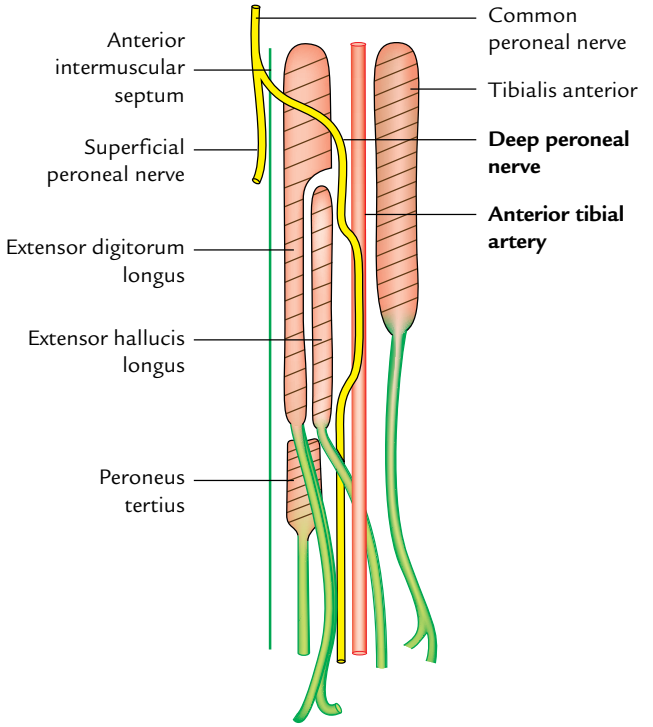


Fig. 27.12 Relation of the anterior tibial artery and deep peroneal nerve.

DORSUM OF THE FOOT

SENSORY INNERVATION OF THE DORSUM OF THE FOOT (Fig. 27.14)

The sensory innervation to the dorsum of the foot is provided by the following four sets of nerves:

1. **Superficial peroneal (musculocutaneous) nerve:** It provides sensory innervation to most of the dorsum of the foot except the skin of the cleft between the first and second toes. It also supplies medial margin of the great toe.
2. **Deep peroneal nerve:** It provides innervation to the cleft between the first and second toes.
3. **Sural nerve:** It supplies lateral margin of the dorsum of the foot and lateral margin of the little toe.
4. **Saphenous nerve:** It supplies medial margins of the dorsum of the foot up to the head of the first metatarsal.

MUSCLES OF THE DORSUM OF THE FOOT

These are as follows:

(a) *Extrinsic tendons* of the muscles of the anterior compartment of the leg (viz., tibialis anterior, extensor hallucis longus, extensor digitorum longus, and peroneus tertius).

(b) *Intrinsic muscle* on the dorsum of the foot is only one—the extensor digitorum brevis.

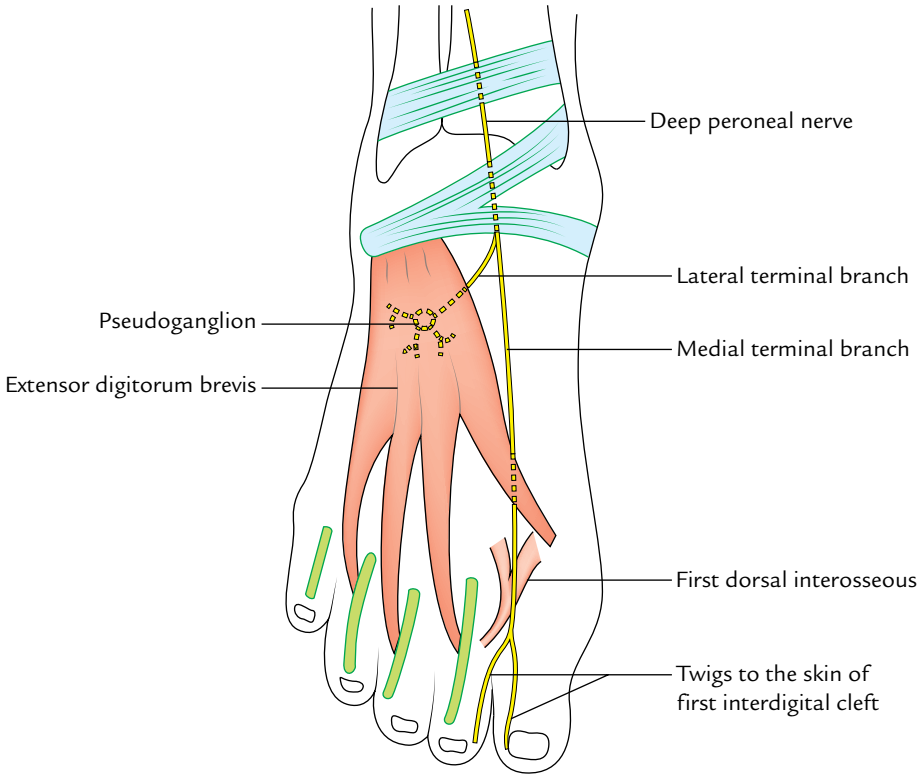


Fig. 27.13 Course and distribution of the deep peroneal nerve on the dorsum of the foot.

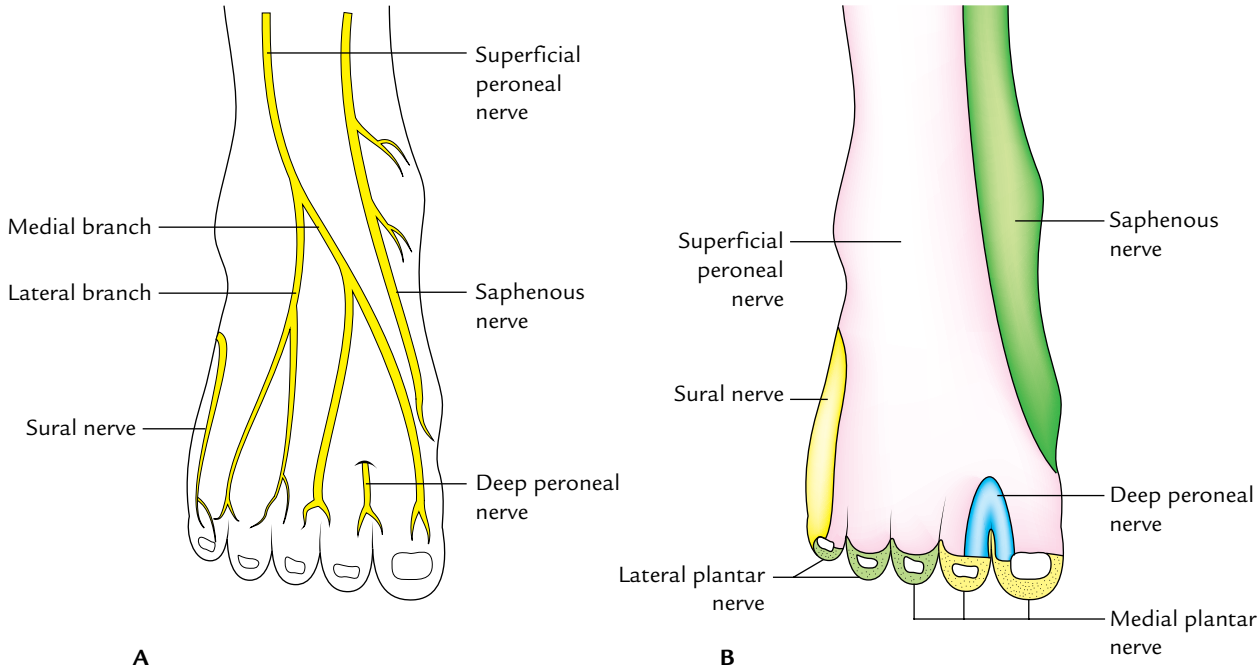


Fig. 27.14 Sensory innervation of the dorsum of the foot: **A**, cutaneous nerves; **B**, territories of the cutaneous nerves.

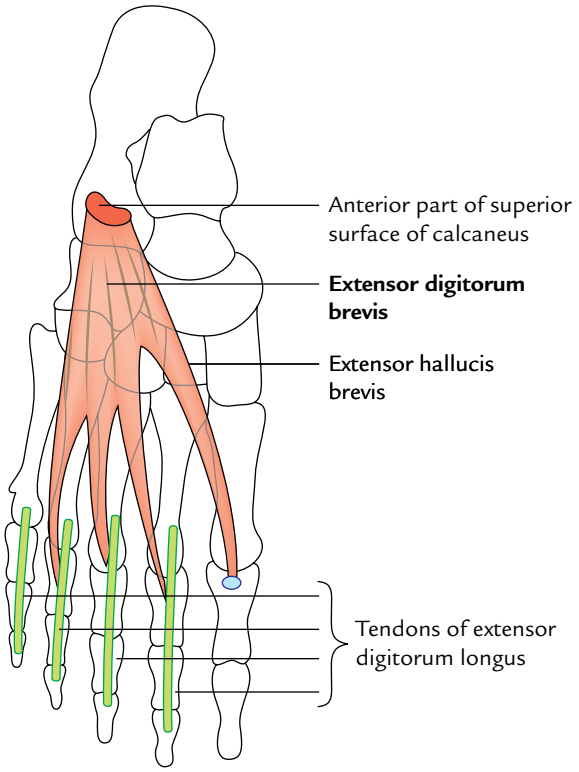


Fig. 27.15 Origin and insertion of extensor digitorum brevis.

Extensor Digitorum Brevis (Fig. 27.15)

It is a small muscle situated on the lateral part of the dorsum of the foot, deep to the tendons of extensor digitorum longus. It is the only muscle on the dorsum of the foot and forms a fleshy swelling anterior to the lateral malleolus. **The young novice doctors sometime diagnose it as a contusion.**

Origin

It arises from the anterior part of the superior surface of calcaneum, medial to the attachment of the stem of inferior extensor retinaculum.

Insertion

The muscle divides into four tendons for the medial four toes. The tendon to the big toe crosses in front of dorsalis pedis artery and inserts on the dorsal surface of the proximal phalanx of the big toe. The lateral three tendons join the lateral side of the tendons of the extensor digitorum longus to the second, third, and fourth toes.

N.B. Medial-most part of the extensor digitorum brevis, which forms the tendon for the big toe, separates or becomes distinct early. It is known as *extensor hallucis brevis.*

Nerve supply

It is by the lateral terminal branch of the deep peroneal nerve.

Actions

1. Extensor hallucis brevis (EHB) extends the metatarsophalangeal joint of the big toe.
2. The other three tendons extend the metatarsophalangeal and interphalangeal joints of second, third, and fourth toes, particularly when the foot is dorsiflexed.

DORSALIS PEDIS ARTERY

It is the chief artery of the dorsum of the foot.

Origin

The dorsalis pedis artery is the direct continuation of the anterior tibial artery in front of the ankle.

Course (Fig. 27.16)

It passes forward along the medial side of the dorsum of the foot to reach the proximal end of the first intermetatarsal space, where it dips downward between the two heads of the first dorsal interosseous muscle to enter the sole of the foot where it ends by anastomosing with the lateral plantar artery.

Relations

Superficial: Extensor hallucis brevis crosses the artery superficially from the lateral to medial side.
Deep: Ankle joint and tarsal bones.
Medial: Tendon of the extensor hallucis longus (EHL).
Lateral: First tendon of the extensor digitorum longus (EDL).

N.B. *Variations of dorsalis pedis artery:* (a) In about 14% of cases, it may be replaced by the perforating branch of the peroneal artery. (b) It may be too large to compensate for the small *lateral plantar artery* of the sole of foot.

Branches

1. and 2. *Lateral and medial tarsal arteries:* They take part in the formation of lateral and medial malleolar arterial networks.

3. *Arcuate artery:* It arises near the base of the second metatarsal, and runs laterally with slight convexity toward the toes, to reach the lateral edge of the foot. It gives **three dorsal metatarsal arteries** (second, third, and fourth), each of which divides into two dorsal arteries for the lateral four toes. The lateral one sends a twig to the lateral side of the little toe.

4. *First dorsal metatarsal artery:* It arises just before the dorsalis pedis artery dips into the sole of the foot. It divides into dorsal digital arteries for the adjacent sides of the first and second toes. It also gives a dorsal digital artery to the medial side of the big toe.

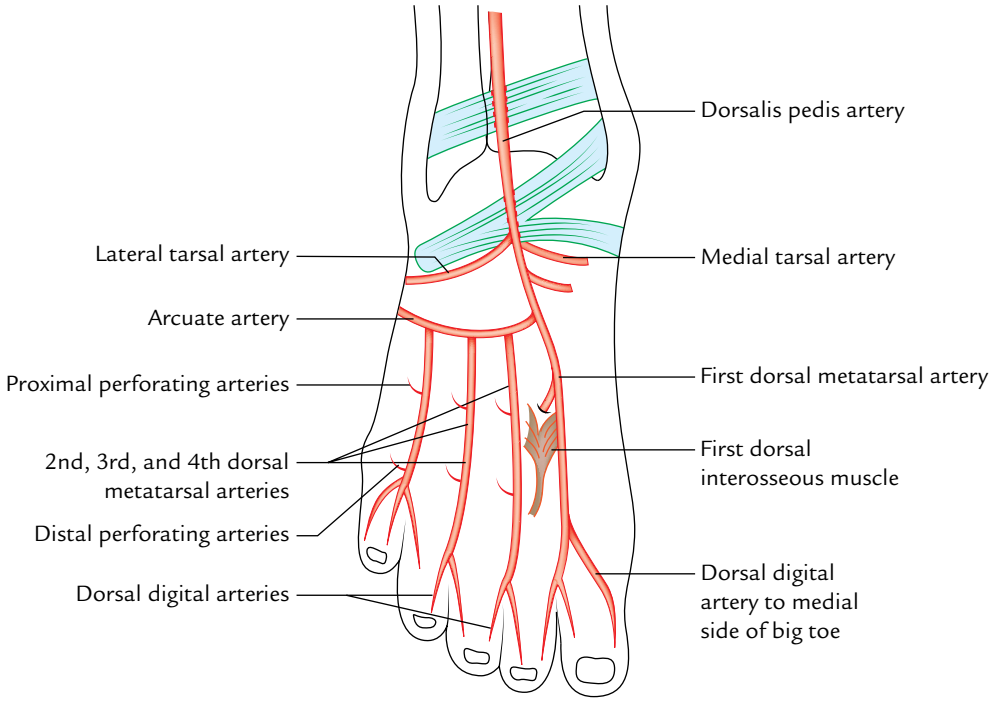


Fig. 27.16 Course and branches of dorsalis pedis artery.

Clinical correlation

Dorsalis pedis artery pulse: It can be easily felt between the tendons of extensor hallucis longus and first tendon of extensor digitorum longus. It is often palpated in patients suffering from vaso-occlusive diseases of the lower limb, viz., **Buerger's disease**.

The clinicians feeling this pulse should know that the dorsalis pedis artery is congenitally absent in about 14% of the cases. In such cases the absence of dorsalis pedis pulse should be confirmed by the *posterior tibial artery pulse*.

Golden Facts to Remember

► Chief muscle of the anterior compartment of the leg	Tibialis anterior
► Most recent muscle of the anterior compartment of the leg from evolutionary point of view	Peroneus tertius
► Only intrinsic muscle on the dorsum of the foot	Extensor digitorum brevis
► All the muscles of the anterior compartment of the leg arise from the fibula *except*	Tibialis anterior which arises from tibia
► Nervus hesitans	Deep peroneal nerve
► Cutaneous innervation of the whole of the dorsum of the foot is from superficial peroneal nerve *except*	Cleft between the first and second toes which is innervated by the deep peroneal nerve
► Most superficial and strongest dorsiflexor of the foot	Tibialis anterior

Clinical Case Study



A 27-year-old healthy individual was recruited in the army. Few days later he was made to run for a long distance as a part of his training. In the night when he was going to sleep he felt severe pain in his legs which was aggravated by movements. He was given a painkiller by the doctor to relieve the pain. Next day morning he could not dorsiflex his feet. He was taken to the army hospital where the doctors found that both of his legs were swollen and tender to pressure. They also noted foot drop on both the sides. He was diagnosed as a case of *"anterior tibial compartment syndrome"/"shin splints."*

Questions

1. What is meant by the anterior tibial compartment syndrome?
2. What is the cause of this syndrome?
3. Why is the anterior compartment of the leg susceptible to this syndrome?
4. Name the nerve and artery which course within the anterior compartment of the leg and supply muscles present within it.
5. Give the cause of foot drop in this syndrome.

Answers

1. It is a clinical condition characterized by pain and swelling of the distal two-third of the leg.
2. Due to overexertion the muscles of the anterior compartment of the leg swell and compress the anterior tibial artery leading to ischemia and pain.
3. Because the anterior compartment is confined mostly by unyielding bones and deep fascia.
4. Deep peroneal nerve and anterior tibial artery.
5. Compression of the deep peroneal nerve.

CHAPTER

28 Lateral and Medial Sides of the Leg

LATERAL SIDE OF THE LEG

The lateral side of the leg deals with the lateral fascial compartment of the leg and its contents.

LATERAL COMPARTMENT OF THE LEG

Boundaries (Fig. 28.1).

Anterior: Anterior intermuscular septum.
Posterior: Posterior intermuscular septum.
Medial: Lateral surface of the fibula.
Lateral: Deep fascia of the leg.

Contents

1. *Muscles:* Peroneus longus and peroneus brevis.
2. *Nerve:* Superficial peroneal nerve.
3. *Artery:* The lateral compartment of the leg does not have its own artery.
4. *Veins:* Small unnamed veins mostly drain into the short saphenous vein.

N.B. The contents of lateral compartment of the leg are supplied by the branches of the peroneal artery (a branch of posterior tibial artery) which reaches the lateral compartment by flexor hallucis longus and posterior intermuscular septum.

PERONEAL RETINACULA (Fig. 28.2)

The peroneal retinacula are two thick bands of deep fascia on the lateral side of ankle which keep the long tendons of peroneus longus and peroneus brevis in position and act as a pulley for them.

SUPERIOR PERONEAL RETINACULUM

The superior peroneal retinaculum is a thickened fibrous band of deep fascia situated just behind the lateral malleolus.

Attachments

Anteriorly: It is attached to the back of the lateral malleolus.
Posteriorly: It is attached to the lateral surface of the calcaneum and superficial transverse fascial septum of the leg.

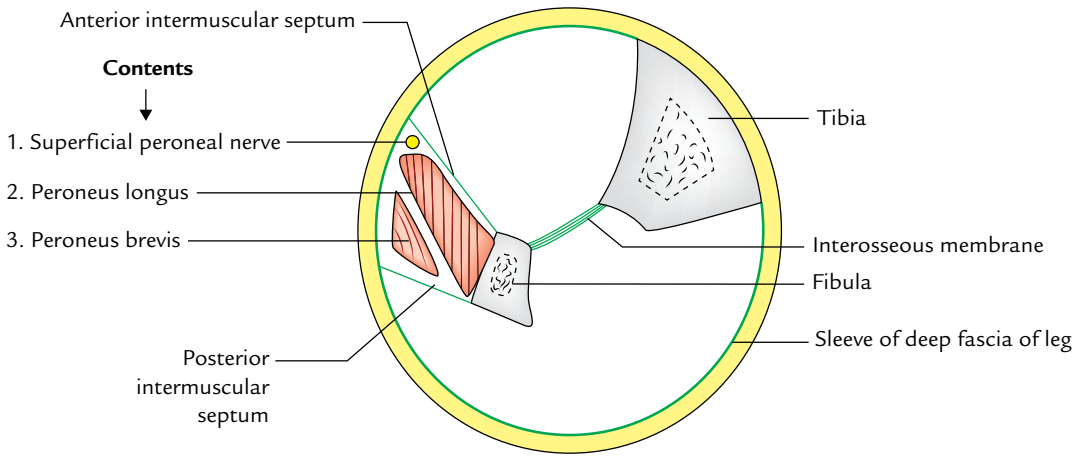


Fig. 28.1 Boundaries and contents of the lateral compartment of the leg.

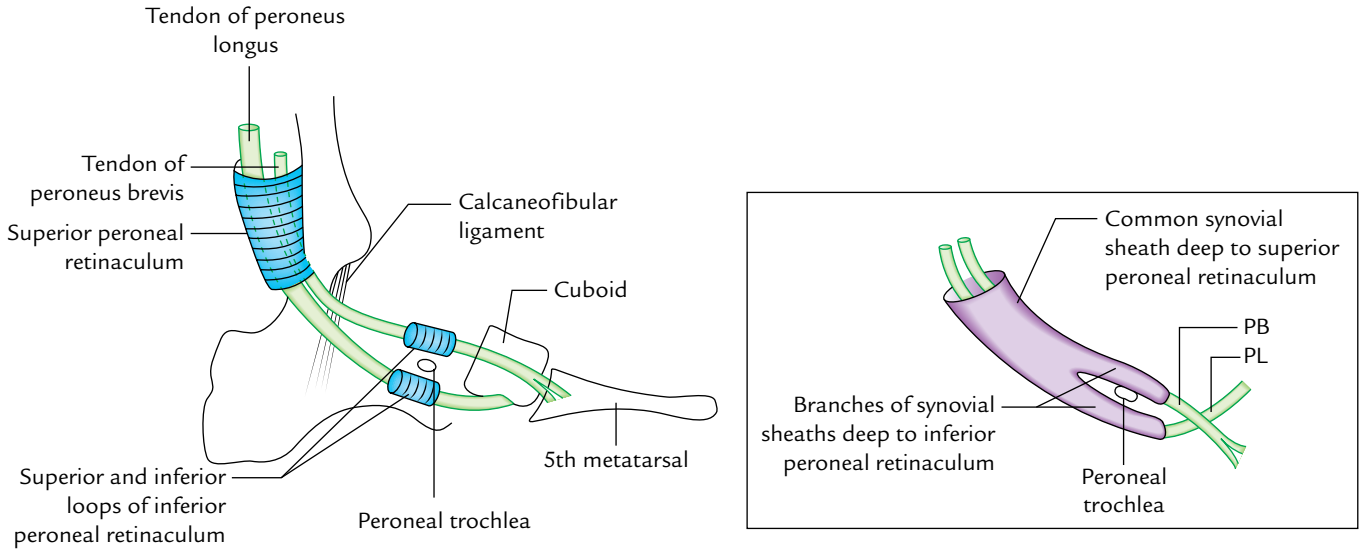


Fig. 28.2 Peroneal trochlea. Figure in the inset shows synovial sheaths around the peroneal tendons (PL = peroneus longus; PB = peroneus brevis).

Relations

The tendons of both peroneus longus and peroneus brevis lie deep to this retinaculum in a single compartment. The tendon of peroneus longus lies superficial to the tendon of peroneus brevis. Both these tendons are enclosed in a common synovial sheath.

INFERIOR PERONEAL RETINACULUM

The inferior peroneal retinaculum is a thickened fibrous band of deep fascia situated anteroinferior to the lateral malleolus.

Attachments

Superiorly: It is attached to the anterior part of superior surface of calcaneum, close to the stem of inferior extensor retinaculum.

Inferiorly: It is attached to the lateral surface of calcaneum.

In between: It is attached to the peroneal trochlea, thus forming two loops, one for the tendon of peroneus brevis and other for the tendon of peroneus longus (Fig. 28.2).

Relations

The tendon of peroneus brevis passes through the superior loop and that of peroneus longus through the inferior loop of the inferior retinaculum. Each tendon is enclosed in a separate synovial sheath, which are the prolongations of the common synovial sheath above, underneath the superior peroneal retinaculum.

Clinical correlation

The synovial sheaths enclosing the tendons of peroneus longus and peroneus brevis are subject to friction and inflammation in athletes who wear tight shoes.

PERONEAL MUSCLES

PERONEUS LONGUS (Fig. 28.3)

The peroneus longus is the longer, larger, and more superficial of the two muscles of the lateral compartment. It is bipennate in the upper part and unipennate in the lower part.

Origin

It arises from:

(a) upper two-third of the lateral surface of the shaft of fibula and adjacent surface of the head of the tibia.
(b) anterior and posterior intermuscular septa of the leg and deep fascia overlying it.

Insertion

The muscle converges below to form a long tendon which lies superficial to the tendon of the peroneus brevis and lodges itself along with the tendon of peroneus brevis into a groove behind the lateral malleolus underneath the superior peroneal retinaculum. After emerging from underneath this retinaculum, the tendon of peroneus longus passes downward and forward through the inferior pulley of the inferior peroneal retinaculum below the peroneal trochlea of the calcaneus.

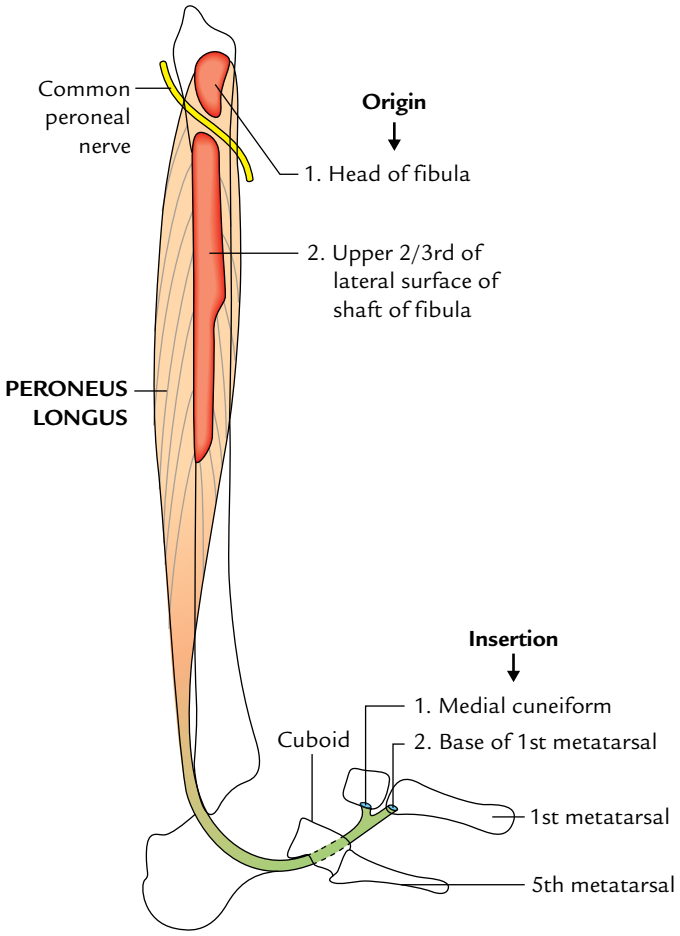


Fig. 28.3 Origin and insertion of the peroneus longus muscle.

On reaching the cuboid bone, the tendon longus changes its direction for the second time to pass through the osseofibrous tunnel on the plantar surface of the cuboid. It then crosses the sole of the foot obliquely from lateral to medial side to be inserted **into the inferolateral surface of the base of the first metatarsal bone and the adjacent part of the medial cuneiform.** The tendon contains a sesamoid bone where it binds around the calcaneus.

Nerve Supply

The peroneus longus is supplied by the superficial peroneal nerve.

Actions

1. It is the chief evertor of the foot.
2. It maintains the lateral longitudinal arch.
3. It also maintains the transverse arches of the foot.

PERONEUS BREVIS (Fig. 28.4)

It is a fusiform bipennate muscle and lies deep to the peroneus longus. As its name indicates, it is shorter than its partner (peroneus longus) in the lateral compartment.

Origin

It arises from:

(a) Lower two-third of the lateral surface of the shaft of the fibula.
(b) Anterior and posterior intermuscular septa of the leg.

Insertion

The muscle converges to form a tendon which passes behind the lateral malleolus underneath the superior peroneal

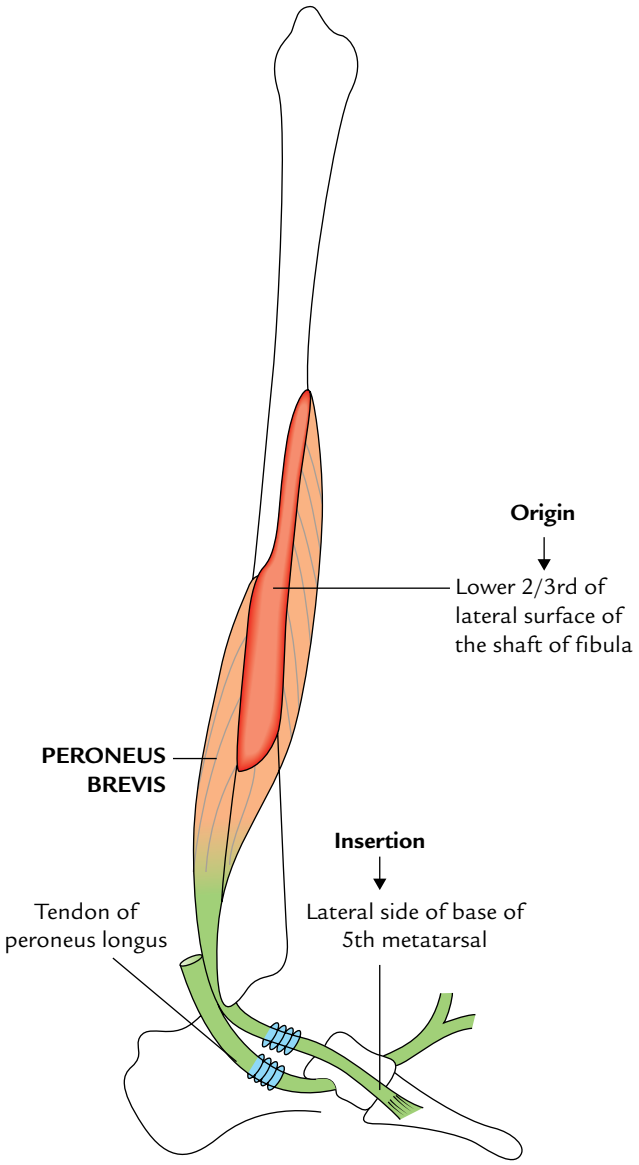


Fig. 28.4 Origin and insertion of the peroneus brevis muscle.

retinaculum. Here it lies in front of the peroneus longus in a common synovial sheath. Then it passes downward and forward above the *peroneal trochlea* of the calcaneus underneath the superior pulley of the inferior peroneal retinaculum. Here it is enclosed in a separate synovial sheath. Finally it passes forward and laterally to be inserted on to the tubercle on the lateral side of the base of the fifth metatarsal.

Nerve Supply

The peroneus brevis muscle is supplied by the superficial peroneal nerve.

Actions

1. It is the evertor of the foot.
2. It maintains the lateral longitudinal arch.

Clinical testing of peroneus longus and brevis

Strongly evert the foot of patient, against resisting; if normal, the tendons of those muscles can be seen and palpated inferior to the lateral malleolus.

The origin, insertion, nerve supply, and actions of the muscles of the lateral compartment of the leg are summarized in Table 28.1.

Clinical correlation

Deformity foot due to overactivity of invertor and evertor muscles of the foot (Fig. 28.5)

- Following an injury to the superficial peroneal nerve, the paralysis of the peroneal muscles and associated overactivity of the invertor muscles of the foot produces a deformity of foot called ***talipes varus***.
- The overactivity of peroneal muscles following paralysis of the anterior tibial muscles (invertors of foot) produces a deformity of foot called ***talipes valgus***.

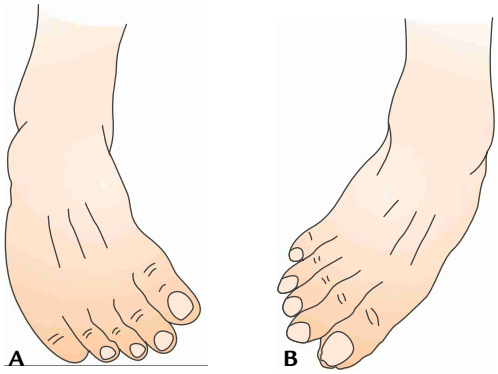


Fig. 28.5 Deformity of foot: A, talipes varus; B, talipes valgus.

SUPERFICIAL PERONEAL NERVE (MUSCULOCUTANEOUS NERVE OF THE LEG)

It is the nerve of the lateral compartment of the leg.

Origin

It is one of the two terminal branches of the common peroneal nerve given at the neck of the fibula (Fig. 28.5). It arises in the substance of peroneus longus on the lateral side of the neck of fibula.

Course and Relations (Fig. 28.6)

It begins on the lateral side of the neck of the fibula and descends for a short distance between the peroneus longus and peroneus brevis, and then lies in a groove between the peroneus brevis and extensor digitorum longus.

At the junction of the upper two-third and lower one-third of the leg, it pierces the deep fascia, and soon divides into a medial and a lateral terminal branches which reach the dorsum of the foot.

Table 28.1 Origin, insertion, nerve supply, and actions of the muscles of the lateral compartment of the leg

Muscles	Origin	Insertion	Nerve supply	Actions
Peroneus longus (larger and lies superficial to the peroneus brevis)	• Upper 2/3rd of the lateral surface of the fibula • Adjoining surface of the lateral condyle of the tibia • Anterior and posterior intermuscular septa	• Lateral side of the base of the 1st metatarsal • Adjoining part of the medial cuneiform	Superficial peroneal nerve (L5; S1, S2)	• Eversion of the foot • Weak plantar flexion of the ankle • Maintains the lateral longitudinal arch
Peroneus brevis (smaller and lies deep to the peroneus longus)	• Lower 2/3rd of the lateral surface of the fibula • Anterior and posterior intermuscular septa	Lateral side of the base of the 5th metatarsal bone	Superficial peroneal nerve (L5; S1, S2)	• Eversion of the foot • Weak plantar flexion of the ankle • Maintains the lateral longitudinal arch

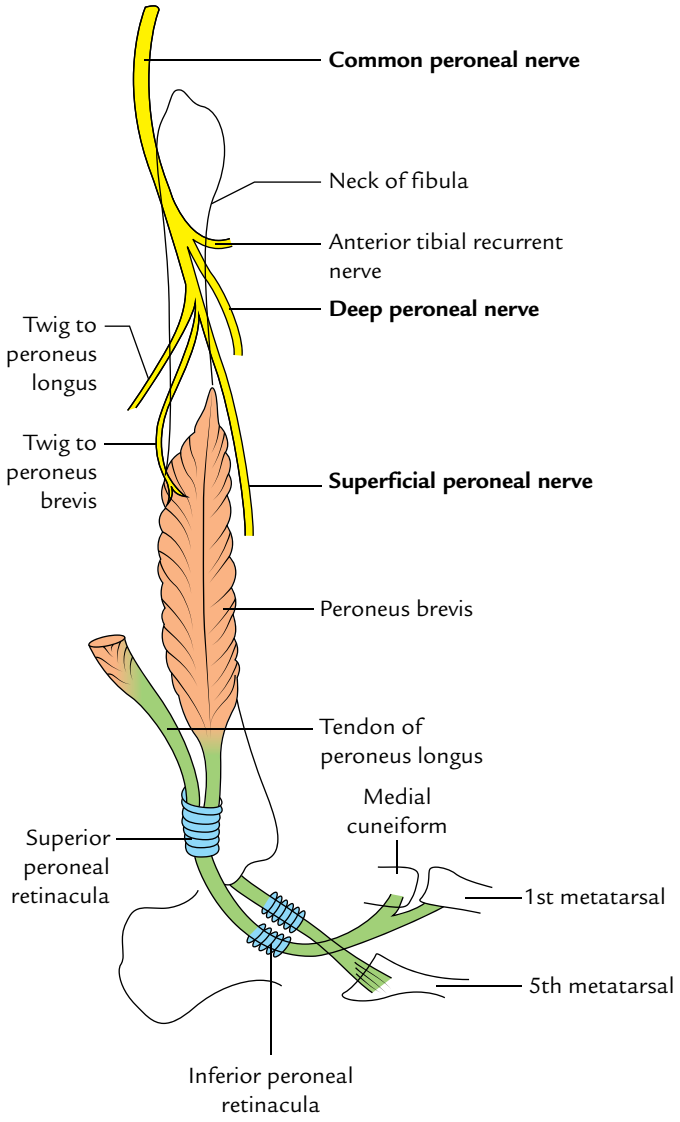


Fig. 28.6 Origin of superficial peroneal nerve from common peroneal nerve on the lateral side of the neck of fibula.

Branches and Distribution

1. **Muscular branches** to peroneus longus and peroneus brevis.
2. **Cutaneous branches** supply the skin of the lower one-third of the lateral side of the leg and dorsum of the foot, except for the territories supplied by the saphenous, sural, and deep peroneal nerves.

The *medial terminal branch* of the superficial peroneal nerve crosses the ankle and divides into two dorsal digital nerves, one for the medial side of the big toe and the other for the second interdigital cleft.

The *lateral terminal branch* of the superficial peroneal nerve also divides into two dorsal digital nerves for the third and fourth interdigital clefts.

N.B. The distribution of the superficial peroneal nerve corresponds to the distribution of radial nerve in the forearm.

Clinical correlation

Injury of superficial peroneal nerve: The injury of superficial peroneal nerve leads to:

(a) Loss of eversion of the foot due to paralysis of peroneus longus and peroneus brevis muscles (evertors of the foot).
(b) Sensory loss on the lateral aspect of the leg.
(c) Sensory loss on the dorsum of the foot except for the web-space between the first and second toe and dorsal aspect of all the toes which is supplied by deep peroneal nerve.

MEDIAL SIDE OF THE LEG

The medial side of the leg consists of the medial surface of the tibia. The greater part of this surface is subcutaneous except for a small upper part, which provides the attachment

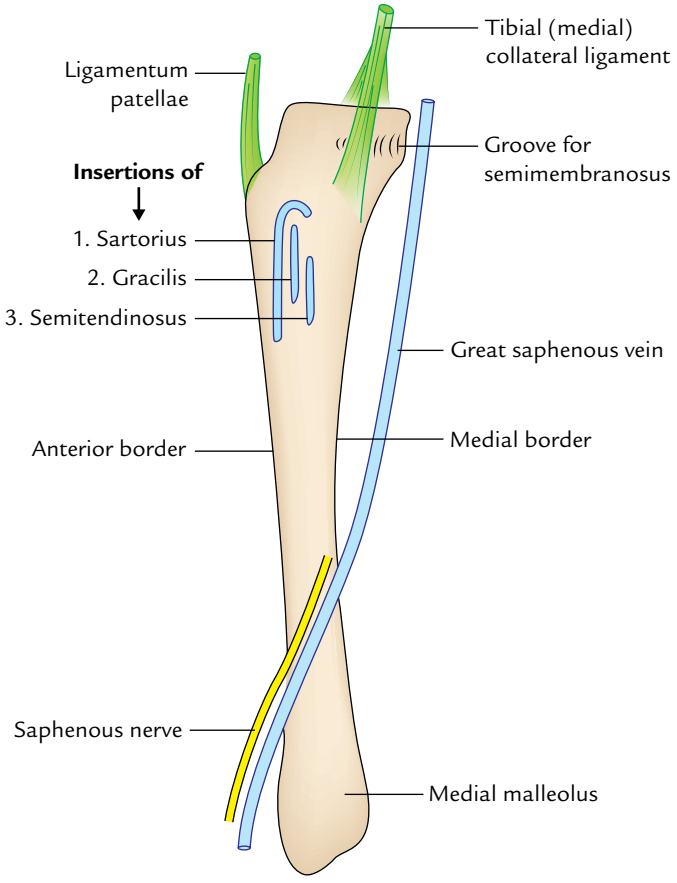


Fig. 28.7 Subcutaneous medial surface of the tibia showing attachment of three tendons of sartorius, gracilis, and semitendinosus; medial collateral ligament and its relation with the great saphenous vein.

to tibial collateral ligament and tendons of sartorius, gracilis, and semitendinosus (Fig. 28.7). All these structures are covered by the deep fascia. The great saphenous vein and the saphenous nerve lie in the superficial fascia as they cross the lower one-third of this surface.

1. ***Saphenous nerve:*** It supplies the skin, fasciae, and periosteum on this surface.
2. ***Tibial collateral ligament:*** Morphologically, it represents the degenerated part of the tendon of hamstring part of adductor magnus. It partly covers the insertion of semimembranosus and is itself crossed superficially by the tendons of sartorius, gracilis, and semitendinosus.
3. ***Tendons of sartorius, gracilis, and semitendinosus:*** The three muscles inserted into the upper part of the medial surface of the tibia represent one muscle from each of the three compartments of the thigh. *Sartorius* belongs to the anterior compartment of the thigh and is supplied by the nerve of that compartment—femoral nerve; *gracilis* belongs to the medial compartment of the thigh and is supplied by the nerve of that compartment—obturator nerve; and *semitendinosus* belongs to the posterior compartment of the thigh and is supplied by the nerve of that compartment—sciatic nerve.

N.B. *Guy ropes:* The lower ends of sartorius, gracilis, and semitendinosus are attached together on the upper part of the medial surface of the tibia but above they are attached wide apart from each other on the hip bone; viz., **sartorius** is attached on the ilium (anterior superior iliac spine), **gracilis** on the pubis (body and inferior ramus), and **semitendinosus** on the ischium (ischial tuberosity). The lower end of these muscles is attached at one point. From this arrangement, it seems that these muscles act as '**guy ropes**' to stabilize the bony pelvis on the femur (Fig. 28.8).

4. ***Bursa anserinus (Fig. 28.9):*** It is a large complicated synovial bursa with several diverticula present on the upper part of the medial surface of the tibia. It not only separates the tendons of sartorius, gracilis, and semitendinosus from each other near their insertion but also from the tibial collateral ligament.
5. ***Great saphenous vein and saphenous nerve:*** The great saphenous vein ascends in front of the medial malleolus and crosses lower one-third of the medial surface of the tibia obliquely with a backward inclination. The saphenous nerve runs downward immediately in front of the great saphenous vein.

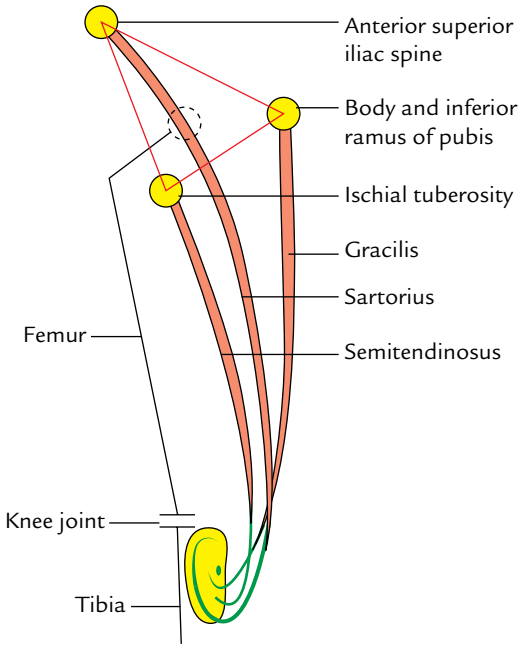


Fig. 28.8 Guy ropes.

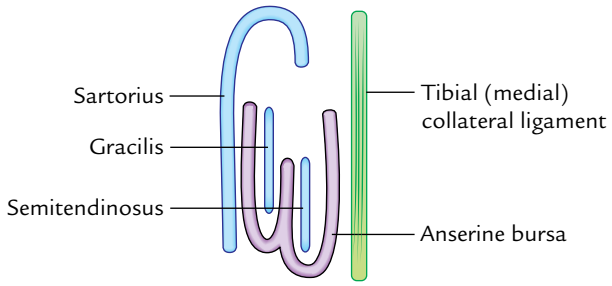


Fig. 28.9 Bursa anserinus.

Clinical correlation

Anserine bursitis: Repeated trauma on the upper part of the medial aspect causes inflammation of anserine bursa (anserine bursitis) which causes pain and swelling at this site.

Golden Facts to Remember

► Most commonly injured nerve in the lower limb	Common peroneal nerve
► Unique feature of the lateral compartment of the leg	It does not have its own artery
► Bursa anserine	Loculated synovial bursa related to tendons of sartorius, gracilis, and semitendinosus at their insertions
► Muscles playing the role of guy ropes in steadying the pelvis	Sartorius, gracilis, and semitendinosus
► Inadvertent inversion of the foot is prevented during the "toe off" of the stance phase by	Eversion of the foot by peroneal muscles
► Grasping muscle of the foot in apes	Peroneus longus

Clinical Case Study



A few days after the removal of the plaster cast for fracture of the upper end of the right fibula, the patient complained of loss of sensation on the lateral aspect of the lower part of the leg and dorsum of the foot. On examination, the doctor found a loss of sensation on the lateral aspect of the lower part of the leg and dorsum of the foot except on the lateral side of the little toe, in the interdigital cleft between the first and second toes and medial margin of the foot up to the head of the first metatarsal bone. Patient was also unable to dorsiflex and evert his right foot.

Questions

1. Name the nerve injured in this patient.
2. Name the site, where common peroneal nerve is easily palpated?
3. Name the deformity in which the patient cannot dorsiflex his foot.
4. Name the chief evertors of the foot.
5. Name the nerve, which innervate the skin of the interdigital cleft between the first and second toes.

Answers

1. Common peroneal nerve.
2. Lateral side of the neck of the fibula.
3. Foot drop.
4. Peroneus longus and peroneus brevis.
5. Deep peroneal nerve.

CHAPTER

29 Back of the Leg

The back of the leg is also called **calf**. It corresponds to the front of the forearm. It deals with the posterior compartment of the leg. It is bulkiest of the three compartments of the leg because it contains large powerful antigravity muscles (e.g., gastrocnemius and soleus), which raise the heel during walking. The posterior compartment of the leg is continuous superiorly with the popliteal fossa and inferiorly with the sole of the foot.

BONY FRAMEWORK (Fig. 29.1)

The bony framework of the posterior compartment of leg is formed by the posterior aspects of the tibia and fibula, which are joined to each other by the interosseous membrane. When the bony framework of the leg is seen from behind, in addition to the tibia and fibula, posterior aspect of the talus, and posterior one-third of the calcaneus are also seen. The posterior third of calcaneus projects posteriorly to form the heel. It ends in a large medial process, which rests on the ground.

The tibial condyle overhangs the shaft of tibia posteriorly and at the sides. The rounded head of the fibula is quite inferior to the knee joint.

The upper part of the posterior surface of the tibia is crossed obliquely by the **soleal line**. The part below the soleal line is subdivided in its upper three-fourth by a vertical line into medial and lateral areas.

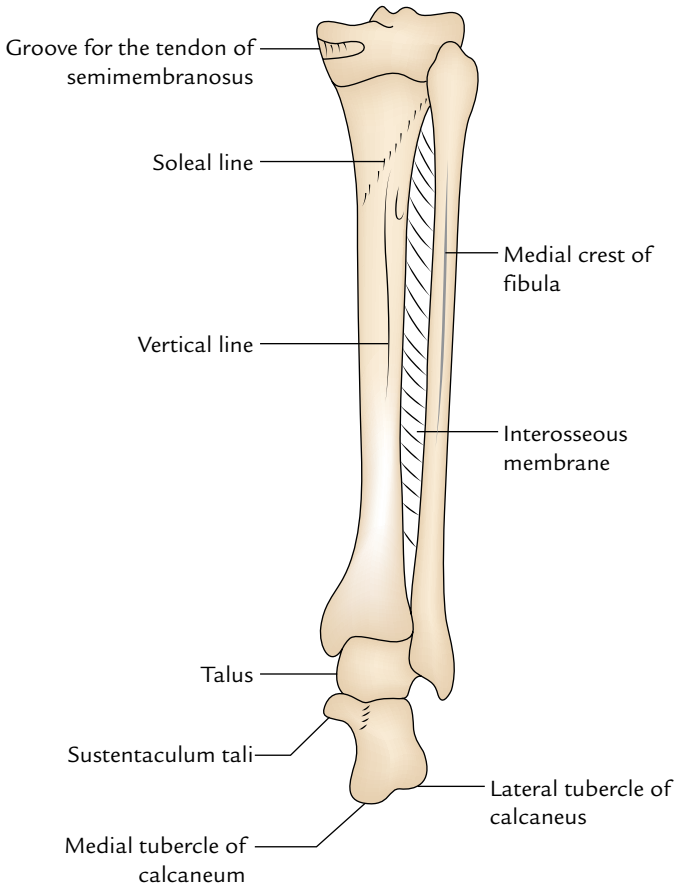


Fig. 29.1 Bony framework of the posterior aspect of the leg.

CONTENTS OF SUPERFICIAL FASCIA

The superficial fascia on the back of the leg contains two superficial veins e.g., short and long saphenous veins) and seven cutaneous nerves.

SUPERFICIAL VEINS (Fig. 29.2)

Short (Small) Saphenous Vein

It is formed at the lateral border of the dorsum of the foot by the union of the lateral end of the *dorsal venous arch* and the *lateral dorsal digital vein* of the little toe. It ascends to reach the back of the leg by passing behind the lateral malleolus. In the leg, it ascends in the mid-line in the lower part of the popliteal fossa; pierces the deep fascia to join the *popliteal vein*. It drains the lateral side of the foot, ankle, and back of the leg. It is connected with the great saphenous veins and the deep veins. It is accompanied by the sural nerve on its lateral side (for details see page 462).

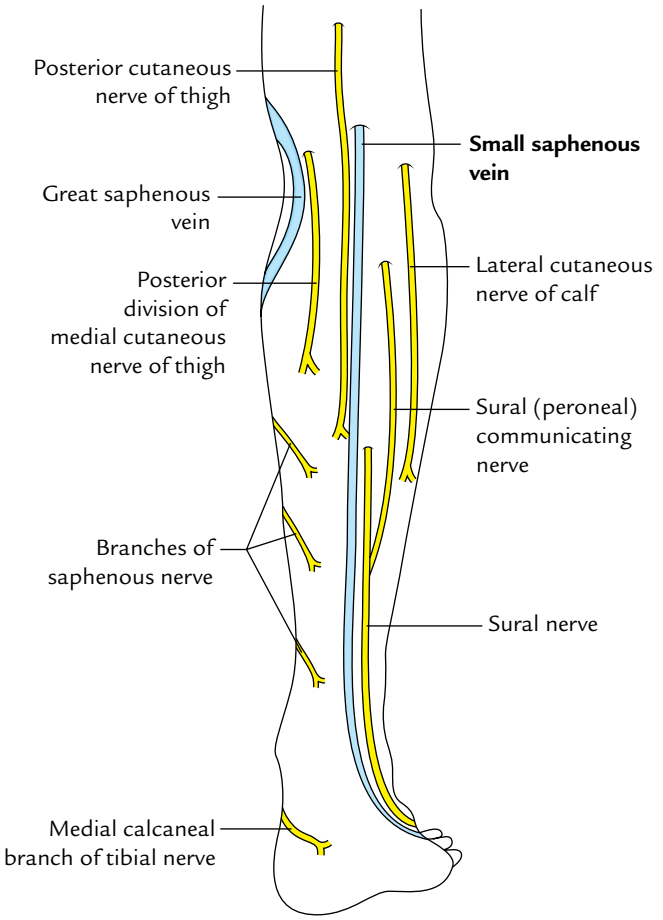


Fig. 29.2 Superficial veins and cutaneous nerves on the back of the leg.

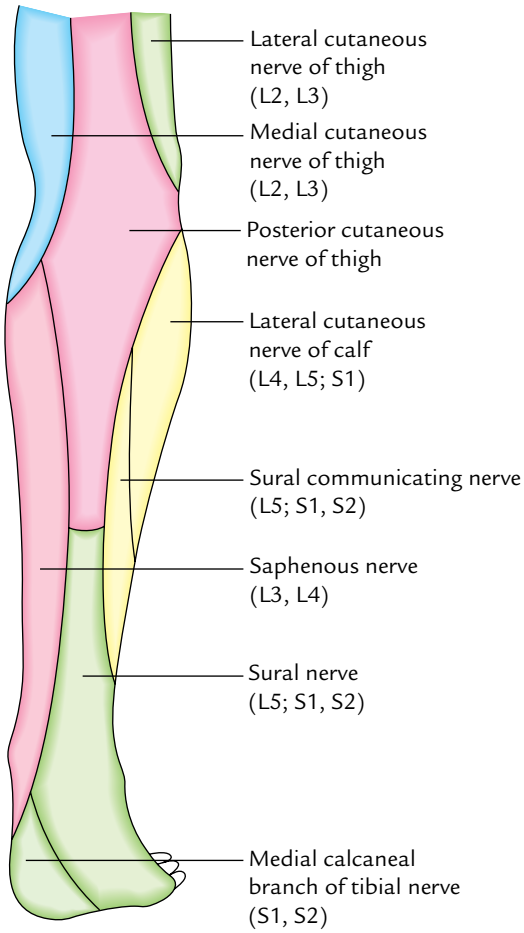


Fig. 29.3 Areas on the back of the leg supplied by different cutaneous nerves.

Great (Long) Saphenous Vein

On the back of the leg it is seen only on the posteromedial side of the knee. It is described in detail in Chapter 33 on pages 458–459.

CUTANEOUS NERVES (Figs 29.2 and 29.3)

The cutaneous nerves as mentioned earlier are seven in number, out of which two supply the skin of the lateral one-third of the back of the leg, two supply the skin of the middle one-third of the leg, two supply the skin of the lateral one-third of the leg, and one nerve supplies the skin of heel.

Saphenous Nerve (L3, L4)

It pierces the deep fascia on the medial side of the knee and accompanies the great saphenous vein, either in front or behind it. It supplies the skin on the medial side of the knee, leg, and medial border of the foot up to the ball of the big toe.

Posterior Division of the Medial Cutaneous Nerve of the Thigh (L2, L3)

It pierces the deep fascia a little above the knee. It supplies the skin of the uppermost part of the medial one-third of the calf.

Posterior Cutaneous Nerve of the Thigh (S1, S2, S3)

It pierces the deep fascia in the middle of the popliteal fossa, and descends with the small saphenous vein to supply the skin of the upper half of the intermediate area of the calf.

Sural Nerve (L5; S1, S2)

It is a branch of the tibial nerve in the popliteal fossa. It pierces deep fascia in the middle of the leg and runs along the short saphenous vein. It is joined by the sural (peroneal) communicating nerve (a branch of common peroneal nerve) about 2 inches above the heel. After passing behind the lateral malleolus, the nerve runs forward along the lateral border of foot, and ends in the skin on the lateral side of the little toe. It supplies the skin of the lower lateral part of the back of the leg, lateral border and adjoining part of the dorsum of the foot, and the lateral side of the little toe.

Lateral Cutaneous Nerve of the Calf (L4, L5; S1)

It is a branch of the common peroneal nerve in the popliteal fossa. It pierces the deep fascia over the lateral head of gastrocnemius and supplies skin of the upper two-third of the lateral area of the leg.

Sural (Peroneal) Communicating Nerve (L5; S1, S2)

It is a branch of the common peroneal nerve. It pierces the deep fascia about 1 inch below the lateral head of gastrocnemius and descends to join the sural nerve about 2 inches above the heel. It supplies skin of the posteromedial part of the lateral area of calf.

Medial Calcaneal Branch (S1, S2)

Medial calcaneal branch of the tibial nerve perforates the flexor retinaculum and supplies the skin of the heel and the adjoining medial side of the sole of the foot.

FLEXOR RETINACULUM

The flexor retinaculum is a thick broad band of the deep fascia (2.5 cm broad) on the medial side of the ankle, behind and below the medial malleolus (Fig. 29.4). It holds the long tendons, vessels, and nerves in position as they curve and pass forward from the back of the leg to the sole of the foot.

Attachments

Anteriorly or *above* : To the posterior border and tip of the medial malleolus.

Posteriorly or *below* : To the medial process of the calcaneal tuberosity.

- *Structures passing deep to flexor retinaculum (Fig. 29.4)*:
- From the medial to lateral side, these are: (a) tendon of **t**ibialis posterior, (b) tendon of flexor **d**igitorum longus, (c) posterior tibial **a**rtery and its branches, (d) posterior tibial **n**erve and its terminal branches, and (e) tendon of flexor **h**allucis longus.
- (*Mnemonic:* **T**om, **D**ick, **A**nd **N**ot **H**arry or **T**he **D**octors **A**re **N**ot **H**ere.)

Clinical correlation

Tarsal tunnel syndrome: If the tibial nerve is compressed deep to the flexor retinaculum (in the osseofibrous tunnel behind the medial malleolus) it leads to a clinical condition called *tarsal tunnel syndrome*. Clinically, it presents as burning, tingling, and pain in the sole of the foot. These symptoms are relieved by dividing the flexor retinaculum surgically.

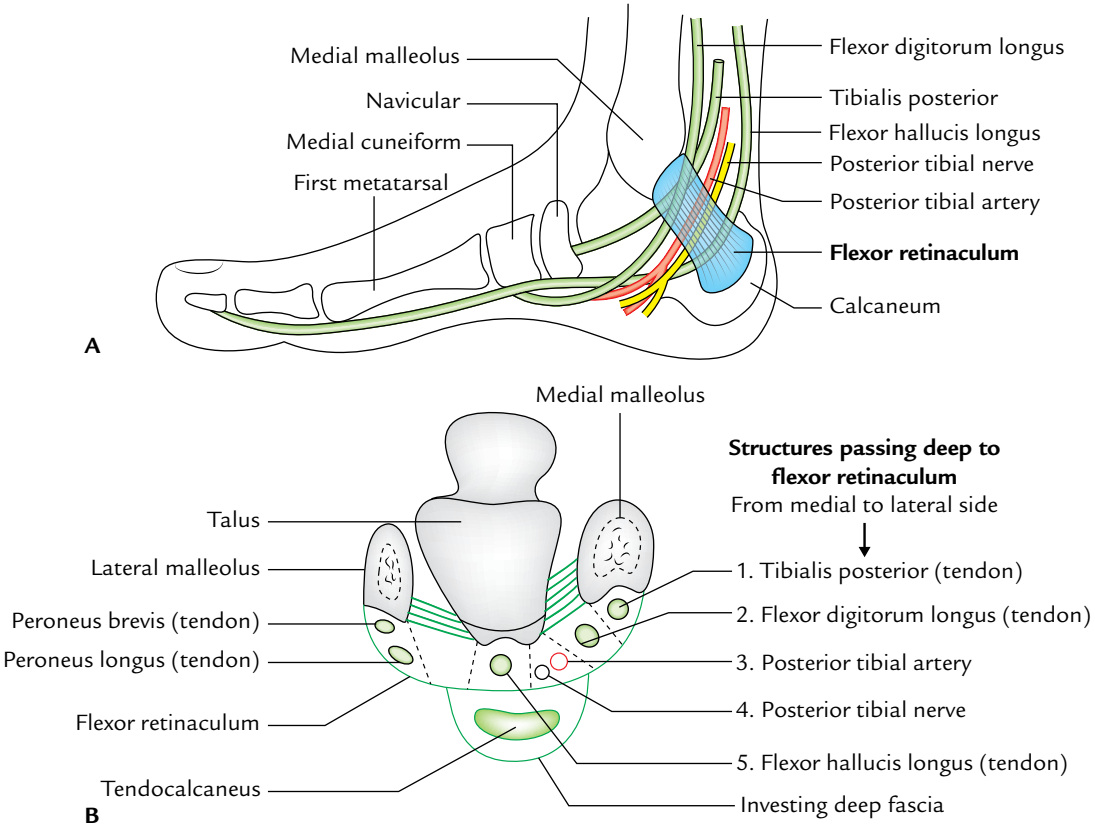


Fig. 29.4 A, Flexor retinaculum of the ankle and the structures passing deep to it; B, transverse section of the ankle showing the flexor retinaculum and the structures passing deep to it.

POSTERIOR COMPARTMENT OF THE LEG

It is the largest osseofascial compartment of the leg.

BOUNDARIES AND SUBDIVISIONS (Fig. 29.5)

Anterior: Posterior surfaces of the tibia, fibula, interosseus membrane, and posterior intermuscular septum.

Posterior: Deep fascia of the leg extending from the medial border of the tibia to the posterior intermuscular septum.

The **posterior compartment of the leg is subdivided** by two strong transverse fascial septa (superficial and deep) into three parts: superficial, middle, and deep.

The **superficial transverse septum** is attached *medially* to the medial border of the tibia and *laterally* to the posterior border of the fibula.

The **deep transverse septum** is attached *medially* to the proximal part of the soleal line and vertical ridge on the posterior surface of the tibia, and *laterally* to the medial crest of the fibula.

- *Superficial part* (between superficial transverse septum and deep fascia) contains gastrocnemius, soleus, and plantaris.
- *Middle part* (between superficial and deep transverse fascial septa) contains flexor digitorum longus, flexor hallucis longus, and posterior tibial nerve and vessels.
- *Deep part* (between deep transverse fascial septum and posterior surfaces of interosseous membrane, tibia, and fibula) contains tibialis posterior.

Contents

- **Muscles:** Superficial and deep groups of the muscles (Table 29.1).
- **Arteries:** Tibial and peroneal arteries.
- **Nerve:** Tibial nerve.

Table 29.1 Superficial and deep muscles of the posterior compartment of the leg

Superficial muscles	Deep muscles
Gastrocnemius	Popliteus
Soleus	Flexor digitorum longus
Plantaris	Flexor hallucis longus Tibialis posterior

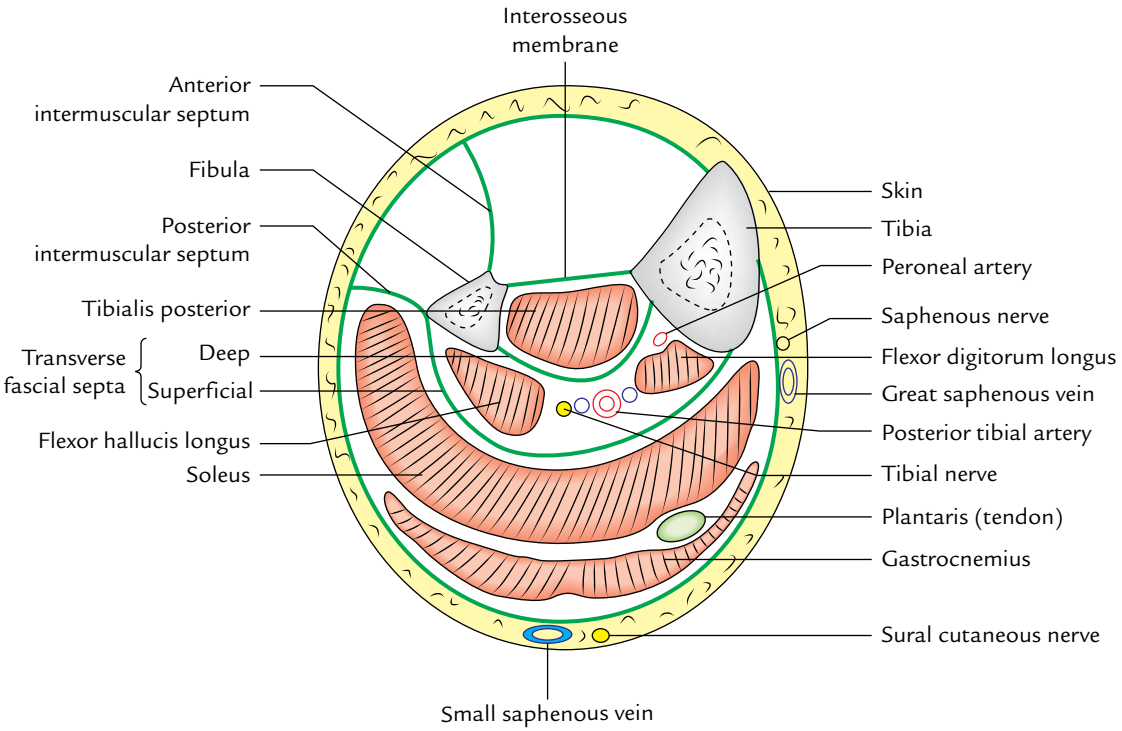


Fig. 29.5 A transverse section of the middle third of the leg showing boundaries and contents of the posterior compartment of the leg.

MUSCLES OF THE POSTERIOR COMPARTMENT OF THE LEG

The bony attachments of the muscles of the back of the leg are shown in Figure 29.6.

Superficial Muscles on the Back of the Leg

These are shown in Figure 29.7.

Gastrocnemius (Fig. 29.8)

It is the largest and most superficial muscle of the posterior compartment. It consists of two heads—medial and lateral.

Origin

1. **Large medial head** arises by a broad flat tendon from the posterosuperior aspect of the medial condyle of the femur behind the adductor tubercle and adjoining part of the posterior (popliteal) surface of the shaft of the femur.
2. **Small lateral head** arises by a broad flat tendon from the lateral surface of the lateral condyle of the femur above the lateral epicondyle and adjoining part of the lateral supracondylar line.

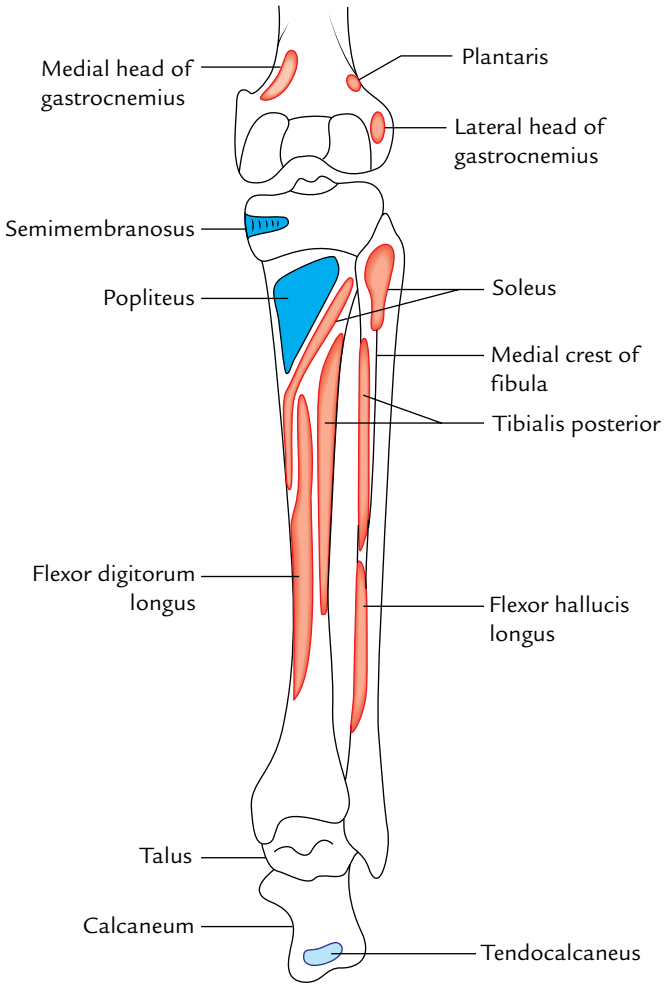


Fig. 29.6 Attachments of the muscles of the back of leg on bony framework of the leg.

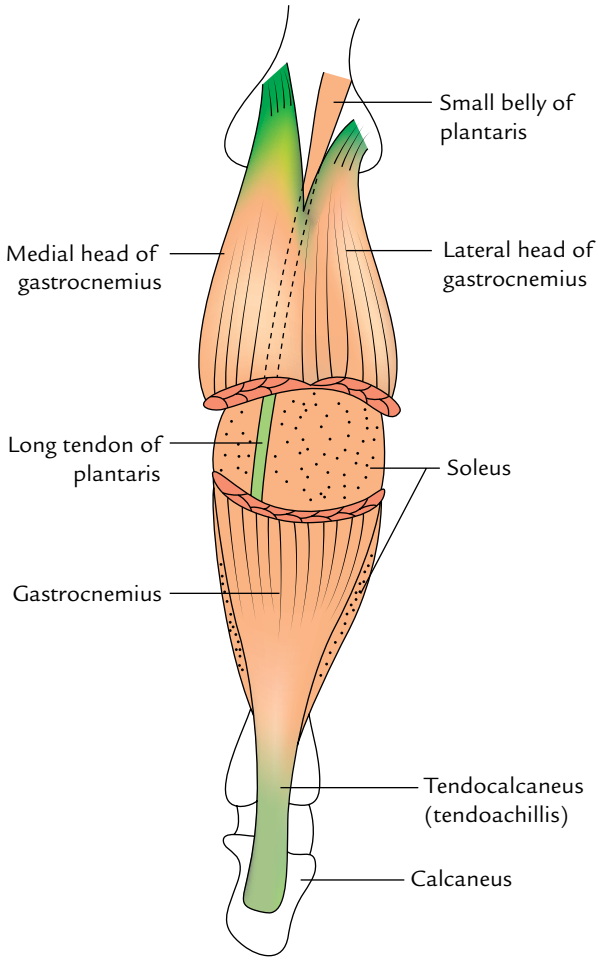


Fig. 29.7 Superficial muscles on the back of the leg (e.g., gastrocnemius, soleus, and plantaris).

Insertion

The fleshy bellies of the two heads descend and unite at the middle of the leg to form broad thin aponeurotic tendon, which unites with the tendon of soleus, a short distance below the middle of the leg to form a long thick tendon—the tendocalcaneus (tendoachillis), which is inserted into the middle of the posterior surface of calcaneum.

Nerve Supply

Both heads are supplied by the tibial nerve in the popliteal fossa.

Actions

1. It is the chief plantar flexor of the foot at the ankle when the knee is extended.
2. It is also a flexor of the knee.
3. It provides rapid movements of the foot during running and jumping.

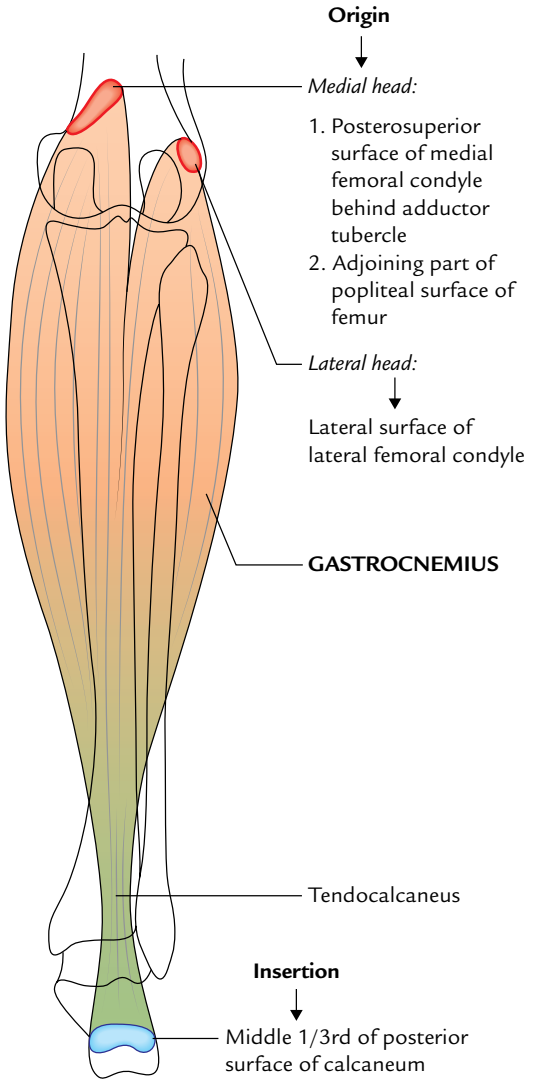


Fig. 29.8 Origin and insertion of the gastrocnemius.

Clinical correlation

Tennis leg: It is painful calf injury in which there is tear or strain of the medial head of gastrocneminus at its musculo-tendinous junction due to overstretching. It commonly occurs in a tennis player who stretches their gastrocnemius too much for a difficult serve.

Soleus (Fig. 29.9)

It is a multipennate muscle lying deep to the gastrocnemius. It is so-named because of its resemblance to a sole/flat fish.

Origin

It has a horseshoe-shaped origin from:

(a) Back of the head and posterior surface of the upper one-fourth of the shaft of the fibula.
(b) Soleal line and middle one-third of the medial border of the tibia.
(c) Tendinous arch between the tibia and fibula.

Insertion

The muscle fibres slope downward to form a massive belly, which gives rise to a tendon. The tendon of soleus fuses with that of gastrocnemius to form **tendocalcaneus,** which is inserted into the middle one-third of the posterior surface of calcaneum.

Nerve Supply

It is supplied by two branches of the tibial nerve:

(a) The first branch supplies the muscle in the popliteal fossa from its superficial surface.
(b) The second branch supplies the muscle in the posterior compartment of the leg from its deep surface.

Actions

1. It is a plantar flexor of the foot at the ankle and steadies the leg on the foot during standing. It is considered as the **"*workhorse*" of the plantar flexion.**
2. It is slow acting but more powerful than gastrocnemius. It acts as a **bottom gear** during strolling. (*Note:* Gastrocnemius acts as a **top gear** during running and jumping.)

N.B.

- The two heads of gastrocnemius and soleus together form the **triceps surae muscle**.
- A small sesamoid bone is found in the tendon of origin of lateral head of gastrocnemius called **fabella.**
- A small synovial bursa (Brodie's bursa) lies deep to medial head to gastrocnemius.
- **Tendocalcaneus (tendoachillis)** is a conjoint tendon of insertion of gastrocnemius and soleus (**triceps surae**). It is the thickest and strongest tendon in the body and is about 15 cm long. It acts as a prime mover of plantar flexion of the foot at the ankle joint.

Clinical Testing

To test gastrocnemius and soleus together (triceps surae), ask the patient to stand on the toes or plantar flex the foot. If normal the tendocalcaneus can be seen and palpated above the heel.

Clinical correlation

Calf muscle pump and peripheral heart: The gastrocnemius and soleus together make a '*calf muscle pump*,' which facilitates the venous return from the lower limb.

The soleus muscle is regarded as the ***peripheral heart*** because it houses large venous sinuses called soleal sinuses which communicate with the superficial veins by perforating veins and with deep veins directly.

Thus contraction of soleus helps in sucking the blood from superficial veins and propelling it into deep veins.

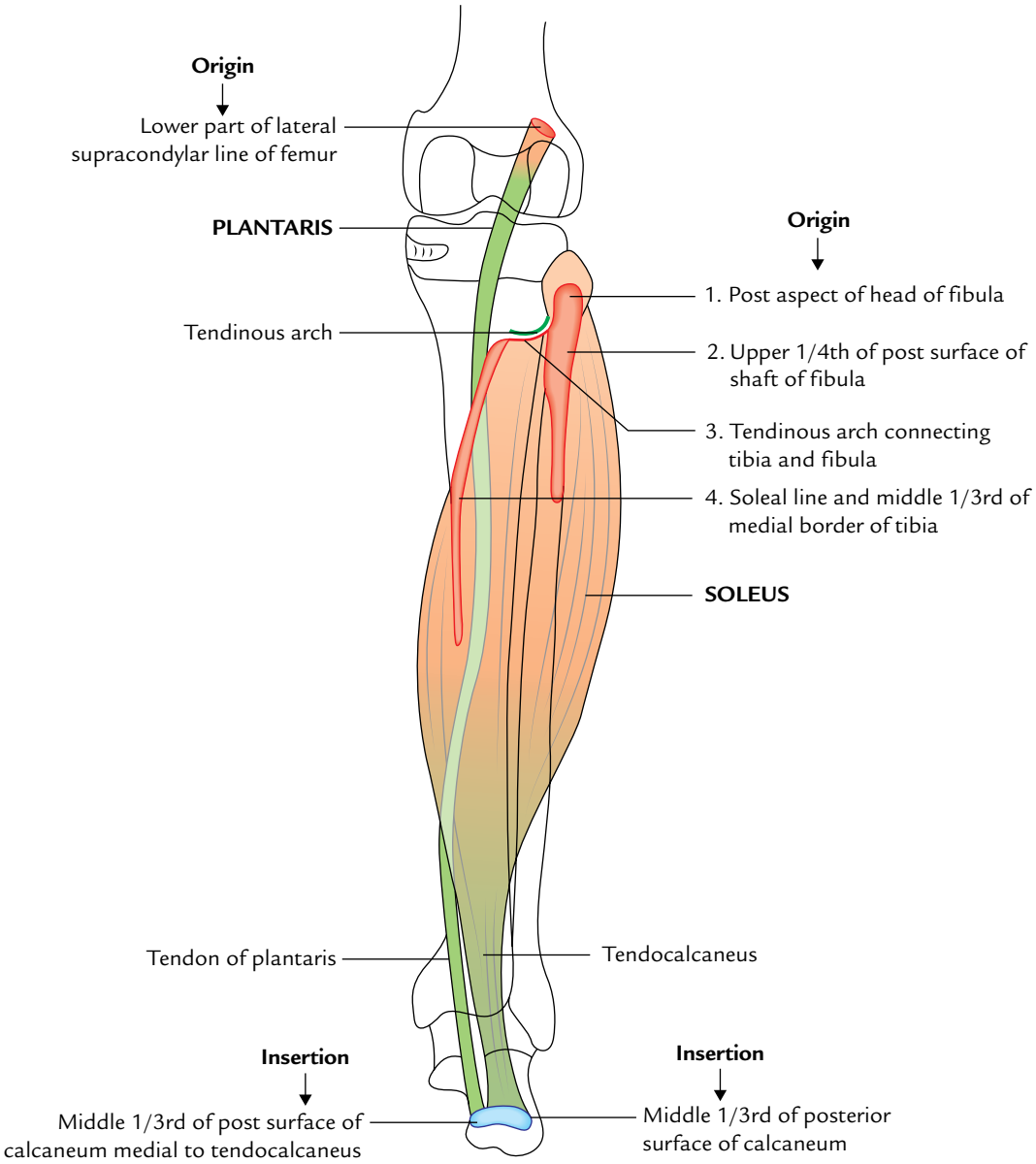


Fig. 29.9 Origin and insertion of the soleus and plantaris.

Plantaris (Fig. 29.9)

It is a small muscle with a short belly and long slender tendon. It lies between the gastrocnemius and soleus.

Origin

It arises from the lower one-third of the lateral supracondylar line and the adjoining part of the oblique popliteal ligament.

Insertion

Its tendon blends with the medial margin of tendocalcaneus and inserted into the middle one-third of the posterior surface of the calcaneum medial to the attachment of tendocalcaneus.

N.B. The long slender tendon of plantaris is easily mistaken by the first year medical students for a nerve; hence, it is often termed *freshman's nerve*.

Clinical correlation

Tendon grafting: The plantaris is a vestigial muscle in human beings and is absent in 5–10% of people. It has insignificant role either as flexor of the knee or as plantar flexor of the ankle; hence, the tendon of plantaris is used for grafting (e.g., reconstructive surgery of the tendons of the hand).

The origin, insertion, nerve supply, and actions of the superficial muscles of the posterior compartment of the leg are summarized in Table 29.2.

Table 29.2 Origin, insertion, nerve supply, and actions of the superficial muscles of the back of the leg

Muscle	Origin	Insertion	Nerve supply	Actions
Gastrocnemius	• *Lateral head* from the lateral surface of the lateral condyle of the femur • *Medial head* from the popliteal surface of the femur and adjoining posterosuperior aspect of the medial condyle of the femur	Middle one-third of the posterior surface of the calcaneus via tendocalcaneus	Tibial nerve (S1, S2)	• Plantar flexion of the ankle when the knee is extended • Flexes the leg at the knee joint
Soleus	• Posterior aspect of the head and upper one-fourth of the posterior surface of the shaft of fibula • Soleal line and middle one-third of the medial border of the tibia • Tendinous soleal arch between the fibula and tibia	Middle one-third of the posterior surface of the calcaneus via tendocalcaneus	Tibial nerve via two branches (S1, S2)	• Plantar flexion of the ankle independent of position of the knee • Steadies the leg on the foot during standing
Plantaris	• Lower part of the lateral supracondylar line • Oblique popliteal ligament	Middle one-third of the posterior surface of the calcaneus	Tibial nerve	Very weak plantar flexion of the ankle

Deep Muscles on the Back of the Leg

The origin, insertion, nerve supply, and actions of the deep muscles on the back of the leg are given in Table 29.3. Only the popliteus muscle is described in detail.

Popliteus (Fig. 29.10)

It is a thin, flat, triangular muscle, which forms the inferior part of the floor of the popliteal fossa.

Origin

Its origin is intracapsular but extrasynovial. It arises by a tendon from:

(a) The deep anterior part of the groove (popliteal groove) on the lateral surface of the lateral condyle of the femur. (*Note:* The posterior part of the groove is occupied by the tendon of popliteus in full flexion.)
(b) Arcuate popliteal ligament.
(c) Outer margin of the lateral meniscus.

Insertion

The tendon passes downward and medially and flares out to form the fleshy part, which is inserted into:

(a) The medial two-third of the triangular area above the soleal line on the posterior surface of the tibia.
(b) The popliteal fascia.

Nerve Supply

The popliteus is supplied by a branch of tibial nerve. It arises in the popliteal fossa, crosses across the superficial aspect of the muscle, winds round its lower border, and supplies the muscle from its deeper aspect.

Actions

1. It unlocks the locked knee by rotating the femur laterally during initial stages of flexion of the knee.
2. It pulls the lateral meniscus backward and prevents it from being trapped and crushed between the condyles of the femur and tibia.
3. It flexes the knee during couching.

Posterior Tibial Artery (Fig. 29.13)

Origin

It is the larger of the two terminal branches of the popliteal artery because its branches not only supply the posterior compartment but also the lateral compartment of the leg and the sole of the foot.

Course and Relations

1. It begins at the lower border of popliteus, between the tibia and fibula, deep to gastrocnemius and enters the back of the leg by passing deep to the *tendinous arch of soleus.*
2. In the leg, it runs downward and slightly medially to reach the posteromedial side of the ankle, midway between the medial malleolus and the medial tubercle of calcaneum.
3. It terminates deep to the flexor retinaculum by dividing into a **large lateral plantar artery** and a **small medial plantar artery**.

Table 29.3 Origin, insertion, nerve supply, and actions of the deep muscles of the back of the leg

Muscle	Origin	Tendon	Insertion	Nerve supply	Actions
Popliteus	• Anterior part of the popliteal groove on the lateral condyle of the femur	Its tendon of origin is 1 inch long, and intracapsular but extrasynovial	• Medial two-third of the triangular area above the soleal line, on the posterior surface of the tibia	Tibial nerve (L4, L5; S1)	• Unlocks the locked knee by rotating the femur laterally during initial stages of flexion of the knee
Flexor digitorum longus (FDL) (Fig. 29.11)	• Upper two-third of the medial part of the posterior surface of the tibia below the soleal line • Fascia covering the tibialis posterior	Its tendon: • Crosses superficially to the tendon of tibialis posterior in the lower part of the leg and tendon of FHL in the sole • Subdivides into four tendons for lateral four toes, accessorius	Plantar surface of the base of distal phalanges of the lateral four toes	Tibial nerve (S2, S3)	• Plantar flexes lateral four toes • Plantar flexes ankle • Maintains the longitudinal arches of the foot
Flexor hallucis longus (FHL) (Fig. 29.11)	• Lower three-fourth of the posterior surface of the shaft of the fibula, behind the medial crest • Adjoining part of the posterior interosseous membrane	Its tendon: • Grooves two bones, talus between its medial and posterior tubercles and the calcaneum underneath sustentaculum tali • In the sole it is crossed superficially by the tendon of FDL	Plantar surface of the base of distal phalanx of great toe	Tibial nerve (S2, S3)	• Plantar flexes great toe • Plantar flexes the ankle weakly • Supports the medial longitudinal arch of the foot
Tibialis posterior (Fig. 29.12)	• Upper 2/3rd of the lateral part of the posterior surface of tibia below the soleal line • Posterior surface of the fibula in front of the medial crest • Upper two-third of the posterior surface of the interosseous membrane	Its tendon: • Crosses superficially by the tendon of FDL in the lower part of the leg • Grooves back of the medial malleolus • Passes deep to flexor retinaculum and superficially to the deltoid ligament	• Chiefly on tuberosity of the navicular bone • Slips pass to all tarsals, except talus, and bases 2nd to 4th metatarsals	Tibial nerve (L4, L5)	• Invertor of the foot • Maintains the medial and longitudinal arches of the foot

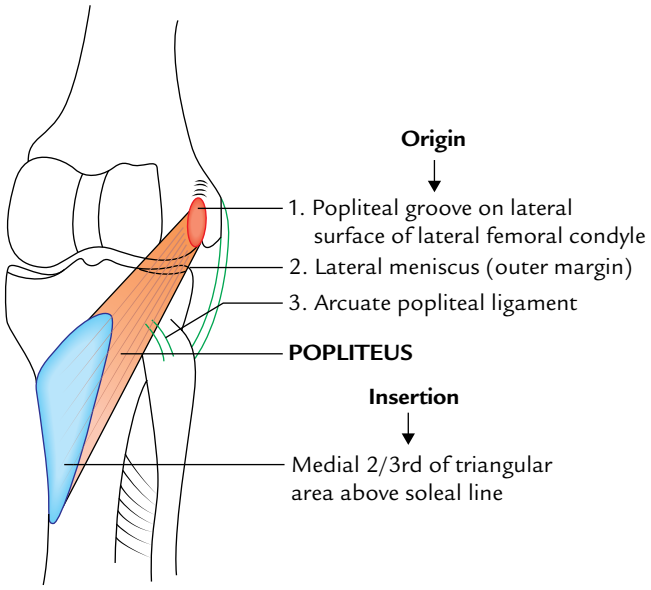


Fig. 29.10 Origin and insertion of the popliteus muscle.

Throughout its course, it is *accompanied* by the tibial nerve, which crosses the artery from the medial to lateral side.

Branches

1. *Peroneal (fibular) artery:* It is the largest and most important branch of the posterior tibial artery. It arises 2.5 cm distal to the inferior border of popliteus.
2. *Muscular branches*: To the muscles of posterior compartment.
3. *Nutrient artery to tibia:* It is the **largest nutrient artery** in the body. It enters the nutrient foramen of tibia below the soleal line.
4. *Circumflex fibular artery:* It encircles the lateral side of the neck of the fibula.
5. *Communicating branch:* It joins with the similar branch of peroneal artery about 5 cm above the ankle.
6. *Medial malleolar branch:* It passes toward the medial malleolus.
7. *Calcaneal branch:* It pierces the flexor retinaculum and supplies soft tissues of the heel.
8. *Terminal branches:* These are medial and lateral plantar arteries of the sole.

Clinical correlation

Posterior tibial pulse: It can be felt against the calcaneum about 2 cm below and behind the medial malleolus, and in front of the medial border of the tendocalcaneus. Since the posterior tibial artery lies deep to the flexor retinaculum, it is important to ask the patient to invert his/her foot to relax the flexor retinaculum. Failure to do so may lead to an erroneous conclusion that this pulse is absent.

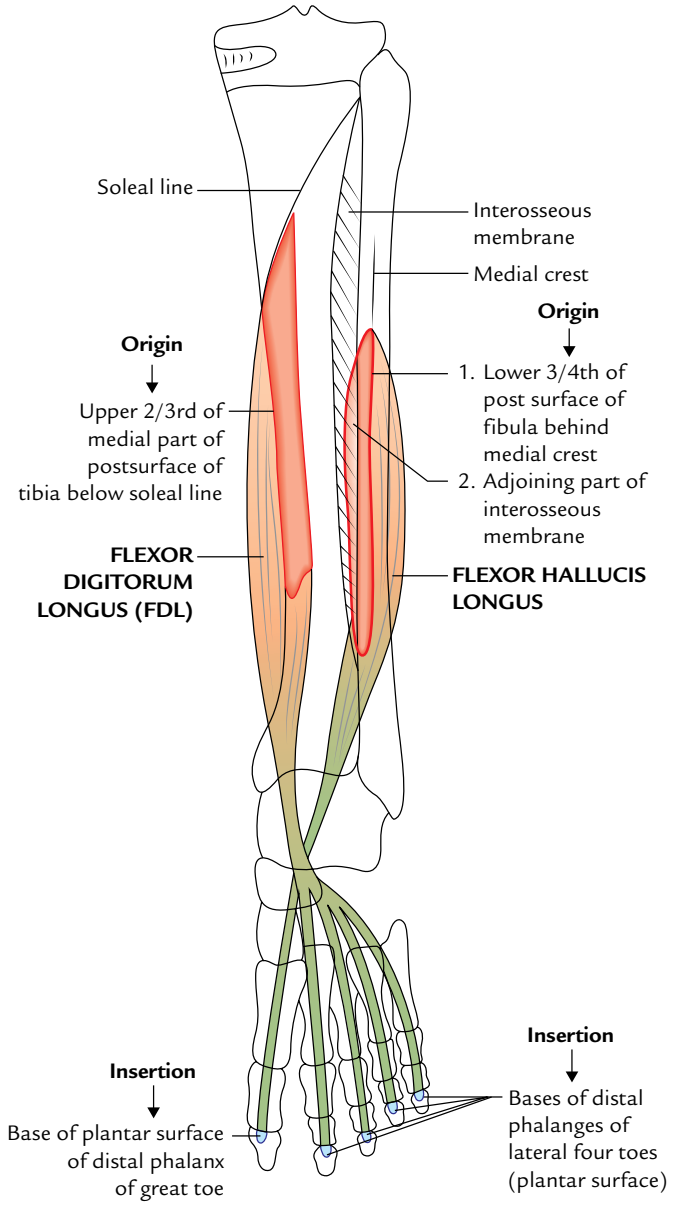


Fig. 29.11 Origin and insertion of the flexor digitorum longus and flexor hallucis longus.

PERONEAL ARTERY (Fig. 29.13)

Origin

It is the largest and most important branch of the posterior tibial artery. It provides blood supply to the posterior and lateral compartments of the leg.

Course and Relations

1. It arises 2.5 cm below the lower border of popliteus.
2. It runs obliquely toward the fibula and then descends along the medial crest of the fibula in a fibrous canal

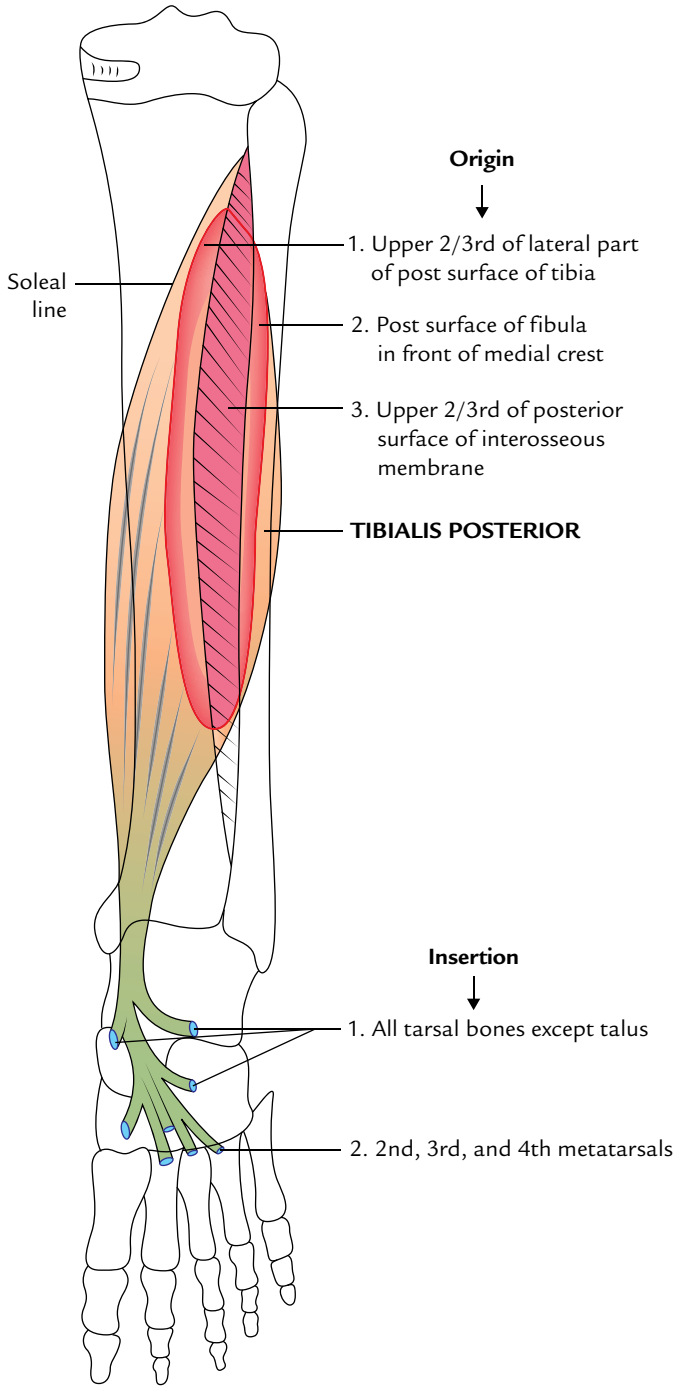


Fig. 29.12 Origin and insertion of the tibialis posterior.

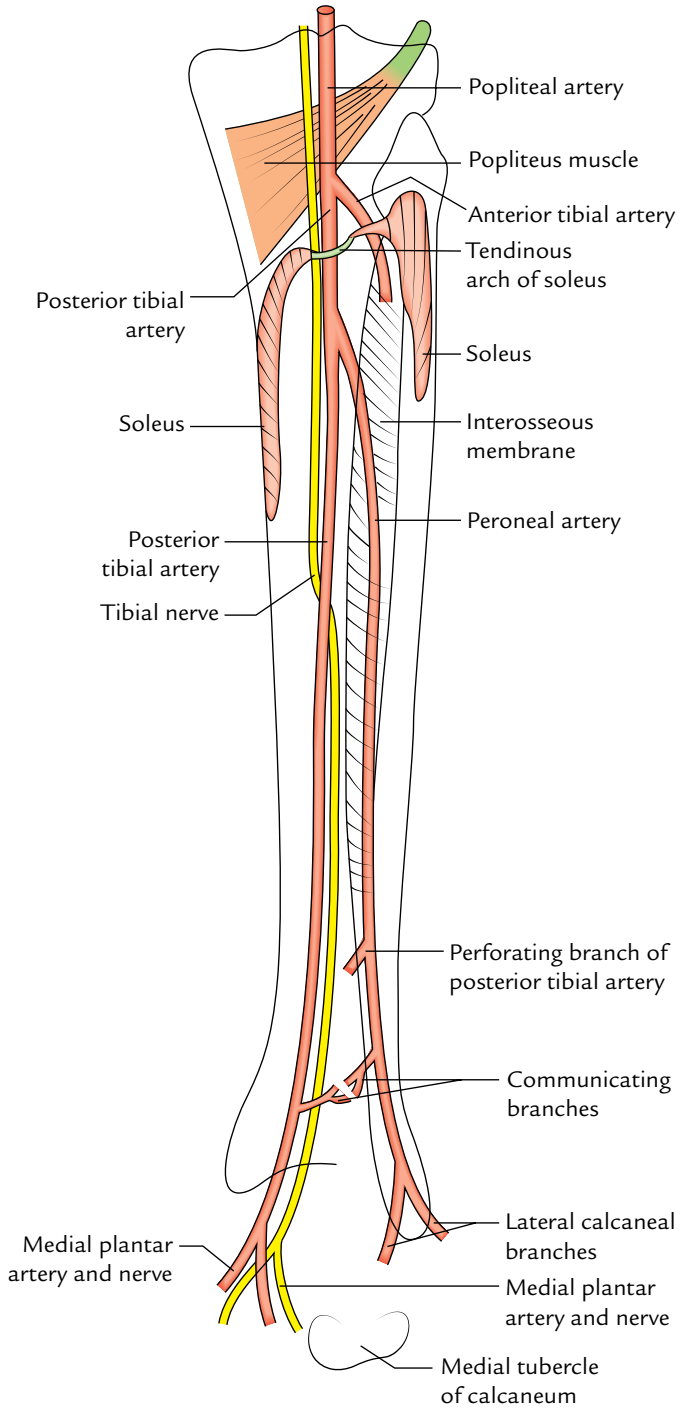


Fig. 29.13 Courses of the posterior tibial and peroneal arteries, and tibial nerve.

between tibialis posterior and flexor hallucis longus. Now it passes behind the inferior tibiofibular and ankle joints, and ends on the lateral surface of calcaneus and terminates by giving the lateral calcaneal arteries.

N.B. The size of peroneal artery is frequently increased. It may join to reinforce or replace the posterior tibial artery in the lower part of the leg.

Branches

1. *Muscular branches* to the posterior and lateral compartments of the leg.
2. *Nutrient artery* to the fibula.
3. *Communicating branch:* It joins with similar branch of the posterior tibial artery about 5 cm above the ankle.
4. *Perforating branch:* It is large and pierces interosseous membrane about 5 cm above the ankle, appears in the

anterior compartment of the leg, and terminates by anastomosing with the lateral malleolar branches of the anterior tibial and dorsalis pedis arteries.

5. *Lateral calcaneal artery:* It is a terminal branch which takes part in the formation of lateral malleolar plexus.

Clinical correlation

Dorsalis pedis artery pulse: It is felt just lateral to the tendon of extensor hallucis longus against the tarsal bones. The clinicians while taking the pulse of the dorsalis pedis artery must keep in mind that the perforating branch of the peroneal artery may reinforce or even replace the dorsalis pedis artery.

TIBIAL NERVE

The tibial nerve is the larger of the two terminal branches of the sciatic nerve.

Origin and Course

1. It arises on the back of the thigh at the junction of the upper two-third and lower one-third and enters the popliteal fossa (see page 374). From the popliteal fossa it enters into the posterior compartment of the leg, by passing deep to the tendinous arch of origin of the soleus along with the posterior tibial vessels.
2. Its course and relations in the leg are similar to that of the posterior tibial artery.
3. In the leg, the nerve is lateral to the artery at first, and then it crosses posterior to the artery from the medial to lateral side and then runs along the lateral side of the artery.
4. It terminates deep to the flexor retinaculum by dividing into medial and lateral plantar nerves.

Branches

1. *Muscular branches:* Popliteus, tibialis posterior, flexor digitorum longus, flexor hallucis longus, and soleus (from its deep surface).
2. *Cutaneous branches:* Medial calcaneal branches which pierce the flexor retinaculum and supply skin of the back and lower surface of the heel—the weight bearing area of the heel.
3. *Articular branches:* To the ankle joint.

Golden Facts to Remember

► Largest compartment of the leg	Posterior compartment
► Peripheral heart	Soleus muscle
► Most powerful tendon in the body	Tendocalcaneus (Achilles tendon)
► Largest nutrient artery in the body	Nutrient artery to tibia (a branch of the posterior tibial artery)
► Muscle functioning as organ of proprioception in the leg	Plantaris
► "Workhorse" of plantar flexion of the foot	Soleus
► Freshman's nerve	Tendon of plantaris
► Largest and most important branch of the posterior tibial artery	Peroneal artery
► Muscle of the leg having maximum concentration of muscle spindles	Plantaris
► All the muscles on the back of the leg are long *except*	Popliteus which is short

Clinical Case Study



A 55-year-old businessman was advised by his physician to reduce the fatty diet and start regular jogging for half an hour daily in the morning.

On one morning while jogging, he heard a sharp snap and felt a sudden pain in his right lower calf. On examination, the examining doctor found that the upper part of the right calf was swollen and a gap was apparent between the swelling and the heel. Diagnosis of "*ruptured tendocalcaneus*" was made.

Questions

1. What is tendocalcaneus and give its chief action?
2. What is the cause of swelling in the upper part of the calf and an apparent gap between it and the heel?
3. Give the site of insertion of tendocalcaneus.
4. What is the commonest cause of rupture of tendocalcaneus?

Answers

1. It is a conjoint tendon of insertion of gastrocnemius and soleus. Its chief action is the plantar flexion of the foot at the ankle.
2. When tendocalcaneus ruptures, the bellies of gastrocnemius and soleus retract upward causing a swelling in the upper part of the calf, leaving a gap between the divided ends of the tendon.
3. Middle one-third of the posterior surface of the calcaneus.
4. An abrupt take off at the start of a 100 m race.

CHAPTER

30 Sole of the Foot

The region/part of the foot that meets the floor or ground is termed **sole of the foot**. In many ways, the structure of the sole of the foot is similar to that of the palm. The major differences arise due to the functional difference between the hand and the foot. The hand is a prehensile organ whereas the foot is concerned with transmission of body weight and locomotion. Note that parts of the foot are in contact with the ground during various phases of standing, walking, and running positions.

SKIN

The skin of the sole presents the following features:

1. It is thick and hairless.
2. It is firmly bound to the underlying deep fascia (plantar aponeurosis) by the numerous fibrous bands.
3. It is creased at the sites of skin movement.
4. It contains large number of sweat glands.

The above features increase the efficiency of grip of the sole on the ground.

N.B. The skin over the major weight-bearing areas of the sole—the heel, lateral margin, and ball of the foot—is very thick.

SUPERFICIAL FASCIA

The superficial fascia is mostly composed of subcutaneous fat in the meshwork of fibrous septa, which anchor the skin with the underlying deep fascia. The superficial fascia is thick and dense over the weight-bearing points to provide fibrofatty cushions at these sites (e.g., posterior tubercles of the calcaneum, metatarsal heads, and pulps of the digits).

CUTANEOUS NERVES (Fig. 30.1)

The skin of the sole of the foot is supplied by three cutaneous nerves, which arise directly or indirectly from the tibial nerve.

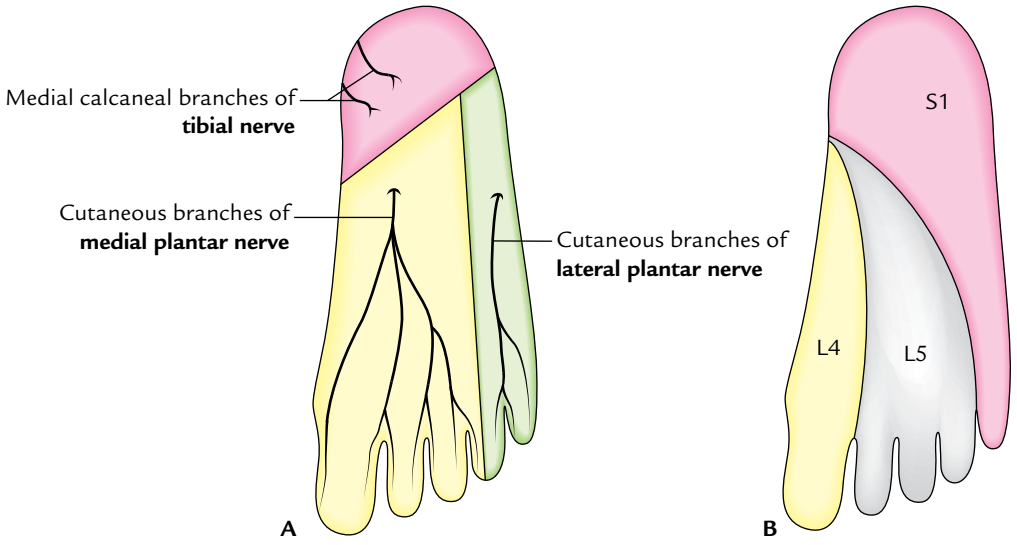


Fig. 30.1 Cutaneous innervation of the sole of the foot: A, areas supplied by three sets of cutaneous nerves; B, the dermatomes.

The areas supplied by the three nerves roughly correspond to the three dermatomes of the sole:

1. **Medial calcaneal branches:** They arise directly from the tibial nerve and supply skin over the posterior and medial portions of the sole—the weight-bearing portion of the heel. The corresponding dermatome is S1.
2. **Cutaneous branches of the medial plantar nerve:** They supply the skin over the larger anteromedial portion of the sole and medial 3½ digits. The corresponding dermatome is L4.
3. **Cutaneous branches of the lateral plantar nerve:** They supply the skin over the smaller anterolateral portion of the sole and lateral 1½ digits. The corresponding dermatome is L5.

DEEP FASCIA

The deep fascia in the region of the sole consists of three parts: central, medial, and lateral. The central part of the deep fascia is very thick and termed *plantar aponeurosis*. The medial and lateral parts are thin and termed medial and lateral plantar fasciae, respectively.

The deep fascia of the sole is composed of compact bundles of collagen fibres which are arranged longitudinally in the plantar aponeurosis and transversely in the medial and lateral plantar fasciae.

The thick central part covers the flexor digitorum brevis. The thin medial and lateral parts cover the abductor hallucis and abductor digiti minimi, respectively.

In the region of toes, the deep fascia forms the *deep transverse metatarsal ligaments* and *fibrous flexor sheaths.*

N.B.

- The deep fascia in the sole is specialized to form three things: (a) plantar aponeurosis, (b) deep transverse metatarsal ligaments, and (c) fibrous flexor sheaths.
- According to some authorities whole of the deep fascia of the sole is termed plantar aponeurosis.

PLANTAR APONEUROSIS (Fig. 30.2)

The plantar aponeurosis is the thickened central part of the deep fascia of the sole.

Features

The plantar aponeurosis is triangular in shape and occupies the central area of the sole. The *apex* of the plantar aponeurosis is attached to the medial tubercle of calcaneum, proximal to the attachment of the flexor digitorum brevis. The *base* of the plantar aponeurosis near the heads of the metatarsals divides into *five bands*, one for each toe. At the point of division, the five processes are bound by the transverse fascial fibres. The digital nerves and vessels pass through the intervals between the processes.

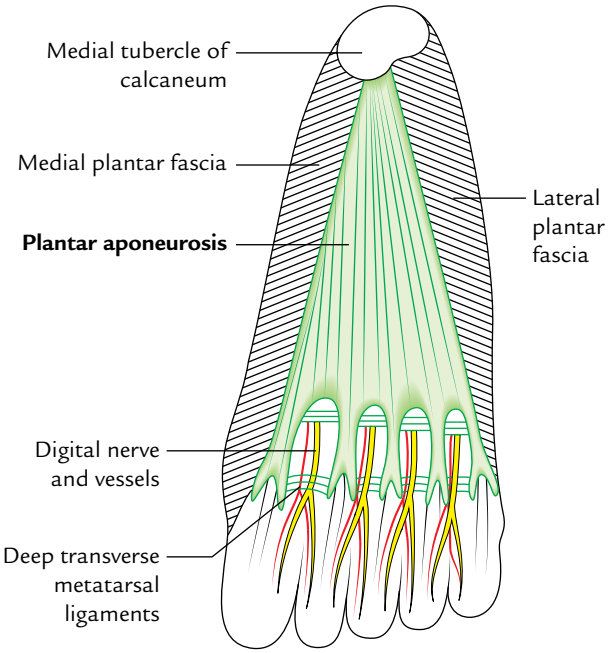


Fig. 30.2 Plantar aponeurosis.

Each band splits opposite the metatarso-phalangeal joints into *a superficial* and *a deep slip*. The superficial slip is attached to the dermis of the skin and blends with the superficial *transverse metatarsal ligaments*. The deep slip divides into two parts, which embrace the flexor tendons, and blend with the *fibrous flexor sheaths* and *deep transverse metatarsal ligaments.*

From the medial and lateral margins of the aponeurosis, the *lateral and medial vertical intermuscular septa* pass deeply, and divide the plantar muscles into three groups—medial, intermediate, and lateral. The thinner *transverse septa* arise from the vertical septa and divide the muscles of the sole into four layers.

N.B. Morphologically the plantar aponeurosis represents the degenerated tendon of plantaris muscle, which has been separated by the enlarging heel during evolution.

Functions

The functions of the plantar aponeurosis are:

1. It firmly fixes the skin of the sole.
2. It provides origin to the muscles of first layer of the sole.
3. It protects the plantar nerve and vessels from compression.
4. It helps to maintain the longitudinal arches of the foot by acting as tie beam.

Clinical correlation

Plantar fasciitis and calcaneal spur: The plantar aponeurosis is stretched during standing position. Hence, tearing or inflammation (*plantar fasciitis*) often occurs in individuals who do a great deal of standing or walking, viz. traffic police personnel. It causes pain and tenderness in the sole of the foot especially underneath the heel during standing. Repeated attack of the plantar fasciitis leads to calcification in the posterior attachment of the plantar aponeurosis forming a *calcaneal spur*.

DEEP TRANSVERSE METATARSAL LIGAMENTS

These are four short, flat bands of fibrous tissue, which connect the plantar ligaments of the adjoining metatarsophalangeal joints. They are related dorsally to interossei, and ventrally to lumbricals and digital nerves and vessels.

FIBROUS FLEXOR SHEATHS

The inferior surface of each toe from the head of metatarsal to the base of distal phalanx is provided with a strong fibrous sheath derived from the deep fascia of the toes. It is attached to the sides of the phalanges.

The proximal end of each sheath receives the deeper part of the slip of plantar aponeurosis. The distal end of the sheath is closed and is attached to the base of the distal phalanx.

The sheath along with the inferior surfaces of the phalanges and interphalangeal joints forms a blind tunnel through which pass long flexor tendon/tendons of the toes.

Their structure and arrangement is similar to that of fibrous flexor sheaths of the fingers. They retain flexor tendons in position during flexion of the toes.

MUSCLES OF THE SOLE OF THE FOOT

There are 18 intrinsic muscles and 4 extrinsic tendons in the sole of the foot. The muscles of the sole are described in four layers from superficial to deep (Table 30.1).

The muscles of the sole are chiefly concerned with supporting the arches of the foot. The short and long muscles of the foot act as synergists.

N.B. *Neurovascular planes of the sole:* There are two neurovascular planes between the muscle layers of the sole:

(a) Superficial neurovascular plane between the first and second layers.

(b) Deep neurovascular plane between the third and fourth layers.

In the *superficial neurovascular plane* lies the trunks of medial and lateral plantar nerves, and the arteries.

In the *deep neurovascular plane* lies the deep branches of the lateral plantar nerve and artery.

The origin, insertion, nerve supply, and actions of the muscles of the sole are given in Table 30.2.

To understand the origin and insertion of the muscles of the foot the student must study the layout of the different bones on the plantar aspect of the skeleton of the foot (Fig. 30.3).

Table 30.1 Muscle layers of the sole of the foot

Layer	Muscles	Features
First layer	• Flexor digitorum brevis • Abductor hallucis • Abductor digiti minimi	They cover whole of the sole
Second layer	• Flexor digitorum accessorius • Four lumbricals • Two tendons (tendon of flexor digitorum longus and tendon of flexor hallucis longus)	Flexor digitorum accessorius and lumbricals are attached to the tendon of flexor digitorum longus
Third layer	• Flexor hallucis brevis • Flexor digiti minimi brevis • Adductor hallucis	• They are confined to the metatarsal region of the sole • Two of these muscles act on the big toe and one on the little toe
Fourth layer	• Interossei (3 plantar interossei and 4 dorsal interossei) • Tendon of tibialis posterior • Tendon of peroneus longus	They fill up the intermetatarsal spaces

Table 30.2 Origin, insertion, nerve supply, and actions of the intrinsic muscles of the sole

Muscles	Origin	Belly/Tendon	Insertion	Nerve supply	Actions
First layer (Fig. 30.4)					
• *Flexor digitorum brevis* (resembles flexor digitorum superficialis of the hand)	Medial tubercle of the calcaneum	• It forms 4 tendons for the lateral 4 toes • Each tendon splits into two slips opposite the bases of proximal phalanges to allow passage of the long flexor tendon	Margins of the middle phalanges of lateral 4 toes	Medial plantar nerve (S2, S3)	Flexor of the lateral toes
• *Abductor hallucis* (lies along the medial border of the foot)	• Medial tubercle of the calcaneum • Flexor retinaculum	Its tendon fuses with medial portion of the tendon of flexor hallucis brevis for a common insertion	Medial side of the base of proximal phalanx of the big toe	Medial plantar nerve (S2, S3)	Flexion and abduction of the big toe
• *Abductor digiti minimi* (lies along the lateral border of the foot)	Medial and lateral tubercles of the calcaneum in a continuous line	Its tendon fuses with the tendon of flexor digiti minimi brevis for a common insertion	Lateral side of the base of proximal phalanx of the little toe	Lateral plantar nerve (S2, S3)	Flexion and abduction of the little toe
Second layer (Fig. 30.5)					
• *Flexor digitorum accessorius*	By two heads: • *Medial head* from the medial concave surface of the calcaneum and adjoining part of the medial tubercle • *Lateral head* from the calcaneum in front of the lateral tubercle	The two heads unite at an acute angle	Tendon of the flexor digitorum longus (FDL)	Lateral plantar nerve (S2, S3)	• Straightens the pull of the long flexor tendons • Assists the flexor digitorum longus in flexing the lateral 4 toes
• *Lumbricals* (4 in number and numbered from medial to lateral side) Tendon of flexor digitorum longus (see Table 29.3) Tendon of flexor hallucis longus (see Table 29.3)	From the tendons of the flexor digitorum longus	Tendons pass forwards on the medial sides of the metatarso-phalangeal joints of the lateral four toes	Into the extensor expansions and bases of proximal phalanges of lateral 4 toes	1st by the medial plantar nerve, and 2nd, 3rd, and 4th by the lateral plantar nerve (S2, S3)	Extension of toes at the interphalangeal joints

(Contd.)

(Contd.)

Muscles	Origin	Belly/Tendon	Insertion	Nerve supply	Actions
Third layer (Fig. 30.6)					
• *Flexor hallucis brevis*	It arises by a Y-shaped tendon: • *Lateral limb* from the medial part of the plantar surface of the cuboid bone • *Medial limb* from the lateral cuneiform	Muscle belly splits in two parts. Each part gives rise to a tendon which contains a sesamoid bone near its insertion. The medial tendon blends with the abductor hallucis, and the lateral tendon blends with the adductor hallucis	Each side of the base of proximal phalanx of the big toe (1st digit)	Medial plantar nerve (S2, S3)	Flexion of the proximal phalanx of big toe
• *Flexor digiti minimi brevis*	Base of the 5th metatarsal bone	Forms narrow tendon which blends with the abductor digiti minimi	By a narrow tendon into the lateral side of the base of proximal phalanx of the little toe	Superficial branch of the lateral plantar nerve (S2, S3)	Flexes the proximal phalanx of the little toe
• *Adductor hallucis*	It arises by two heads: • *Large oblique head* from bases of the 2nd, 3rd, and 4th metatarsals • Small *transverse head* from the plantar ligaments of the metatarso-phalangeal joints of 3rd, 4th, and 5th toes	The common tendon of both heads fuses with the lateral tendon of the flexor hallucis brevis	By the composite tendon on the lateral side of the base of proximal phalanx of the big toe	Deep branch of the lateral plantar nerve (S2, S3)	• Adduction of the big toe • Maintains transverse arches of the foot
Fourth layer (Fig. 30.7)					
• *Dorsal interossei* * (4 in number, and lies between the metatarsal bones; Fig. 30.8A)	They are bipennate and arises from the adjacent sides of the metatarsal bones (1–5)	Tendons are arranged on the abductor sides of the toes	• 1st on the medial side of the proximal phalanx of 2nd digit • 2nd–4th on the lateral sides of 2nd–4th digits	Lateral plantar nerve (S2, S3)	• Abducts digits (2nd–4th) • Flexes the metatarso-phalangeal joints
• *Plantar interossei* * (3 in number and lies rather below the metatarsals Fig. 30.8B)	They are unipennate and arises from the medial sides of the metatarsal bones (3rd–5th)	Tendons are arranged on the adductor sides of the toes	Medial sides of bases of the phalanges of 3rd–5th digits	Lateral plantar nerve (S2, S3)	• Adducts the digits (2nd–4th) • Flexes the metatarso-phalangeal joints

Mnemonic:* Plantar interossei **ADduct (**PAD**) the toes and arise from single metatarsal as unipennate muscles; whereas **D**orsal interossei **AB**duct (**DAB**) the toes and arise from two metatarsals as bipennate muscles.

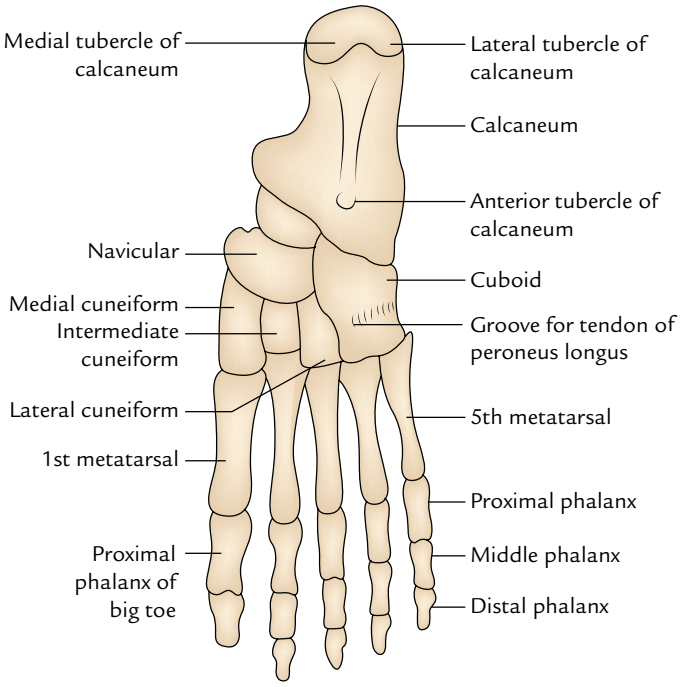


Fig. 30.3 Layout of different bones on the plantar aspect of the skeleton of the foot.

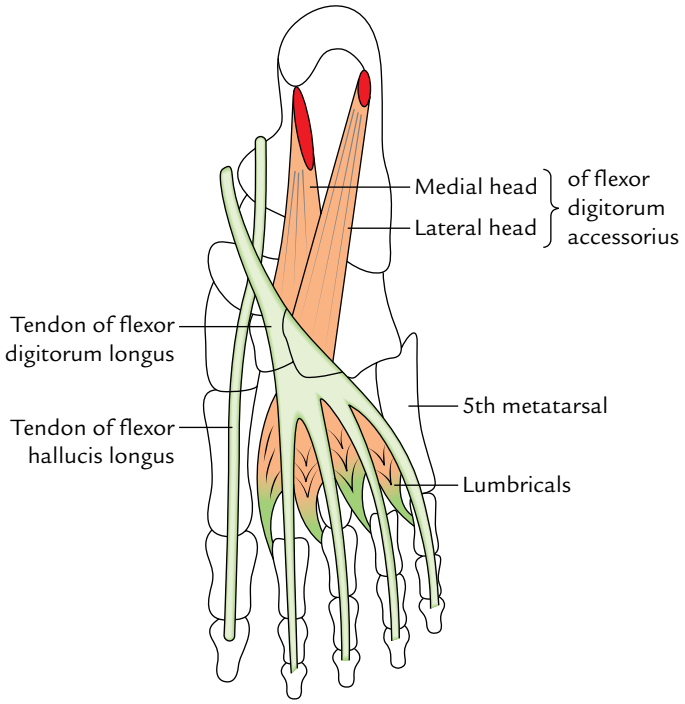


Fig. 30.5 Muscles of the second layer of the sole.

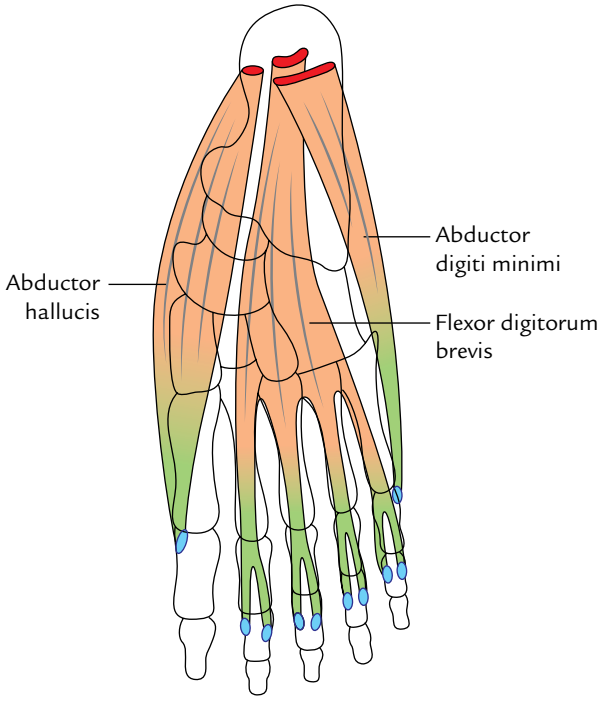


Fig. 30.4 Muscles of the first layer of the sole.

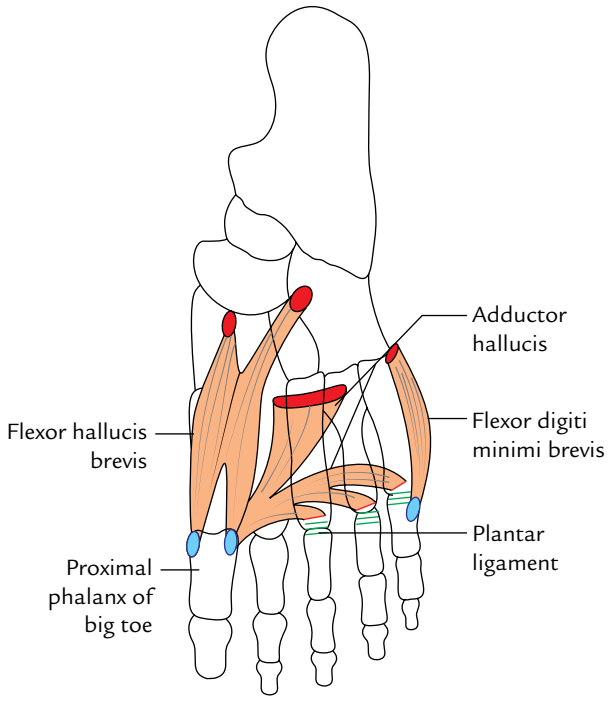


Fig. 30.6 Muscles of the third layer of the sole.

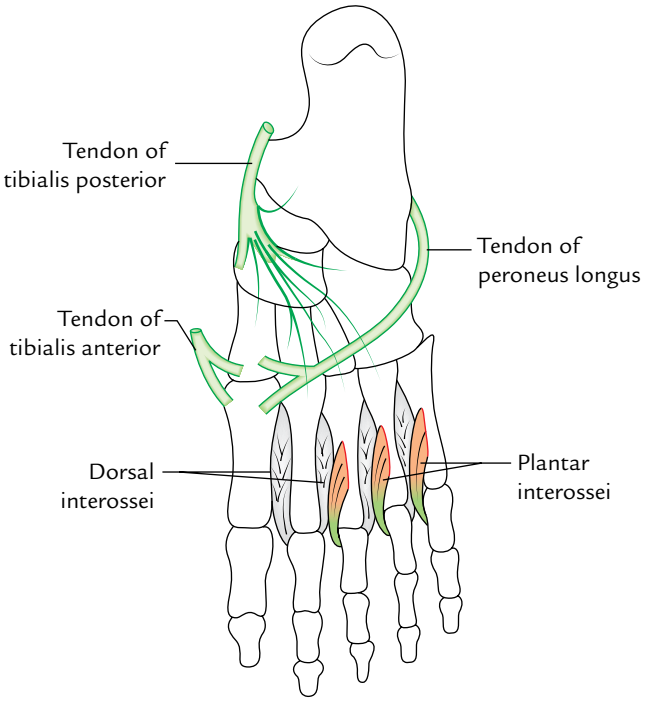


Fig. 30.7 Muscles of the fourth layer of the sole.

PLANTAR NERVES

There are two plantar nerves—medial and lateral.

MEDIAL PLANTAR NERVE

The medial plantar nerve is the *pre-axial nerve* of the foot.

Origin and Course

It is the larger terminal branch of the tibial nerve and begins deep to flexor retinaculum. It passes forward between the abductor hallucis and flexor digitorum brevis. It is accompanied by the medial plantar artery on its medial side.

Branches and Distribution (Fig. 30.9)

It gives rise to the following branches:

1. **Muscular branches to four muscles:**
   (a) Abductor hallucis
   (b) Flexor digitorum brevis } by the main trunk.
   (c) Flexor hallucis brevis, by first digital nerve to the medial side of the big toe.
   (d) First lumbrical, by the second digital nerve.

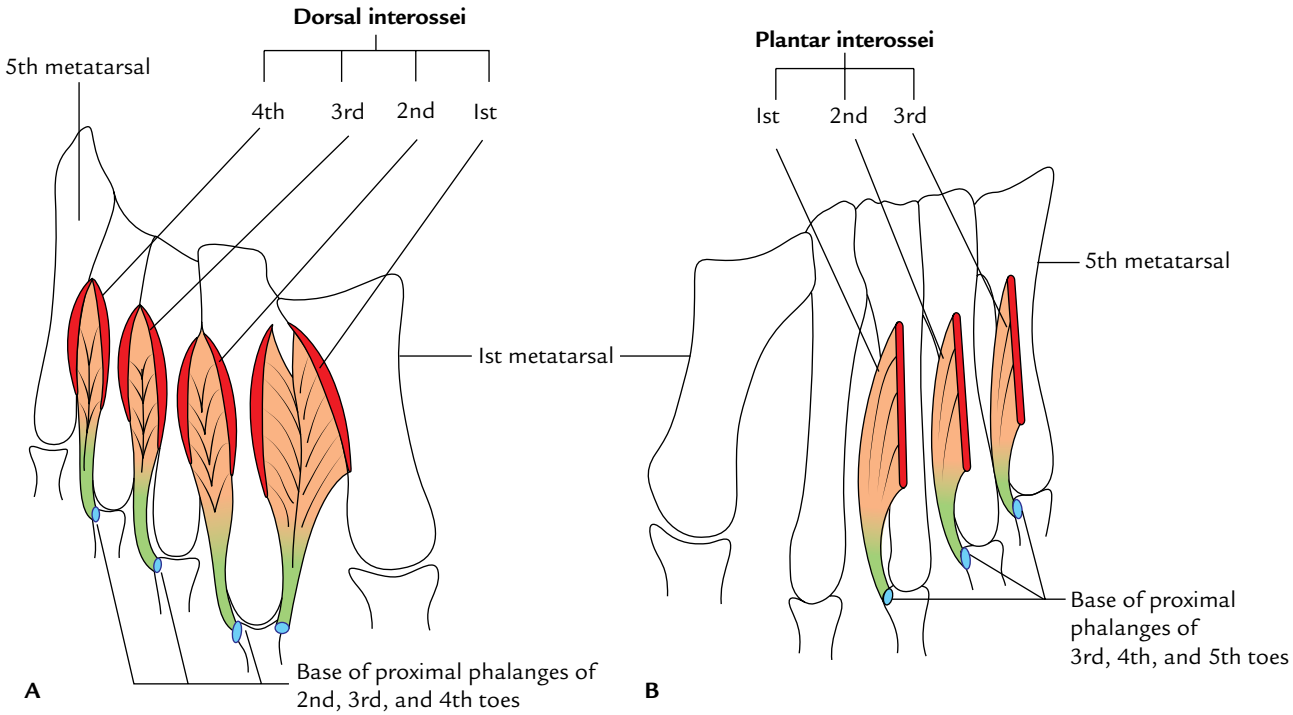


Fig. 30.8 Origin and insertion of interosseous muscles of the foot: A, dorsal interossei; B, plantar interossei.

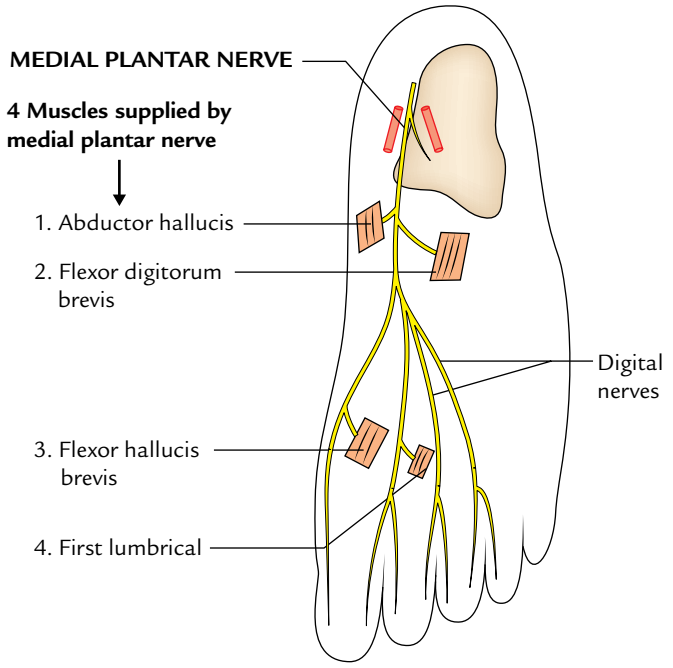


Fig. 30.9 Course and motor distribution of medial plantar nerves.

2. **Cutaneous branches** (four digital nerves) supply skin of the medial 3½ toes and larger medial part of the sole. Each digital nerve also supplies dorsum of the terminal phalanx.
3. **Articular branches** supply the joints of tarsus and metatarsus.

N.B. The distribution of *medial plantar nerve* in the sole corresponds to the distribution of *median nerve in the hand.*

Clinical correlation

Medial plantar nerve entrapment: The medial plantar nerve may be compressed either deep to flexor retinaculum or deep to abductor hallucis due to the repetitive eversion of the foot (e.g., during gymnastics and running). Clinically, it presents as burning, numbness, and tingling (paraesthesia) on the medial side of the sole and in the region of navicular tuberosity.

Since it commonly occurs in runners, this condition is referred to as **"jogger's foot."**

LATERAL PLANTAR NERVE

The lateral plantar nerve is the *post-axial nerve* of the foot.

Origin and Course

It is the smaller terminal branch of the tibial nerve and begins deep to the flexor retinaculum. It appears in the sole deep to the abductor hallucis. It passes forward and laterally to the base of the fifth metatarsal in between the first and second layers of the muscles. On reaching between flexor digitorum brevis and abductor digiti minimi, it divides into the superficial and deep branches.

Branches and Distribution (Fig. 30.10)

It gives rise to the following branches:

1. **Branches from the main trunk:**
   (a) *Cutaneous* to lateral part of the sole.
   (b) *Muscular* to supply two muscles: abductor digiti minimi and flexor digitorum accessorius.
2. **Branches from the superficial branch:** The *superficial branch* divides into two branches—lateral and medial.
   The *lateral branch* supplies three muscles: flexor digiti minimi brevis, third plantar interosseous, and fourth dorsal interosseous. It also supplies the skin on the lateral side of little toe.
   The *medial branch* communicates with the medial plantar nerve and supplies the skin of the fourth cleft.
3. **Branches of the deep branch:** It courses medially lying between the third and fourth layers of the sole. It supplies

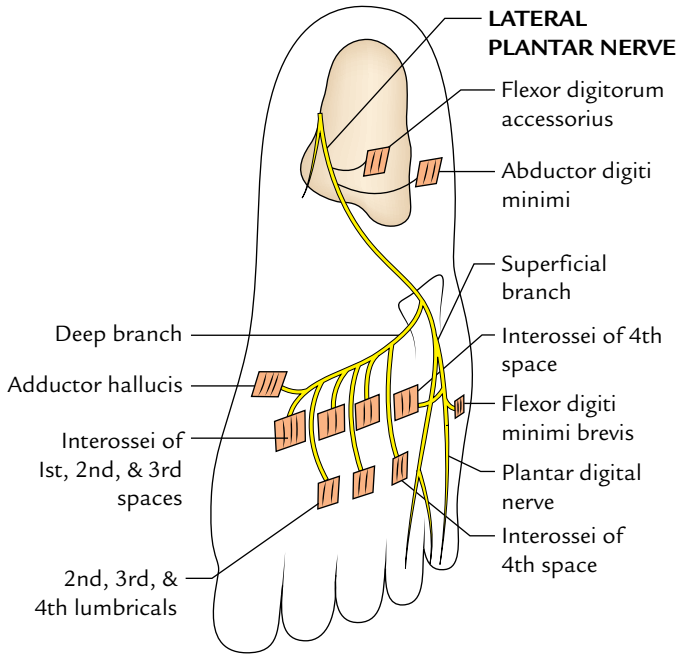


Fig. 30.10 Course and motor distribution of lateral plantar nerves.

Table 30.3 Motor innervation of the medial and lateral plantar nerves

Medial plantar nerve	Lateral plantar nerve
Four muscles • Abductor hallucis • Flexor digitorum brevis • Flexor hallucis brevis • First lumbrical	• Flexor digitorum accessorius • Abductor digiti minimi • Abductor hallucis • Flexor digiti minimi brevis • All interossei • Second, third, and fourth lumbricals

Table 30.4 Distribution of pre-axial and post-axial nerves in the palm and sole

Pre-axial nerves		Post-axial nerves	
Median nerve	**Medial plantar nerve**	**Ulnar nerve**	**Lateral plantar nerve**
Abductor pollicis	Abductor hallucis	Abductor digiti minimi	Abductor digiti minimi
Flexor pollicis brevis	Flexor hallucis brevis	Flexor digiti minimi	Flexor digiti minimi brevis
Opponens pollicis	No corresponding muscle	Opponens digiti minimi	No corresponding muscle
Flexor digitorum superficialis	Flexor digitorum brevis	Abductor pollicis All interossei	Abductor hallucis All interossei Flexor digitorum accessorius
Lumbricals 1 and 2	Lumbrical 1	Lumbricals 3 and 4	Lumbricals 2, 3, and 4
3½ Digits	3½ Digits	1½ Digit	1½ Digit

nine muscles: abductor hallucis, three lumbricals (second, third, and fourth), and five interossei of the first three intermetatarsal spaces.

The motor innervation of the medial and lateral plantar nerves is summarized in Table 30.3.

N.B. The distribution of pre-axial and post-axial nerves in the palm and sole is given in Table 30.4.

PLANTAR ARTERIES (Figs 30.11 and 30.12)

There are two plantar arteries—medial and lateral.

MEDIAL PLANTAR ARTERY

The medial plantar artery is the smaller terminal branch of the posterior tibial artery. It arises beneath the flexor retinaculum and appears in the sole deep to abductor hallucis accompanied by the medial plantar nerve on its lateral side. It terminates on the medial side of big toe by dividing into following two branches:

1. A small branch, which passes distally along the medial side of the big toe and anastomoses with the digital branch of first plantar metatarsal artery, on the lateral side of the big toe.
2. A large branch, which splits into three *superficial digital branches* and anastomoses with the first to third plantar metatarsal arteries.

Distribution

It gives rise to the following branches:

1. **Muscular branches** to the adjoining muscles.
2. **Cutaneous branches** to the medial side of the sole.
3. **Digital arteries** (vide supra).

LATERAL PLANTAR ARTERY

The lateral plantar artery is the larger terminal branch of the posterior tibial artery. It arises beneath the flexor retinaculum and runs forward towards the base of the fifth metatarsal. The lateral plantar nerve lies on its medial side. At the base of the fifth metatarsal bone, it curves medially with concavity facing proximally and runs towards the first inter-digital space to join the dorsalis pedis artery and thus forms the *plantar arch.*

It gives rise to the following branches:

1. **Muscular branches** to supply the adjoining muscles.
2. **Superficial branches** to supply skin and fasciae laterally.
3. **Anastomotic branch** to anastomose at the lateral border of foot with the lateral tarsal and arcuate arteries.
4. **Calcaneal branch** is occasional to the heel.

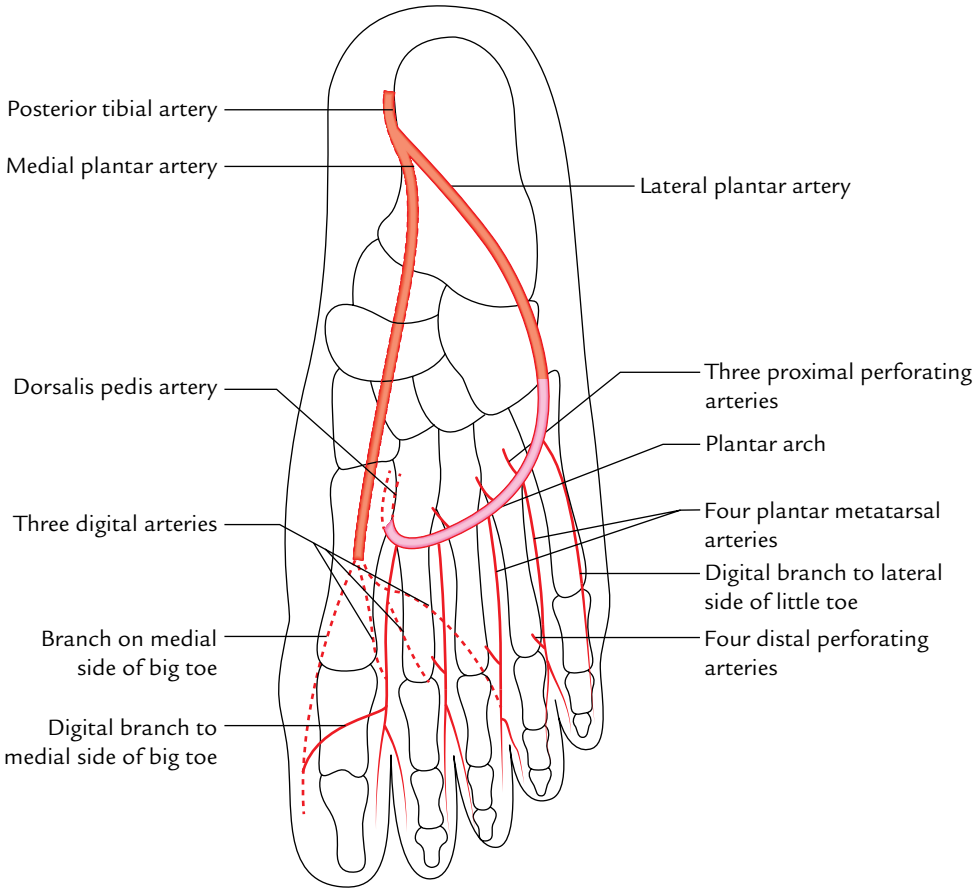


Fig. 30.11 Course and branches of the medial and lateral plantar arteries.

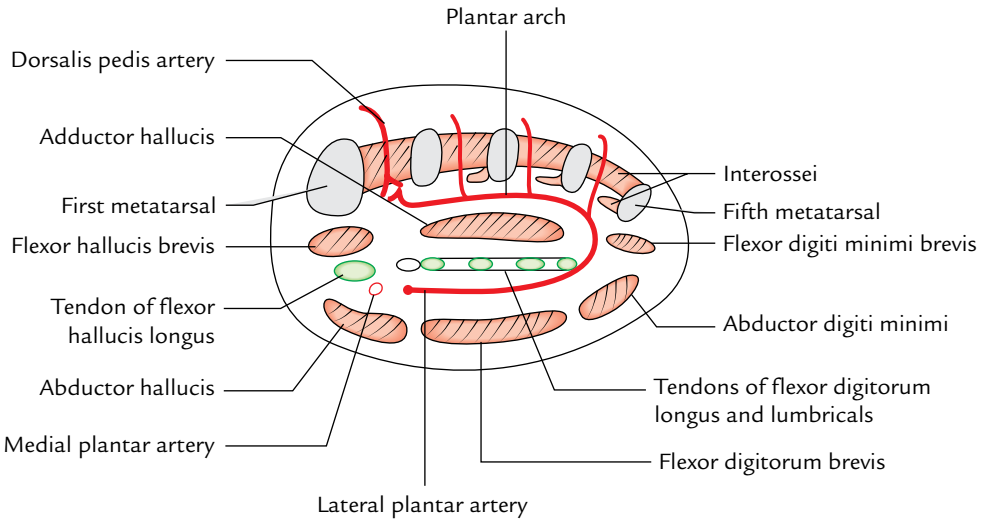


Fig. 30.12 Transverse section of the right foot through the metatarsal showing arrangements of structures in the sole.

N.B. *Plantar arch:* It is an arterial arch formed by the direct continuation of the lateral plantar artery from the base of the fifth metatarsal bone to the first intermetatarsal space. It is completed medially by anastomosing with the terminal part of the dorsalis pedis artery. *It extends from the base of the fifth metatarsal bone to the proximal part of the first intermetatarsal space.* It lies between the third and fourth layers of the sole. It is convex distally across the bases of fourth, third, and second metatarsals. It is accompanied by a deep branch of lateral plantar nerve, which lies in its concavity.

BRANCHES

It gives rise to the following branches:

1. **Four plantar metatarsal arteries:** They pass forward in the intermetatarsal spaces and end by dividing into a pair of plantar digital arteries. Near their point of division, each plantar metatarsal artery gives a *distal perforating branch* to join the corresponding **dorsal metatarsal artery.** The first plantar metatarsal artery sends a digital branch to the medial side of the big toe. The digital branch to the lateral side of little toe arises from the lateral plantar artery.
2. **Three proximal perforating arteries:** They ascend through the proximal parts of the second, third, and fourth intermetatarsal spaces, and anastomoses with the corresponding dorsal metatarsal arteries.

N.B.

- All the structures (e.g., main neurovascular bundle and long flexor tendons) enter the sole through a gap between the flexor retinaculum and the calcaneus called **porta pedis** except the tendon of peroneus longus, which enters the sole through a groove beneath the cuboid called **side gate of the sole**.
- The passages for proximal and distal perforating arteries are called **windows of the sole**.

Golden Facts to Remember

► Podiatry	Field that deals with the study and care of the foot
► Most common hind foot clinical problem	Plantar fasciitis
► All the interossei of sole are supplied by the deep branch of the lateral plantar nerve *except*	Those in the fourth intermetatarsal space, which are supplied by the superficial branch of the lateral plantar nerve
► Porta pedis	Osseofibrous tunnel between flexor retinaculum and calcaneus
► Policeman's heel	Plantar fasciitis

Clinical Case Study



A 55-year-old woman complained of pain on the plantar surface of her heel and on the medial aspect of the foot. The pain often became severe while beginning to walk in the morning but disappeared after 5–10 minutes of rest after activity. However, it again recurred on walking after rest.

On examination, the doctor located the point of tenderness at the medial tubercle of the calcaneum and on the medial surface of this bone. Pain increased with passive extension of the big toe and was further exacerbated by dorsiflexion of the foot and/or weight bearing. A radiograph of the foot revealed an abnormal bony process protruding from the medial tubercle of the calcaneum. A diagnosis of the plantar fasciitis was made.

Questions

1. What is the meaning of plantar fasciitis? Give its possible cause.
2. What is the plantar fascia? Give its attachments.
3. What is calcaneal spur?
4. Give the cause of tenderness in the region of calcaneal spur.

Answers

1. It is the inflammation of plantar aponeurosis at its proximal attachment to the medial tubercle of the calcaneum due to repeated minor trauma. It occurs in individuals who do a great deal of standing and walking, particularly if inappropriate footwear is worn.
2. See pages 420–421.
3. Repeated attacks of the plantar fasciitis induce ossification in the proximal attachment of the plantar aponeurosis forming a calcaneal spur.
4. Usually a synovial bursa develops at the end of the spur that may become inflamed and tender.

CHAPTER

31 Arches of the Foot

The human foot is described as an architectural marvel of the nature. Its construction is the best example of the structural adaptation to the function.

The foot performs two major functions:

1. It acts as a pliable platform to support the body weight during standing position.
2. It acts as a lever to propel the body forward during walking, running, and jumping.

To fulfill the first function, the foot is designed in the form of elastic arches. These arches are segmented so that they can sustain the stress of weight and thrusts at the optimum level.

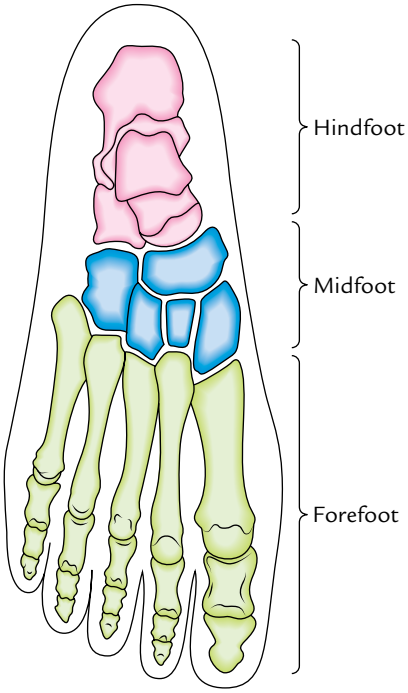


Fig. 31.1 Division of the foot and its bones into three anatomical and functional segments/parts of the skeleton of the foot.

To fulfill the second function, the foot is so constructed that it is transformable in a lever. The segmented arched lever converts the foot into a spring, which is ideally suited for its functions.

The foot and its bones are divided into the following three anatomical and functional segments (Fig. 31.1):

1. The *hindfoot* consists of talus and calcaneus.
2. The *midfoot* consists of navicular, cuboid, and cuneiforms.
3. The *forefoot* consists of metatarsals and phalanges.

The skeleton of the foot is arched, both longitudinally and transversely, with the concavity directed towards the plantar surface (i.e., the bones of the foot are arranged to form the **transverse and longitudinal arches**). The presence of arches makes the sole concave both anteroposteriorly and transversely. This is best reflected in the footprint showing the weight-bearing points of the sole (Fig. 31.2).

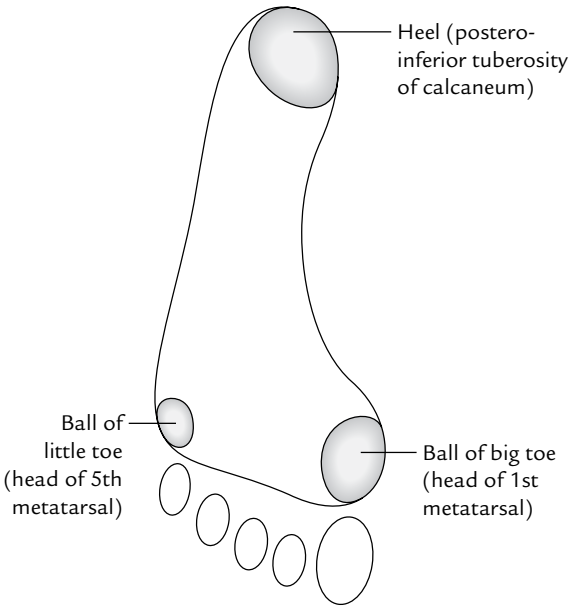


Fig. 31.2 Right foot print showing weight-bearing points.

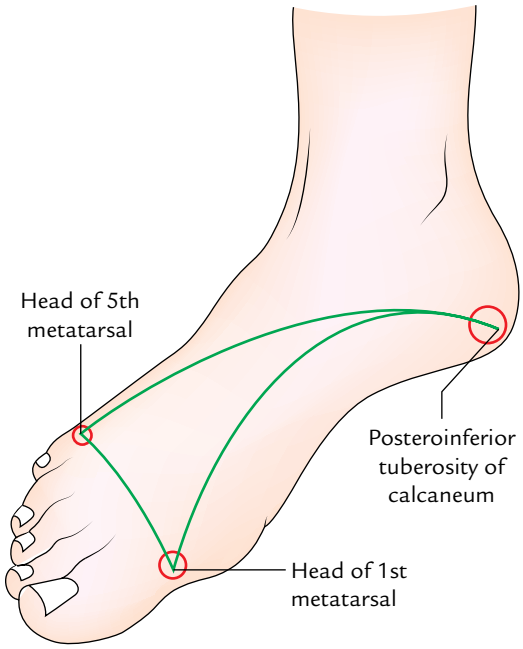


Fig. 31.3 Distribution of the body weight among three points.

During the standing position, the weight of the body is distributed among three points: (a) posteroinferior tuberosity of the calcaneum (heel), (b) head of first metatarsal, and (c) head of fifth metatarsal (Fig. 31.3).

The arches of the foot are present right from birth, but due to the presence of excessive subcutaneous fat in the soles, they are not apparent (i.e., masked) during infancy and childhood.

N.B.

- An arched foot is a distinctive feature of man, which distinguishes him from other primates.

Factors Maintaining the Arches of the Foot

The students can easily appreciate the factors maintaining the arches of the foot by applying the engineering device used to support a stone bridge. The devices used to support a stone bridge are (Fig. 31.4):

1. Shape of stones.
2. Intersegmental ties (staples).
3. Slings.
4. Tie beams.

TYPES OF ARCHES

There are two types of arches of the foot—longitudinal and transverse.

1. There are two longitudinal arches in each foot: (a) medial and (b) lateral.

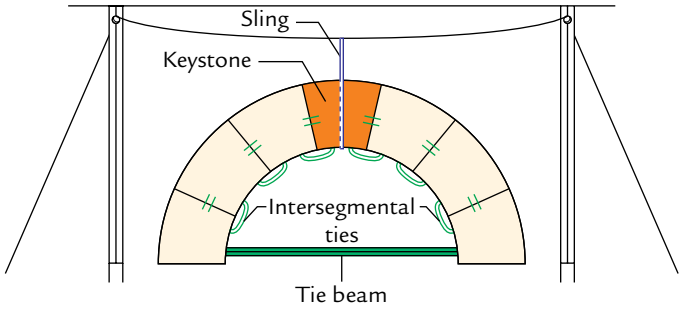
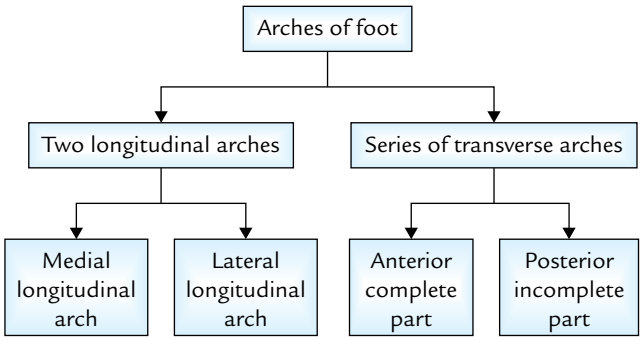


Fig. 31.4 Supports of a stone bridge.



Flowchart 31.1 Classification of the arches of the foot.

2. There are a series of transverse arches in each foot. At the heads of metatarsals, the transverse arch is complete but posteriorly it forms a half dome, which is completed by its counterpart in the opposite foot. Thus, in the posterior part of the foot the transverse arch becomes complete when the both feet are held close to each other.

The classification of arches is summarized in Flowchart 31.1.

LONGITUDINAL ARCHES (Fig. 31.5)

Each longitudinal arch has: (a) two pillars, (b) a summit, and (c) joints.

Medial Longitudinal Arch (Fig. 31.5A and B)

The medial longitudinal arch is formed by the calcaneum, talus, navicular, three cuneiforms, and medial three metatarsals.

Pillars

1. The medial half of the calcaneum forms the **posterior pillar** of the medial longitudinal arch.
2. The heads of the medial three metatarsals form the **anterior pillar** of the medial longitudinal arch.

Summit

The talus lies at the summit of this arch. Therefore, the **talus is the keystone of this arch.**

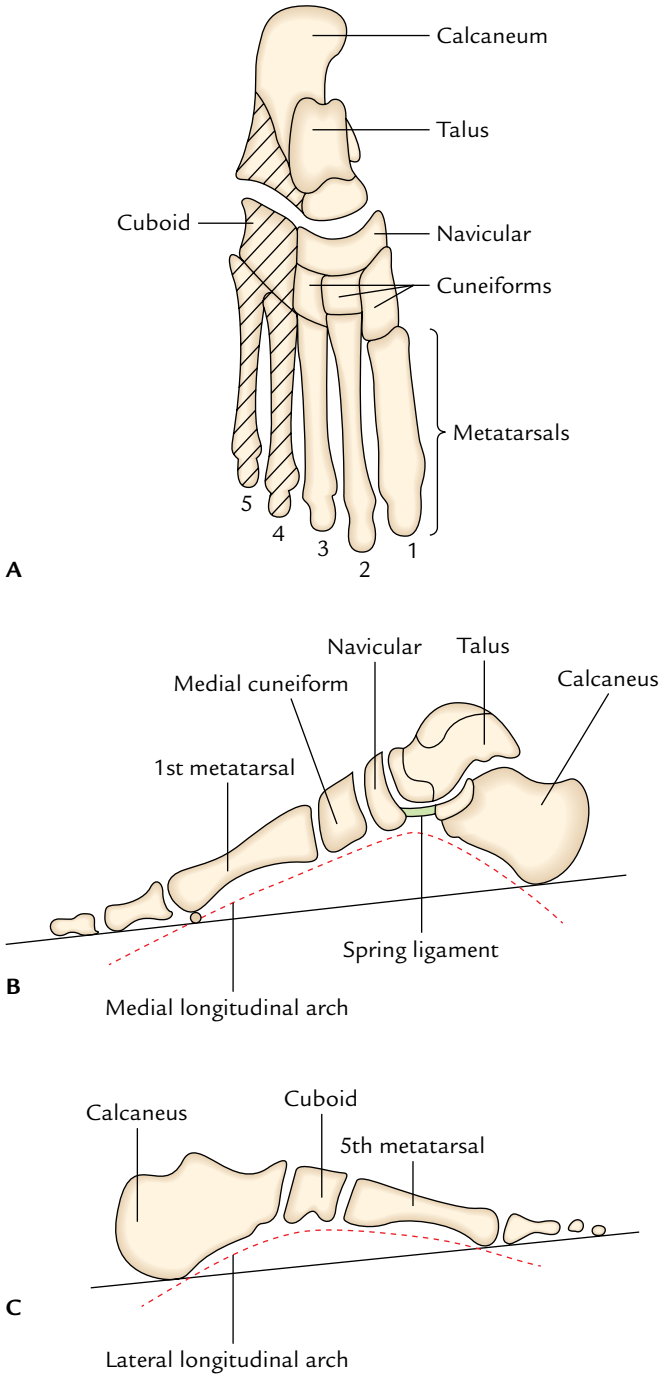


Fig. 31.5 Formation of the longitudinal arches of the foot: A, bones forming medial longitudinal arch are unhatched and bones forming lateral longitudinal arch are hatched. Note calcaneum is common to both but primarily it forms the lateral longitudinal arch; B, medial longitudinal arch—side view; C, lateral longitudinal arch—side view. Note the medial longitudinal arch is higher than the lateral arch. It is more pliant due to the presence of talocalcaneonavicular and subtalar joints. It is more involved in the propulsion, i.e., initiating the next step during walking.

Joints

The main joints of the medial longitudinal arch are talocalcaneonavicular and subtalar joints.

Lateral Longitudinal Arch

The lateral longitudinal arch is formed by the calcaneum cuboid and lateral two metatarsals. It is characteristically low and almost touches the ground (Fig. 31.5C). It is involved in receiving and supporting the body weight during walking and running.

Pillars

The **posterior pillar** of the lateral longitudinal arch is formed by the lateral tubercle of the calcaneum and the **anterior pillar** is formed by the heads of the lateral two metatarsals.

Summit

The summit of the lateral longitudinal arch lies at the level of articular facets on the superior surface of the calcaneum (i.e., at the level of subtalar joint).

Joints

The main joint of the lateral longitudinal arch is **calcaneocuboid joint.**

The lateral longitudinal arch being lower and less mobile than the medial longitudinal arch is adapted for transmission of weight and thrusts.

FACTORS MAINTAINING THE LONGITUDINAL ARCHES

A. Factors Maintaining the Medial Longitudinal Arch (Fig. 31.6)

Bones

The sustentaculum tali partly support the head of talus.

Ligaments

The important ligaments which help to maintain the medial longitudinal arch are: (a) plantar calcaneonavicular ligament (spring ligament) which provides dynamic support to the head of talus, (b) interosseous ligaments connecting the adjacent bones, and (c) interosseous talocalcanean ligament, connecting these bones. These ligaments act as **intersegmental ties.**

Muscles, tendons, and aponeurosis

1. **Acting as slings (i.e., suspending arch from above):** The tendon of **tibialis posterior** lying underneath the spring ligament provides dynamic supports to the head of talus and suspends the arch from above. In this endeavor, it is aptly supported by the tendons of flexor hallucis longus.

   The **flexor hallucis longus** is the bulkiest and strongest muscle to support the medial longitudinal arch. This muscle has three functions with respect to the medial longitudinal arch:

   (a) It stretches the arch like the string of a bow.

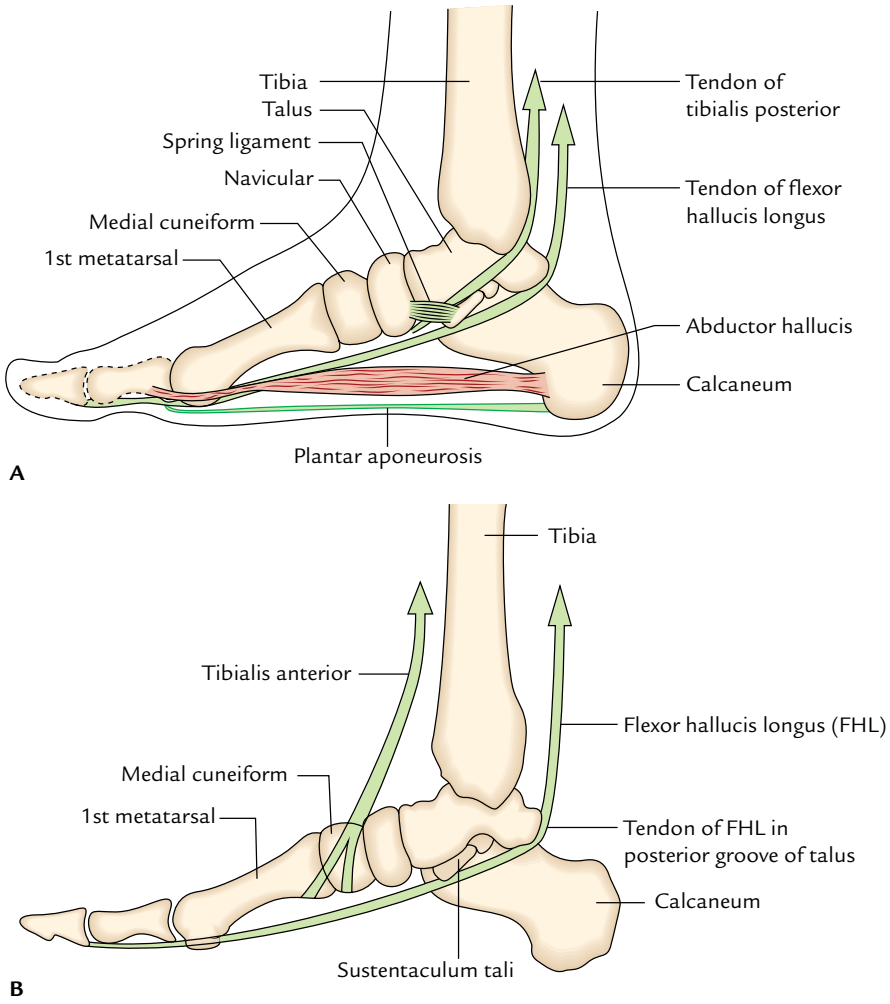


Fig. 31.6 Factors maintaining the medial longitudinal arch: A, supports of the head of talus; B, slings.

(b) It supports the calcaneus by passing underneath the sustentaculum tali.
(c) It supports the talus by passing along its posterior groove.
The tendon of **tibialis anterior** also exerts a sling action.

2. **Acting as tie beams (i.e., structures which prevent separation of the pillars):** The medial part of the plantar aponeurosis and abductor hallucis assisted by the flexor hallucis brevis act as tie beam to maintain the height of the medial longitudinal arch.

B. Factors Maintaining the Lateral Longitudinal Arch (Fig. 31.7)

Bones

The proper shaping of the distal end of calcaneus and proximal end of cuboid. The cuboid is the **keystone of longitudinal arch**.

Ligaments

The important ligaments which help to maintain the lateral longitudinal arch are as follows:

1. **Short plantar ligament:** The short plantar ligament is broad and thick. It lies deep to the long plantar ligament and supports the calcaneocuboid joint from below.
2. **Long plantar ligament:** The long plantar ligament is quite long and supports the joints between the calcaneum, cuboid, and related metatarsals.

These ligaments act as **intersegmental ties**.

Muscles, tendons, and aponeurosis

1. **Acting as tie beams:** The lateral part of the **plantar aponeurosis** and the intrinsic muscles of the little toe (e.g., lateral part of the flexor digitorum brevis, abductor digiti minimi brevis, and flexor digiti minimi brevis) function as **tie beams** of this arch.

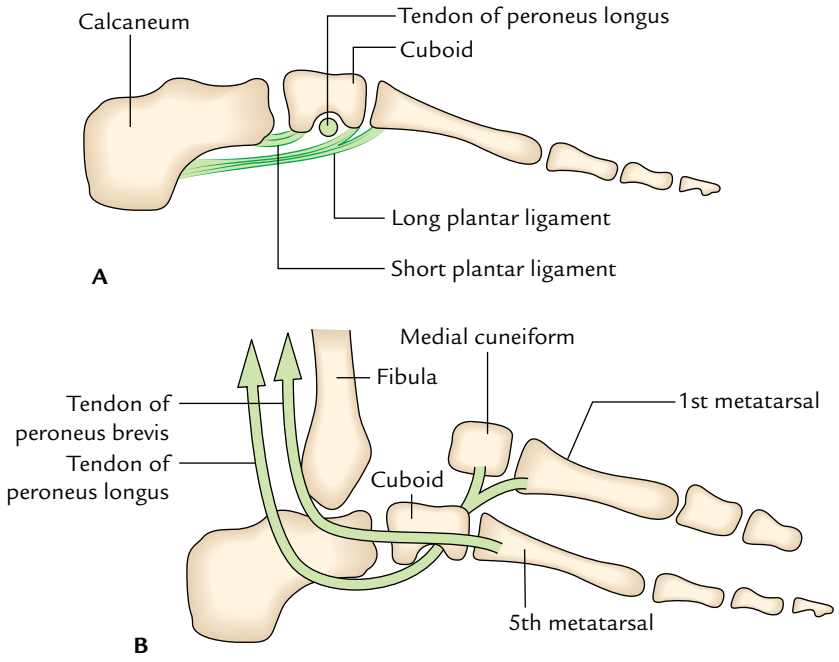


Fig. 31.7 Factors maintaining the lateral longitudinal arch: **A**, short and long plantar ligaments; **B**, tendons of the peroneus brevis and peroneus longus.

Table 31.1 Differences between the medial and lateral longitudinal arches

Medial longitudinal arch	Lateral longitudinal arch
• Formed by more bones and more joints	• Formed by less bones and less joints
• Characteristic feature is **resiliency**	• Characteristic feature is **rigidity**
• Higher and more mobile	• Lower and less mobile
• Involved in propulsion during locomotion (i.e., initiating the next step during walking)	• Involved in receiving and supporting the body weight
• Summit is formed by the talus	• Summit is formed by the calcaneum
• Main joint is talocalcaneonavicular joint (the most vulnerable part of the arch)	• Main joint is calcaneocuboid (the most vulnerable part of the arch)

2. **Acting as slings:** The **tendons of peroneus brevis** and **peroneus tertius**, which are inserted on the base of the fifth metatarsal, act as weak **slings** from above.

The **tendon of peroneus longus**, which grooves the plantar aspect of cuboid and courses transversely across the sole to be inserted on the base of first metatarsal and adjoining part of medial cuneiform, supports the cuboid bone from above through its **pulley-like action.**

The important differences between the medial and lateral longitudinal arches are given in Table 31.1.

TRANSVERSE ARCHES

Anterior Transverse Arch

The heads of the metatarsals form the **anterior transverse arch.** It is a complete arch because during standing position the heads of first and fifth metatarsals come into contact to the ground and form the two ends of the arch.

Posterior Transverse Arch

The posterior transverse arch is formed by greater parts of the tarsus and metatarsus. It is an incomplete arch because only its lateral end comes into contact with the ground during standing position. It forms only half of the dome in one foot. The complete dome is formed when the two feet are brought together (Fig. 31.8).

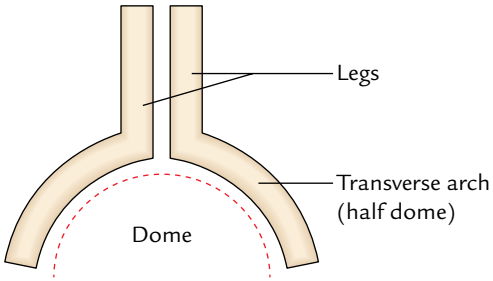


Fig. 31.8 Half dome of the transverse arch is completed when the two feet are kept together.

Factors Maintaining the Transverse Arches

Bones

Most of the tarsal and metatarsal bones have larger dorsal and smaller plantar surfaces (i.e., wedge-shaped), which help to form and maintain the concavity on the plantar aspect of the foot skeleton.

Ligaments

These are small ligaments, which bind together the cuneiform bones and metatarsals. Superficial and deep transverse metatarsal ligaments at the heads of metatarsals function as intersegmental ties to maintain the shallow arch at the heads of metatarsals.

Muscles and tendons

1. **Acting as tie beams:** The tendons of peroneus longus and tibialis posterior support the transverse arch as tie beam.
2. **Acting as slings:** The peroneus tertius and peroneus brevis on the lateral side and tibialis anterior on the medial side support the transverse arch as slings.
3. **Acting as intersegmental ties:** The dorsal interossei act as intersegmental ties.

FUNCTIONS OF THE ARCHES

The functions of the arches of the foot are as follows:

1. Distribute the body weight to the weight-bearing points of the sole (e.g., heel; balls of the toes, mainly those of first and fifth toes and lateral border of the sole).
2. Act as shock absorber during jumping by their spring-like action.
3. The medial longitudinal arch provides a propulsive force during locomotion.
4. The lateral longitudinal arch functions as a static organ of support and weight transmission.
5. The concavity of the arches protects the nerves and vessels of the sole.

Clinical correlation

- **Flat foot (pes planus):** The flat foot is the commonest of all foot problems. It occurs due to the collapse of medial longitudinal arch. During long periods of standing the plantar aponeurosis and spring ligament are overstretched. As a result, the support of the head of talus is lost and is pushed downward between the calcaneus and the navicular bones. This leads to flattening of the medial longitudinal arch with lateral deviation of the foot.

  The effects of the flat foot are:

  (a) The person usually has clumsy shuffling gait due to the loss of spring in the foot.

  (b) Makes the foot more liable to trauma due to loss of the shock absorbing function.

  (c) The compression of the nerves and vessels of the sole is due to the loss of concavity of the sole.

  The compression of the communication between the medial and lateral plantar nerves leads to neuralgic pain in the forefoot (**metatarsalgia**).
- **High arched foot (pes cavus):** The exaggeration of the longitudinal arch of the foot causes *pes cavus*. This usually occurs because of a contracture (plantar flexion) at the transverse tarsal joint. When the patient walks with a high arched foot there is dorsiflexion of the metatarsophalangeal joints and the plantar flexion of the interphalangeal joints of the toes.
- **Club foot/Talipes:** (Latin *talipes* 5 clubfoot). The club foot may be congenital or acquired. There are five types of clubfoot as under:

  (a) **Talipes equinus (horse like):** In this condition, the foot is plantar flexed and person walks on the toes with heel raised.

  (b) **Talipes calcaneus:** In this condition, the person walks on the heel with forefoot raised.

  (c) **Talipes varus:** In this condition, the foot is inverted and adducted. The person walks on the outer border of the foot.

  (d) **Talipes valgus:** In this condition, the foot is everted and abducted. The person walks on the inner border of his foot.

  (e) **Talipes equinovarus:** It is the commonest deformity of the foot. In this condition, the foot is inverted, adducted, and plantar flexed.
- **Hallux valgus:** In this condition, the big toe is deviated laterally at the metatarsophalangeal joint. It usually occurs due to constant wearing of pointed shoes with high heel. The head of the first metatarsal bone becomes prominent and rubs on the shoe. This leads to the formation of protective **adventitious bursa** called **bunion** on the medial side of the big toe.
- **Hammer toe:** It is a deformity of the toe in which metatarsophalangeal and distal interphalangeal joints are hyper-extended but the proximal interphalangeal joint is acutely flexed. This deformity usually affects the 2nd and 3rd toes.

Golden Facts to Remember

► Most important factors in maintaining the transverse arch of the foot	Tendons of peroneus longus and tibialis posterior
► Primary weight bearing point of the foot when standing	Posteroinferior tuberosity of the calcaneum
► Most vulnerable part of the medial longitudinal arch	Talocalcaneonavicular joint
► Most vulnerable part of the lateral longitudinal arch	Calcaneocuboid joint
► Keystone of medial longitudinal arch	Talus
► Keystone of lateral longitudinal arch	Cuboid
► Most pronounced site of the transverse arch	Middle of the metatarsals
► Commonest deformity of the foot	Talipes equinovarus
► Rocker-bottom foot	Plantar concavity is replaced by plantar convexity (e.g., in trisomy 18, or in Edward syndrome)

Clinical Case Study



A 20-year-old healthy individual went for recruitment in the Army. He fulfilled all the physical requirements except that he was having flat feet. Consequently he was rejected.

Questions

1. What is flat foot (pes planus)?
2. Give the anatomical basis of the collapse of the medial longitudinal arch.
3. What is the keystone of the medial longitudinal arch?
4. What are the effects of flat foot?
5. Define inversion and eversion, and name the joints where these movements take place.

Answers

1. It is a condition in which there is a collapse of the medial longitudinal arch.
2. (a) The loss of support to the head of talus, for details see page 437.
   (b) Overstretching of ligaments supporting the medial longitudinal arch (e.g., spring ligament, short and long plantar ligaments, and plantar aponeurosis).
3. Talus.
4. See page 437.
5. (a) *Inversion* is the movement in which the medial margin of the foot is raised and the sole faces medially.
   Eversion is the movement in which the lateral margin of the foot is raised and the sole faces laterally.
   (b) Movements of inversion and eversion take place at subtalar, talocalcaneonavicular, and midtarsal joints.

CHAPTER

32 Joints of the Lower Limb

The joints of the lower limb include:

1. Hip joint.
2. Knee joint.
3. Tibiofibular joints.
4. Ankle joint.
5. Foot joints.

The hip joint is already described in detail in Chapter 26. The remaining joints are described in this chapter.

KNEE JOINT

The knee joint is the largest and most complicated joint in the body. It is the major weight-bearing joint in the body. It is prone to undergo degenerative changes with advanced age leading to osteoarthritis. Hence it should be studied thoroughly.

TYPE

It is a synovial joint of **modified hinge** variety. It is not a typical hinge joint because it undergoes some degree of automatic (conjunct) rotation during flexion and extension of the knee.

Actually, it is a compound joint consisting of three articulations: right and left *condylar joints* between the condyles of the femur and tibia, and one *saddle joint* between the femur and patella (Figs 32.1 and 32.2).

ARTICULAR SURFACES (Figs 32.1–32.3)

The articular surfaces of the knee joints are:

1. Articular surfaces of medial and lateral condyles of the femur.
2. Trochlear surface of the femur.
3. Articular surface of the patella.
4. Articular surfaces of medial and lateral condyles of the tibia.

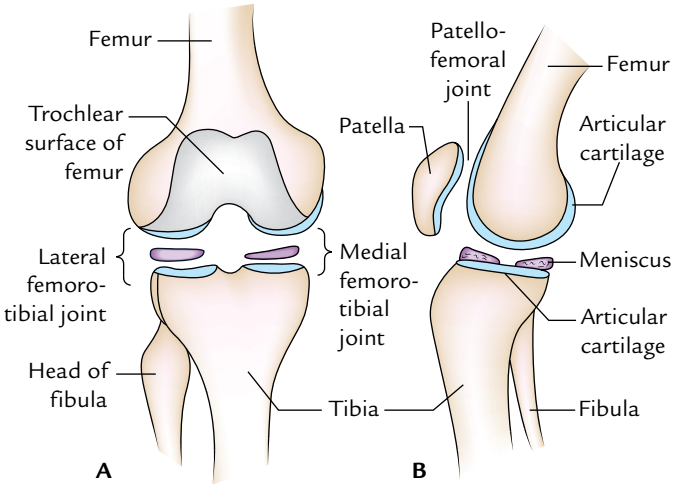


Fig. 32.1 Three primitive joints included in the knee joint: A, condylar type of medial and lateral femorotibial joints; B, saddle type of patellofemoral joint.

The articular surfaces of the knee joints are described in detail in the following text:

1. *Articular surfaces of medial and lateral condyles of the femur occupy the anterior, inferior, and posterior surfaces of these condyles respectively.* They are convex anteroposteriorly and from side to side. The medial condylar surface is longer anteroposteriorly and narrower or mediolaterally than that of lateral condyle. Anteriorly they are continuous with each other through trochlear surface of the femur but posteriorly they are separated from each other by an intercondylar notch.
2. *Trochlear surface of the femur* is located on the anterior aspect of the lower end of the femur. It articulates with the posterior surface of the patella. It is pulley-shaped, consisting of medial and lateral sloping surfaces meeting with each other in a median vertical groove. The lateral sloping surface is longer than that of medial.

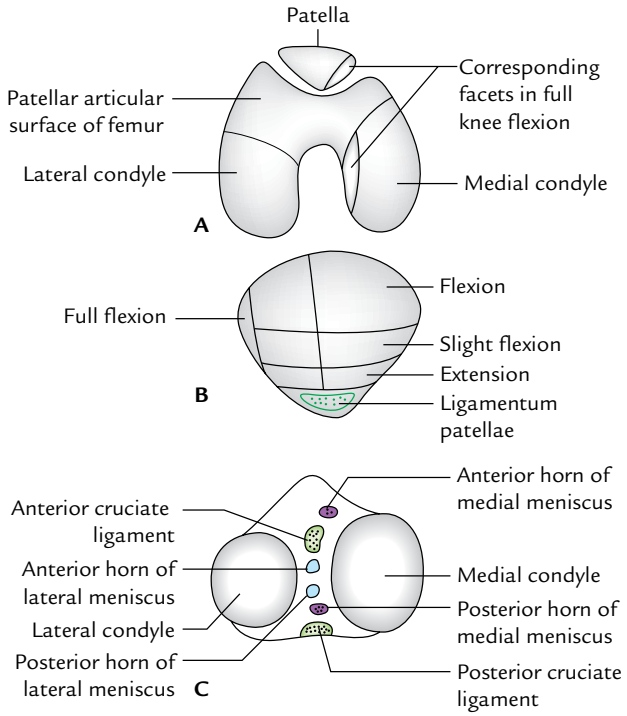


Fig. 32.2 Articular surfaces of the knee joint: **A,** inferior aspect of the patella and lower end of the femur; **B,** posterior aspect of the patella; **C,** superior aspect of the tibia.

3. *Articular surface of the patella is* on the posterior aspect of patella and articulates with the trochlear surface of the femur. It has a larger lateral area and a smaller medial area.

   Near the medial margin of patella, there is a narrow semilunar strip which comes in contact with a similar strip on the medial condyle of femur in full flexion.
4. Articular surfaces of the medial and lateral condyles of tibia are on the upper surfaces of these condyles. They are separated from each other by a rough intercondylar area.
   - The articular surface on medial tibial condyle is oval and larger. Its anteroposterior diameter is more than the transverse diameter.
   - The articular surface on the lateral tibial condyle is circular.

The articular surfaces on the upper surfaces of the medial and lateral condyles of the tibia are slightly concave centrally and flat at the periphery where they are covered by the corresponding menisci.

STABILITY OF THE KNEE JOINT

Structurally, the knee joint is relatively weak because of the incongruence of its articular surfaces. The tibial condyles are

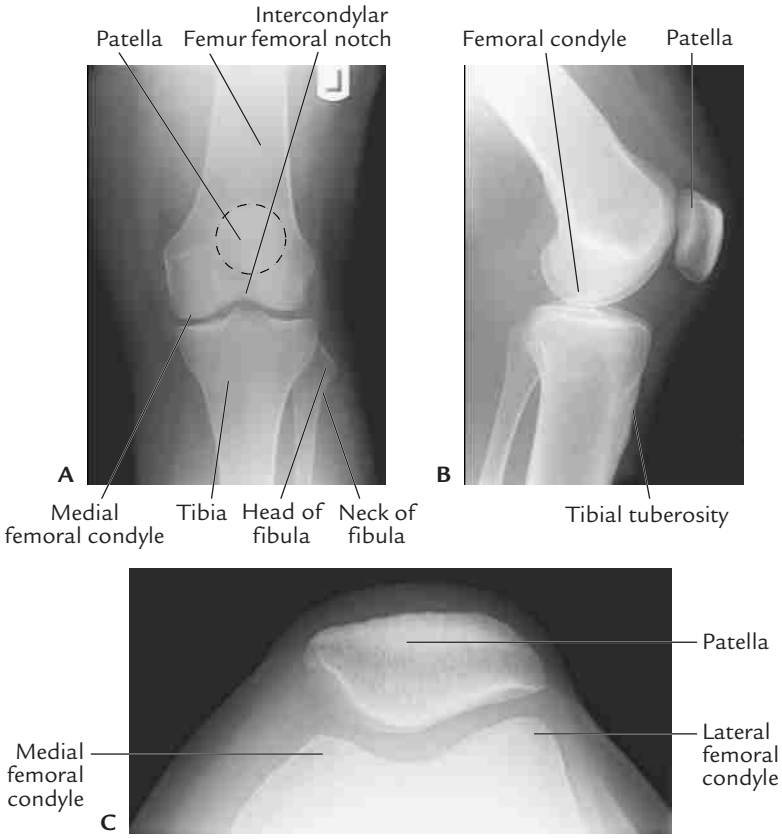


Fig. 32.3 Radiograph of the knee joint: **A,** AP view; **B,** lateral view; **C,** sky view. (*Source:* Fig. 8.2, Page 220, *Integrated Anatomy,* Heylings, David JA; Spence, Roy JA; Kelly, Barry E. Oxford: Churchill Livingstone, 2007, All rights reserved.)

too small and shallow to hold the large convex femoral condyles. The femoropatellar articulation is also not quite stable because of their shallow articular surfaces and due to an outward angulation between the long axes of the femur and tibia.

Factors Maintaining the Stability of the Knee Joint

The stability of the knee joint is maintained by the following factors:

1. Strength and actions of the surrounding muscles and tendons.
2. Medial and lateral collateral ligaments maintain side-to-side stability.
3. Cruciate ligaments maintain anteroposterior stability.
4. Iliotibial tract helps in stabilizing a partly flexed knee.

LIGAMENTS

The important ligaments of the knee joint are as follows (Fig. 32.4):

1. Capsular ligament.
2. Ligamentum patellae.
3. Tibial and fibular collateral ligaments.
4. Anterior and posterior cruciate ligaments.
5. Medial and lateral menisci.

The other secondary ligaments of the knee joint are as follows:

1. Oblique popliteal ligament.
2. Arcuate popliteal ligament.
3. Transverse ligament.
4. Coronary ligaments.

Capsular Ligament

It is a thin **fibrous sac** (Fig. 32.4) which surrounds the joint. It is deficient anteriorly, where it is replaced by the patella, quadriceps femoris, medial and lateral patellar retinacula, and ligamentum patellae.

Femoral attachment

It is attached about 1/2 to 1 cm beyond the articular margins with the following three special features.

Anteriorly it is deficient in the middle where it is pierced by the suprapatellar bursa.

Posteriorly it is attached to the intercondylar line.

Laterally it encloses the origin of popliteus.

Tibial attachment

It is attached about 1/2 to 1 cm beyond the articular margins with the following three special features.

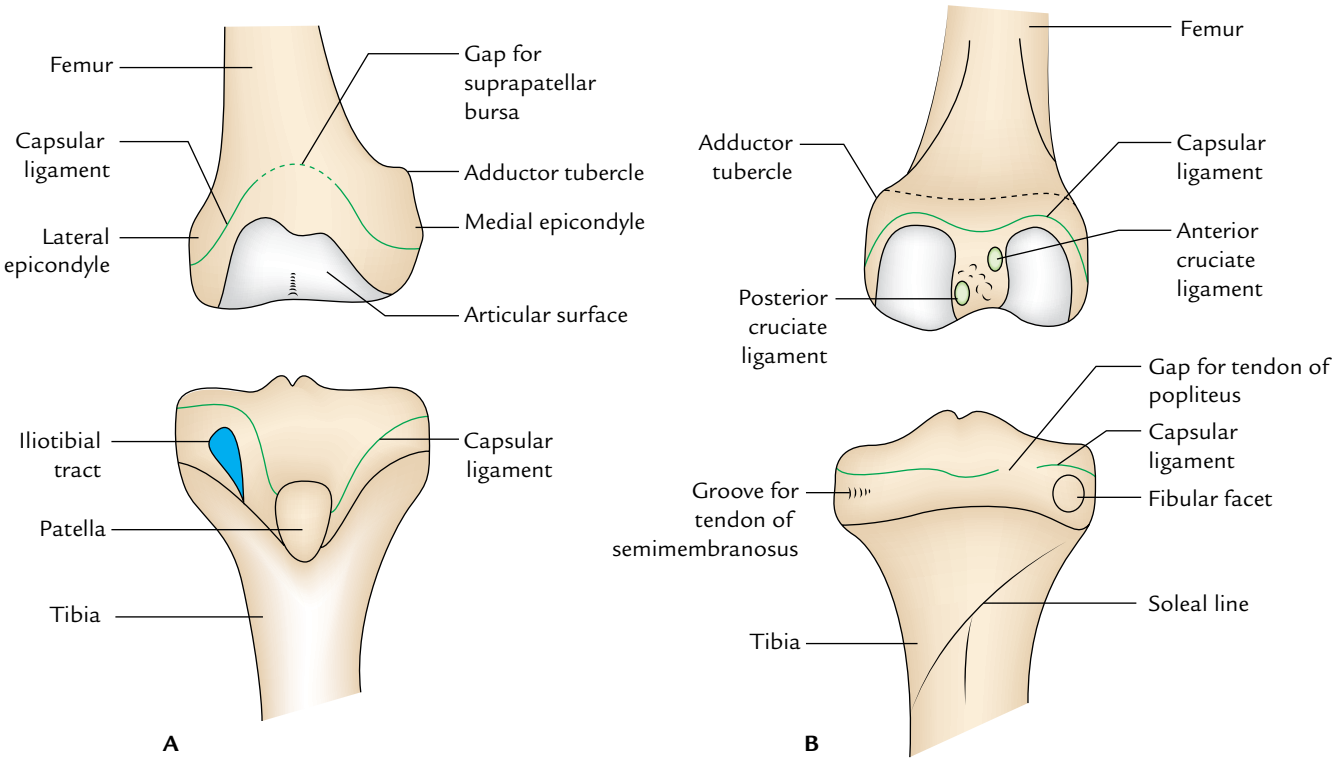


Fig. 32.4 Attachment of capsular ligament of the knee joint: A, anterior aspects of the femur and tibia; B, posterior aspects of the femur and tibia. The arrows indicate gaps in the capsule.

Anteriorly it descends along the margins of the condyles to tibial tuberosity, *where it is deficient* for the attachment of the ligamentum patellae.

Posteriorly it presents a gap behind the lateral condyle for the passage of the tendon of popliteus muscle.

Synovial Membrane

The synovial membrane (Fig. 32.5) lines the inner aspect of the fibrous capsule and the portions of the bones enclosed within it but ceases at the periphery of the articular cartilages the medial and lateral menisci.

In front above the patella, it is prolonged as *suprapatellar bursa,* and below the patella, it covers the deep surface of the infrapatellar pad of fat, which separates it from the ligamentum patellae. The apex of suprapatellar bursa is attached to the *articularis genu muscle.*

A median triangular fold of the synovial membrane called **infrapatellar fold** extends upward and backward from the fat pad to the intercondylar fossa of the femur.

The lateral margins of the infrapatellar synovial fold are free and form the *alar folds,* which contain the fibrofatty tissue.

From the posterior aspect of the fibrous capsule, the synovial membrane projects forward in the intercondylar region as cul-de-sac to envelope the sides of both cruciate ligaments and in front of the anterior cruciate ligament.

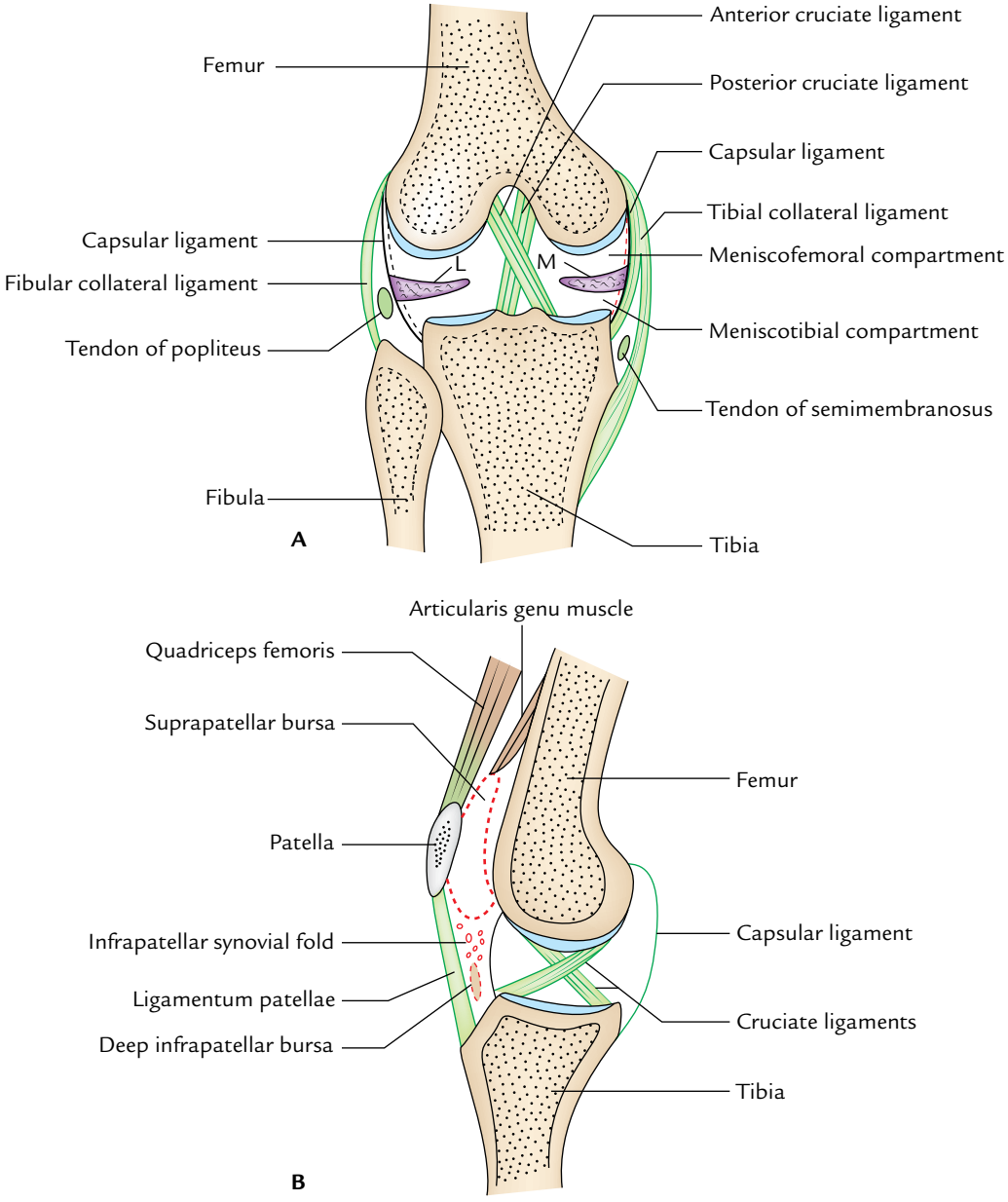


Fig. 32.5 Knee joint: **A,** coronal section; **B,** sagittal section (L = lateral meniscus, M = medial meniscus).

Ligamentum Patellae

It is actually the tendon of insertion of quadriceps femoris, which extends from the apex of the patella to the upper part of the tibial tuberosity.

Ligamentum patellae is about 7.5 cm long and 2.5 cm broad. It is *attached above to the margins and rough posterior surface of the apex of patella, and below it is attached to the smooth, upper part of tibial tuberosity*. It is related to the subcutaneous and deep infrapatellar bursae, and infrapatellar pad of fat.

Tibial (Medial) Collateral Ligament (Fig. 32.6A)

It is a strong, long (about 10 cm), thick, and flat band of fibrous tissue. It consists of **superficial and deep parts**. Both parts are attached above to the medial epicondyle of the femur just below the adductor tubercle.

The **superficial part** of the ligament is long and attached below to the upper part of the medial border and adjoining posterior part of the medial surface of the tibia. It covers the inferior medial genicular nerve and vessels, and anterior part of the tendon of semimembranosus. The tendons of sartorius, gracilis, and semitendinosus cross its lower part superficially.

The deep part of the ligament is short and blends with the fibrous capsule, and with the peripheral margin of the medial meniscus. It is attached below to the medial condyle of the tibia above the groove for the tendon of semimembranosus.

Fibular (Lateral) Collateral Ligament (Fig. 32.6B)

It is short (about 5 cm long) and cord-like ligament. Above it is attached to the lateral epicondyle of the femur just above the popliteal groove. Below it is embraced by the tendon of biceps femoris and attached to the head of fibula in front of its apex. Its deep surface is not adherent to the fibrous capsule. It is separated from the capsule and lateral meniscus by the tendon of popliteus. Its lower part is separated from the capsule by the *inferior lateral genicular nerve and vessels*.

N.B.

- Morphologically, the medial collateral ligament represents the degenerated tendon of insertion of the ischial head of the adductor magnus, and fibular ligament represents the degenerated tendon of the peroneus longus.
- The medial collateral ligament is much thicker and stronger than the lateral collateral ligament because the knee joint gaps more on the medial side, hence the need for stronger stabilization on that side.

Cruciate Ligaments

These are two thick, strong fibrous bands, which act as direct bonds of union between the femur and tibia. They are present inside the knee joint. They represent the collateral ligaments of the primitive femorotibial joints. They maintain anteroposterior stability of the knee joint. They are named anterior and posterior according to their site of attachment to the tibia. The cruciate ligaments are intracapsular but extrasynovial.

The ligaments cross each other like the letter "X" hence the name cruciate.

Anterior Cruciate Ligament

The anterior cruciate ligament is attached below to the anterior part of the intercondylar area of the tibia. It runs upward, backward, and laterally and is attached to the

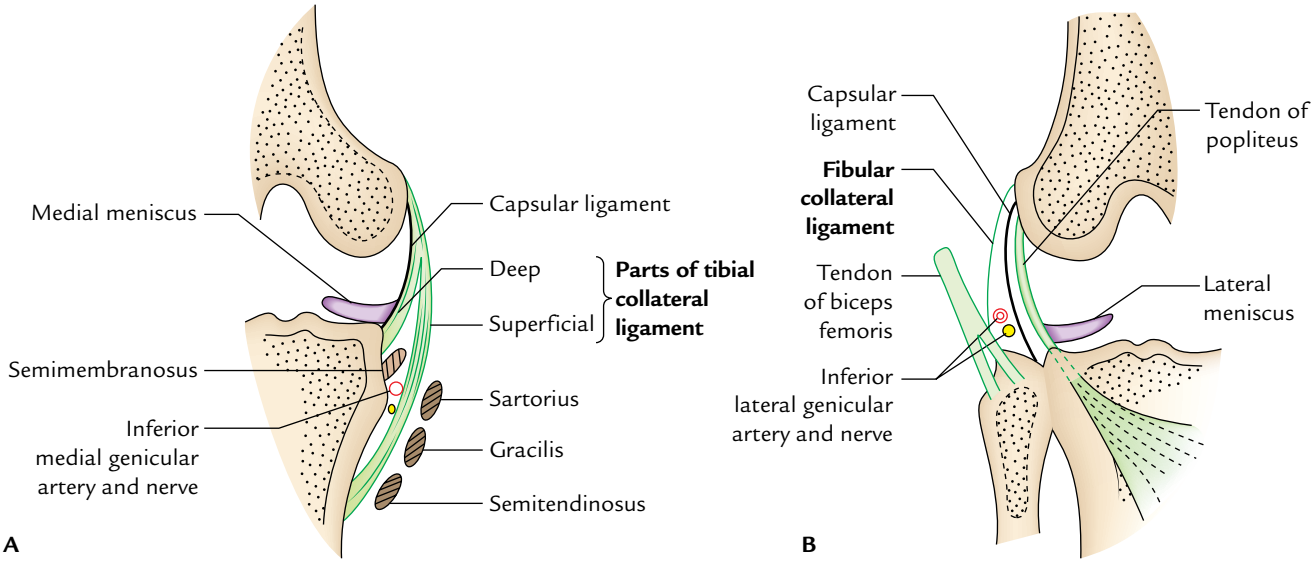


Fig. 32.6 Attachments and relations of collateral ligaments of the knee joint: A, tibial (medial) collateral ligament; B, fibular (lateral) collateral ligament.

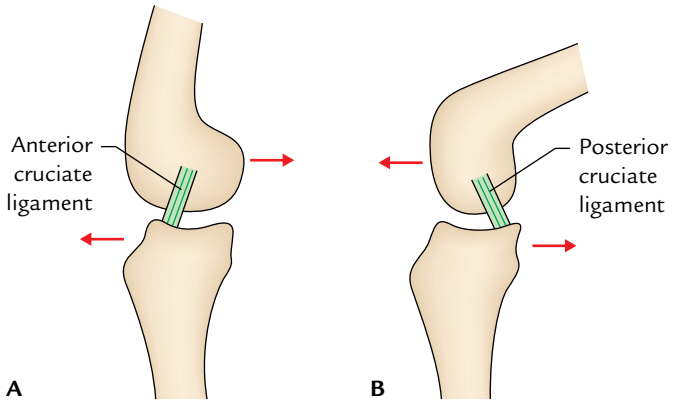


Fig. 32.7 Anterior and posterior cruciate ligaments. Note anterior cruciate ligament is taut during knee extension and posterior cruciate ligament during knee flexion.

posterior part of the medial surface of the lateral condyle of femur. *It is taut during extension of the knee and prevents the posterior dislocation of femur on tibia and anterior dislocation of tibia on femur* (Fig. 32.7A).

Posterior Cruciate Ligament

The posterior cruciate ligament is attached below to the posterior part of the intercondylar area of the tibia. It runs upward, forward, and medially and is attached to the anterior part of the lateral surface of the medial condyle of the femur. *It is taut during flexion of the knee and prevents the anterior dislocation of femur on tibia and posterior dislocation of tibia on femur* (Fig. 32.7B).

Medial and Lateral Menisci (Semilunar Cartilages, Fig. 32.8)

These two crescent-shaped intra-articular discs are made up of fibrocartilage. They have thick peripheral border and thin inner border. They deepen the articular surfaces of the condyles of tibia, and partially divide the joint cavity into the upper (meniscofemoral) and lower (meniscotibial) compartments. Flexion and extension of the knee take place in the upper compartment, whereas the rotation of the knee occurs in the lower compartment.

Each meniscus has: (a) *two ends* (anterior and posterior), which are attached to tibia; *two borders*—the thick outer border is fixed to the fibrous capsule and the thin inner border which is free; and (b) *two thin surfaces*—the upper surface is concave for the femur; the lower surface is flat for the peripheral 2/3rd of tibial condyles.

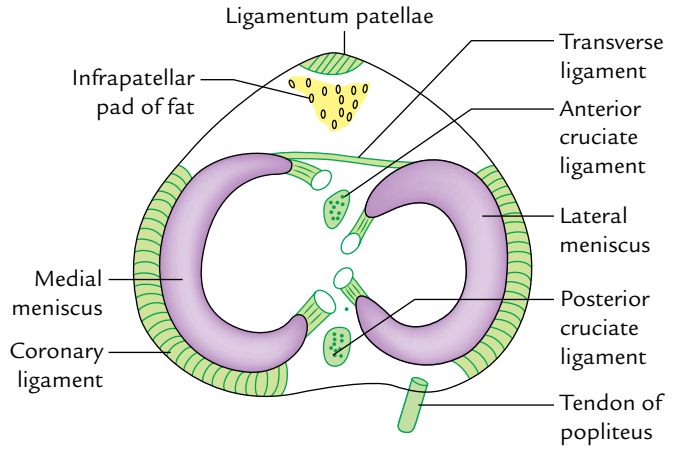


Fig. 32.8 Medial and lateral menisci of the knee joint.

Medial Meniscus

It is nearly semilunar in shape and wider behind than in front. It presents anterior posterior ends or horns, which are attached to the intercondylar area of the tibia. It is adherent to the deep part of tibial collateral ligament. It is firmly attached to the tibial plateau by coronary ligaments.

Lateral Meniscus

It is nearly circular in shape with more or less a uniform width. It also presents anterior and posterior horns, which are also attached to the intercondylar area of the tibia.

The posterior horn of the lateral meniscus is attached to the medial condyle of femur by the anterior and posterior **meniscofemoral ligaments**. The anterior meniscofemoral ligament (**ligament of Humphrey**) passes in front of the posterior cruciate ligament whereas the posterior meniscofemoral ligament (**ligament of Wrisberg**) passes behind the posterior cruciate ligament.

These ligaments play an important role in regulating the movements of lateral meniscus during the extension of the knee joint.

The lateral meniscus is attached to the medial part of the tendon of popliteus and thus the mobility of its posterior horn is controlled by the popliteus and two meniscofemoral ligaments.

The differences between medial and lateral menisci are given in Table 32.1.

Table 32.1 Differences between the medial and lateral menisci

Medial meniscus	Lateral meniscus
• C-shaped/semilunar in shape	• "O"-shaped/circular in shape
• Attached to the medial collateral ligament	• Attached to the tendon popliteus muscle
• More prone to injury	• Less prone to injury

Functions of the menisci

These are as follows:

1. The menisci increase the concavities of the tibial condyles for better congruence with the femoral condyles.
2. They act as swabs to lubricate the joint cavity.
3. They act as shock absorber to protect the articular cartilage during weight transmission.
4. They adapt to the varying curvatures of the different parts of the femoral condyles.

Clinical correlation

Meniscal tears: The *injuries to menisci* are commonly caused by the twisting strains in a slightly flexed knee, as in kicking a football. The meniscus may get separated from the capsule, or it may be torn longitudinally (**bucket-handle tear**) or transversely.

The medial meniscus is more prone to injury than the lateral because of its firm fixity to tibial collateral ligament, and greater excursion during the rotatory movements. The lateral meniscus is protected by the popliteus muscle because its medial fibres pulls the posterior horn of meniscus backward, so that it is not crushed between the articular surfaces.

Pain on the medial rotation of tibia on the femur indicates injury of the medial meniscus; while pain on the lateral rotation of tibia on the femur indicates injury of the lateral meniscus.

Oblique Popliteal Ligament

It is an expansion from the tendon of semimembranosus muscle. It runs upward and laterally superficial to the capsule to be attached to the intercondylar line of the femur. It blends with and strengthens the capsule of knee joint posteriorly. It is intimately related to the popliteal artery and pierced by: (a) middle genicular nerve, (b) middle genicular vessels, and (c) posterior division of the obturator nerve.

Arcuate Popliteal Ligament

It is a Y-shaped fibrous band. The stem of the band is attached to the styloid process of the fibula. The large posterior or limb of the band arches over the tendon of popliteus and is attached to the posterior border of the intercondylar area of the tibia. The small anterior limb (often deficient) passes deep to the fibular collateral ligament and is attached to the lateral condyle of the femur.

Transverse Ligament

It extends transversely and connects the anterior ends of the medial and lateral menisci. It is present only in about 40% of the individuals.

Coronary Ligaments

They are parts of the fibrous capsule, which provide attachment to the peripheral margins of the medial and lateral menisci to the tibia.

BURSAE AROUND THE KNEE (Fig. 32.9)

There are about 12 bursae around the knee, four anterior, three lateral, three medial, and two posterior.

Anterior Bursae

These are:

1. **Subcutaneous prepatellar bursa** (**bursa of housemaid's knee**). It lies deep to the skin in front of lower half of the

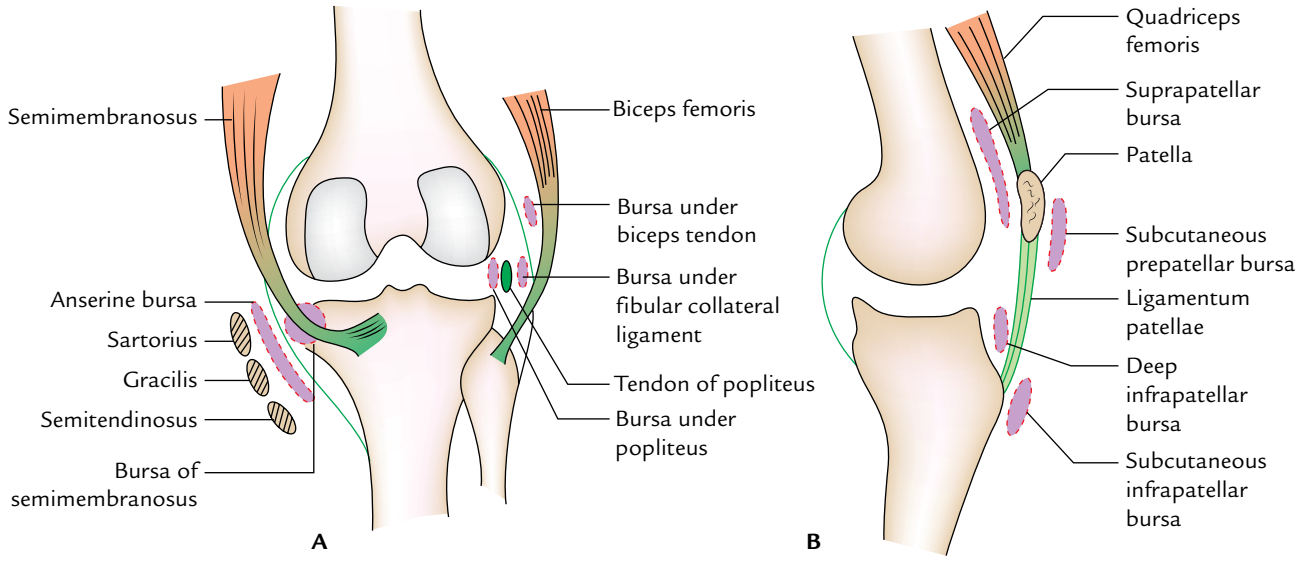


Fig. 32.9 Bursae around the knee joint: A, bursae on the medial and lateral aspects of the knee; B, bursae on the front of the knee.

patella and upper half of the ligamentum patellae and tibial tuberosity.
2. **Subcutaneous infrapatellar bursa** between the skin and smooth lower part of the tibial tuberosity.
3. **Deep infrapatellar bursa**, between ligamentum patellae and tibial tuberosity.
4. **Suprapatellar bursa** between the anterior surface of lower part of the femur and deep surface of the quadriceps femoris.

Lateral Bursae

These are:

1. The bursa between the fibular collateral ligament and tendon of biceps femoris.
2. The bursa between the fibular collateral ligament and tendon of popliteus.
3. The bursa between the tendon of popliteus and lateral condyle of femur. This bursa is really a synovial tube around the tendon of popliteus; hence it communicates with the joint cavity.

Medial Bursae

These are:

1. The bursa, which separates the tendons of sartorius, gracilis, and semitendinosus from each other and from the tibial collateral ligament (**bursa anserine**).
2. The bursa between the tendon of semimembranosus and medial collateral ligament.
3. The bursa between the tendon of semimembranosus and medial condyle of the tibia. It may communicate with the knee joint.

Posterior Bursae

These are:

1. The bursa between the lateral head of gastrocnemius and capsule of the joint.
2. The bursa between the medial head of gastrocnemius and capsule of the joint (**Brodie's bursa**).

Clinical correlation

- **Housemaid's knee:** It is the inflammation of the prepatellar bursa (*prepatellar bursitis*). It occurs due to friction of bursa against the patella when it comes in contact with the ground during scrubbing of the floor by the housemaid. Such a bursa may get very large and drop by its weight to much below to its original position.
- **Clergyman's knee:** It is the inflammation of subcutaneous infrapatellar bursa (*subcutaneous infrapatellar bursitis*). It occurs due to the friction of bursa against the tibial tuberosity due to kneeling (e.g., kneeling during prayer by Christian priests/clergymen, roofers, and floor tilers.
- **Baker's cyst:** The chronic inflammation of bursa deep to semimembranosus may present as cystic swelling in the medial part of the popliteal fossa called *Baker's cyst.*

RELATIONS OF THE KNEE JOINT (Fig. 32.10)

They are as follows:

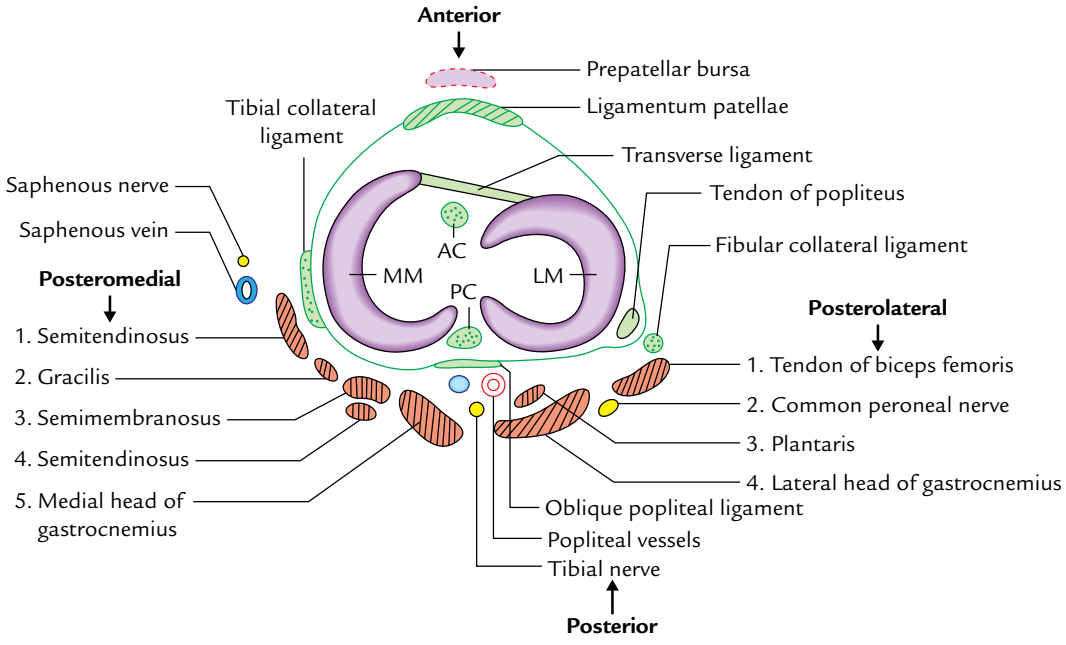


Fig. 32.10 Relations of the knee joint (transverse section of right knee joint) (MM = medial meniscus, LM = lateral meniscus, AC = anterior cruciate ligament, PC = posterior cruciate ligament).

Anteriorly: Tendon of the quadriceps femoris, patella, ligamentum patellae, patellar plexus of the nerves, and prepatellar synovial bursa.
Anteromedially: Medial patellar retinaculum.
Anterolaterally: Lateral patellar retinaculum and iliotibial tract.
Posteriorly: Popliteal vessels, tibial nerve, and oblique popliteal ligament.
Posterolaterally: In the upper part, tendon of biceps femoris and common peroneal nerve; in the lower part, lateral head of gastrocnemius and plantaris.
Posteromedially: In the upper part, sartorius, gracilis, semimembranosus, and semitendinosus.

In the lower part, medial head of gastrocnemius and popliteus.

BLOOD SUPPLY

The knee joint is richly supplied by the blood through the arterial anastomosis around the knee, which is formed by: (a) five genicular branches of popliteal artery, (b) descending genicular branch of femoral artery, (c) descending branch of the lateral circumflex femoral artery, (d) two recurrent branches of the anterior tibial artery, and (e) circumflex fibular branch of the posterior tibial artery.

NERVE SUPPLY

The knee joint has rich nerve supply by:

(a) *Femoral nerve* through its branches to vasti, especially to vastus medialis.
(b) *Tibial and common peroneal nerves* through their genicular branches.
(c) *Obturator nerve* through its posterior division.

MOVEMENTS

The following movements occur at the knee joints:

1. Flexion
2. Extension

} Main/active movements.

3. Medial rotation
4. Lateral rotation

} Conjunct movements.

Flexion and Extension

These movements occur in the upper meniscofemoral compartment of the joint, i.e., above the menisci.

In flexion the angle between the posterior thigh and leg is decreased whereas in extension the angle between the posterior thigh and leg is increased (i.e., return from the flexion back to the anatomical position).

The movements of the flexion and extension at knee differ from ordinary hinge movements in following two ways:

(a) The *transverse axis* around which these movements takes place is not fixed (e.g., during extension the axis moves forward and upward and during flexion it moves backward and downward).
(b) During these movements, there is an **automatic (conjunct) rotation** of the knee (viz., medial rotation of the femur during last 30° of extension; and lateral rotation of the femur during initial stages of the flexion). When foot is off the ground, tibia rotates instead of femur but in the opposite direction.

Medial and Lateral Rotation

These movements take place in the lower meniscotibial compartment of the joint, i.e., below the menisci. These movements occur around the vertical axis.

The medial and lateral rotations usually occur with flexion and extension (**conjunct rotations**) but may occur independently if the knee is flexed (**adjunct rotations**).

The conjunct rotations play an important role in locking and unlocking of the knee.

N.B. Locking and unlocking of the knee:

(a) **Locking of the knee:** When the foot is on the ground, the locking is defined as the medial rotation of femur on the tibia during the terminal phase of extension of the knee. When the knee is locked it becomes absolutely rigid and all the ligaments of the joint are taut. This is known as **"screw home mechanism".**

(b) **Unlocking of the knee:** When the foot is on the ground, the unlocking is defined as the lateral rotation of the femur on the tibia during initial phase of the flexion. The unlocking is brought about by the popliteus muscle. When the knee is unlocked, it can be further flexed by the hamstring muscles.

The locking of knee is essential for bearing load during erect posture. The locked joint must be unlocked to facilitate progress of locomotion. Hence, during locomotion locking and unlocking of the knee takes place alternatively and rhythmically.

The differences between the locking and unlocking of the knee are as summarized in the box below:

Locking of the knee joint	Unlocking of the knee joint
• Medial rotation of the femur on tibia during terminal phase of extension	• Lateral rotation of the femur on tibia during initial phase of the flexion
• It is brought about by quadriceps femoris	• It is brought about by the popliteus muscle
• Locked knee becomes absolutely rigid	• Unlocked knee can be further flexed
• All ligaments are taut	• All ligaments are relaxed

Table 32.2 Movements of the knee joint

	Muscles producing movements	
Movements	**Chief muscles**	**Accessory muscles**
Flexion	• Semimembranosus • Semitendinosus • Biceps femoris	• Popliteus (initiates flexion) • Sartorius • Gracilis • Gastrocnemius • Plantaris
Extension	Quadriceps femoris	Tensor fasciae latae
Medial rotation	• Semitendinosus • Semimembranosus • Popliteus	• Sartorius • Gracilis
Lateral rotation	Biceps femoris	• Gluteus maximus • Tensor fasciae latae

The movements of the knee joint and muscles producing them are given in Table 32.2.

N.B. The range of motion (ROM) of flexion is 130° whereas the range of motion of extension is 0–50°.

The ROM for flexion is greater when the hip joint is fixed. The ROM for extension is greater when the hip is extended and limited when the hip is flexed.

Clinical correlation

- **Osteoarthritis:** Being a weight-bearing joint, the knee joint is commonly involved in *osteoarthritis* (degenerative wear and tear of articular cartilages). The movements may be painful, limited, and produce grating. Radiographs of the knee region reveal osteophytes, i.e., peripheral lipping of the articular ends.
- **Injuries to cruciate ligaments (Fig. 32.11):** *The anterior cruciate ligament is more commonly damaged than the posterior ligament*. The anterior cruciate ligament is injured in the anterior dislocation of the tibia; whereas, the posterior ligament is injured in the posterior dislocation of the tibia. Tear of the cruciate ligaments leads to abnormal anteroposterior mobility.

  If the anterior cruciate ligament is torn the tibia is pulled excessively forward on the femur (**anterior drawer sign**) and if the posterior cruciate ligament is torn the tibia is pulled excessively backward (**posterior drawer sign**; Fig 32.11).
- **Aspiration of the knee joint:** The collections of fluid are common in the knee joint. It gives rise to swelling above and at the sides of the patella. In such cases, patellar tap often demonstrates a floating patella. Aspiration of the fluid can be done on either side of the ligamentum patellae. But the joint is usually approached from its lateral side using three bony points as landmarks for the needle insertion: (a) tibial tuberosity, (b) lateral epicondyle of the femur, and (c) apex of patella. This triangular area is also used for drug injection in treating the knee pathology.

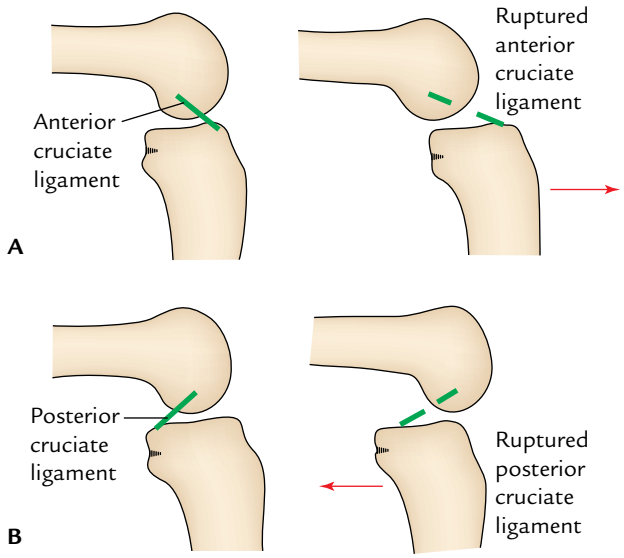


Fig. 32.11 Injury of cruciate ligaments of the knee joint: A, rupture of anterior cruciate ligament (positive anterior Drawer's sign); B, rupture of the posterior cruciate ligament (positive posterior Drawer's sign).

- **Arthroscopy of the knee joint:** It is an endoscopic examination (visualization) of the interior of the knee joint cavity with minimal disruption of the tissues. The ligament repair or replacement can also be performed by using an *arthroscope*.
- **Knee replacement:** If the knee joint is badly damaged by the osteoarthritis, an artificial joint consisting of plastic tibial component and metal femoral component is connected to the tibial and femoral bone ends after removal of the damaged areas.
- **Unhappy triad of the knee joint:** A combination of injury of the (a) tibial collateral ligament, (b) medial meniscus, and (c) anterior cruciate ligament is called "unhappy triad" of the knee joint.

TIBIOFIBULAR JOINTS

There are three joints between the tibia and fibula: (a) superior tibiofibular, (b) middle tibiofibular, and (c) inferior tibiofibular.

SUPERIOR TIBIOFIBULAR JOINT

The **superior tibiofibular joint** is a small plane type of synovial joint between the head of fibula and the lateral condyle of tibia. It may communicate with the knee joint through the popliteal bursa. It permits some gliding or rotatory movements for adjusting the lateral malleolus during movements at the ankle joint (e.g., during dorsiflexion).

MIDDLE TIBIOFIBULAR JOINT

The **middle tibiofibular joint** is a fibrous joint formed by the interosseous membrane connecting the interosseous borders of the shafts of tibia and fibula. Its fibres are directed downward and laterally. It is wide above and narrow below where it blends with the **interosseous ligament of the inferior tibiofibular joint.** A large opening above the upper free margin of interosseous membrane provides passage to the anterior tibial vessels. The interosseous membrane presents a small opening near its lower end for the passage of perforating branch of the peroneal artery.

Functions of the interosseous membrane

These are:

1. Provides additional surface for the attachment of muscles.
2. Binds tibia and fibula.
3. Resists downward movement of fibula by the powerful fibular muscles.

N.B. Most fibres of the interosseous membrane (vide supra) are directed downward and laterally except in the upper part where they are directed downward and medially.

INFERIOR TIBIOFIBULAR JOINT

The **inferior tibiofibular joint** is a *syndesmosis* variety of the fibrous joint. *It is the strongest of all the three tibiofibular joints, because the strength of ankle joint largely depends upon its integrity.*

The roughened opposed surfaces of the lower ends of tibia and fibula are connected by a very strong *interosseous ligament*, which forms the chief bond of union between the lower ends of these bones. The interosseous ligament is covered both in front and behind by the *anterior and posterior tibiofibular ligaments*, respectively. The posterior tibiofibular ligament is stronger than the anterior tibiofibular ligament. Its lower and deep portion forms the *inferior transverse tibiofibular ligament*, which is a strong thick band of yellowish elastic fibres passing transversely from the upper part of malleolar fossa to the posterior border of the articular surface of tibia. It accentuates the concavity of the tibiofibular mortise of the ankle joint.

The inferior tibiofibular joint permits slight movements, to allow the lateral malleolus to rotate laterally during dorsiflexion of the ankle.

A brief comparison of the superior, middle, and inferior tibiofibular joints is given in Table 32.3.

ANKLE JOINT (TALOCRURAL JOINT)

The ankle joint is a strong weight-bearing joint of the lower limb (Fig. 32.12).

TYPE

It is a synovial joint of hinge variety.

ARTICULAR SURFACES (Fig. 32.13)

1. **Proximal articular surface** of the ankle joint is formed by the articular facets of the:
   (a) Lower end of tibia including its medial malleolus.
   (b) Lateral malleolus.
   (c) Inferior transverse tibiofibular ligament.

These three together form a deep **tibiofibular socket** (also called **"tibiofibular mortise"**).

2. **Distal articular surface** of the ankle joint is formed by the: articular facets on the upper, medial, and lateral aspects of the body of the talus.

Table 32.3 Comparison of the superior, middle, and inferior tibiofibular joints

Superior tibiofibular joint	Middle tibiofibular joint	Inferior tibiofibular joint
Plane type of the synovial joint	Fibrous joint	Syndesmosis variety of the fibrous joint
Formed by articulation between the oval articular facet on the lateral condyle of tibia and similar facet on the head of fibula	Formed by the union of interosseous border of tibia and fibula by an interosseous membrane	Formed by the union of rough triangular surface of the lower ends of tibia and fibula by a strong interosseous, and anterior and posterior tibiofibular ligaments
Permits some gliding movements	Permits rotation of fibula during dorsiflexion of the knee	Accentuates the "*tibiofibular mortise*," and allows lateral rotation of the lateral malleolus during dorsiflexion
Innervated by the nerve to popliteus and recurrent genicular nerve	Innervated by the nerve to popliteus	Innervated by the deep peroneal, tibial, and saphenous nerves

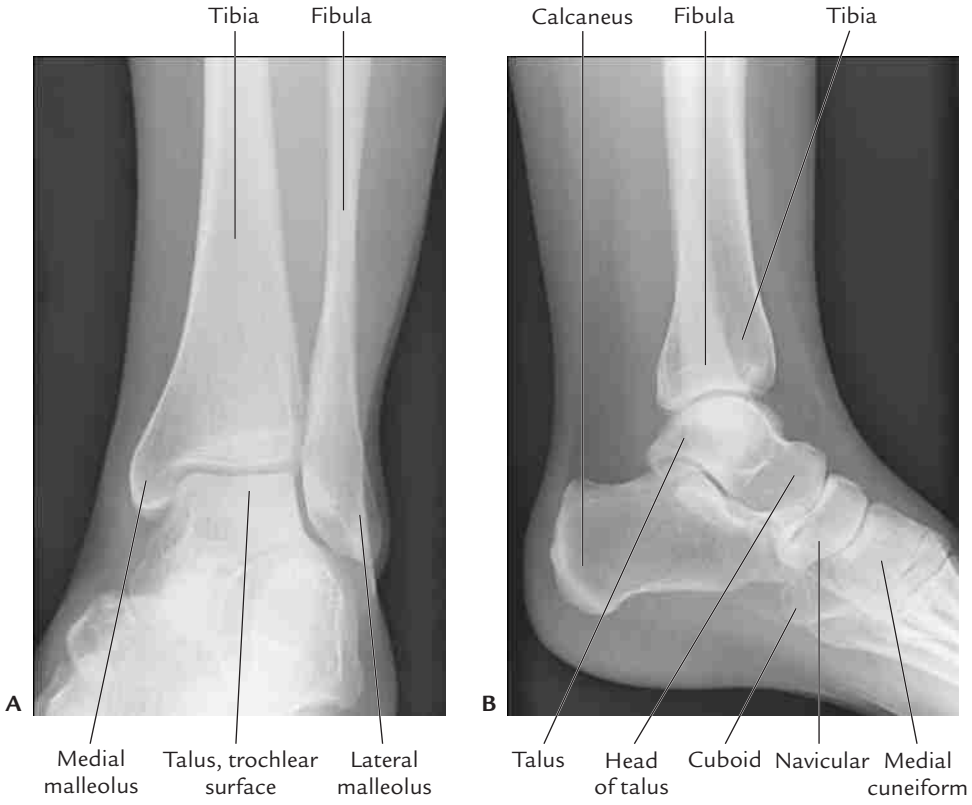


Fig. 32.12 Radiograph of the ankle joint: **A,** AP view; **B,** side view. (*Source:* Fig. 8.4, Page 222, *Integrated Anatomy,* Heylings, David JA; Spence, Roy JA; Kelly, Barry E. Oxford: Churchill Livingstone, 2007, All rights reserved.)

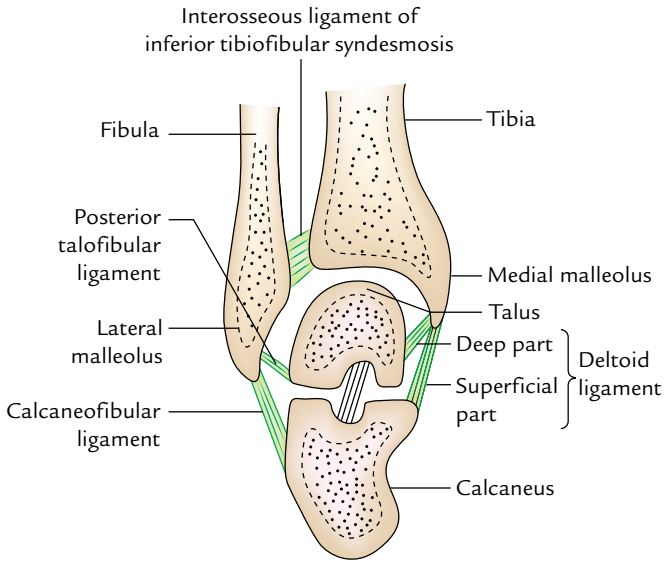


Fig. 32.13 Articular surfaces of the ankle and talocalcanean joints.

The body of talus presents three articular surfaces:

(a) Superior pulley-shaped articular surface (trochlear surface).

(b) Medial comma-shaped articular surface.

(c) Lateral triangular articular surface.

The wedge-shaped body of the talus fits into the socket above.

The articular surface on the inferior aspect of the lower end of tibia articulates with trochlear surface of the talus. The articular surface on the lateral aspect of medial malleolus articulates with the comma-shaped articular surface on the medial side of the talus. The articular surface on the medial aspect of lateral malleolus articulates with the large triangular articular surface on the lateral side of the body of talus.

N.B. The ankle joint resembles a pincer or monkey wrench gripping a section of hemisphere. (cf. tibiofibular mortise gripping the wedge-shaped body of the talus.)

STABILITY OF THE ANKLE JOINT

The trochlear surface on the superior aspect of the body of talus is wider in front than behind.

During dorsiflexion, ankle joint of the anterior wider part of the trochlea moves posteriorly and fits properly into the tibiofibular mortise (pincer), hence joint is stable. During plantar flexion, the narrow posterior part of the trochlea

does not fit properly in the tibiofibular mortise (pincer), hence the joint is unstable during plantar flexion.

Factors Maintaining the Stability of the Ankle Joint

1. Close interlocking of its articular surfaces.
2. Strong, medial, and lateral collateral ligaments.
3. Deepening of tibiofibular socket posteriorly by the inferior transverse tibiofibular ligament.
4. Tendons (four in front and five behind) crossing the ankle joint.
5. Other ligaments of this joint.

LIGAMENTS

The important ligaments of ankle joint are:

1. Capsular ligament.
2. Medial and lateral collateral ligaments.

Fibrous Capsule

It surrounds the joint completely. It is attached to the articular margins of the joint all around with two exceptions:

Posterosuperiorly it is attached to the inferior transverse tibiofibular ligament.

Anteroinferiorly it is attached to the dorsum of the neck of talus at some distance from the trochlear surface.

The joint capsule is thin in front and behind to allow hinge movements and thick on either side where it blends with the collateral ligaments.

The **synovial membrane** lines the inner surface of the joint capsule, but ceases at the periphery of the articular cartilages. A small synovial process extends upward into the inferior tibiofibular syndesmosis.

Deltoid or Medial Ligament (Fig. 32.14A)

The deltoid ligament is a very strong triangular ligament on the medial side of the ankle. It is divided into two parts: superficial and deep. Above, both the parts have a common attachment to the apex and margins of the medial malleolus. Below, the attachment of superficial and deep parts differs as under:

Superficial part: Its fibres are divided into three parts: anterior, middle, and posterior.

(a) Anterior fibres (*tibionavicular*) are attached to the tuberosity of navicular bone and the medial margin of spring ligament.
(b) Middle fibres (*tibiocalcanean*) are attached to the whole length of sustentaculum tali.
(c) Posterior fibres (*posterior tibiotalar*) to the medial tubercle and adjoining part of the medial surface of talus.

Deep part (*anterior tibiotalar*) is attached to the anterior part of the medial surface of talus.

Lateral Ligament (Fig. 32.14B)

The lateral ligament consists of three parts: anterior talofibular, posterior talofibular, and calcaneofibular.

1. *Anterior talofibular ligament* is a weak flat band, which extends forward and medially from the anterior margin of lateral malleolus, to the neck of talus just in front of the fibular facet.
2. *Posterior talofibular ligament* is a strong band, which extends backward and medially from the posterior margin of the lateral malleolus to the posterior tubercle of the talus.
3. *Calcaneofibular ligament* is a long rounded cord, which runs downward and backward from the notch on the lower border of lateral malleolus to the tubercle on the lateral surface of the calcaneum.

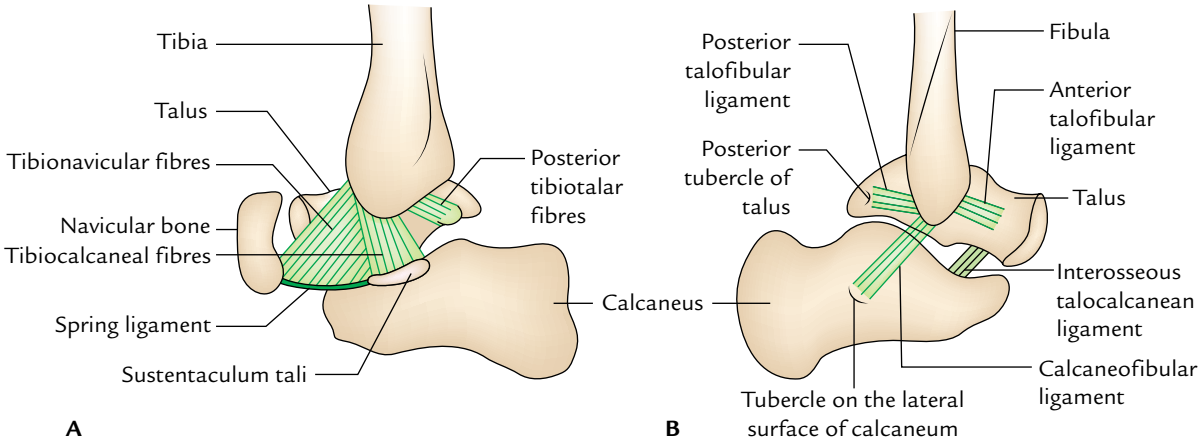


Fig. 32.14 Collateral ligaments of the ankle joint: A, deltoid ligament; B, lateral ligament.

N.B. Superficially, the *deltoid ligament* is crossed by the tendons of tibialis posterior and flexor digitorum longus whereas *lateral ligament* is crossed superficially by the tendons of peroneus longus and brevis.

RELATIONS OF THE ANKLE JOINT (Fig. 32.15)

Anterior: *Anteriorly* from medial to lateral side the ankle joint is related to the following structures:

1. Tibialis anterior.
2. Extensor Hallucis longus.
3. Anterior tibial Artery.
4. Deep peroneal Nerve.
5. Extensor Digitorum longus.
6. Peroneus tertius.

Mnemonic: The Himalayas Are Not Dry Plateaus.

Posterior: *Posteriorly* from medial to lateral side the ankle joint is related to the following structures:

1. Tibialis posterior.
2. Flexion Digitorum longus.
3. Posterior tibial Artery.
4. Posterior tibial Nerve.
5. Flexor Hallucis longus.

Mnemonic: The Doctors Are Not Here.

ARTERIAL SUPPLY

It is by the malleolar branches of anterior tibial, posterior tibial, and peroneal arteries.

NERVE SUPPLY

It is by the branches of deep peroneal and tibial nerves. (The segmental innervations is by L4, L5; S1, S2 spinal segments.)

Movements

The following movements take place at the ankle joint:

1. Dorsiflexion.
2. Plantar flexion.

N.B. When the foot is plantar flexed, the ankle joint also permits some degree of side-to-side gliding, rotation, adduction, and abduction.

Dorsiflexion

In *dorsiflexion*, the forefoot is raised and the angle between front of the leg and dorsum of the foot is diminished. In this movement, the wider anterior part of trochlea is forced posteriorly between the malleoli. It is a **close-packed position of the ankle joint** with maximum congruence of the articular surfaces and tension of ligaments. The ankle joint is most stable in dorsiflexion.

Plantar Flexion

In *plantar flexion*, the forefoot is depressed, and the angle between the leg and foot is increased. In this position, the tibiofibular socket encloses the narrower posterior part of the trochlear surface of talus and some joint space is

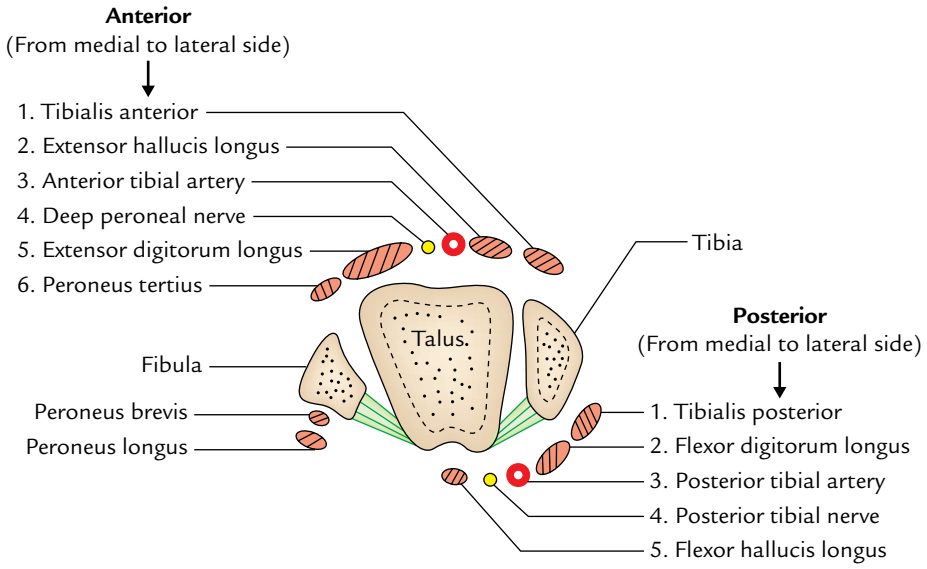


Fig. 32.15 Relations of the ankle joint (transverse section of the ankle joint).

Table 32.4 Movements of the ankle joint

	Muscles producing movements	
Movements	**Principal muscles**	**Accessory muscles**
Dorsiflexion	*Tibialis anterior*	• Extensor digitorum longus • Extensor hallucis longus • Peroneus tertius
Plantar flexion	• Gastrocnemius • Soleus	• Plantaris • Tibialis posterior • Flexor hallucis longus • Flexor digitorum longus

available between the tibiofibular mortise and the narrow posterior part of the trochlea. It is a **loose-packed position** of the ankle joint. The joint is unstable in plantar flexion.

N.B. The ankle joint is stable in dorsiflexion and unstable in plantar flexion.

The range of motion (ROM) for dorsiflexion is 20° and for plantar flexion is 45°.

The dorsiflexion is controlled by the L4, L5 spinal segments and plantar flexion by the S1, S2 spinal segments.

Movements and muscles producing them are given in Table 32.4.

Clinical correlation

- **Ankle sprains (Figs 32.16 and 32.17):** The excessive stretching and/or tearing of ligaments of the ankle joint is called the *ankle sprain*. The ankle sprains are usually caused by the falls from height or twists of ankle.

  When the plantar-flexed foot is excessively inverted, the anterior and posterior talofibular and calcaneofibular ligaments are stretched and torn. The *anterior talofibular ligament is most commonly torn*.

  When the plantar-flexed foot is excessively everted, the deltoid ligament is not torn; instead there is an avulsion fracture of medial malleolus.

  The inversion sprains are more common than eversion sprains.
- **Dislocation of the ankle:** The dislocations of ankle joint are rare because it is a very stable joint due to tibiofibular mortise. However, whenever dislocation occurs it is always accompanied by the fracture of one of the malleoli.
- **Pott's fracture (fracture dislocation of the ankle; Fig. 31.18):** It occurs when the foot is caught in the rabbit hole and everted forcibly. In this condition, the following sequence of events takes place:
  1. *Oblique fracture of the lateral malleolus* due to internal rotation of the tibia.
  2. *Transverse fracture of the medial malleolus* due to pull by strong deltoid ligament.

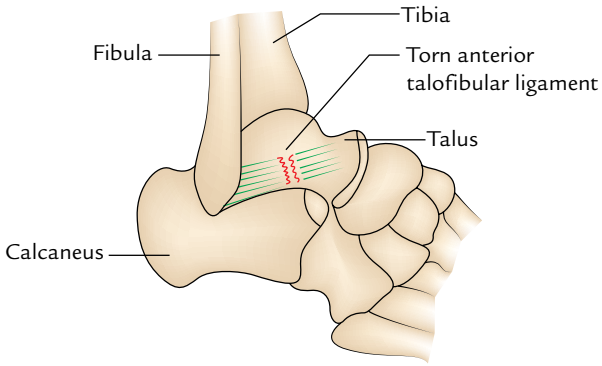


Fig. 32.16 Ankle sprain (note torn fibres of the anterior talofibular ligament).

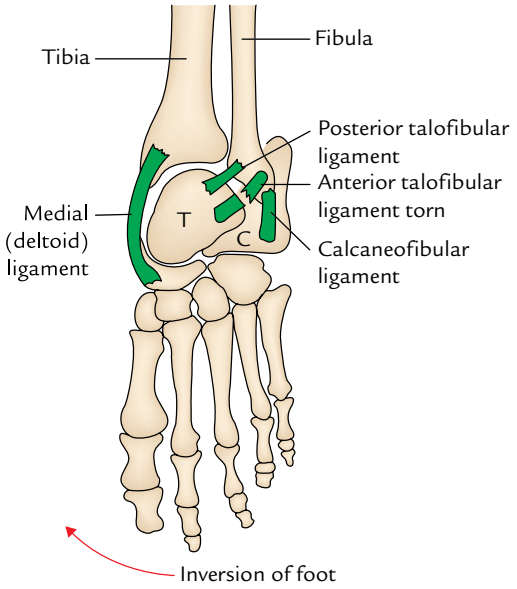


Fig. 32.17 Inversion and eversion injuries of the foot (T = talus, C = calcaneus).

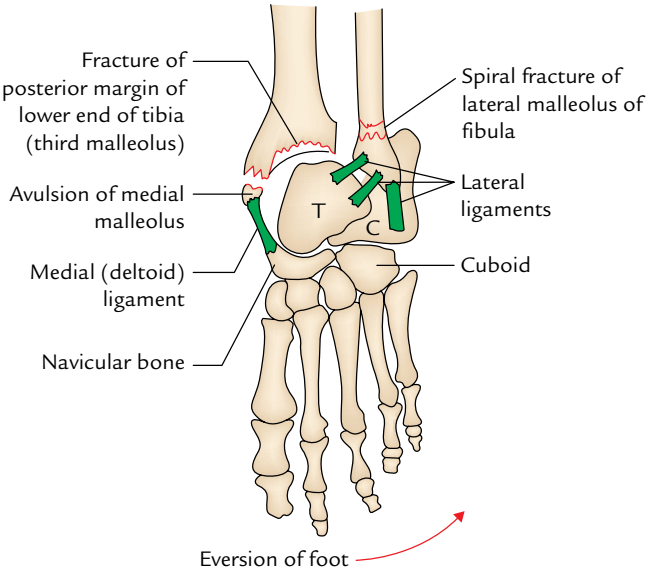


Fig. 32.18 Pott's fracture (T = talus, C = calcaneus).

3. *Fracture of the posterior margin of the lower end of tibia* (**third malleolus**) because it is carried forward.

These stages are also termed **first, second, and third degree of Pott's fracture,** respectively. The third degree of Potts fracture is also called trimalleolar fracture.

- **Optimum position of the ankle:** The optimum position of the ankle is one in which ankle joint is in slight plantar flexion. The knowledge of position is essential for applying plaster cast in the ankle region.

JOINTS OF THE FOOT

There are numerous joints in the foot formed between tarsal, metatarsal, and phalangeal bones. The joints of the foot include intertarsal, tarsometatarsal, intermetatarsal, metatarsophalangeal, and interphalangeal joints. Those joints which are of more functional value are discussed.

INTERTARSAL JOINTS

They are of two types: major and minor. Only major intertarsal joints are discussed in the following text.

Subtalar (Talocalcanean) Joints (Fig. 32.19)

There are two joints between the talus and calcanean: posterior talocalcanean joint and anterior talocalcaneonavicular joint. The posterior talocalcanean joint is often designated as **subtalar joint**. It is a plane type of synovial joint. It is formed between the concave facet on the inferior surface of the body of talus and convex facet on the middle one-third of the superior surface of the calcaneum.

Ligaments

These are: (a) fibrous capsule, (b) lateral and medial talocalcanean ligaments, (c) interosseous talocalcanean ligament, and (d) cervical ligament.

The *interosseous talocalcanean ligament* is thick and very strong. It forms the chief bond of union between the talus and calcaneum. It occupies sinus tarsi and separates the *talocalcanean joint* from the *talocalcaneonavicular joint*. It extends obliquely from the sulcus tali to the sulcus calcanei. *It becomes taut in eversion.* The *cervical ligament* is lateral to sinus tarsi. It extends upward and medially from upper surface of the calcaneum to the tubercle on the inferolateral aspect of the neck of talus. It becomes taut in inversion.

Talocalcaneonavicular Joint (Fig. 32.19)

It is a compound articulation consisting of anterior talocalcanean and talonavicular joints. *It is roughly a ball and socket type of synovial joint.* The articular surface on the rounded head of the talus fits into the socket formed by the calcaneum, navicular, and spring ligament.

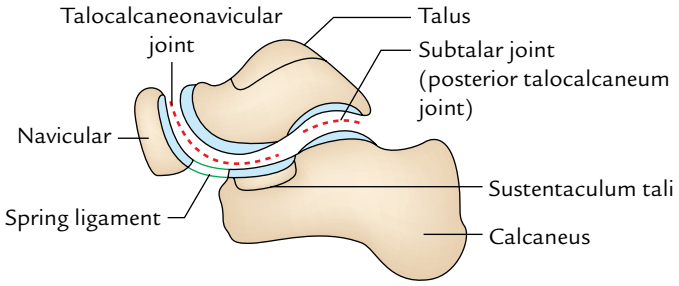


Fig. 32.19 Subtalar and talocalcaneonavicular joints.

Ligaments

These are: (a) fibrous capsule, (b) spring ligament, and (c) medial limb (calcaneonavicular part) of *bifurcate ligament.*

The *spring ligament (plantar calcaneonavicular ligament)* is a powerful fibrocartilaginous band, which extends from the anterior margin of the sustentaculum tali to the plantar surface of navicular bone between its tuberosity and articular margin. It takes part in forming the socket for the head of the talus. It is the most important ligament to maintain the medial longitudinal arch. Its upper surface has a triangular fibrocartilagenous facet for the head of the talus. Its plantar surface is supported by the tendon of tibialis posterior medially, and by the tendons of flexor hallucis longus and flexor digitorum longus laterally.

The talocalcaneonavicular joint permits the movements of inversion and eversion.

Calcaneocuboid Joint (Fig. 32.20)

It is a saddle type of synovial joint. The opposed articular surfaces of the calcaneum and cuboid are reciprocally concavoconvex.

Ligaments

The ligaments of the joint are: (a) fibrous capsule, (b) lateral limb (calcaneocuboid part) of bifurcate ligament, (c) long plantar ligament, and (d) short plantar ligament.

The *bifurcate ligament* is Y-shaped. Its stem is attached to the anterolateral part of the sulcus calcanei. Its medial limb (calcaneonavicular part) is attached to the dorsolateral surface of the navicular bone and its lateral limb (calcaneocuboid part) to the dorsomedial surface of the cuboid bone. The two limbs provide support to the joints.

The long planar ligament is the longest ligament. It is strong and its importance in maintaining the arches of foot is surpassed only by the spring ligament. It *extends from triangular plantar surface of the calcaneum to the lips of the groove on cuboid and beyond it to the bases of the middle three metatarsals (second to fourth).* It converts the groove on the

plantar surface of cuboid into a tunnel for the passage of tendon of peroneus longus. Morphologically, it represents the divorced tendon of the gastrocnemius.

The *short plantar ligament (plantar calcaneocuboid ligament)* lies deep to long plantar ligament. It is broad and extends from the anterior tubercle of calcaneum to the plantar surface of the cuboid behind its ridge.

Transverse Tarsal (Midtarsal) Joint (Fig. 32.20)

It is a compound joint consisting of calcaneocuboid and talonavicular joints. Talonavicular is also a part of talocalcaneonavicular joint. The calcaneocuboid and talonavicular joints are grouped together because both are placed nearly in the same transverse plane. However, the two joints have different axes of movements. The movements of midtarsal joint help in inversion and eversion of the foot.

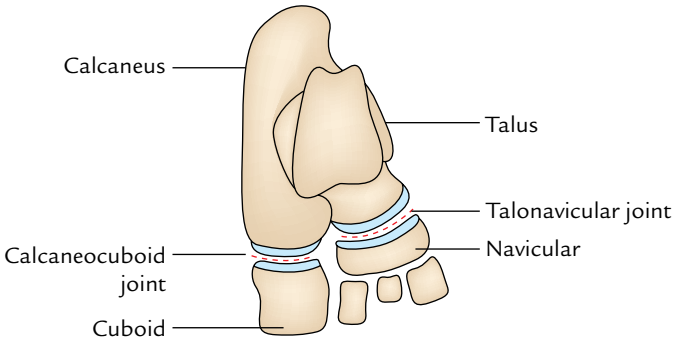


Fig. 32.20 Transverse tarsal (midtarsal) joint.

INVERSION AND EVERSION OF THE FOOT

The inversion and eversion and rotational movements of the foot on the talus.

The **inversion** is a movement in which the medial border of the foot is raised so that the sole faces medially.

The **eversion** is a movement in which the lateral border of the foot is raised so that the sole faces laterally.

N.B. There is difference in the relational movements associated with inversion and eversion when the foot is off the ground and on the ground as given in Table 32.5.

Joints Taking Part

1. Subtalar joint.
2. Talocalcaneonavicular joint } Main joints.
3. Transverse tarsal/midtarsal joint—Accessory joint.

Axis of Movements (Fig. 32.21)

The movements of *inversion* and *eversion* take place around an **oblique axis** which runs forward, upward, and medially passing from the back of calcaneum through the sinus tarsi to emerge at the superomedial aspect of the neck of talus.

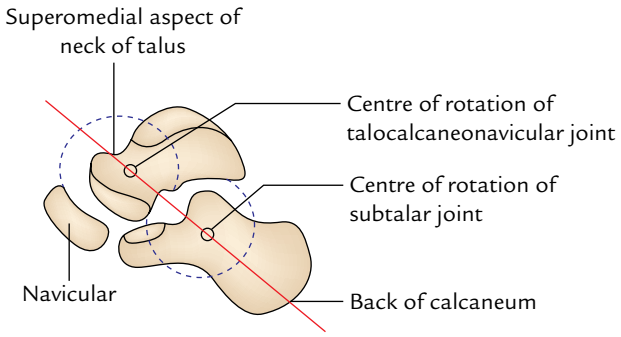


Fig. 32.21 Axis of inversion and eversion (red line).

Range of Movements (ROM)

1. The range of movement of inversion is much more than that of eversion (inversion = 30°, eversion = 20°).
2. The range of these movements is appreciably increased in plantar flexion of the foot because in this position, the narrow posterior part of the trochlear surface of talus occupies the tibiofibular socket (mortise), which permits some degree of side to side movement of the talus.

Table 32.5 Movements of inversion and eversion when the foot is off the ground and when the foot is on the ground

Movement	Inversion	Eversion
When foot is off the ground	• Range of motion is more • Inversion consists of adduction of the forefoot, lateral rotation (supination) of the forefoot, and plantar flexion of the ankle	• Range of motion is more • Eversion consists of abduction of the forefoot, medial rotation (pronation) of the forefoot, and dorsiflexion of the ankle
When foot is on the ground	• Range of motion is less • Inversion consists of only lateral rotation (supination) of the forefoot • Heads of the medial 2 metatarsals are raised	• Range of motion is less • Eversion consists of only medial rotation (pronation) of the forefoot • Heads of the lateral 3 metatarsals are raised

Table 32.6 Movements of inversion and eversion

	Muscles	
Movements	**Principal muscles**	**Accessory muscles**
Inversion (ROM = 30°)	• Tibialis anterior • Tibialis posterior	• Flexor hallucis longus • Flexor digitorum longus
Eversion (ROM = 20°)	• Peroneus longus • Peroneus brevis	• Peroneus tertius

Muscles Producing Movements

The movements of inversion and eversion and muscles producing them are given in Table 32.6.

Functional Significance

The movements of inversion and eversion are necessary for walking on uneven and sloping grounds. They greatly help the foot in adjusting it to such grounds. When feet are supporting the body weight, these movements occur in a modified form called supination and pronation.

TARSOMETATARSAL JOINTS

These are **plane type** of synovial joints. They are connected by dorsal, plantar, and interosseous tarsometatarsal ligaments. They are five in number. The first joint possesses a separate cavity, the second and third together have one cavity. These joints permit only limited gliding movements.

METATARSOPHALANGEAL JOINTS

These are **ellipsoidal type** of the synovial joints. They are connected by the capsular, collateral, plantar, and deep transverse metatarsal ligaments. Two collateral ligaments strengthen the sides of each joint. These joints permit the slight gliding movements. The deep transverse ligaments (four in number) connect the plantar ligaments of adjacent metatarsophalangeal joints.

Movements

These joints permit dorsiflexion, plantar flexion, adduction, and abduction.

- The range of dorsiflexion is more (50–60°) than that of plantar flexion (30–40°).
- The *axis of adduction and abduction of the toes passes through the least mobile second metatarsal bone.* (cf. The axis of adduction and abduction of fingers passes through the third metacarpal bone.)

INTERPHALANGEAL JOINTS

These are typical **hinge joints**. They are connected by the capsular and collateral ligaments.

Movements

These joints permit dorsiflexion and plantar flexion of distal phalanges.

Golden Facts to Remember

► Most stable position of the knee joint	Erect extended position
► Most important muscle to stabilize the knee joint	Quadriceps femoris
► Only mammal who does not have menisci and popliteus	Fruit bat
► Most frequently injured joint in the lower limb	Ankle joint
► Key muscle of the knee joint	Popliteus
► Unhappy triad of knee joint	Injury of (a) tibial collateral ligament, (b) medial meniscus, and (c) anterior cruciate ligament
► Most commonly injured meniscus of the knee joint	Medial meniscus
► Strongest tibiofibular joint	Inferior tibiofibular joint
► Most important ligament for maintaining the arches of the foot	Spring ligament

Clinical Case Study



A final year medical student, while playing football received a blow on the lateral side of his right knee and fell on the ground. He felt sharp pain on the medial aspect of the right knee and was not able to extend the leg on the same side. He was taken to the hospital where the examining doctor found that his right knee was swollen especially above the patella. **Drawer's sign** was negative. The radiograph of the knee did not reveal any fracture. He was diagnosed as a case of *torn medial meniscus of the knee joint.*

Questions

1. Enumerate the main intracapsular structures of the knee joint?
2. Why is the injury of medial meniscus more common than that of a lateral meniscus?
3. What is the cause of swelling of the knee especially above the patella in this case?
4. What type of meniscal tear is commonly seen in football players?
5. What is the "*unhappy triad*" of the knee joint?

Answers

1. The main intracapsular structures of the knee joint are: (a) medial and lateral menisci, (b) anterior and posterior cruciate ligaments, and (c) suprapatellar and deep infrapatellar bursae.
2. Because it is firmly attached with the tibial collateral ligament. A sudden blow on the lateral side of a flexed weight-bearing knee can cause rupture of tibial collateral ligament and concomitant tear of the medial meniscus.
3. The accumulation of synovial fluid within the joint due to traumatic synovitis. The distension of suprapatellar bursa led to the large amount of swelling above the patella.
4. Bucket handle tear (longitudinal split of the meniscus).
5. It is a combination of injury to the (a) tibial collateral ligament, (b) medial meniscus, and (c) anterior cruciate ligament.

CHAPTER

33

Venous and Lymphatic Drainage of the Lower Limb

VENOUS DRAINAGE OF THE LOWER LIMB

The **venous drainage of the lower limb** is of immense clinical and surgical importance. The venous blood of the lower limb is drained against gravity. However, a number of factors help to facilitate its drainage (vide infra). If these factors fail to help the drainage, the stagnation of venous blood in the superficial veins cause **varicose veins** and in the deep veins lead to **deep vein thrombosis**.

Factors Helping the Venous Drainage of the Lower Limb

1. The *contraction of the calf muscles* (chief factor) squeezes the blood upward along the deep veins. Note the calf muscles act as "**calf pump (peripheral heart)**."
2. *Transmitted pulsations from the adjacent arteries.*
3. *Presence of valves in the perforating veins* prevents the reflux of blood into the superficial veins during contraction of the calf muscles.
4. *Presence of valves in the deep veins* supports the column of blood and maintains unidirectional upward flow of the blood.
5. *Negative intrathoracic pressure* becomes more negative during inspiration and yawning.
6. In recumbent position, the **"vis-a-tergo"** is produced by the contraction of the heart and suction action of the diaphragm.

CLASSIFICATION OF THE VEINS

The veins of the lower limb are classified anatomically and functionally into the following three types (Fig. 33.1):

1. Superficial veins.
2. Deep veins.
3. Perforating veins.

The **superficial veins** essentially include the great and small saphenous veins, which represent pre-axial and post-axial veins of the developing lower limb, respectively. They lie in the superficial fascia on the surface of the deep fascia and are *thick walled* because of the presence of the smooth muscle. They possess valves, which are more numerous in their distal part than in the proximal part. A large proportion of their blood is drained into the deep veins through the perforating veins.

The **deep veins** include the anterior tibial, posterior tibial, peroneal, popliteal, and femoral veins. They are surrounded and supported by the powerful muscles. They possess more valves. They accompany the arteries. Below the knee, they are arranged as a pair of **venae comitantes** along the arteries but above the knee, they form single large vein. All the veins from muscles draining into deep veins also possess valves except those in the soleus where they are arranged in the form of **venous sinuses (soleal sinuses).**

The **perforating veins (perforators)** pierce the deep fascia and connect the superficial veins with the deep veins. Their valves permit only one-way flow of the blood, from the superficial veins to the deep veins. There are about five perforators along the great saphenous vein, and one perforator along the small saphenous vein.

N.B. The *venous blood* of the thigh and leg flows from the superficial to deep veins (being directed by the valves of perforating veins). However, in foot, the venous blood flows from the deep veins (in the sole) to the superficial veins (on the dorsum of the foot).

SUPERFICIAL VEINS

Great Saphenous Vein

The great saphenous vein lies in the superficial fascia and is easily seen (Greek *saphenous* = easily seen). The great saphenous vein is the **longest vein** of the body and represents the **pre-axial vein** of the lower limb. It is also called long saphenous vein (Fig. 33.2).

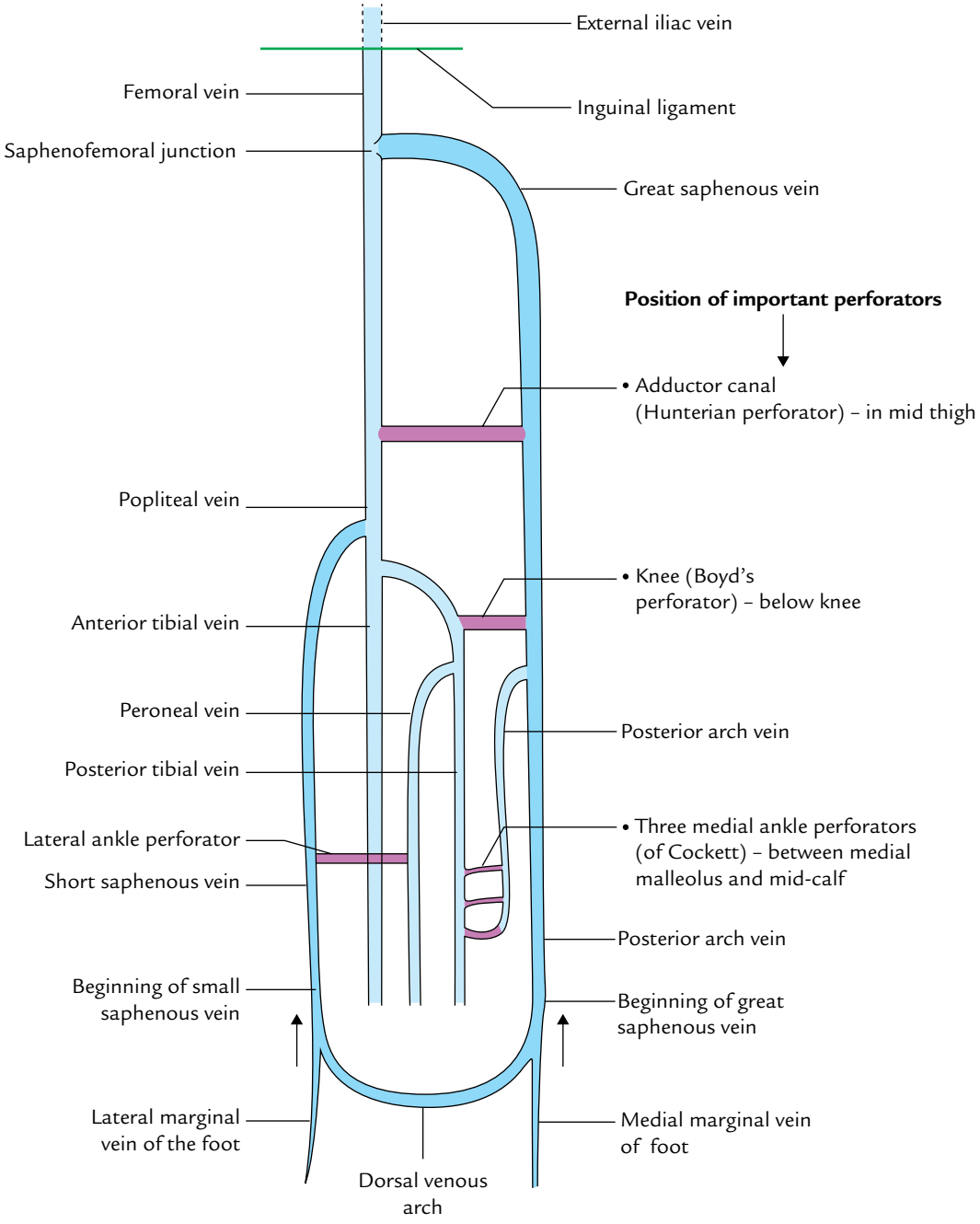


Fig. 33.1 Schematic representation of the veins of the lower limb (deep vein = light blue colour, superficial veins = deep blue colour, perforating vein = violet).

Course

It is formed on the dorsum of foot by the union of the medial end of the **dorsal venous arch of the foot** and **medial marginal vein of the foot**. The vein runs upward about 2.5 cm in front of the medial malleolus, crosses obliquely the medial surface of the lower third of tibia, and then ascends a little behind the medial border of tibia to reach the knee, where it lies on the posteromedial aspect of the knee joint, about one hand-breadth posterior to the patella; from here it runs upward along the medial side of the thigh to reach the saphenous opening (fossa ovalis).

It passes through the saphenous opening after piercing the cribriform fascia and drains into the **femoral vein** after piercing the femoral sheath.

Tributaries

1. *At the commencement:* Medial marginal vein of the big toe.

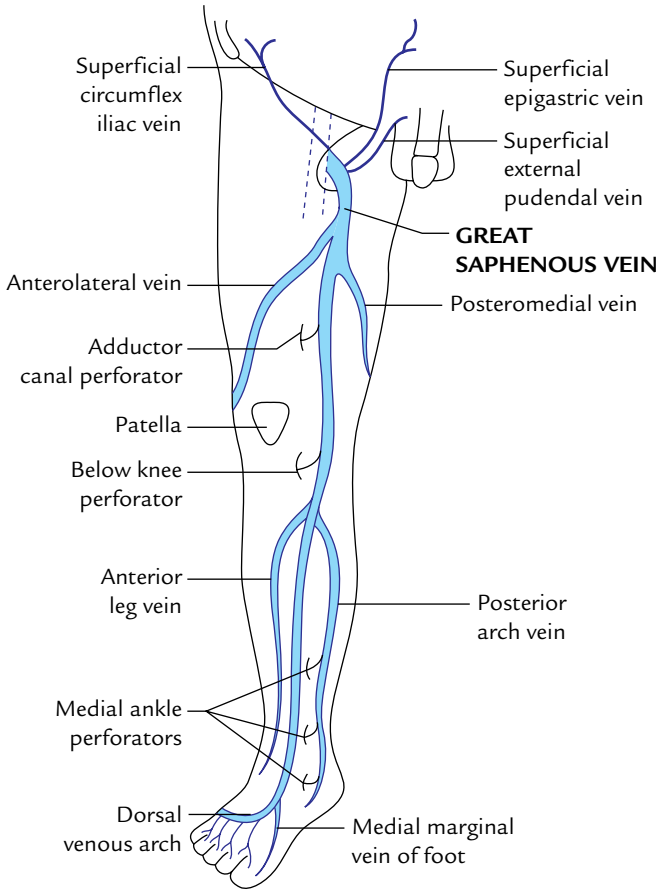


Fig. 33.2 Formation, course, termination, and tributaries of the great saphenous vein.

2. *In the leg:*
   (a) *Communicating veins* between small saphenous and deep veins.
   (b) *Posterior arch vein.* It is fairly large and constant. It collects the blood from the posteromedial aspect of the calf and begins of a series of small venous arches connecting the three medial ankle-perforating veins (perforators).
3. *Just below the knee:*
   (a) *Anterior veins of the leg.* They extend diagonally (upward, forward, and medially) across the shin and join the great saphenous vein.
   (b) *A few veins from the calf* which communicates with the small saphenous vein.
4. *In the thigh:*
   (a) *Anterolateral vein.* It commences in the lower part of the front of thigh, crosses the apex of femoral triangle, and joins the great saphenous vein in the upper part of the thigh.
   (b) *Posteromedial vein (accessory saphenous vein).* It commences from the posteromedial aspects of the thigh and joins with the great saphenous vein; sometimes it may communicate below with the small saphenous vein.
5. *Just before piercing the cribriform fascia:*
   (a) *Superficial epigastric vein.*
   (b) *Superficial circumflex iliac vein.*
   (c) *Superficial external pudendal vein.*
   These veins accompany the corresponding superficial branches of the femoral artery.
6. *Just before the termination in the femoral vein: Deep external pudendal vein* (last tributary) drains the blood from the anterior part of the perineum.

N.B. *Thoraco-epigastric vein:* It runs along the anterolateral wall of the trunk. It connects the superficial epigastric vein with the lateral thoracic vein. Thus, it establishes an important communication between the femoral and axillary veins (i.e., upper and lower limbs).

Valves in the Great Saphenous Vein

There are about 10 to 20 valves in the great saphenous vein, out of which the location of two needs special mention here: (a) one, which lies just before it pierces the cribriform fascia and (b) the other, which lies at its junction with the femoral vein (**saphenofemoral valve**). The saphenofemoral valve is of great functional significance. It lies about 3.5 to 4 cm inferolateral to the pubic tubercle. In about 80% individuals, the external iliac vein possesses a valve, which protects the *saphenofemoral valve* against high venous pressure. The remaining 20% cases who do not have this valve become the victim of high venous pressure and develop **varicose vein**, which commences at the saphenofemoral junction and gradually extends downward.

Surface Marking of the Great Saphenous Vein

1. *At ankle*, it lies 2.5 cm anterior to the medial malleolus.
2. *In leg*, it ascends by crossing the medial surface and medial border of the tibia.
3. *At knee*, it lies about a hand's breadth posterior to the medial margin of the patella.
4. *In thigh*, it ascends obliquely on the medial aspect of the thigh to reach a point 3.5–4 cm inferolateral to the pubic tubercle (saphenofemoral junction).

Clinical correlation

- **Venesection of the great saphenous vein:** The great saphenous vein in front of medial malleolus at ankle is the most preferred site of venesection (cut-down) in emergency situation when the superficial veins elsewhere in the body are collapsed and invisible, to insert the canula for prolonged administration of intravenous fluids.

  Note that, in front of medial malleolus, saphenous nerve lies in front of the vein. Hence, during cut-down procedure, the saphenous nerve should be recognized to avoid its injury.

- **Great saphenous vein graft:** In coronary by pass surgery to relieve the ischemia of the heart, a segment of great saphenous vein is removed and used for aortocoronary grafting to by pass an arterial obstruction. Due to the presence of valves, the vein has to be reversed so that its valves do not obstruct the blood flow.

Small (Short) Saphenous Vein (Fig. 33.3)

It is formed below and behind the lateral malleolus by the union of the lateral end of the dorsal venous arch, and the lateral dorsal digital vein of the little toe. It runs upward behind the lateral malleolus, along the lateral edge of tendocalcaneus, and is accompanied by the *sural nerve* on its lateral side. Thereafter it runs in the middle of the back of the leg, pierces the deep fascia, and undergoes a subfascial course between the two heads of the gastrocnemius until it reaches the middle of the popliteal fossa. Here it turns inward to terminate into the popliteal vein. The *posterior femoral cutaneous nerve* accompanies the upper part of the vein, while passing from deep to superficial (see Fig. 29.2B).

The small saphenous vein contains 7–13 valves.

N.B.

- *Variations of small saphenous vein at its termination:*
  1. Just before piercing the popliteal fascia, it may give rise to a communicating branch to the *accessory saphenous vein.*
  2. Occasionally, it ends below the knee and terminates in the great saphenous vein or in the deep muscular veins of the leg.
  3. It may bifurcate, with one limb terminating in the great saphenous vein and other in the popliteal vein.
- *Pseudo-short saphenous veins:* These large muscular veins drain venous blood from two heads of gastrocnemius into the popliteal vein. The varicosity of these veins may mimic the varicosity of short saphenous vein; hence, they are often termed **pseudo-short saphenous veins**.

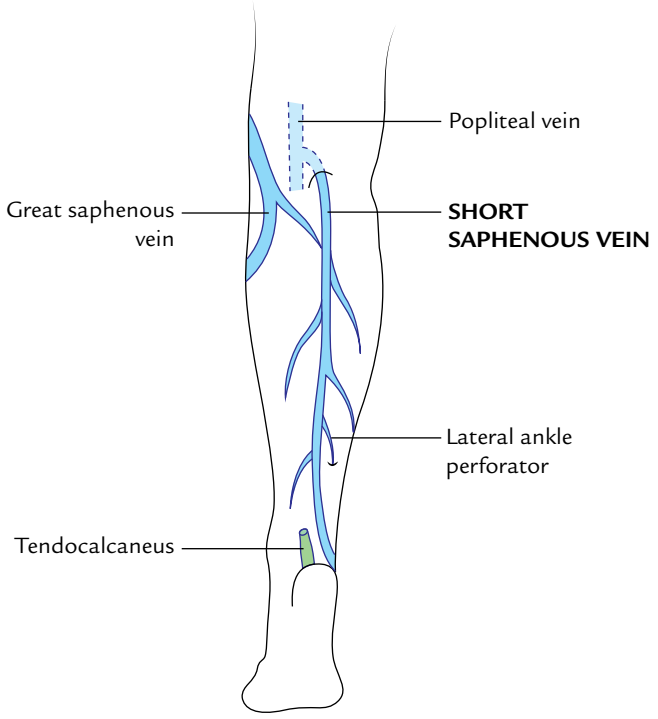


Fig. 33.3 Small (short saphenous) veins.

PERFORATING VEINS (PERFORATORS)

As described earlier they are communicating venous channels between the superficial and deep veins. These veins are called perforators because they perforate the deep fascia. The perforators are classified into two types: indirect and direct (Fig. 33.4).

1. *Indirect perforators:* They connect the superficial veins with the deep veins through muscular veins.
2. *Direct perforators:* They connect the superficial veins with the deep veins directly.

Location of Perforators

The position of five or six perforators is fairly constant, as mentioned below (Fig. 33.1):

1. **An adductor canal (Hunterian) perforator:** It connects the great saphenous vein with the femoral vein in the lower part of the adductor (Hunter's) canal.
2. **A knee perforator (Boyd's perforator):** It connects the great saphenous vein with the posterior tibial vein just below the knee and close to the medial border of tibia.
3. **A lateral ankle perforator:** It communicates the short saphenous vein with the peroneal vein. It is situated at the junction of middle and lower third of the leg.
4. **Three medial ankle perforators (of Cockett):** These are situated close to the medial border of the lower third of tibia between the medial malleolus and mid-calf and connect the great saphenous vein with the posterior tibial veins.
   (a) *Upper medial ankle perforator:* It lies at the junction of the middle and lower third of the leg.
   (b) *Middle medial ankle perforator:* It lies about 4 cm above the medial malleolus.
   (c) *Lower medial ankle perforator:* It lies posteroinferior to the medial malleolus.

N.B. The three medial ankle perforators join with one another (by a series of venous arcades) to form the *posterior arch vein.*

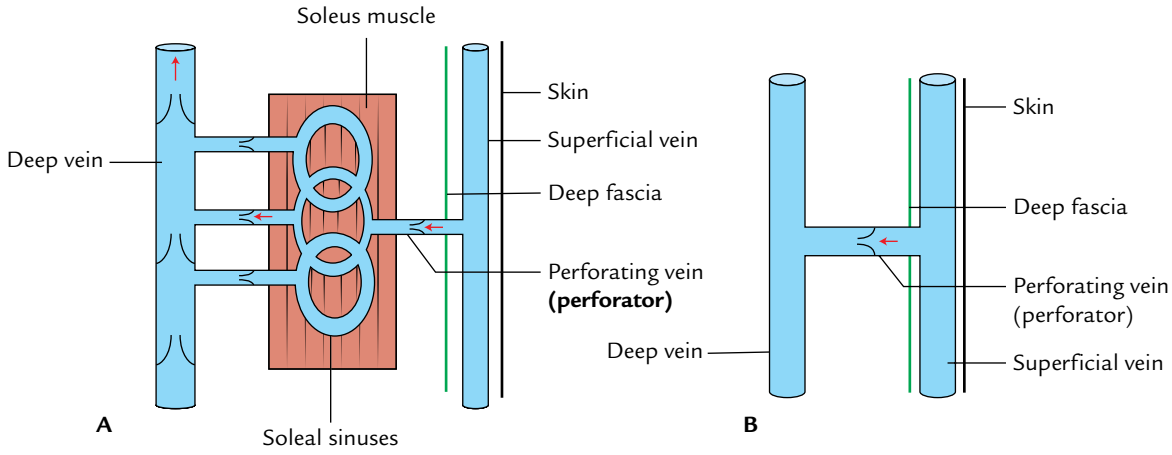


Fig. 33.4 Perforators: A, indirect perforators; B, direct perforators.

Clinical correlation

Calf pump and peripheral heart: In upright position, the venous return from the lower limb occurs against gravity and depends largely on the contraction of calf muscles. Therefore, these muscles are termed **calf pump**.

The soleus muscle contains venous sinuses filled with blood. When soleus muscle contracts, it pumps the blood from its large venous sinuses into the deep veins, and when it is relaxed it sucks the blood from the superficial veins, and the venous sinuses within it are refilled. The unidirectional blood flow is maintained by the valves in the perforating veins. Hence, the soleus is sometimes termed **peripheral heart**.

The soleal sinuses are common site for *thrombosis* and source of pulmonary embolism in sedentary individuals. The *phlebitis* of soleal sinus may be dangerous because the spread of infection from here may damage the valves in the perforators.

DEEP VEINS

The deep veins of the leg lie in the tight fascial compartment along the arteries.

The major deep veins of the lower limb are as follows:

1. Deep veins of the sole (e.g., medial and lateral plantar veins).
2. Venae comitantes accompanying the dorsalis pedis, anterior tibial, and posterior tibial arteries.
3. Popliteal vein.
4. Femoral vein.

The features of the deep veins are already described on page 458.

Clinical correlation

Deep vein thrombosis: The veins of the lower limb possess **valves**, which direct the blood flow to the heart particularly against gravity in upright posture. The valves are more numerous in the deep veins than in the superficial veins. The veins from the muscles draining into the deep veins have valves except those in the **soleus**, which are arranged in the form of **venous sinuses**. The blood flow is sluggish in the soleal sinuses particularly when the muscles are put to rest. Prolonged rest in bed by some patients after surgery is unwise because it may develop **deep vein thrombosis**, which may culminate into life-threatening complication of the **pulmonary embolism**.

VARICOSE VEINS AND VARICOSE ULCERS OF THE LOWER LIMB

When the veins become dilated and tortuous they are called **varicose veins** (Fig. 33.5). The superficial veins of the lower limbs commonly become varicosed due to incompetency of the valves, following prolonged standing (e.g., bus conductors, traffic police personnel, nurses).

1. *Incompetency of valves in the perforating veins:* If the valves in the perforating veins become incompetent, the defective veins become **"high pressure leaks"** during muscular contraction, whereby the high pressure of the deep veins is transmitted to the superficial veins. As a result, the superficial veins become dilated and tortuous.
2. *Incompetency of valves at the termination of the superficial veins:* For example, if *saphenofemoral valve* becomes incompetent, the great saphenous vein becomes dilated and tortuous. The varicosity commences from the

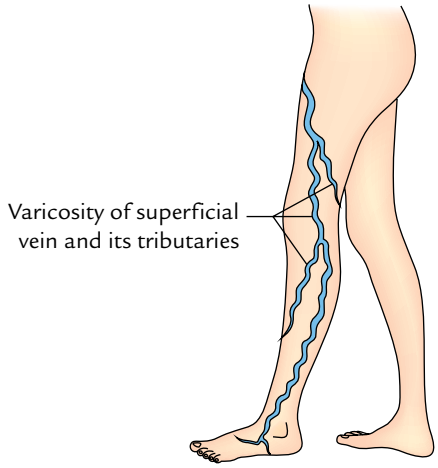


Fig. 33.5 Varicose veins in the left lower limb.

saphenofemoral junction and extends gradually downward.

The dilatation of superficial veins and gradual degeneration of their walls may lead to *varicose ulcers*.

Recognition of Sites of Incompetent Valves

The clinicians recognize the sites of incompetent valves by the following two tests: Trendelenburg test and Perthes' test (tourniquet test). Note, by these tests only superficial veins and perforators can be tested, but not the deep veins.

1. *Trendelenburg test:* The patient is asked to lie down, and the veins are emptied by raising the lower limb and stroking the varicose veins proximally. Now pressure is applied with the thumb at the saphenofemoral junction, and the patient is asked to stand up quickly.

   To test the superficial veins, the pressure is released. If the varicose veins are filled quickly from above, it indicates incompetency of the superficial veins, and test is positive. If the veins are not filled, the test is negative.

   To test the perforating veins, the pressure at the saphenofemoral junction is not released, but maintained for about a minute. Gradual filling of the superficial veins indicates incompetency of valves of the perforating veins, allowing the blood from deep to superficial veins and test is positive.
2. *Perthes' test* (tourniquet test): It is employed to test the incompetence of the deep veins. A tourniquet is tied around the upper part of thigh, tight enough to occlude the saphenous vein but not the femoral vein. The patient is asked to do gentle exercise, i.e., walk quickly for a while, with tourniquet in place. If the perforating and deep veins are normal, the varicose veins will shrink, whereas if they are blocked, the varicose veins become more distended.

N.B. The identification of exact sites of the defective perforators is essential before ligating such perforators. When numerous perforators are involved, the entire length of great saphenous vein is removed by **stripping operation**.

LYMPHATIC DRAINAGE OF THE LOWER LIMB

The knowledge of lymphatic drainage of the lower limb is of great clinical importance because inflammatory lesions of limb cause painful enlargement of the lymph nodes whereas blockage of lymphatics by microfilarial parasites leads to massive edema of the lower limb.

Most of the lymph from lower limb is drained into the inguinal lymph nodes, either directly (mostly) or indirectly (partly) through the popliteal and anterior tibial nodes. However, the deep structures of the gluteal region and upper part of the back of thigh are drained into the internal iliac nodes.

LYMPH NODES

The lymph nodes are classified into two types: superficial and deep.

1. *Superficial lymph nodes* include the superficial inguinal nodes.
2. *Deep lymph nodes* include the deep inguinal nodes, popliteal nodes, and anterior tibial nodes.

SUPERFICIAL LYMPH NODES (Fig. 33.6A)

Superficial Inguinal Nodes

These are present in the superficial fascia of the inguinal region. They are arranged in two groups: upper and lower, resembling the letter "T."

The **upper horizontal group** contains five or six nodes, which lie below the inguinal ligament. The **lateral members of upper group** (2 or 3 nodes) receive **afferent** from:

(a) Gluteal region.
(b) Upper part of the lateral side of the thigh.
(c) Flank and back of the abdominal wall below the umbilical plane.

The **medial members of the upper group** receive **afferent** from:

(a) Subcutaneous tissue of the anterior abdominal wall below the umbilicus.
(b) Penis including prepuce and scrotum in male, vulva and vagina below the hymen in female.

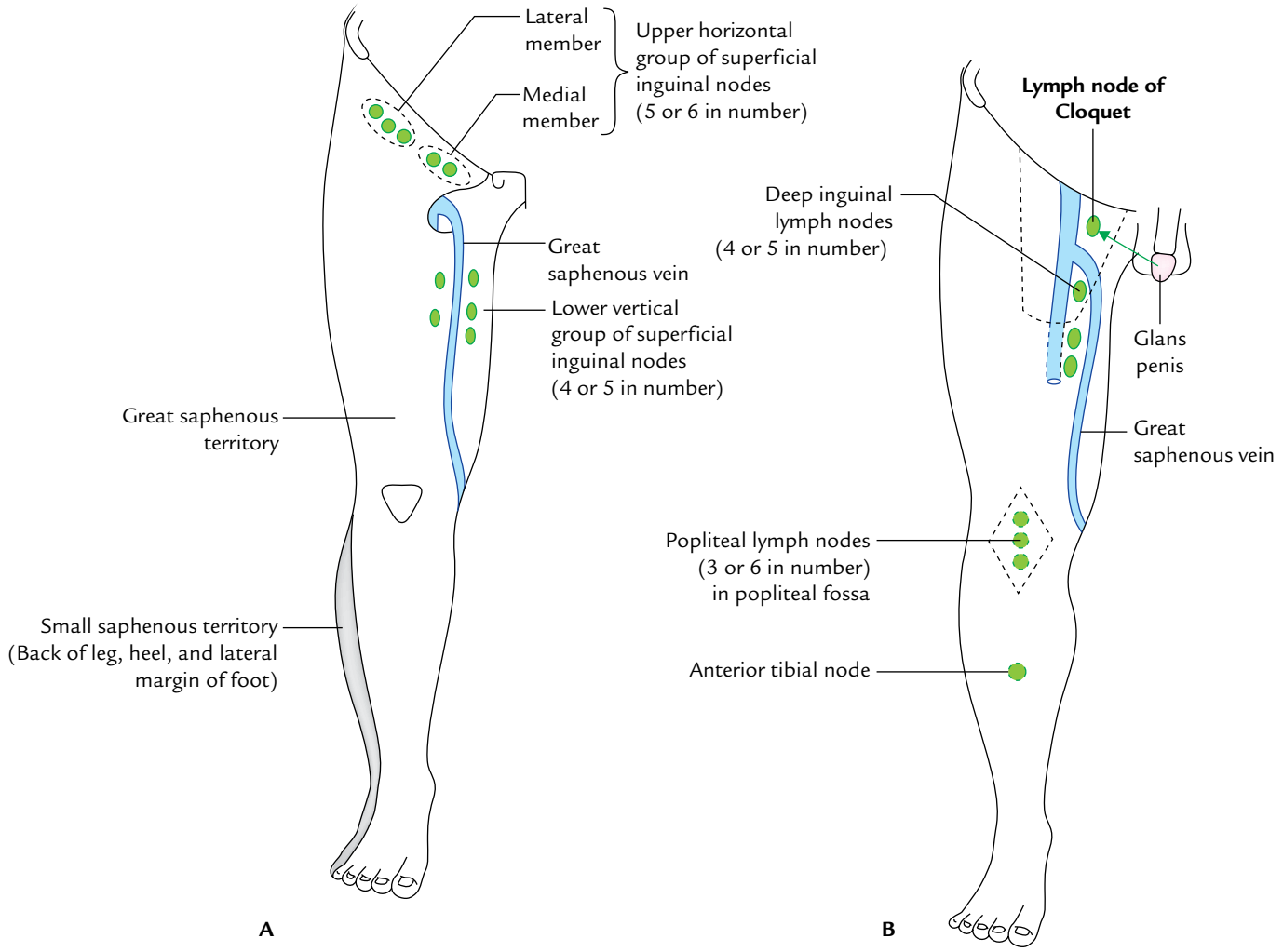


Fig. 33.6 Lymph node groups of the lower limb: A, superficial inguinal lymph nodes; B, deep inguinal lymph nodes.

(c) Perineum and lower part of the anal canal below the pectinate line.

(d) Few lymphatics from the superolateral angle of uterus which accompany the round ligament of uterus.

The **lower vertical group** consists of four or five nodes. They are placed along both sides of the terminal part of great saphenous vein. They receive afferent from the skin and fasciae of the most of the lower limb (great saphenous territory), except from buttock, which drain into upper lateral group and the short saphenous territory, which drain into popliteal nodes.

N.B.

- The efferents from all the superficial inguinal nodes pierce the cribriform fascia and terminate into the deep inguinal nodes.

DEEP LYMPH NODES (Fig. 33.6B)

Deep Inguinal Lymph Nodes

These are about four to five in number, and lie on the medial side of the upper part of the femoral vein in the femoral triangle. The most proximal node of this group (**gland of Cloquet or Rosenmüller**) lies in the femoral canal. These nodes receive afferents from: (a) the superficial inguinal nodes, (b) popliteal nodes, (c) glans of penis/clitoris, and (d) deep lymphatics of the lower limb accompanying femoral vessels.

Their efferent vessels from the deep inguinal lymph nodes drain into the external iliac nodes after piercing the femoral septum, which closes the femoral ring.

Popliteal Lymph Nodes

The **popliteal lymph nodes** (about three to six in number) lie embedded in the popliteal pad of fat near the termination

of the small saphenous vein; one node lies between the popliteal artery and the oblique popliteal ligament. The popliteal nodes receive **afferents** from: (a) the territory of small saphenous vein (i.e., lateral side of the foot, heel, and lateral half of the back of leg), (b) deep parts of the leg, running along the anterior and posterior tibial vessels, and (c) the knee joint.

The efferents from the popliteal nodes run along the popliteal and femoral vessels to terminate into the deep inguinal nodes.

N.B. The **popliteal lymph nodes** are unique in the sense that they are the only deep nodes, which receive both the superficial and deep lymph vessels.

Anterior Tibial Lymph Node

The **anterior tibial lymph node** is an inconstant node found along the upper part of anterior tibial artery. When present, it receives the afferents from the anterior compartment of the leg, and its efferents pass into the popliteal nodes.

LYMPHATICS

Like lymph nodes, the lymph vessels of lower limb are also classified into two groups: superficial and deep.

SUPERFICIAL LYMPHATICS

The superficial lymphatics are larger and more numerous than the deep lymphatics. They run in the superficial fascia. They form two main streams: (a) most of them *(the main stream)* follow the great saphenous vein, and drain into the lower vertical group of superficial inguinal lymph nodes, and (b) the remaining one *(accessory stream)* follows the small saphenous vein, and drains into the popliteal lymph nodes.

DEEP LYMPHATICS

The deep lymphatics are smaller and fewer than the superficial lymphatics. They drain all the structures lying deep to the deep fascia. They run along the main blood vessels of the lower limb and drain into the deep inguinal nodes, either directly or indirectly through the popliteal nodes.

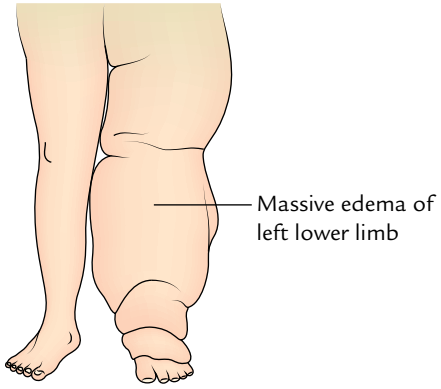


Fig. 33.7 Elephantiasis of the left lower limb.

N.B. The deep lymphatics from the gluteal region and upper part of the back of thigh accompany the gluteal vessels and drain into the internal iliac nodes.

Clinical correlation

- **Elephantiasis (Fig. 33.7):** The lymph vessels of the lower limb are often blocked, particularly in the endemic areas, by the microfilarial parasites (*Wuchereria bancrofti*). This causes massive edema of the lower limb producing a clinical condition called **elephantiasis**. In this condition, there is hypertrophy of the skin and subcutaneous tissue to an unusual proportion.
- **Enlargement of the inguinal lymph nodes:** It is the commonest cause of swelling in the subinguinal region. The common causes of enlargement of these lymph nodes is infection, boil or abscess in the drainage area. The other causes are filariasis, Hodgkin's disease, etc.

  The enlarged inguinal nodes should be differentiated from *ectopic testis*, *femoral hernia*, and *psoas abscess*.

Golden Facts to Remember

► Longest vein in the body	Great saphenous vein
► Most preferred site of venesection (cut-down)	Great saphenous vein in front of medial malleolus
► Varicose vein	Dilated tortuous vein
► Commonest cause of swelling in the subinguinal area	Enlargement of inguinal lymph nodes
► Lymph node of Cloquet or Rosenmüller	Deep inguinal lymph node present in the femoral canal
► Most of the lymph from the lower limb is drained into	Lower vertical group of superficial inguinal lymph nodes
► Elephantiasis	Massive edema (lymph edema) of lower limb due to blockage of its lymphatics by microfilarial parasites
► Vein of Leonardo da Vinci	Posterior arch vein, a tributary of great saphenous vein
► Commonest cause of varicosities of veins in the lower limb	Incompetency of valves in perforating veins, or superficial veins, or both

Clinical Case Study



A 42-year-old man visited hospital with complaints of chronic dull pain in his both legs. On examination, the doctor found dilated tortuous veins on the medial sides of both lower limbs. The skin in front of medial malleolus was discoloured, dry, and scaly. He was diagnosed as a case of "**varicose veins**."

Questions

1. Define varicose veins.
2. Which is the vein commonly involved in varicosity?
3. Give two important causes of varicosity of great saphenous vein.
4. Give the cause for discoloured, dry, and scaly skin in front of medial malleolus.
5. What are the complications of varicose veins?
6. Give the anatomical basis of varicose ulcers.
7. What will be the effects if valves in perforating veins, and saphenofemoral valve become incompetent?

Answers

1. Dilated tortuous veins.
2. Great saphenous veins.
3. (a) Incompetency of valves in the perforating veins (perforators), and (b) incompetency of valve at the saphenofemoral junction.
4. Due to less oxygen supply to the skin. The venous stasis in the varicose veins in front of medial malleolus, lowers the oxygen supply to the skin.
5. Thrombophlebitis and venous (varicose) ulcers.
6. The venous stasis affects the oxygen supply of the skin, which initially becomes dry, discoloured, and later sloughs off producing a varicose ulcer.
7. If valves in perforating veins become incompetent, there will be reversal of blood flow from deep to superficial veins leading to the dilatation and tortuosity of these veins and if saphenofemoral valve becomes incompetent, the blood will flow from femoral vein into the great saphenous vein making it dilated and tortuous.

CHAPTER

34 Innervation of the Lower Limb

In view of high incidence of neurological disorders of the lower limb, the innervation of the lower limb is discussed in a separate chapter for easy comprehension by the students and clinicians. This includes the description of major nerves, cutaneous innervation, segmental innervation, and sympathetic innervation of the lower limb.

NERVES OF THE LOWER LIMB

The nerves of the lower limb are derived from the ventral (anterior primary) rami of the lumbar and sacral nerves forming the **lumbar plexus** (L1–L4) in the posterior abdominal wall and the **sacral plexus** (L4–S4) in the pelvis.

The main nerves of the lower limb are as follows:

1. Femoral nerve.
2. Obturator nerve.
3. Sciatic nerve.
4. Tibial nerve.
5. Common peroneal nerve.
6. Superficial peroneal nerve.
7. Deep peroneal nerve.

The last four nerves are actually direct or indirect branches of the sciatic nerve.

The study of these nerves is important because of their frequent involvement in various injuries and peripheral neuropathies.

FEMORAL NERVE (Fig. 34.1)

The **femoral nerve** is the nerve of anterior compartment of the thigh. It arises within the psoas major muscle from the posterior divisions of the L2–L4 ventral rami in the abdomen. It descends through the psoas major and emerges on its lateral border to pass between the psoas and iliacus to enter the thigh behind the inguinal ligament and lateral to the femoral sheath. In femoral triangle, it splits into anterior and posterior divisions, 2 cm distal to the inguinal ligament. The two divisions straddle the lateral circumflex femoral artery. Its motor branches supply iliacus in the abdomen and all the muscles of anterior compartment of the thigh. Its cutaneous branches supply the large cutaneous area on the anterior and medial aspect of the thigh, medial side of leg, and foot. It also gives articular branches to the hip and knee joints. The femoral nerve is described in detail in Chapter 22, p. 340.

Clinical correlation

- **Injury of the femoral nerve:** It is rare but may be injured by a stab, gunshot wounds, or a pelvic fracture. The following are the characteristic clinical features:
  - (a) *Motor loss*
    - – Weak flexion of the thigh, due to paralysis of the iliacus and sartorius muscles.
    - – Inability to extend the knee, due to paralysis of the quadriceps femoris.
  - (b) *Sensory loss*
    - – Sensory loss over the anterior and medial aspects of the thigh, due to involvement of the intermediate and lateral cutaneous nerves of the thigh.
    - – Sensory loss on the medial side of the leg and foot up to the ball of the great toe (first metatarsophalangeal joint), due to involvement of the saphenous nerve.
- **Femoral nerve neuropathy:** The main trunk of the femoral nerve is not subject to an *entrapment neuropathy* but it may be compressed by the retroperitoneal tumors. A localized neuropathy of the femoral nerve may occur in diabetes mellitus. The following are the characteristic clinical features:
  - (a) Wasting and weakness of quadriceps leading to considerable difficulty in walking.
  - (b) Pain and paraesthesia on the anterior and medial aspects of the thigh extending down along the medial aspect of the leg and foot along the distribution of the saphenous nerve.

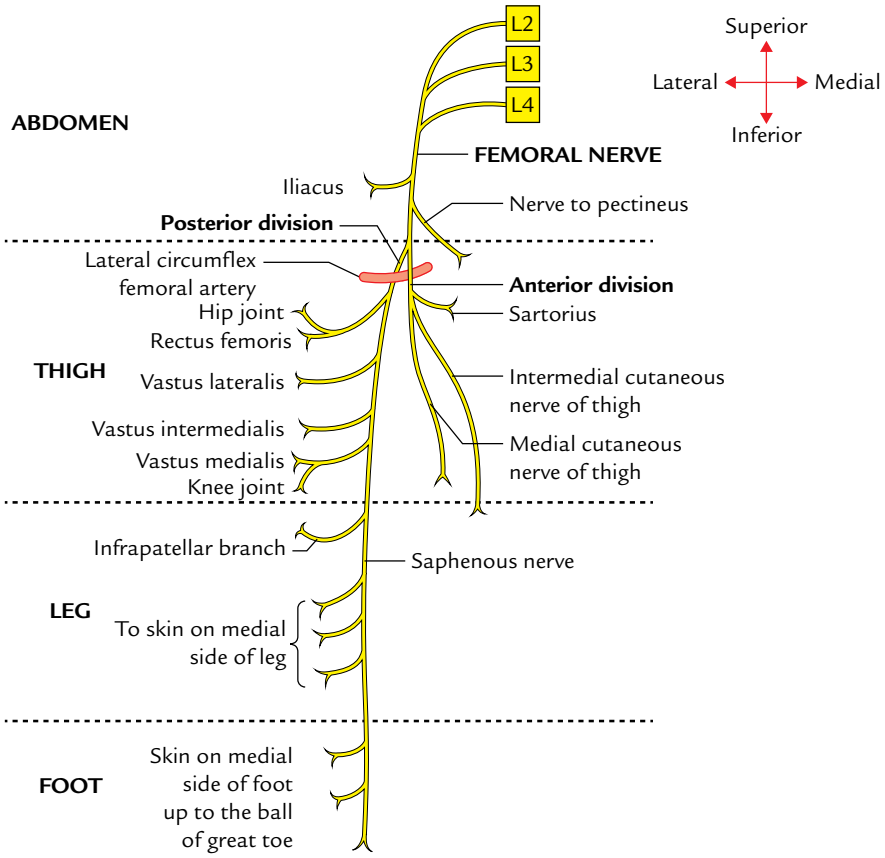


Fig. 34.1 Summary of branches of the femoral nerve. Note all the muscles on the front of thigh are supplied by posterior division except sartorius which is supplied by the anterior division.

OBTURATOR NERVE

The obturator nerve is the nerve of adductor compartment of the thigh. It arises within the psoas major from anterior divisions of the ventral rami of L2–L4 spinal nerves (Fig. 34.2). The nerve descends in the psoas major and emerges from its medial border on the ala of sacrum. It then descends along the lateral wall of the lesser pelvis on the obturator internus and passes through the upper anterior part of the obturator foramen to enter the adductor (medial) compartment of the thigh. Near the obturator foramen it divides into anterior and posterior divisions which straddle the adductor brevis muscle. Its motor branches supply all the muscles of the adductor compartment of the thigh. Its sensory branches supply cutaneous area on the lower-half of the medial aspect of the thigh. It also gives the articular branches to the hip and knee joints. The obturator nerve is described in detail in Chapter 23, p. 347.

Clinical correlation

- **Injury of the obturator nerve:** The obturator nerve may be injured in the anterior dislocation of the hip joint, or during radical retropubic prostatectomy. The following are the characteristic clinical features:
  (a) *Motor loss:* Loss of adduction of the thigh, due to paralysis of adductor muscles of the thigh.
  (b) *Sensory loss:* Sensory loss on the medial aspect of thigh, due to involvement of the cutaneous branch of the anterior division of the obturator nerve.
- **Obturator nerve neuropathy:** The syndrome of an obturator nerve entrapment causing the medial thigh pain is described in athletes with large adductor muscles.
- **Surgical division of the obturator nerve:** It is sometimes done to relieve the spasm of adductor muscles in the *spastic paralysis*.
- **Irritation of the obturator nerve:** The inflammation of the ovary causes localized peritonitis in the region of ovarian fossa which may cause irritation of the obturator nerve. In such a case, the pain may be referred to the hip, knee, and medial side of the thigh.
- **Referred pain:** In diseases of the hip joint, the pain may be referred to the medial side of the thigh.

SCIATIC NERVE (Fig. 34.3)

The sciatic nerve (thickest nerve in the body) is the nerve of posterior compartment of the thigh. It arises in the pelvis

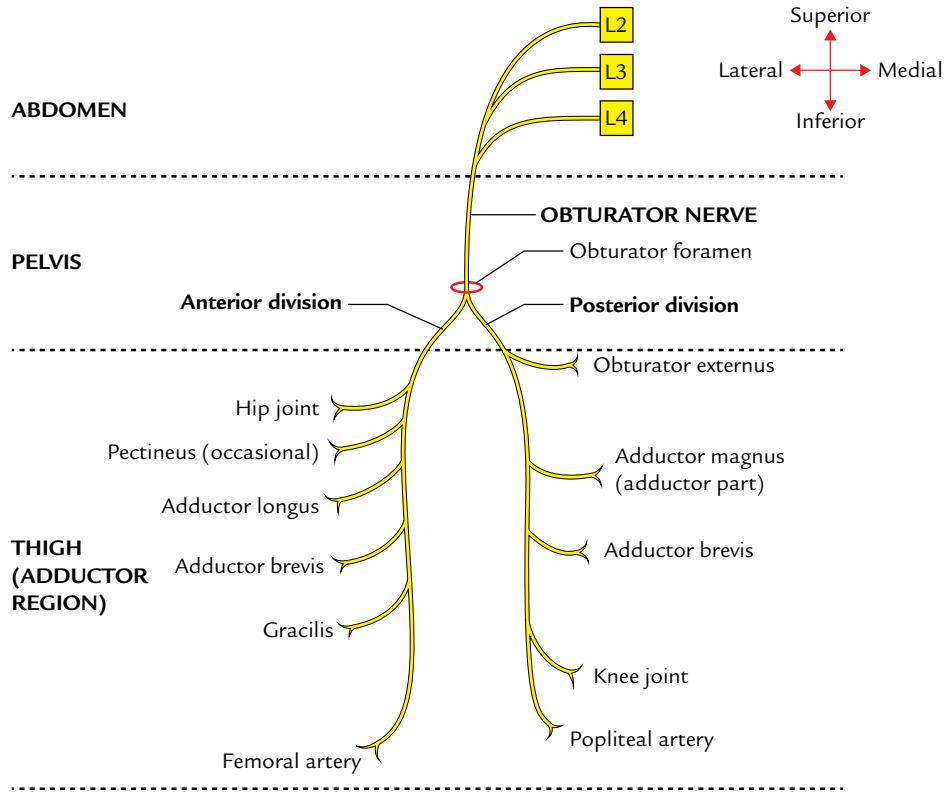


Fig. 34.2 Summary of main branches of the obturator nerve.

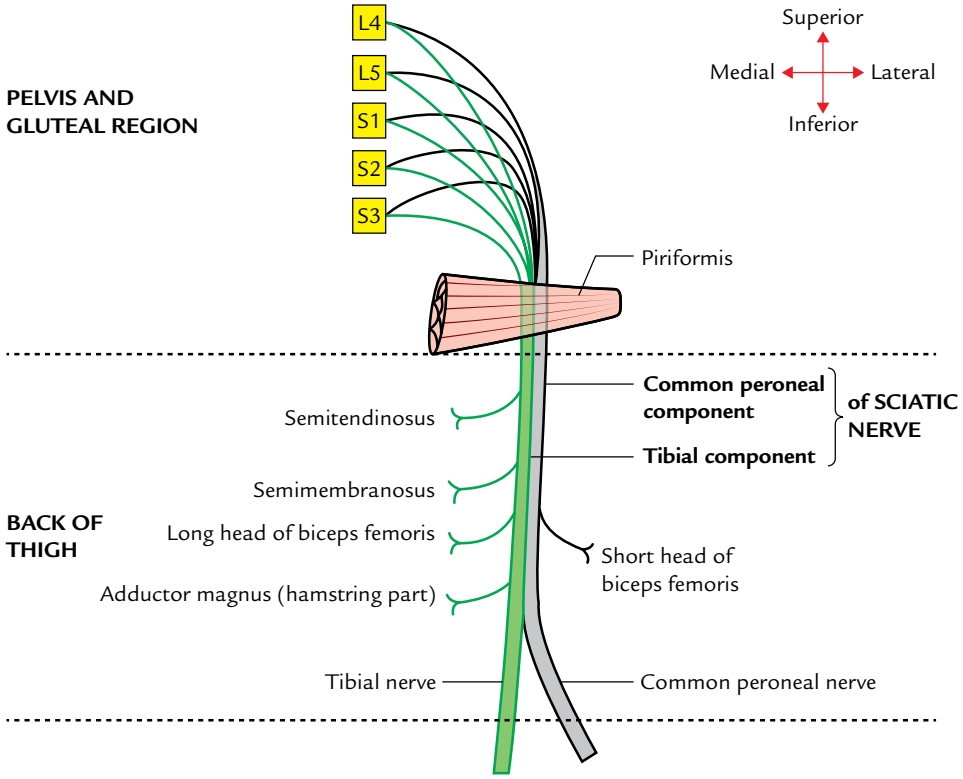


Fig. 34.3 Summary of branches of the sciatic nerve in the thigh. Ventral divisions of ventral rami are shown by green colour and dorsal division of ventral rami by grey colour. Note all the muscles on the back of thigh are supplied by tibial component of sciatic nerve except short head of biceps femoris which is supplied by its common peroneal component.

from ventral rami of L4–S3 spinal nerves. It leaves the pelvis through greater sciatic foramen below piriformis to enter the gluteal region. Here it descends between the greater trochanter and ischial tuberosity along the back of the thigh. A little above the popliteal fossa (junction of middle and lower thirds of the thigh), it divides into terminal tibial and common peroneal nerves.

Distribution

The distribution of sciatic nerves is as follows:

1. **Motor**
   (a) The motor branches from undivided trunk supply all the muscles of the back of thigh, which extend the hip and flex the knee.
   (b) The motor branches of the tibial nerve supply the muscles of the leg, which flex the leg and muscles of the foot, which plantar flex the foot.
   (c) The motor branches of the common peroneal nerve supply muscles of the leg which dorsiflex the foot.
2. **Sensory**
   (a) The sensory branches of the tibial nerves supply the skin on the back of the calf, medial and lateral sides of the heel, lateral border of the foot, and whole of the sole.
   (b) The sensory branches of the common peroneal nerve supply the skin on the anterolateral and lateral surfaces of the leg and whole of dorsum of the foot except the area which is supplied by the deep peroneal nerve.

Clinical correlation

- **Injury of the sciatic nerve:** The sciatic nerve may be injured by penetrating wounds, posterior dislocation of the hip, fracture of the pelvis, total hip replacement surgery (1%), or misplaced therapeutic injection in the gluteal region (most common cause). The following are the characteristic clinical features:
  (a) *Motor loss*
    – Inability to extend the thigh and flex the knee, due to paralysis of the hamstring muscles.
    – Loss of all movements below the knee with foot drop, due to paralysis of all the muscles of the leg and foot.
    The motor loss leads to flail foot which leads to great difficulty in walking. The patient walks with high stepping gait.
  (b) *Sensory loss:* The sensory loss on the back of the thigh and whole of the leg and foot except the area innervated by the saphenous nerve, due to involvement of the cutaneous nerves derived from the tibial and common peroneal nerves.
- **Sciatic nerve neuropathy:** As the sciatic nerve leaves the pelvis, sometimes, it passes through the piriformis muscle and at that point, it may become entrapped leading to **piriformis syndrome** (see page 362). It is a common anatomical variant but an extremely rare entrapment neuropathy.
- **Sciatica:** It is a term applied to a clinical condition characterized by shooting pain felt along the course of distribution of the sciatic nerve (e.g., buttock, posterior aspect of thigh, lateral aspect of leg, and dorsum of the foot). It occurs due to compression and irritation of L4–S3 spinal nerve roots by herniated intervertebral disc of the lumbar vertebrae.

TIBIAL NERVE (Fig. 34.4)

The **tibial nerve** is the larger terminal branch of the sciatic nerve. It arises above the popliteal fossa and passes downward successively through the middle of popliteal fossa and posterior compartment of the leg, and then enters the sole of the foot by passing deep to the flexor retinaculum where it divides into the **medial enters the side of the foot by passing and lateral plantar nerves.** Its motor branches supply all the muscles of the posterior compartment of the leg directly and all the muscles of the sole through its terminal branches—the medial and lateral plantar nerves. Its sensory branches through the medial and lateral plantar nerves supply whole of the skin of the sole of foot and toes including dorsal aspects of their last phalanges.

Clinical correlation

- **Effects of injury of the tibial nerve:** The tibial nerve may be injured by a lacerated wound in the popliteal fossa or posterior dislocation of the knee joint. The characteristic clinical features are as follows:
  (a) *Motor loss:*
    – Foot is held dorsiflexed and everted, due to paralysis of the muscles of posterior compartment of the leg.
    – Loss of prominence of calf and tendocalcaneus, due to paralysis of the triceps surae muscle (gastrocnemius and soleus).
    – Loss of plantar flexion of foot, due to paralysis of the flexors of ankle.
    – Inability to stand on the toes, due to loss of plantarflexion of foot.
  (b) *Sensory loss:* The loss of sensation in the sole and plantar aspects of the toes including the dorsal aspects of their distal phalanges, due to involvement of the cutaneous branches.
- **Tarsal tunnel syndrome:** It occurs due to compression of the tibial nerve in the osseofibrous tunnel under the flexor retinaculum of the ankle. It clinically presents as pain and paresthesia in the sole of the foot, which often becomes worse at night.
- **Morton's metatarsalgia (also called plantar digital neuroma):** It occurs due to the formation of a neuroma following pressure on one of the plantar digital nerves just

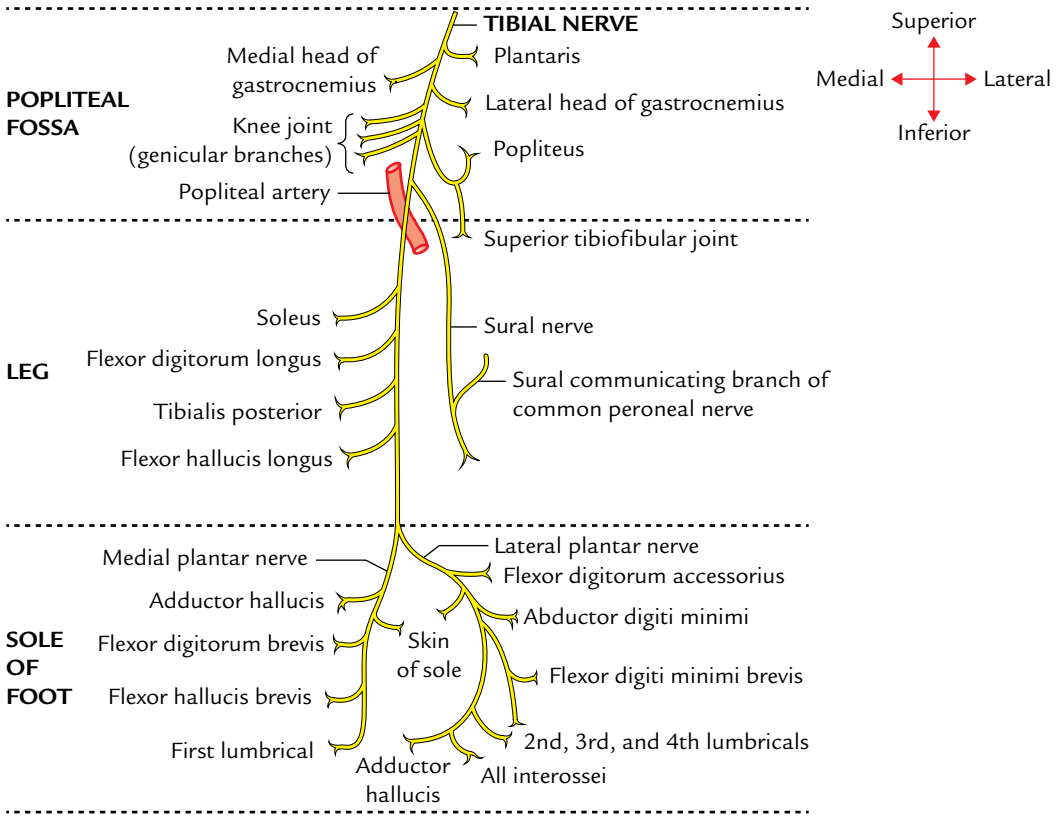


Fig. 34.4 Summary of main branches of the *tibial nerve*. Note all the muscular branches in the popliteal fossa arise from lateral side except for the medial head of gastrocnemius.

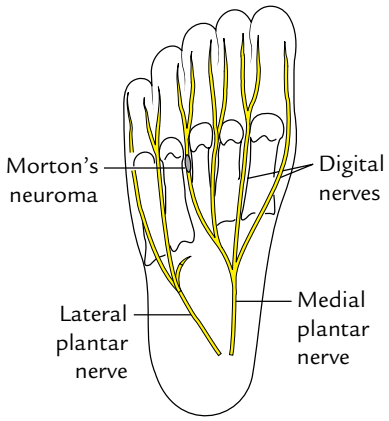


Fig. 34.5 Morton's neuroma leading to Morton's neuralgia.

prior to its bifurcation at one of the toe clefts. It most commonly affects the plantar digital nerve running between the third and fourth metatarsal heads to the third web-space (Fig. 34.5). Clinically, it presents as intermittent pain on the plantar aspect of the forefoot usually between the third and fourth metatarsals.

COMMON PERONEAL NERVE (Fig. 34.6)

The **common peroneal nerve** is the smaller terminal branch of the sciatic nerve. It arises in the lower third of the thigh just above the popliteal fossa. It passes into the popliteal fossa along its upper lateral boundary just beneath the edge of the biceps tendon. Now, it runs over plantaris and the lateral head of gastrocnemius. It runs over the fibular attachment of the soleus to wind around the lateral aspect of the neck of fibula to reach deep to peroneus longus where it divides into two terminal branches—**deep** and **superficial peroneal nerves.**

The **motor branches of the deep peroneal nerve** supply all the muscles of the anterior compartment of the leg including extensor digitorum brevis on the dorsum of the foot.

The **motor branches of the superficial peroneal nerve** supply all the muscles of the lateral compartment of the leg.

The sensory branches of the deep peroneal nerve supply the skin of cleft between the great and second toes.

The sensory branches of the superficial peroneal nerve supply most of the skin on the dorsum of foot except in the cleft between the great and second toes being supplied by the deep peroneal nerve. The lateral margin of the dorsum of foot including the lateral margin of little toe is supplied by the sural nerve. The medial margin of the dorsum of foot up

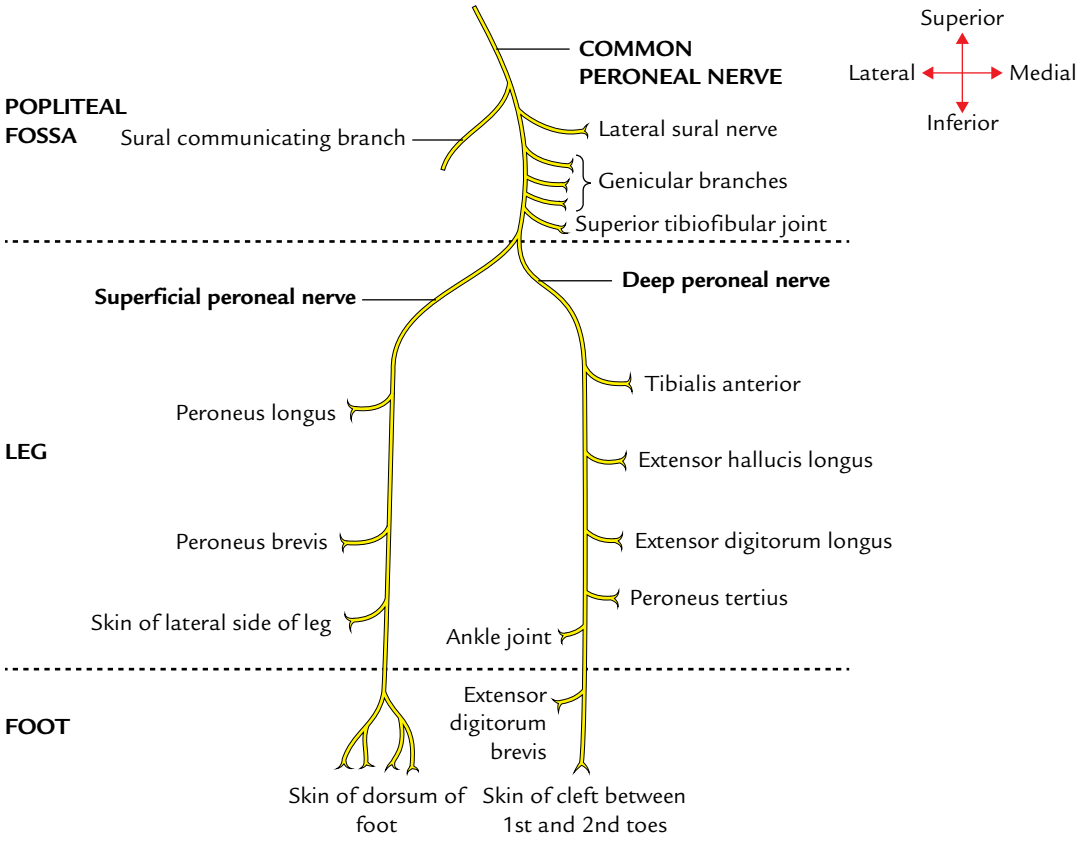


Fig. 34.6 Summary of main branches of the common peroneal nerve.

to the ball of the great toe is being supplied by the saphenous nerve.

Clinical correlation

- **Effects of injury to the common peroneal nerve:** The common peroneal nerve is extremely vulnerable to injury as it winds around the posterolateral aspect of the neck of the fibula. At this site it may be injured by the direct trauma, fracture neck of fibula, or tightly applied plaster cast. The characteristic clinical features are as follows (Fig. 34.7):
  (a) *Motor loss:*
    – **Foot drop**, due to the paralysis of muscles of the anterior compartment of the leg (dorsiflexors of the foot).
    – **Loss of extension of toes**, due to the paralysis of extensor digitorum longus and extensor hallucis longus.
    – **Loss of eversion of foot,** due to the paralysis of peroneus longus and peroneus brevis (evertors of the foot).
  (b) *Sensory loss:* The sensory loss due to involvement of the cutaneous branches, on the anterolateral aspect of the leg, and whole of dorsum foot except the areas supplied by the saphenous and sural nerves.

N.B. Due to paralysis of the dorsiflexors and evertors of the foot, the patient cannot stand on the heel. He has *high stepping gait*, in which foot is raised higher than the normal so that the toes do not hit the ground. In addition, if the foot is put down on the ground suddenly, it produces a slapping sound called *foot slap*.

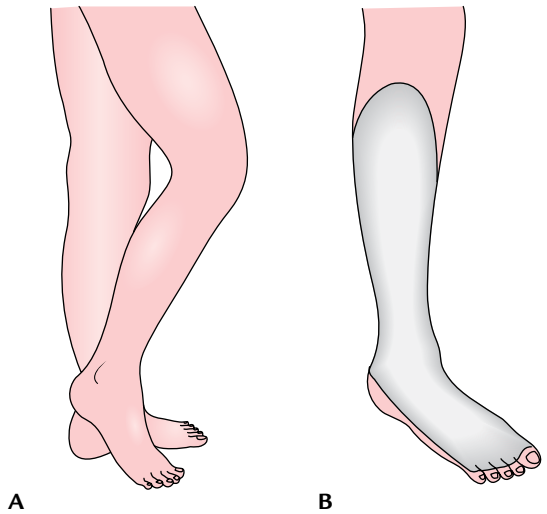


Fig. 34.7 Effects of injury of the common peroneal nerve: A, foot drop; B, sensory loss.

CUTANEOUS INNERVATION OF THE LOWER LIMB (Figs 34.8 and 34.9)

The cutaneous nerves supplying the lower limb are derived from the branches of nerves of lumbar and sacral plexuses except for some proximal unisegmental nerves arising from T12 or L1 spinal nerves. The most of the cutaneous innervation of the thigh is provided by the lateral and posterior cutaneous nerves of the thigh and cutaneous branches of the femoral nerve, the names of which describe their distribution. The anterior cutaneous nerves from the femoral nerve in addition to the anterior aspect of the thigh also supply the most of the medial aspect of the thigh. The cutaneous innervation of the leg on its anteromedial aspect is provided by the saphenous nerve, the posterolateral aspect by the sural nerve, and the anterolateral aspect by the superficial peroneal nerve. The cutaneous innervation of dorsum of the foot is mostly provided by the superficial peroneal nerves (Fig. 34.9A). The cutaneous innervation of the sole of the foot is provided by the cutaneous branches of the medial and lateral plantar nerves (Fig. 34.9B). The cutaneous nerves are described in detail with different regions of the lower limb.

SEGMENTAL INNERVATION OF THE LOWER LIMB (Figs 34.10 and 34.11)

SEGMENTAL INNERVATION OF THE SKIN (DERMATOMES)

The area of the skin supplied by a spinal nerve is termed **dermatome**.

The lower limb bud develops on the ventrolateral aspect of the body wall opposite L1–S3 spinal segments; hence, the skin of the lower limb is innervated by these segments of the spinal cord.

Since the lower limb develops from the ventrolateral aspect of the body wall, it is innervated only by the ventral rami of spinal nerves. The exception to this rule is the superomedial quadrant of the gluteal region, which is

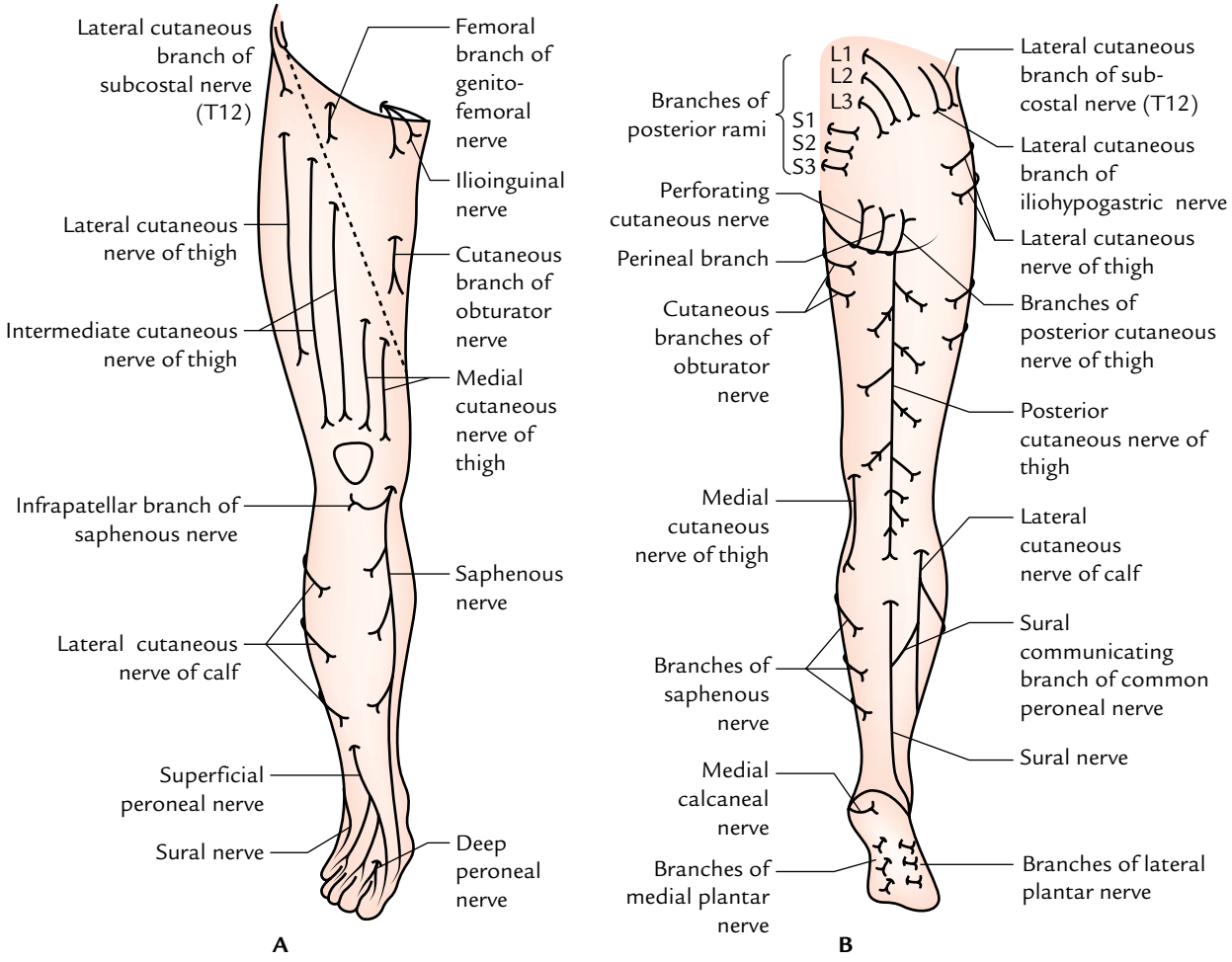


Fig. 34.8 Cutaneous nerves of the lower limb: A, on the anterior aspect; B, on the posterior aspect.

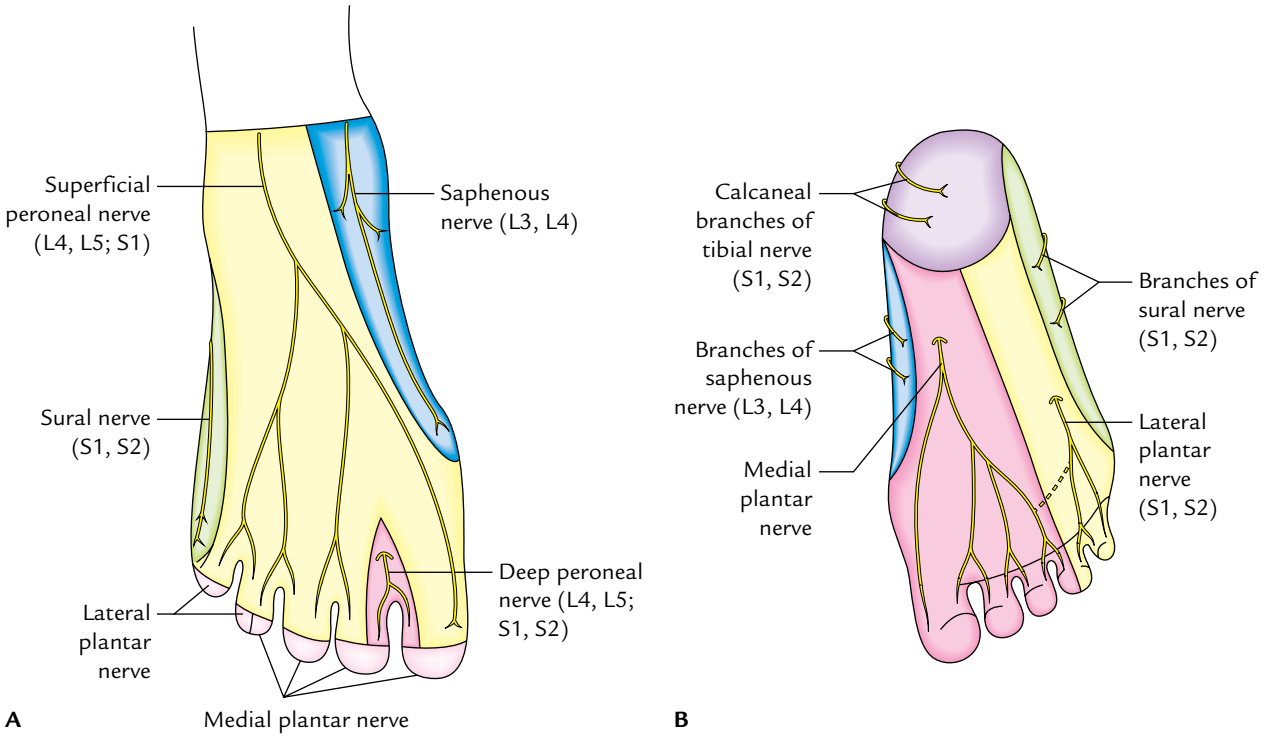


Fig. 34.9 Cutaneous nerves of the foot and their areas of distribution: **A,** dorsum of the foot; **B,** sole of the foot.

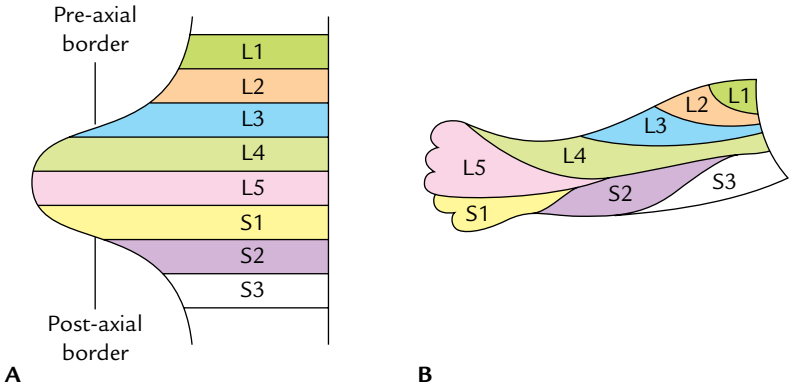


Fig. 34.10 Arrangement of dermatomes in the developing lower limb: **A,** simple dermatomal pattern, L1 spinal nerve supplying the pre-axial strip and S3 spinal nerve supplying the posterior strip; **B,** definitive dermatomal pattern of the lower limb bud.

innervated by the dorsal rami of L1–L3 and S1–S3 spinal nerves.

Initially the limb bud has **cephalic** and **caudal borders** called **pre-axial** and **post-axial borders,** respectively. The great toe is along the pre-axial border and the little toe is along the post-axial border. The arrangement of the dermatome pattern is simple at this stage; L1 spinal nerve supplies pre-axial strip of the skin and S3 spinal nerve supplies the post-axial strip of the skin (Fig. 34.10A).

Later the lower limb bud rotates medially by 90°. Therefore, the great toe comes to lie medially and the little toe lies laterally. As a result, the dermatomes are arranged in a sequence, above downward (L1–L4) along the *pre-axial border*, and below upward (S1–S3) along the *post-axial border* (Fig. 34.10B).

The middle three toes and adjoining area of the dorsum of foot and lateral side of the leg are supplied by L5 segment.

As the limb elongates the central dermatomes (L4, L5; S1) are pulled in such a way that they are represented only in the

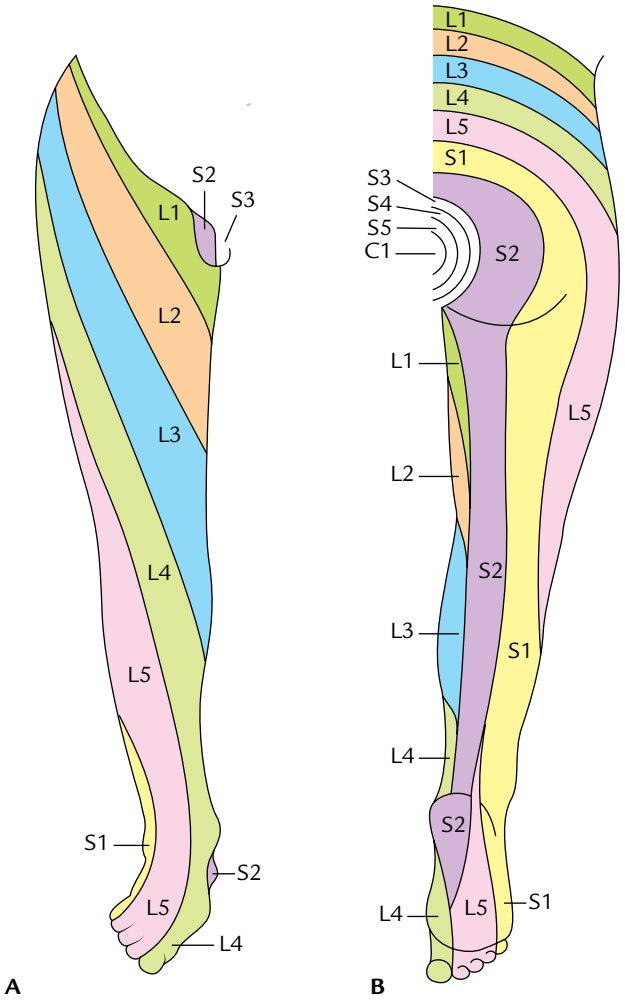


Fig. 34.11 Dermatomes of the lower limb: A, anterior aspect; B, posterior aspect.

distal part of the limb and are buried proximally. The line along which the central dermatomes are buried is called an **axial line**.

The segmental innervation of the skin in adults is shown in Figure 34.11 and is given in Table 34.1.

Table 34.1 Segmental innervation of the skin of the lower limb

Segments	Area supplied
L1, L2, L3	Anterior aspect of the thigh in sequence from above downward
L4	Medial aspect of the leg
L5	Lateral aspect of the leg and medial side of the foot
S1	Lateral side of the foot and sole
S2	Middle of back of the thigh and leg
S3, S4	Buttocks

Clinical correlation

In cases of the **paraplegia** (paralysis of lower limbs) clinicians can determine the site of lesion in the spinal cord (segments L1–S3) by performing the sensory examination for pain, touch, and temperature. This is so because sensory loss of the skin following injury to the spinal cord conforms to the dermatome. The upper limit of the sensory loss (dermatome) indicates the level of lesion.

SEGMENTAL INNERVATION OF THE MUSCLES

Most of the muscles are supplied by more than one segment of the spinal cord but only a few segments innervate the muscle predominantly. The damage of the predominant segment causes maximum paralysis. The segmental innervation of the muscles of the lower limb is given in Table 34.2.

The knowledge of these segmental values is of clinical importance in diagnosis of injuries of the nerves or to the spinal cord segments from which they arise.

SEGMENTAL INNERVATION OF THE JOINT MOVEMENTS

The four consecutive spinal segments regulate the movements of a particular joint in the limb. The upper two segments regulate the one movement and the lower two segments regulate the opposite movement, for a joint one segment distal in the limb, the centre (four consecutive spinal segments) lies in block one segment lower in the spinal cord.

The segmental innervation of the joints of lower limb is given in Table 34.3.

Table 34.2 Segmental innervation of the muscles of the lower limb

Segments	Muscles innervated
L1	Psoas major
L2, L3	Gracilis, pectineus, sartorius
L3, L4	Quadriceps femoris, adductors of the thigh
L4, L5	Tensor fasciae latae, tibialis anterior, tibialis posterior
L5; S1	Gluteus medius, gluteus minimus, extensor hallucis longus, extensor digitorum longus, peroneus longus, peroneus brevis, semitendinosus, semimembranosus
S1, S2	Gluteus maximus, biceps femoris, gastrocnemius, soleus
S2, S3	Flexor digitorum longus, flexor hallucis longus, interossei

Table 34.3 Segmental innervation of the joints of the lower limb

Joint	Four consecutive spinal segments (centre)	Movements
Hip joint	L2, L3, L4, L5	• Flexion, adduction, and medial rotation (L2, L3) • Extension, abduction, and lateral rotation (L4, L5)
Knee joint	L3, L4, L5; S1	• Extension (L3, L4) • Flexion (L5; S1)
Ankle joint	L4, L5; S1, S2	• Dorsiflexion (L4, L5) • Plantar flexion (S1, S2)

SYMPATHETIC INNERVATION OF THE LOWER LIMB

The sympathetic innervation of the lower limb is derived from the lower three thoracic and upper two lumbar (T10–L2) segments of the spinal cord.

The **preganglionic fibres** arising from the lateral horn cells pass out through ventral roots. Then they pass down in the sympathetic chain to relay in the lumbar and upper two or three sacral ganglia.

The **postganglionic fibres** arising from the lumbar ganglia pass through the femoral nerve to supply the femoral artery and its branches.

The postganglionic fibres arising from the sacral ganglia (S2–S3) pass through the tibial nerve to supply the popliteal artery and its branches.

The sympathetic stimulation causes dilatation of the blood vessels supplying skeletal muscles. However, the sympathetic fibres are vasomotor, sudomotor, and pilomotor to the skin.

N.B. The lower limbs do not have parasympathetic innervation.

Clinical correlation

Buerger's disease: It is an obliterative disorder of the lower limb arteries distal to the knee. It typically occurs in young male smokers. Clinically, it presents as **intermittent claudication**, digital ischaemia, and loss of the ankle pulses. The progressive ischaemia of digits may eventually require amputation of the foot. The treatment includes stopping smoking and sympathectomy.

The sympathetic denervation of the lower limb is achieved by removing second, third, and fourth lumbar ganglia with the intermediate chain (lumbar sympathectomy). It is important to note that the first lumbar ganglion is preserved because it controls the internal urethral sphincter. Its inadvertent removal leads to dry coitus.

Golden Facts to Remember

► Thickest nerve in the body	Sciatic nerve
► Commonest cause of the sciatic nerve injury	Iatrogenic (i.e., wrongly given injection in the gluteal region by the health professionals)
► Most commonly injured nerve in the lower limb	Common peroneal nerve
► Commonest site of the injury of common peroneal nerve	Neck of the fibula

Clinical Case Study



A 57-year-old patient was given a course of antibiotics by the intramuscular injections in his left gluteal region during his admission in the hospital. A week after discharge from the hospital, he experienced numbness and tingling sensations in his left lower limb down along the posterior aspect of the thigh and along the lateral sides of the leg and dorsum of the foot. On examination, the doctors found impaired skin sensation along the lateral side of the leg and the dorsum of the foot. When asked to walk his left foot tended to catch on steps. While sitting on the chair with feet off the ground his left foot assumed a plantar-flexed position (**foot drop**).

Questions

1. Name the nerve that is injured in this patient.
2. What is the commonest cause of injury to the sciatic nerve?
3. What are the two components of the sciatic nerve and tell which component is mostly affected by injury and why?
4. Give the anatomical basis of foot drop following the sciatic nerve injury?
5. What is sciatica? Give its common cause?

Answers

1. Sciatic nerve.
2. Misplaced intramuscular injection in the gluteal region.
3. The tibial and common peroneal components. The common peroneal component is most affected probably because its fibres lie most superficial in the sciatic nerve.
4. In sciatic nerve injury, all the muscles below the knee are paralyzed and the weight of foot causes it to assume the plantar-flexed position (foot drop).
5. Sciatica is a clinical condition in which patient complains of pain on the posterior aspect of the thigh, posterior and lateral sides of the leg, and lateral part of the dorsum of foot. It commonly occurs due to prolapse of an intervertebral disc pressing one or more roots of the sciatic nerve.

Multiple Choice Questions

CHAPTER 1

1. **Select the *incorrect statement* about the boundaries of the abdominal cavity proper:**
   (a) It is bounded superiorly by the diaphragm
   (b) It is bounded inferiorly by the perineum
   (c) It is bounded posteriorly by the posterior abdominal wall
   (d) It is open inferiorly

2. **All of the following abdominal viscera are protected by the thoracic cage *except*:**
   (a) Liver
   (b) Spleen
   (c) Pancreas
   (d) Stomach

3. **Select the *incorrect statement* about the abdominal cavity proper:**
   (a) It is under cover of the thoracic cage superiorly
   (b) It is continuous inferiorly with the pelvic cavity
   (c) It is partially protected inferiorly by the false pelvis
   (d) It is located between the diaphragm and pelvic outlet

4. **All of the following abdominal viscera are partly protected by greater pelvis *except*:**
   (a) Ileum
   (b) Caecum
   (c) Sigmoid colon
   (d) Rectum

5. **The abdominal cavity extends superiorly into the osseocartilaginous thoracic cage up to:**
   (a) 3rd intercostal space
   (b) 4th intercostal space
   (c) 5th intercostal space
   (d) 6th intercostal space

Answers

1. *b*, 2. *c*, 3. *d*, 4. *d*, 5. *b*

CHAPTER 2

1. **The angle between the anterior surface of L5 vertebra and pelvic surface of S1 vertebra (lumbosacral angle) is:**
   (a) 90°
   (b) 120°
   (c) 180°
   (d) 210°

2. **The smooth medial part of ala of sacrum is related to all *except*:**
   (a) Lumbosacral trunk
   (b) Femoral nerve
   (c) Iliolumbar artery
   (d) Sympathetic chain

3. **Regarding anatomical position of hip bone all are correct *except*:**
   (a) Symphyseal surfaces of pubic bones lie in the coronal plane
   (b) Pubic tubercle and anterior superior iliac spine lie in the same coronal plane
   (c) Linea terminalis of pelvis lies at an angle of about 70° to the horizontal plane
   (d) Anterior superior iliac spine lies at the level of the sacral promontory

4. **All are the components of linea terminalis *except*:**
   (a) Pubic crest
   (b) Pubic tubercle
   (c) Arcuate line of ilium
   (d) Promontory of sacrum

5. **The median relations of pelvic surface of the sacrum include all *except*:**
   (a) Median sacral vessels

(b) Sympathetic trunks
(c) Middle rectal artery
(d) Rectum

6. **All are contents of sacral canal *except*:**
   (a) Conus medullaris
   (b) Filum terminale
   (c) Lower part of cauda equina
   (d) Spinal meninges

7. **All structures emerge through the sacral hiatus *except*:**
   (a) Fifth sacral nerves
   (b) Coccygeal nerves
   (c) Filum terminale
   (d) Spinal meninges

8. **Select the *incorrect statement* about the coccyx:**
   (a) Its pelvic surface is related to ganglion impar
   (b) Its dorsal surface provides attachment to filum terminale
   (c) Its base articulates with the apex of the sacrum
   (d) Its pelvic surface provides attachment to sphincter ani externus muscle

Answers

1. *b*, 2. *b*, 3. *a*, 4. *b*, 5. *c*, 6. *a*, 7. *d*, 8. *d*

CHAPTER 3

1. **The membranous layer of superficial fascia of anterior abdominal wall is continuous with all *except*:**
   (a) Buck's fascia of penis
   (b) Dartos muscle of scrotum
   (c) Fascia of Gallaudet
   (d) Colles' fascia of perineum

2. **Which of the following structures lie in between the superficial and deep layers of superficial fascia of the anterior abdominal wall?**
   (a) Superior epigastric artery
   (b) Inferior epigastric artery
   (c) Superficial epigastric artery
   (d) Lower five intercostal nerves

3. **All the statements about external oblique muscle are correct *except*:**
   (a) Its lower-free border forms the inguinal ligament
   (b) Its aponeurosis takes part in the formation of rectus sheath
   (c) It forms the boundary of lumbar (Petit's) triangle
   (d) It takes origin from the outer lip of the iliac crest

4. **The neurovascular plane in the anterior abdominal wall lies between:**
   (a) External and internal oblique muscles
   (b) Internal and transversus abdominis muscles
   (c) Transversus abdominis muscle and fascia transversalis
   (d) Fascia transversalis and parietal peritoneum

5. **Anterior wall of rectus sheath below the arcuate line is formed by the aponeurosis of:**
   (a) External oblique
   (b) Internal oblique
   (c) Transversus abdominis
   (d) All of the above

6. **Muscle whose aponeurosis forms both anterior and posterior walls of rectus sheath is:**
   (a) External oblique
   (b) Internal oblique
   (c) Transversus abdominis
   (d) None of the above

7. **All are the contents of rectus sheath *except*:**
   (a) Superior epigastric artery
   (b) Rectus abdominis
   (c) Superficial epigastric artery
   (d) Inferior epigastric artery

8. **Select the *incorrect statement* about the pyramidalis:**
   (a) It lies superficial to the rectus abdominis
   (b) It is tensor of linea alba
   (c) It arises from the body of pubis
   (d) It is supplied by the genital branch of the genitofemoral nerve

Answers

1. *c*, 2. *c*, 3. *d*, 4. *b*, 5. *d*, 6. *b*, 7. *c*, 8. *d*

CHAPTER 4

1. **All of the following ligaments are derived from inguinal ligament *except*:**
   (a) Lacunar ligament
   (b) Interfoveolar ligament
   (c) Pectineal ligament
   (d) Reflex inguinal ligament

2. **All the statements about cremaster muscle are correct *except*:**
   (a) It consists of loops of skeletal muscle fibre, derived from internal oblique muscle
   (b) Its function is to pull the testis up toward the superficial inguinal ring
   (c) It is supplied by ilioinguinal nerve
   (d) Its action is not under voluntary control

3. **Cremasteric reflex tests the integrity of spinal segments:**
   (a) T11 and T12
   (b) L1 and L2
   (c) L3 and L4
   (d) S1 and S2

4. **Select the *incorrect statement* about the conjoint tendon:**
   (a) It is formed by the fusion of aponeurosis of internal oblique and transversus abdominis
   (b) It forms the medial half of the posterior wall of the inguinal canal
   (c) It is attached to the pubic crest and pecten pubis
   (d) It is formed by the fusion of aponeurosis of external and internal oblique muscles

5. **All the statements about Hesselbach's triangle are correct *except*:**
   (a) It lies superficial to the inguinal canal
   (b) It is bounded laterally by inferior epigastric artery
   (c) It is bounded medially by lateral border of rectus abdominis
   (d) It is bounded inferiorly by the inguinal ligament

6. **The midinguinal point is a point midway between:**
   (a) Pubic tubercle and anterior superior iliac spine
   (b) Pubic tubercle and iliac tubercle
   (c) Pubic symphysis and anterior superior iliac spine
   (d) Pubic symphysis and iliac tubercle

7. **Regarding inguinal canal all the statements are true *except*:**
   (a) It is an intermuscular slit in the lower part of the anterior abdominal wall
   (b) It is about 4 cm long
   (c) It provides passage to spermatic cord
   (d) The ilioinguinal nerve enters it through deep inguinal ring

8. **All the statements about indirect inguinal hernia are true *except*:**
   (a) The hernial sac enters the inguinal canal through deep inguinal ring
   (b) It commonly occurs due to persistence of the processus vaginalis
   (c) The neck of hernial sac lies medial to the inferior epigastric vessels
   (d) It is more in young adults than in elderly individuals

9. **Select the *incorrect statement* about the direct inguinal hernia:**
   (a) The hernial sac enters the inguinal canal pushing the posterior wall of the inguinal canal
   (b) The neck of hernial sac lies medial to the inferior epigastric vessels
   (c) The neck of direct inguinal hernia is narrow as compared to that of indirect inguinal hernia
   (d) It is less commonly strangulated as compared to the indirect inguinal hernia

Answers

1. *b*, 2. *c*, 3. *b*, 4. *d*, 5. *a*, 6. *c*, 7. *d*, 8. *c*, 9. *c*

CHAPTER 5

1. **Select the *incorrect statement* about the penis:**
   (a) It is a copulatory organ in male
   (b) Its root lies in the superficial perineal pouch
   (c) It is traversed by the urethra
   (d) It is enclosed in the deep fascia

2. **All are true about the corpus spongiosum of penis *except*:**
   (a) It forms the glans penis
   (b) It forms the bulb of penis
   (c) It lacks cavernous spaces
   (d) It is traversed by the urethra

3. **Lymph from the glans penis is drained into:**
   (a) External iliac lymph nodes
   (b) Internal iliac lymph nodes
   (c) Deep inguinal lymph nodes
   (d) Superficial inguinal lymph nodes

4. **All of the following factors play a role in the erection of penis *except*:**
   (a) Psychogenic/cutaneus stimulation
   (b) Parasympathetic stimulation
   (c) Filling of blood in cavernous spaces of corpora cavernosa by the helicine arteries
   (d) Sympathetic stimulation

5. **Layer of the anterior abdominal wall that does not continue in the scrotal wall is:**
   (a) Skin
   (b) External oblique muscle
   (c) Internal oblique muscle
   (d) Transversus abdominis muscle

6. **The innermost covering of the testis is formed by:**
   (a) Parietal layer of tunica vaginalis
   (b) Visceral layer of tunica vaginalis
   (c) Tunica vasculosa
   (d) Tunica albuginea

7. **All of the following statements about testis are true *except*:**
   (a) It develops in the abdominal cavity from a genital ridge derived from the intermediate mesoderm

(b) Testicular pain is carried to T10 and T11 spinal segments through sympathetic fibres
(c) Lymph from the testis is drained into pre- and para-aortic lymph nodes
(d) Testicular veins drain into the renal veins of the corresponding side

8. **Testis normally descends into the lower part of the scrotum at the age of:**
(a) Seventh month of intrauterine life
(b) Eighth month of intrauterine life
(c) Ninth month of intrauterine life
(d) First year of postnatal life

9. **All of the following statements about epididymis are correct *except*:**
(a) It consists of head, body, and tail
(b) Its duct develops from the mesonephric duct
(c) Its duct continues below with the ductus deferens
(d) Its body and tail are applied on to the posteromedial aspect of the testis

10. **The embryological remnants found in relation to the testis and epididymis include all *except*:**
(a) Appendix of the testis
(b) Organ of Giraldes
(c) Paraoöphoron
(d) Appendix of the epididymis

Answers

1. *d*, 2. *c*, 3. *c*, 4. *d*, 5. *d*, 6. *c*, 7. *d*, 8. *c*, 9. *d*, 10. *c*

CHAPTER 6

1. **Select the *incorrect statement* about the peritoneum:**
(a) It is the largest serous membrane in the body
(b) It is lined by endothelial cells
(c) In males it surrounds a closed sac, the peritoneal cavity
(d) In females it surrounds a sac, which has two openings at the ends of uterine tubes

2. **Regarding parietal peritoneum all the statements are correct *except*:**
(a) It forms the innermost layer of the abdominal wall
(b) It is composed of the mesothelial serous membrane
(c) It is insensitive to pain
(d) It develops from the somatopleuric mesoderm

3. **The epiploic foramen (of Winslow) is bounded by:**
(a) Caudate lobe of the liver superiorly
(b) First part of the duodenum inferiorly
(c) Portal vein posteriorly
(d) Right free margin of the lesser omentum anteriorly

4. **Select the *incorrect statement* about the mesentery:**
(a) It suspends the loops of jejunum and ileum from the posterior abdominal wall
(b) Its root extends from the duodenojejunal flexure to the ileocaecal junction
(c) Its root is 6 m long
(d) It crosses in front of the abdominal aorta

5. **Regarding lesser omentum, all the statements are correct *except*:**
(a) It extends between the lesser curvature of stomach and the inferior surface of the liver
(b) Superiorly it is attached to the margins of grooves for ligamentum teres and ligamentum venosum
(c) Its right free margin forms the anterior boundary of epiploic foramen
(d) It contains right and left gastric artery

6. **Select the *incorrect statement* about the sigmoid mesocolon:**
(a) It is a triangular fold of peritoneum suspending the sigmoid colon from pelvic wall
(b) Its apex lies at the division of left common iliac artery
(c) Its attachment extends inferiorly up to the level of S2 vertebra
(d) The left ureter lies behind its apex

7. **Regarding falciform ligament all the statements are correct *except*:**
(a) It is a sickle-shaped fold of peritoneum
(b) It connects the inferior surface of the liver to the anterior abdominal wall
(c) Its concave inferior margin contains ligamentum teres
(d) Its convex superior margin is attached to the inferior surface of the diaphragm and anterior abdominal wall

8. **All are retroperitoneal/extraperitoneal organs *except*:**
(a) Kidneys
(b) Suprarenal glands
(c) Spleen
(d) Descending colon

Answers

1. *b*, 2. *c*, 3. *c*, 4. *c*, 5. *b*, 6. *c*, 7. *b*, 8. *c*

CHAPTER 7

1. **All statements regarding the stomach are correct *except*:**
(a) It is the most dilated part of the alimentary tract
(b) It lies mainly in the left hypochondriac, epigastric, and umbilical regions

(c) Its most fixed part is the pylorus (gastroduodenal junction)
(d) Its most fixed part is the cardiac end (gastroesophageal junction)

2. **All of the following arteries supply the stomach *except*:**
   (a) Left gastric artery
   (b) Left epiploic artery
   (c) Right gastric artery
   (d) Short gastric artery

3. **All of the following structures form the stomach bed *except*:**
   (a) Body of pancreas
   (b) Splenic artery
   (c) Right crus and dome of the diaphragm
   (d) Transverse mesocolon

4. **All the structures forming the stomach bed are separated from the stomach by the lesser sac *except*:**
   (a) Left kidney
   (b) Left suprarenal gland
   (c) Left colic flexure
   (d) Spleen

5. **Select the *incorrect statement* about the gastric canal:**
   (a) It is located along the lesser curvature of the stomach
   (b) It is also called Magenstrasse
   (c) It is produced due to the absence of longitudinal fold of gastric mucosa
   (d) It is prone to ulceration

6. **Select the *incorrect statement* about the spleen:**
   (a) It is the largest lymphoid organ in the body
   (b) It filters the microbial agents from the circulation
   (c) It develops in the ventral mesogastrium
   (d) It is located in the left hypochondrium between the fundus of the stomach and diaphragm

7. **Regarding spleen all statements are true *except*:**
   (a) Its upper pole (medial end) lies 3/4 cm lateral to the spine of T10 vertebra
   (b) Its long axis corresponds to the long axis of the 10th rib
   (c) Its lower pole (lateral end) extends forward up to the midclavicular line
   (d) Its superior border presents one or two notches near its lateral end

8. **The hilum of spleen is located on its visceral surface in the region of:**
   (a) Renal impression
   (b) Gastric impression
   (c) Colic impression
   (d) Pancreatic impression

Answers

1. *c*, 2. *b*, 3. *c*, 4. *d*, 5. *c*, 6. *c*, 7. *c*, 8. *b*

CHAPTER 8

1. **Select the *incorrect statement* about the liver:**
   (a) It almost completely occupies right hypochondrium
   (b) It forms about 1/18th of total body weight in a newborn
   (c) It lies largely outside the thoracic cage
   (d) It weighs about 1–2 kg

2. **Factors keeping the liver in position include all *except*:**
   (a) Hepatic veins
   (b) Intra-abdominal pressure
   (c) Ligaments of liver
   (d) Hepatic pedicle

3. **All the statements about the bare area of the liver are correct *except*:**
   (a) It is the largest non-peritoneal area of the liver
   (b) It is located on the posterior surface of the right lobe
   (c) It is related to the right spleen
   (d) It is one of the sites of porta systemic anastomosis

4. **Select the *incorrect statement* about porta hepatis:**
   (a) It is a horizontal fissure on the inferior surface of the liver
   (b) The neck of gallbladder lies at its right end
   (c) The junction of the fissures for ligamentum teres and ligamentum lies at its left end
   (d) The right and left hepatic ducts enter the liver through it

5. **Which of the following structures lie anteriorly in the hepatic pedicle just above the termination of the cystic duct?**
   (a) Portal vein
   (b) Common hepatic duct
   (c) Right hepatic artery
   (d) Left hepatic artery

6. **Select the *incorrect statement* about the gallbladder:**
   (a) It lies in an elongated fossa on the inferior surface of the right lobe of liver
   (b) It can store up to 50 ml of bile
   (c) Its body is covered with peritoneum on all sides
   (d) It is distended in extrinsic obstruction of the common bile duct

7. **The total number of hepatic segments is:**
   (a) 6
   (b) 7

(c) 8
(d) 9

8. **The notable features on the visceral surface of the liver are all *except*:**
   (a) Gallbladder
   (b) Bare area of liver
   (c) Porta hepatis
   (d) Fissure of ligamentum teres hepatis

Answers

1. *c*, 2. *d*, 3. *c*, 4. *d*, 5. *b*, 6. *c*, 7. *c*, 8. *b*

CHAPTER 9

1. **All the statements about the duodenum are correct *except*:**
   (a) It is the most fixed part of the small intestine
   (b) It is about 25 cm long
   (c) It extends below the level of umbilicus
   (d) It produces an impression on the visceral surface of the liver

2. **Select the *incorrect statement* about the duodenum:**
   (a) It is a proximal C-shaped loop of the small intestine
   (b) It lies opposite L2, L3, and L4 vertebrae
   (c) It extends from the pylorus to the duodenojejunal flexure
   (d) It is the most deeply placed part of the gastrointestinal tract

3. **Which part of the duodenum is called duodenal cap by the radiologists?**
   (a) First part
   (b) Second part
   (c) Third part
   (d) Fourth part

4. **Regarding duodenum which of the following statements is *not correct?***
   (a) The gastroduodenal artery runs behind its first part
   (b) Above the opening of major duodenal papilla, it develops from foregut
   (c) Below the opening of major duodenal papilla, it develops from hindgut
   (d) The common bile duct opens into its second part

5. **All the statements about the pancreas are correct *except*:**
   (a) It lies transversely across the posterior abdominal wall behind the lesser sac
   (b) It is an exo-endocrine gland
   (c) Its ducts open into the second part of the duodenum
   (d) It is mostly intraperitoneal

6. **The posterior surface of the head of pancreas is related to all *except*:**
   (a) Inferior vena cava
   (b) Aorta
   (c) Terminal parts of renal veins
   (d) Bile duct

7. **Select the *incorrect statement* about the tail of pancreas:**
   (a) It lies within the lienorenal ligament
   (b) It comes in contact with the surface of the spleen
   (c) It has maximum concentration of islets of Langerhans
   (d) It is the most fixed part of the pancreas

8. **All the statements about the neck of pancreas are correct *except*:**
   (a) Its anterior surface is related to the pylorus
   (b) The portal vein is formed behind it
   (c) It lies behind the lesser sac
   (d) It presents an upward projection called tuber omentale

Answers

1. *c*, 2. *b*, 3. *a*, 4. *c*, 5. *d*, 6. *b*, 7. *d*, 8. *d*

CHAPTER 10

1. **Select the *incorrect statement* about jejunum:**
   (a) It forms the upper two-fifth of the mobile part of the small intestine
   (b) It begins at the duodenojejunal flexure
   (c) It has more Peyer's patches in its wall than ileum
   (d) Its mesentery presents windows

2. **All the statements regarding ileum are correct *except*:**
   (a) It forms distal three-fifth of the mobile part of the small intestine
   (b) It terminates at the ileocaecal junction
   (c) Its mesentery does not present windows
   (d) Its proximal part has maximum concentration of Peyer's patches

3. **Meckel's diverticulum is due to persistence of:**
   (a) Allantois
   (b) Vitello-intestinal duct
   (c) Left umbilical vein
   (d) Ductus venosus

4. **Select the *incorrect statement* about Meckel's diverticulum:**
   (a) It is located about 2 feet proximal to the ileocaecal junction
   (b) It is attached to the antimesenteric border
   (c) It is usually 20 cm long
   (d) It may cause intestinal obstruction

5. **The length of the large intestine is about:**
   (a) 1 m
   (b) 1.5 m
   (c) 2 m
   (d) 2.5 m

6. **All are characteristic features of the large intestine *except*:**
   (a) Appendices epiploicae
   (b) Sacculations
   (c) Villi
   (d) Taenia coli

7. **Select the *incorrect statement* about the caecum:**
   (a) It forms the commencement of the large intestine
   (b) It is ampullary type in most of the cases
   (c) It has greater breadth than length
   (d) It develops from the hindgut

8. **The caecum is supplied by:**
   (a) Ileocolic artery
   (b) Middle colic artery
   (c) Left colic artery
   (d) Right colic artery

9. **The appendix is devoid of all of the following *except*:**
   (a) Appendices epiploicae
   (b) Taenia coli
   (c) Sacculations
   (d) Mesentery

10. **Select the *incorrect statement* about the appendix:**
    (a) The base of appendix corresponds to McBurney's point
    (b) The appendicular artery is a branch of ileocolic artery
    (c) The appendix is innervated by T10 spinal segment
    (d) The commonest position of appendix is subcaecal

Answers

1. *c*, **2.** *d*, 3. *b*, 4. *c*, 5. *a*, 6. *c*, **7.** *d*, 8. *a*, 9. *d*, 10. *d*

CHAPTER 11

1. **Select the *incorrect statement* about the kidneys:**
   (a) They are major excretory organs of the body
   (b) They lie on the posterior abdominal wall opposite T12–L3 vertebrae
   (c) Their upper poles are about 5 cm away from the midline
   (d) Their hila are crossed by the transpyloric plane

2. **All the muscles are related to the posterior surface of the kidney *except*:**
   (a) Transversus abdominis
   (b) Internal oblique
   (c) Psoas major
   (d) Quadratus lumborum

3. **Anterior surface of the left kidney is related to all *except*:**
   (a) Stomach
   (b) Pancreas
   (c) Left colic flexure
   (d) Left crus of diaphragm

4. **In renal colic, the pain is referred from the loin to groin mainly through the spinal segments:**
   (a) T9–T12
   (b) T10–L1
   (c) T11–L2
   (d) T12–L3

5. **The length of each ureter is about:**
   (a) 15 cm
   (b) 20 cm
   (c) 25 cm
   (d) 30 cm

6. **All structures form anterior relations of the left ureter *except*:**
   (a) Left colic vessels
   (b) Ileocolic vessels
   (c) Left vessels
   (d) Sigmoid mesocolon

7. **All the statements are true about renal pelvis *except*:**
   (a) It is the upper expanded funnel-shaped part of the ureter
   (b) It receives major calyces
   (c) It lies anterior to renal vessels
   (d) It develops from the ureteric bud

8. **All the statements about the right suprarenal gland are true *except*:**
   (a) It lies at a lower level than left suprarenal gland
   (b) Partly it lies behind the inferior vena cava
   (c) Superiorly it is related to the bare area of liver
   (d) Right suprarenal vein drains into the inferior vena cava

Answers

1. *c*, 2. *b*, 3. *d*, 4. *c*, 5. *c*, 6. *b*, 7. *c*, 8. *a*

CHAPTER 12

1. **All of the following are the muscles of the posterior abdominal wall *except*:**
   (a) Psoas major
   (b) Quadratus lumborum

(c) Transversus abdominis
(d) Diaphragm

2. **All the statements about psoas major are correct *except*:**
   (a) It arises from all the lumbar vertebrae
   (b) It contains the lumbar plexus within its substance
   (c) It is pierced by the genitofemoral nerve
   (d) It is the chief extensor of the hip joint

3. **Select the *incorrect statement* about the thoraco-lumbar fascia:**
   (a) It consists of three layers
   (b) It encloses three muscles between its layers
   (c) It provides origin to psoas major muscle
   (d) It provides attachment to latissimus dorsi muscle

4. **Select the *incorrect statement* about the abdominal aorta:**
   (a) It extends from T12–L4 vertebrae
   (b) It lies to the right of the inferior vena cava
   (c) It is crossed anteriorly by the pancreas
   (d) Its dorsal aspect gives origin to the median sacral artery

5. **All the statements about lumbar plexus are correct *except*:**
   (a) It is formed by the ventral rami of upper four lumbar nerves
   (b) It lies within the substance of psoas major muscle
   (c) It communicates with the subcostal nerve
   (d) It divides the psoas major muscle into two planes

6. **All of the following branches of lumbar plexus emerge from the lateral border of the psoas major *except*:**
   (a) Iliohypogastric
   (b) Ilioinguinal
   (c) Obturator
   (d) Femoral

7. **Select the *incorrect statement* about the inferior vena cava:**
   (a) It is the largest and widest vein of the body
   (b) It is formed in front of the L5 vertebra
   (c) It drains both gonadal veins
   (d) It is crossed anteriorly by the root of mesentery

8. **All of the following veins are tributaries of the inferior vena cava *except*:**
   (a) Right suprarenal
   (b) Left suprarenal
   (c) Right renal
   (d) Left renal

Answers

1. *d*, **2.** *d*, **3.** *c*, **4.** *b*, **5.** *d*, **6.** *c*, **7.** *c*, **8.** *b*

CHAPTER 13

1. **All of the following structures form the boundaries of pelvic inlet *except*:**
   (a) Sacral promontory
   (b) Anterior borders of alae of sacrum
   (c) Pecten pubis
   (d) Pubic tubercle

2. **Which of the following conjugate diameters of the true pelvis is most important clinically?**
   (a) External conjugate
   (b) True conjugate
   (c) Diagonal conjugate
   (d) Obstetrical conjugate

3. **Commonest type of female pelvis is:**
   (a) Platypelloid type
   (b) Gynecoid type
   (c) Android type
   (d) Anthropoid type

4. **The percentage of gynecoid pelvic in females is about:**
   (a) 60%
   (b) 42%
   (c) 32%
   (d) 22%

5. **Select the *incorrect statement* about the pelvic outlet:**
   (a) It is bounded anteriorly by the lower margin of pubic symphysis
   (b) It is bounded anterolaterally by the conjoint ischiopubic rami
   (c) It is bounded posterolaterally by the sacrotuberous ligaments
   (d) Its transverse diameter is maximum

6. **Which of the following statements about the true pelvis is incorrect?**
   (a) The maximum diameter at its inlet is transverse diameter
   (b) The maximum diameter at its outlet is anteroposterior diameter
   (c) The maximum diameter at its inlet is anteroposterior diameter
   (d) The minimum diameter at its outlet is transverse diameter

7. **Which of the following joints is of synovial variety?**
   (a) Lumbosacral joint
   (b) Sacroiliac joint
   (c) Pubic symphysis
   (d) Sacrococcygeal joint

8. **Which of the following ligaments is the strongest?**
   (a) Iliolumbar ligament
   (b) Interosseous sacroiliac ligament
   (c) Sacrospinous ligament
   (d) Sacrotuberous ligament

Answers

1. *d*, 2. *c*, 3. *b*, 4. *b*, 5. *d*, 6. *c*, 7. *b*, 8. *b*

CHAPTER 14

1. **The muscles of the true pelvis include all *except*:**
   (a) Piriformis
   (b) Obturator internum
   (c) Ischiocavernosus
   (d) Levator ani

2. **All are the parts of the levator ani *except*:**
   (a) Levator prostate
   (b) Pubococcygeus
   (c) Iliococcygeus
   (d) Coccygeus

3. **All are the constituents of pelvic diaphragm *except*:**
   (a) Levator ani
   (b) Obturator internus
   (c) Coccygeus
   (d) Pelvic fascia

4. **All are the branches of the posterior division of the internal iliac artery *except*:**
   (a) Iliolumbar artery
   (b) Lateral sacral arteries
   (c) Superior gluteal artery
   (d) Obturator artery

5. **The root value of sacral plexus is:**
   (a) L4, L5; S1, S2, S3
   (b) L3, L4, L5; S1, S2
   (c) L5; S1, S2, S3, S4
   (d) L2, L3, L4; S1, S2

6. **All the statements about the sacral plexus are correct *except*:**
   (a) It is formed by the anterior primary rami of L4 to S4 nerves
   (b) It lies deep to pelvic fascia
   (c) It gives off sciatic nerve
   (d) It gives off femoral nerve

7. **All of the following arteries of the pelvis are branches of the internal iliac artery *except*:**
   (a) Umbilical artery
   (b) Uterine artery
   (c) Ovarian artery
   (d) Vaginal artery

8. **All are the tributaries of internal iliac vein *except*:**
   (a) Uterine veins
   (b) Vaginal veins
   (c) Obturator veins
   (d) Umbilical veins

Answers

1. *c*, 2. *d*, 3. *b*, 4. *d*, 5. *a*, 6. *d*, 7. *c*, 8. *d*

CHAPTER 15

1. **All of the following structures form deep boundaries of perineum *except*:**
   (a) Ischiopubic rami
   (b) Ischial tuberosities
   (c) Sacrospinous ligaments
   (d) Sacrotuberous ligaments

2. **All are muscles of superficial perineal pouch *except*:**
   (a) Bulbospongiosus
   (b) Ischiocavernosus
   (c) Sphincter urethrae
   (d) Superficial transverse perineal

3. **All are the contents of deep perineal pouch in male *except*:**
   (a) Membranous urethra
   (b) Cowper's glands
   (c) Root of penis
   (d) Sphincter urethrae muscle

4. **Select the *incorrect statement* about the perineal membrane:**
   (a) It forms the boundary of both superficial and deep perineal pouches
   (b) Its anterior border thickens to form transverse perineal ligament
   (c) Its posterior border provides attachment to the perineal body in the midline
   (d) It is pierced by the pudendal nerve

5. **The gap between the arcuate pubic ligament and transverse perineal ligament in male transmits:**
   (a) Spongy urethra
   (b) Dorsal nerves of penis
   (c) Deep dorsal vein of penis
   (d) (b) and (c)

6. **All of the following are contents of ischiorectal fossa *except*:**
   (a) Posterior scrotal nerve and vessels

(b) Perineal branch of fourth sacral nerve
(c) Sphincter ani externus
(d) Inferior rectal nerve and vessels

7. **All of the following are attached to the perineal body *except*:**
(a) Bulbospongiosus
(b) Externus sphincter of anal canal
(c) External sphincter of urethra
(d) Ischiocavernosus

8. **All of the following are contents of superficial perineal pouch *except*:**
(a) Bulbs of the vestibule of vagina
(b) Bartholin glands
(c) Bulbourethral glands
(d) Superficial transverse perineal muscles

Answers

1. *c*, 2. *c*, 3. *c*, 4. *d*, 5. *d*, 6. *c*, 7. *d*, 8. *c*

CHAPTER 16

1. **Select the *incorrect statement* about the urinary bladder:**
(a) Its neck is its most fixed part
(b) Its superior surface is the most movable part
(c) Its trigone is derived from the absorbed parts of the mesonephric ducts
(d) Its position is abdominopelvic at birth

2. **The bladder structure which serves as a guide to locate the ureteric orifice is:**
(a) Interureteric crest
(b) Uvula vesicae
(c) Trigone
(d) Uretero-urethral ridge

3. **The female bladder is supplied by all arteries *except*:**
(a) Superior vesical
(b) Inferior vesical
(c) Uterine
(d) Vaginal

4. **The length of the male urethra is about:**
(a) 10 cm
(b) 12 cm
(c) 16 cm
(d) 20 cm

5. **All the structures open into the posterior wall of prostatic urethra *except*:**
(a) Prostatic utricle
(b) Lacunae of Morgagni
(c) Ejaculatory ducts
(d) Prostatic glands

6. **All the statements about the prostatic part of urethra are correct *except*:**
(a) It is about 3 cm in length
(b) It is the widest and most dilatable part
(c) Its lumen is lined by the stratified squamous epithelium
(d) It receives the openings of ejaculatory ducts

7. **Select the *incorrect statement* about the female urethra:**
(a) It is about 4 cm long
(b) It is embedded in the posterior wall of the vagina
(c) It pierces the perineal membrane
(d) It opens into the vestibule of the vagina

8. **All the structures open in the female urethra *except*:**
(a) Lacunae of Morgagni
(b) Urethral glands
(c) Bartholin's glands
(d) Glands of Skene

9. **All the statements about the female urethra are correct *except*:**
(a) It is about 4 cm long
(b) It lies immediately anterior to the vagina
(c) Its external orifice is situated about 2 inches (5 cm) behind the clitoris
(d) It pierces the urogenital diaphragm

Answers

1. *d*, 2. *a*, 3. *b*, 4. *d*, 5. *b*, 6. *c*, 7. *b*, 8. *a*, 9. *c*

CHAPTER 17

1. **Select the *incorrect statement* about the prostate gland:**
(a) It is located in the lesser pelvis below the neck of urinary bladder
(b) It is devoid of muscular tissue
(c) Its anterior lobe is devoid of glandular tissue
(d) Its median lobe is the common site of benign hypertrophy

2. **All the following structures lie within the prostate gland *except*:**
(a) Ejaculatory ducts
(b) Prostatic utricle
(c) Prostatic venous plexus
(d) Urethra

3. **Which lobe of prostate is responsible for producing *uvula vesicae*?**
(a) Anterior
(b) Posterior

(c) Lateral
(d) Median

4. **The surgical approaches of prostatectomy include all *except*:**
   (a) Suprapubic
   (b) Retropubic
   (c) Transrectal
   (d) Transurethral

5. **The length of vas deferens is about:**
   (a) 15 cm
   (b) 25 cm
   (c) 35 cm
   (d) 45 cm

6. **A *true statement* regarding the vas deferens is:**
   (a) It is the principal content of the spermatic cord
   (b) It develops from the mesonephric duct
   (c) It opens in the membranous part of the urethra
   (d) It is extraperitoneal throughout the abdominopelvic part of its course

7. **The secretion of seminal vesicle is rich in:**
   (a) Fructose
   (b) Prostaglandin
   (c) Alkaline phosphatase
   (d) (a) and (b)

8. **Select the *incorrect statement* about rectovesical fascia of Denonvilliers:**
   (a) It separates the prostate from the rectum
   (b) It is formed by the fusion of two layers of the peritoneum intervening between the prostate and the rectum in fetus
   (c) It is not adherent to the posterior aspect of the prostate gland
   (d) It prevents the backward spread of the prostatic cancer

Answers

1. *b*, 2. *c*, 3. *d*, 4. *c*, 5. *d*, 6. *c*, 7. *d*, 8. *c*

CHAPTER 18

1. **The forward angulations between the long axes of vagina and cervix is called angle of:**
   (a) Anteflexion
   (b) Anteversion
   (c) Retroflexion
   (d) Retroversion

2. **When the urinary bladder is empty the angle of anteflexion is about:**
   (a) 60°
   (b) 90°
   (c) 120°
   (d) 150°

3. **All are the true ligaments of uterus *except*:**
   (a) Transverse cervical ligaments
   (b) Round ligaments
   (c) Broad ligaments
   (d) Uterosacral ligaments

4. **Most important true ligaments providing support to the uterus are:**
   (a) Transverse cervical ligaments
   (b) Pubocervical ligaments
   (c) Uterosacral ligaments
   (d) Round ligaments

5. **The lymph from the cervix is drained in all of the following lymph nodes *except*:**
   (a) External iliac
   (b) Internal iliac
   (c) Superficial inguinal
   (d) Sacral

6. **Select the *incorrect statement* about the vagina:**
   (a) Its anterior wall is shorter than the posterior wall
   (b) Its posterior fornix is related to the rectouterine pouch
   (c) It is lined by the stratified columnar epithelium
   (d) It is lubricated by the cervical mucous

7. **Select the *incorrect statement* about the uterine tube:**
   (a) It is about 10 cm long
   (b) It conveys ova from the ovary to the uterine cavity
   (c) It lies in the attached border of the broad ligament
   (d) It provides the site for fertilization of the ovum

8. **All the statements about the ovary are true *except*:**
   (a) It is connected to the broad ligament through the mesovarium
   (b) It is covered by the peritoneum
   (c) Left ovarian vein drains into left renal vein
   (d) Lymphatics from the ovary drain into pre-aortic and para-aortic lymph nodes

9. **Select the *incorrect statement* about the clitoris:**
   (a) It is homologous to the penis in male
   (b) It does not consist of corpus spongiosum
   (c) It transmits urethra
   (d) It develops from the genital tubercle

Answers

1. *b*, **2.** *c*, **3.** *c*, **4.** *a*, **5.** *c*, **6.** *c*, **7.** *c*, **8.** *b*, **9.** *c*

CHAPTER 19

1. **Select the *true statement* about the rectum:**
   (a) It is about 15 cm long
   (b) It is straight in its course
   (c) Its posterior surface is covered by peritoneum
   (d) It presents anteroposterior and lateral curvatures
2. **The rectum is characterized by the presence of:**
   (a) Taenia coli
   (b) Haustrations
   (c) Permanent transverse mucosal folds within the lumen
   (d) Appendices epiploicae
3. **The length of the rectum is about:**
   (a) 5 cm
   (b) 7.5 cm
   (c) 12 cm
   (d) 15 cm
4. **The middle one-third of the rectum is covered by the peritoneum:**
   (a) On all sides
   (b) On the front
   (c) On each side
   (d) On the front and sides
5. **The internal hemorrhoids/piles are due to dilatation of the radicles of:**
   (a) Superior rectal vein
   (b) Middle rectal vein
   (c) Inferior rectal vein
   (d) (b) and (c)
6. **The chief artery supplying the rectum is:**
   (a) Superior rectal
   (b) Middle rectal
   (c) Inferior rectal
   (d) Median sacral
7. **Select the *incorrect statement* about the anal canal:**
   (a) It is situated in the perineum
   (b) It is about 4 cm long
   (c) It is related on each side to the ischiorectal fossa
   (d) It is entirely lined by the stratified squamous epithelium
8. **Anorectal ring is formed by all of the following *except*:**
   (a) External sphincter
   (b) Internal sphincter
   (c) Coccygeus
   (d) Puborectalis

Answers

1. *d*, 2. *c*, 3. *c*, 4. *b*, 5. *a*, 6. *b*, 7. *d*, 8. *c*

CHAPTER 20

1. **All are antigravity muscles of the lower limb *except*:**
   (a) Gluteus maximus
   (b) Quadriceps femoris
   (c) Adductor muscles of the thigh
   (d) Calf muscles
2. **Line of gravity in the lower limb passes anterior to all *except*:**
   (a) Sacrum
   (b) Hip joint
   (c) Knee joint
   (d) Ankle joint
3. **Arterial pulses can be felt in the lower limb at all of the following sites *except*:**
   (a) At midinguinal point
   (b) In the popliteal fossa
   (c) On the dorsum of the foot
   (d) In the sole of the foot
4. **Most preferred site for the intramuscular injection in the lower limb is:**
   (a) Gluteal region
   (b) Thigh
   (c) Leg
   (d) Foot

Answers

1. *c*, 2. *b*, 3. *d*, 4. *d*

CHAPTER 21

1. **All the statements about the hip bone are correct *except*:**
   (a) It consists of three parts: ilium, ischium, and pubis
   (b) In early childhood, three parts are separated from each other by the triradiate fibrocartilage
   (c) Primary centre ossification for the ilium appears first
   (d) It presents a cup-shaped hollow on the lateral aspect of its middle constricted part
2. **Ventral segment of the iliac crest provides attachment to all of the following muscles *except*:**
   (a) Tensor fascia latae
   (b) External oblique muscle of the abdomen
   (c) Latissimus dorsi
   (d) Erector spinae
3. **Medial part of pectin pubis provides attachment to all the structures *except*:**
   (a) Lacunar ligament
   (b) Reflected part of the inguinal ligament

(c) Pectineal ligament (of Cooper)
(d) Conjoint tendon

4. **All the statements about the femur are correct *except*:**
   (a) It is the largest and strongest bone of the body
   (b) Its nutrient artery is the largest nutrient artery in the body
   (c) Its angle of anteversion varies from 10° to 15°
   (d) Its neck–shaft angle is less in female

5. **Select the *incorrect statement* about the patella:**
   (a) It is the largest sesamoid bone in the body
   (b) It develops in the tendon of quadriceps femoris
   (c) It has a natural tendency to dislocate medially
   (d) The articular cartilage covering its articular surface is the thickest articular cartilage in the body

6. **Select the *incorrect statement* about the fibula:**
   (a) It is the lateral bone of the leg
   (b) It corresponds to the radius bone of the upper limb
   (c) The law of union of epiphyses is violated in it
   (d) It is commonly used for bone grafting

7. **All the statements about calcaneum are correct *except*:**
   (a) It is the largest tarsal bone
   (b) It starts ossifying after birth
   (c) Anteriorly it articulates with cuboid
   (d) It provides attachment to tendo achillis

8. **Talus provides attachment to:**
   (a) Tibialis posterior
   (b) Tibialis anterior
   (c) Extensor digitorum brevis
   (d) None of the above

9. **Select the *incorrect statement* about the metatarsals:**
   (a) They are miniature long bones
   (b) They ossify from two centres
   (c) Epiphyseal centres of all metatarsals appear at their distal ends
   (d) Second metatarsal is most fixed metatarsal bone

Answers

1. *b*, 2. *d*, 3. *c*, 4. *b*, 5. *c*, 6. *b*, 7. *b*, 8. *d*, 9. *c*

CHAPTER 22

1. **The centre of saphenous opening lies about 3–4 cm below and lateral to:**
   (a) Midinguinal point
   (b) Midpoint of the inguinal ligament
   (c) Pubic tubercle
   (d) Pubic symphysis

2. **Select the *correct statement* about the cribriform fascia:**
   (a) It covers femoral ring
   (b) It covers saphenous opening
   (c) It pierces by the femoral vein
   (d) It forms anterior wall of the femoral sheath

3. **Iliotibial tract provides attachment to:**
   (a) Gluteus maximus
   (b) Gluteus medius
   (c) Tensor fasciae latae
   (d) All the above
   (e) (a) and (c)

4. **Part of quadriceps femoris not attached to femur is:**
   (a) Vastus lateralis
   (b) Vastus medialis
   (c) Vastus intermedius
   (d) Rectus femoris

5. **Select the *incorrect statement* about femoral triangle:**
   (a) It is bounded laterally by the medial border of sartorius
   (b) It is bounded medially by lateral border of adductor longus
   (c) Its base is formed by the inguinal ligament
   (d) Its floor is formed by the muscles

6. **The principal artery supplying the thigh is:**
   (a) Femoral
   (b) Profunda femoris
   (c) Lateral circumflex femoral
   (d) Medial circumflex femoral

7. **All are contents of adductor canal *except*:**
   (a) Femoral artery
   (b) Femoral vein
   (c) Nerve to vastus lateralis
   (d) Saphenous nerve

8. **A stab wound through the apex of femoral triangular can injure all *except*:**
   (a) Femoral artery
   (b) Profunda femoris artery
   (c) Adductor brevis
   (d) Medial cutaneous nerve of the thigh

Answers

1. *c*, 2. *b*, 3. *e*, 4. *d*, 5. *b*, 6. *b*, 7. *c*, 8. *c*

CHAPTER 23

1. **Adductor magnus is supplied by:**
   (a) Femoral nerve
   (b) Obturator nerve

(c) Sciatic nerve
(d) All of the above
(e) (a) and (c)

2. **Muscle of the adductor group of thigh not attached to the femur is:**
   (a) Adductor longus
   (b) Adductor brevis
   (c) Adductor magnus
   (d) Gracilis

3. **Adductor of thigh innervated by two different nerves is:**
   (a) Adductor longus
   (b) Adductor brevis
   (c) Adductor magnus
   (d) Gracilis

4. **Which muscle helps in the extension of the hip?**
   (a) Pectineus
   (b) Adductor longus
   (c) Adductor brevis
   (d) Adductor magnus

5. **Most medial muscle of the adductor compartment of thigh is:**
   (a) Adductor longus
   (b) Adductor brevis
   (c) Gracilis
   (d) Adductor magnus

6. **Select the *incorrect statement* about the adductor brevis:**
   (a) It lies deep to pectineus and adductor longus
   (b) It is usually supplied by the anterior division of obturator nerve
   (c) It intervenes between anterior and posterior divisions of the obturator nerve
   (d) It usually forms the floor of femoral triangle

7. **Select the *incorrect statement* about the obturator nerve:**
   (a) It is derived from L2, L3, L4 spinal nerves
   (b) It contributes in the formation of subsartorial plexus of the nerves
   (c) It enters the thigh deep to the medial part of inguinal ligament
   (d) Its posterior division gives an articular twig to the knee joint

8. **All forms cruciate arterial anastomosis on the back of the thigh *except*:**
   (a) Transverse cervical branch of medial circumference femoral artery
   (b) Ascending branch of first perforating artery
   (c) Inferior branch of deep division of the inferior gluteal artery
   (d) Transverse branch of the lateral circumflex femoral artery

Answers

1. *e*, 2. *d*, 3. *c*, 4. *d*, 5. *c*, 6. *d*, 7. *c*, 8. *c*

CHAPTER 24

1. **The key muscle of the gluteal region is:**
   (a) Gluteus maximus
   (b) Gluteus medius
   (c) Gluteus minimus
   (d) Piriformis

2. **All structures enter the gluteal region through greater sciatic foramen *except*:**
   (a) Sciatic nerve
   (b) Pudendal nerve
   (c) Tendon of obturator internus
   (d) Internal pudendal artery

3. **Safe quadrant of the gluteal region for intramuscular injection is:**
   (a) Upper and inner
   (b) Upper and outer
   (c) Lower and inner
   (d) Lower and outer

4. **The gluteal muscle used for intramuscular injection in the gluteal region is:**
   (a) Gluteus maximus
   (b) Gluteus medius
   (c) Gluteus minimus
   (d) Tensor fasciae latae

5. **All the muscles of the gluteal region are supplied by the superior gluteal nerve *except*:**
   (a) Gluteus maximus
   (b) Gluteus medius
   (c) Gluteus minimus
   (d) Tensor fasciae latae

6. **All the statements about sciatic nerve are correct *except*:**
   (a) It arises from sacral plexus
   (b) It enters the gluteal region through greater sciatic foramen
   (c) It is related to upper and inner quadrant of the gluteal region
   (d) It does not supply any muscle of the gluteal region

7. **When standing on one foot the tilting of the pelvis on opposite side is prevented by the contraction of:**
   (a) Gluteus maximus
   (b) Gluteus medius
   (c) Gluteus minimus
   (d) (b) and (c)

8. **All the structures pass through the lesser sciatic foramen *except*:**
   (a) Pudendal nerve
   (b) Nerve to quadratus femoris
   (c) Nerve to obturator internus
   (d) Internal pudendal vessels

Answers

1. *d*, 2. *c*, 3. *b*, 4. *b*, 5. *a*, 6. *c*, 7. *d*, 8. *b*

CHAPTER 25

1. **All the hamstring muscles:**
   (a) Take origin from the ilium
   (b) Are inserted into one of the bones of the leg
   (c) Are supplied by tibial part of the sciatic nerve
   (d) Are extensor of the hip and flexor of the knee
2. **Muscle supplied by both common peroneal and tibial parts of the sciatic nerve is:**
   (a) Semitendinosus
   (b) Semimembranosus
   (c) Biceps femoris
   (d) Adductor magnus
3. **Which of the following is the most deeply located structure in the popliteal fossa?**
   (a) Popliteal artery
   (b) Popliteal vein
   (c) Tibial nerve
   (d) Common peroneal nerve
4. **Sural nerve is a branch of:**
   (a) Sciatic nerve
   (b) Tibial nerve
   (c) Common peroneal nerve
   (d) None of the above
5. **Which of the following is not a branch of the common peroneal nerve?**
   (a) Sural nerve
   (b) Lateral sural nerve
   (c) Sural communicating nerve
   (d) Lateral inferior genicular nerve
6. **Which of the following is not a content of the popliteal fossa?**
   (a) Tibial nerve
   (b) Common peroneal nerve
   (c) Posterior cutaneous nerve of the thigh
   (d) Saphenous nerve
7. **Select the *incorrect statement* about the oblique popliteal ligament of the knee:**
   (a) It is an expansion of the tendon of semitendinosus
   (b) It is attached to the lateral condyle of femur
   (c) It is pierced by the middle genicular artery
   (d) It is pierced by the genicular branch of the posterior division of obturator nerve
8. **Sciatic nerve usually terminates:**
   (a) At the junction of upper and middle third of the thigh
   (b) At the junction of middle and lower third of the thigh
   (c) In the middle of the thigh
   (d) In the middle of popliteal fossa

Answers

1. *a*, 2. *c*, 3. *a*, 4. *b*, 5. *a*, 6. *d*, 7. *a*, 8. *b*

CHAPTER 26

1. **Y-shaped ligament of the hip joint is:**
   (a) Ischiofemoral
   (b) Iliofemoral
   (c) Pubofemoral
   (d) Ligament of the head of femur
2. **Synovial bursa which communicates with the synovial cavity of the hip joint lies deep to:**
   (a) Gluteus maximus
   (b) Gluteal minimus
   (c) Psoas major
   (d) Gluteus medius
3. **The pubofemoral ligament of the hip joint becomes tense during:**
   (a) Flexion of the thigh
   (b) Extension of the thigh
   (c) Adduction of the thigh
   (d) Abduction of the thigh
4. **Nerve likely to be injured in posterior dislocation of the hip joint is:**
   (a) Superior gluteal
   (b) Inferior gluteal
   (c) Sciatic
   (d) Pudendal
5. **Aseptic avascular necrosis of the head of femur usually occurs due to:**
   (a) Subcapital fracture of the neck of femur
   (b) Cervical fracture of the neck of femur
   (c) Basal fracture of the neck of femur
   (d) Intertrochanteric fracture
6. **Select the *incorrect statement* about the congenital dislocation of the hip joint:**
   (a) It is more common than acquired dislocation

(b) It is usually unilateral
(c) It affects girls more than boys
(d) Its incidence is about 1.5 per 1000 live births

7. **Select the *incorrect statement* about the neck shaft angle of the femur:**
(a) It is about 125° in adults
(b) It is about 95° in children
(c) Its decrease leads to *coxa vara*
(d) Its increase leads to *coxa valga*

8. **The chief source of blood supply to the head and neck of the femur is derived from:**
(a) Acetabular branches of the obturator artery
(b) Acetabular branches of the medial circumflex femoral artery
(c) Retinacular branches of the medial circumflex femoral artery
(d) Nutrient artery of the femur

Answers

1. *b*, 2. *c*, 3. *d*, 4. *c*, 5. *a*, 6. *b*, 7. *b*, 8. *c*

CHAPTER 27

1. **All are muscles of the anterior compartment of the leg *except*:**
(a) Tibialis anterior
(b) Extensor hallucis longus
(c) Peroneus tertius
(d) Peroneus brevis

2. **Most medial and superficial muscle of the anterior compartment of the leg is:**
(a) Tibialis anterior
(b) Extensor hallucis longus
(c) Extensor digitorum longus
(d) Peroneus tertius

3. **Select the *incorrect statement* about the muscles of the anterior compartment of the leg:**
(a) They are dorsiflexor of the foot at the ankle
(b) They are supplied by the superficial peroneal nerve
(c) They lie in a tight osseofascial compartment
(d) Most of them arise from the fibula

4. **All of the following structures pass beneath the superior extensor retinaculum of the foot *except*:**
(a) Anterior tibial artery
(b) Deep peroneal nerve
(c) Dorsal pedis artery
(d) Peroneus brevis

5. **All statements regarding the sensory innervation of the dorsum of the foot are correct *except*:**
(a) Superficial peroneal nerve supplies most of the dorsum of the foot
(b) Cleft between the first and second toes is supplied by the deep peroneal nerve
(c) Lateral margin of the little toe is supplied by the sural nerve
(d) Medial margin of the big toe is supplied by the saphenous nerve

6. **Skin on the medial side of the great toe is supplied by:**
(a) Saphenous nerve
(b) Superficial peroneal nerve
(c) Sural nerve
(d) Deep peroneal nerve

7. **Select the *incorrect statement* about the deep peroneal nerve:**
(a) It is one of the two terminal branches of the sciatic nerve
(b) It is the nerve of the anterior compartment of the leg
(c) It provides sensory innervation to skin in the cleft between the first and second toes
(d) Its lesion results in foot drop

8. **All statements regarding dorsalis pedis artery are correct *except*:**
(a) It is the continuation of anterior tibial artery at the ankle joint
(b) It helps to form plantar arch by joining with the deep branch of the lateral plantar artery
(c) It lies lateral to the tendon of extensor hallucis longus
(d) It lies medial to the medial terminal branch of the superficial peroneal nerve

Answers

1. *d*, 2. *a*, 3. *b*, 4. *d*, 5. *d*, 6. *b*, 7. *a*, 8. *c*

CHAPTER 28

1. **Tendon of which muscle traverses the sole of foot from the lateral to medial side:**
(a) Tibialis anterior
(b) Peroneus longus
(c) Peroneus brevis
(d) Peroneus tertius

2. **Regarding peroneus longus, which *statement is not correct?***
(a) It arises from the fibula
(b) It passes deep to both superior and inferior peroneal retinacula

(c) It grooves the cuboid bone
(d) It is inserted into the base of the 5th metatarsal

3. **Muscles of the lateral compartment are supplied by:**
   (a) Deep peroneal nerve
   (b) Superficial peroneal nerve
   (c) Tibial nerve
   (d) Sural nerve

4. **Dorsiflexion of the foot is produced by:**
   (a) Peroneus longus
   (b) Peroneus brevis
   (c) Peroneus tertius
   (d) None of the above

5. **Bursa anserine (anserine bursa) is related to tendons of all of the following muscles *except*:**
   (a) Sartorius
   (b) Gracilis
   (c) Semitendinosus
   (d) Semimembranosus

Answers

1. *b*, 2. *d*, 3. *b*, 4. *c*, 5. *d*

CHAPTER 29

1. **The muscle "*triceps surae*" is formed by:**
   (a) Popliteus and soleus
   (b) Soleus and gastrocnemius
   (c) Plantaris and gastrocnemius
   (d) Flexor digitorum longus and flexor hallucis longus

2. **Which muscle is regarded as the *'peripheral heart'*?**
   (a) Soleus
   (b) Gastrocnemius
   (c) Tibialis posterior
   (d) Flexor digitorum longus

3 **The most deeply placed muscle in the posterior compartment of the leg is:**
   (a) Flexor digitorum longus
   (b) Flexor hallucis longus
   (c) Tibialis posterior
   (d) Soleus

4. **All the statements about plantaris is correct *except*:**
   (a) It lies between soleus and gastrocnemius
   (b) It is absent in about 15–30% of people
   (c) Its tendon is sometimes dubbed as *"freshman nerve"*
   (d) Its tendon can be removed for grafting without causing functional disability

5. **Select the *true statement* about posterior tibial artery:**
   (a) It is the smaller of the two terminal branches of the popliteal artery
   (b) It runs between gastrocnemius and soleus
   (c) It supplies the muscles of both posterior and lateral compartments of the leg
   (d) Its pulsations can be felt near its origin

6. **The tibialis posterior is inserted into all of the following tarsal bones *except*:**
   (a) Calcaneum
   (b) Talus
   (c) Navicular
   (d) Cuboid

7. **Injury of the tibial nerve can cause loss of:**
   (a) Inversion of the foot
   (b) Eversion of the foot
   (c) Plantar flexion of the foot
   (d) Dorsiflexion of the foot

8. **Largest and most important branch of the posterior tibial artery is:**
   (a) Circumflex fibular artery
   (b) Peroneal artery
   (c) Medial plantar artery
   (d) Lateral plantar artery

Answers

1. *b*, 2. *a*, 3. *c*, 4. *b*, 5. *c*, 6. *b*, 7. *c*, 8. *b*

CHAPTER 30

1. **Select the *incorrect statement* about the plantar aponeurosis:**
   (a) It is thickening of the central part of the deep fascia of the sole of foot
   (b) It provides insertion to the tendon of plantaris
   (c) It helps to maintain the longitudinal arches of the foot
   (d) It sends slip to all the 5 digits

2. **All are muscles of first layer of sole *except*:**
   (a) Abductor hallucis
   (b) Flexor hallucis brevis
   (c) Abductor digiti minimi
   (d) Flexor digitorum brevis

3. **The medial plantar nerve supply all muscles *except*:**
   (a) First lumbrical
   (b) Abductor hallucis
   (c) Flexor digitorum brevis
   (d) Flexor digitorum accessories

4. **Select the *incorrect statement* about the interossei of the sole of foot:**
   (a) They are seven in number
   (b) Dorsal interossei produce abduction of the toes
   (c) Plantar interossei produce adduction of the toes
   (d) They produce extension of the metatarsophalangeal joints and flexion of the interphalangeal joints

5. **All are the muscles of the third layer of sole *except*:**
   (a) Flexor hallucis brevis
   (b) Adductor hallucis
   (c) Abductor hallucis
   (d) Flexor digiti minimi brevis

6. **Fourth layer of the sole contains:**
   (a) Tendon of peroneus longus
   (b) Tendon of flexor hallucis longus
   (c) Tendon of tibialis posterior
   (d) (a) and (c)

7. **Which lumbrical muscle in the sole of foot is not supplied by the lateral plantar nerve?**
   (a) First lumbrical
   (b) Second lumbrical
   (c) Third lumbrical
   (d) Fourth lumbrical

8. **Which muscle of the sole contains two sesamoid bones?**
   (a) Abductor hallucis
   (b) Flexor digiti minimi brevis
   (c) Flexor hallucis brevis
   (d) Adductor hallucis

Answers

1. *b*, 2. *b*, 3. *d*, 4. *d*, 5. *c*, 6. *d*, 7. *a*, 8. *c*

CHAPTER 31

1. **Bone which is considered as the *keystone* of medial longitudinal arch of foot is:**
   (a) Calcaneum
   (b) Cuboid
   (c) Talus
   (d) Navicular

2. **Which of the following structures acts as a tie beam of the longitudinal arches of the foot?**
   (a) Tendon of tibialis posterior
   (b) Plantar aponeurosis
   (c) Plantar calcaneonavicular ligament
   (d) Interosseous talocalcaneal ligament

3. **The most important factor for maintaining the arches of the foot is:**
   (a) Shapes of the bones
   (b) Ligaments and plantar aponeurosis
   (c) Tendons of muscles
   (d) Intertarsal joints

4. **The main joints of the medial longitudinal arch of foot is:**
   (a) Midtarsal
   (b) Talocalcaneonavicular
   (c) Calcaneocuboid
   (d) Cuneonavicular

5. **The main joint of the lateral longitudinal arch of foot is:**
   (a) Subtalar
   (b) Cuneocuboid
   (c) Calcaneocuboid
   (d) Talocalcaneonavicular

6. **In talipes equinus, the patient walks on:**
   (a) Toes
   (b) Heel
   (c) Lateral margin of the foot
   (d) Medial margin of the foot

7. **All are features of the talipes equinovarus *except*:**
   (a) Inversion
   (b) Adduction
   (c) Plantar flexion
   (d) Lateral rotation of the foot

8. **The muscle that does not maintain the transverse arch of the foot is:**
   (a) Tibialis posterior
   (b) Peroneus longus
   (c) Adductor hallucis
   (d) Extensor digitorum brevis

Answers

1. *c*, 2. *b*, 3. *b*, 4. *b*, 5. *c*, 6. *a*, 7. *d*, 8. *d*

CHAPTER 32

1. **All the statements about capsule of the knee join are correct *except*:**
   (a) Anteriorly it is replaced by the quadriceps femoris, patella, and ligamentum patellae
   (b) Posteriorly it is strengthened by oblique popliteal ligament
   (c) Posterolaterally it provides passage to the tendon of popliteus
   (d) Inferiorly it encloses the soleus muscle

2. **Most important factor to stabilize the knee joint is:**
   (a) Muscle
   (b) Ligaments
   (c) Articular surfaces
   (d) Menisci

3. **Anteroposterior stability of the knee joint is maintained by:**
   (a) Oblique popliteal ligament
   (b) Arcuate popliteal ligament
   (c) Cruciate ligaments
   (d) Medial and lateral collateral ligaments

4. **Medial meniscus is more vulnerable to the injury because:**
   (a) It is attached to the tibial collateral ligament and allows greater excursion during rotatory movements
   (b) Posterior fibres of its anterior end are continuous with the transverse ligament
   (c) It is larger than the medial lemniscus
   (d) Its anterior posterior ends are attached to the intercondylar area of the tibia

5. **Locking of the knee joint is caused by the contraction of:**
   (a) Popliteus
   (b) Biceps femoris
   (c) Semimembranosus
   (d) Quadriceps femoris

6. **Select the *incorrect statement* about the locking and unlocking of the knee joint:**
   (a) Locking occurs at the end of extension
   (b) There is medial rotation of the tibia on femur in locking
   (c) Unlocking occurs during initial phase of flexion
   (d) Locking is essential for standing erect and locomotion

7. **Ankle joint is most stable in:**
   (a) Plantar flexion
   (b) Dorsiflexion
   (c) Inversion
   (d) Eversion

8. **What type of ankle joint is:**
   (a) Ellipsoid
   (b) Saddle
   (c) Modified hinge
   (d) Pivot

Answers

1. *d*, 2. *a*, 3. *c*, 4. *a*, 5. *d*, 6. *b*, 7. *b*, 8. *c*

CHAPTER 33

1. **The approximate number of superficial inguinal lymph nodes is:**
   (a) 2 to 3
   (b) 4 to 5
   (c) 7 to 9
   (d) 15 to 20

2. **The commonest cause of swelling in the subinguinal region is:**
   (a) Enlargement of inguinal lymph nodes
   (b) Femoral hernia
   (c) Psoas abscess
   (d) Ectopic testis

3. **Select the *incorrect statement* about the deep lymphatics of the lower limb:**
   (a) They are smaller and fewer in number than the superficial lymphatics
   (b) They drain lymph from all structures deep to deep fascia
   (c) They run along the principal vessels of the lower limb
   (d) They drain mostly into external iliac lymph nodes

4. **All the statements about great saphenous vein are correct *except*:**
   (a) It is the longest vein in the body
   (b) It is thin-walled and contains only few valves
   (c) It is connected to the deep veins of the leg and thigh by perforating veins (perforators)
   (d) It terminates in the femoral vein after piercing the anterior wall of the femoral sheath

5. **All the statements about small saphenous vein are correct *except*:**
   (a) It begins as the continuation of the lateral end of the dorsal venous arch
   (b) It possess numerous valves
   (c) It is accompanied by the sural nerve in the leg
   (d) It drains into the great saphenous vein

Answers

1. *a*, 2. *b*, 3. *d*, 4. *b*, 5. *d*

CHAPTER 34

1. **Select the *incorrect statement* about the sciatic nerve:**
   (a) It arises from the anterior primary rami of L4–S3 spinal nerves
   (b) It is the thickest nerve in the body

(c) It lies superficial to biceps femoris
(d) It undergoes a subfascial course in the angle between the gluteus maximus and long head of the biceps femoris

2. Complete injury of the sciatic nerve in the gluteal region will not produce:

(a) Weakness in the flexion of thigh
(b) Foot drop
(c) Complete loss of the skin sensation in the foot
(d) Paralysis of all the muscles of the leg and foot

3. All the statements about the femoral nerve are correct *expect*:

(a) It is the largest branch of the lumbar plexus
(b) It arises from the ventral branches of the anterior primary rami of L2, L3, L4 spinal nerves
(c) It enters the femoral triangle deep to the inguinal ligament
(d) It lies in the groove between psoas major and iliacus

4. The saphenous nerve is:

(a) A branch of the femoral nerve
(b) The longest cutaneous nerve in the body
(c) Closely related to great saphenous vein at the ankle
(d) Thickest cutaneous nerve in the body

5. All the statements about the obturator nerve are correct *except*:

(a) It arises from the ventral branches of the anterior primary rami, of L2, L3, L4 spinal nerves
(b) It enters the thigh deep to the inguinal ligament
(c) It supplies all the muscles of the adductor compartment
(d) It takes part in the formation of subsartorial plexus of the nerves

Answers

1. *c*, 2. *a*, 3. *b*, 4. *d*, 5. *c*

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